

Cross-linguistic Variation in the Expression of Heterogeneous Plural Marking and its Theoretical Consequences

Beccy Lewis, Ph.D.

University of Connecticut, 2025

ABSTRACT

This dissertation investigates the morphological expression of associative, similitive and approximative plurals (henceforth *heterogeneous* plurals) cross-linguistically, establishing a number of novel generalizations regarding the use of the regular plural marker as (or as part of) a heterogeneous plural. A typological examination of these plurals reveals that using regular plural morphology for a heterogeneous plural is not arbitrary – as has been previously assumed – but rather tightly constrained by independent aspects of the nominal domain in the relevant cases.

Chapter 2 investigates those languages that use the regular plural as (or as part of) the associative plural (associative plurals attach to a noun, X, to generate an expression meaning ‘X and their family/friends/associates’). I show that there are a number of ways in which the regular plural can be recruited for associative plural marking cross-linguistically, and establish the novel typological generalization that the relevant languages also lack free-standing definite articles (they either lack definite articles or have affixal definite articles).

Beccy Lewis,
University of Connecticut, 2025

Chapter 3 offers a new analysis of associative plurals in the relevant languages that deduces the typological generalization. Furthermore, I show that a number of empirical puzzles regarding the interaction between associative plural morphology and other aspects of the nominal system receive straightforward explanations on the present analysis.

Chapter 4 discusses a prevalent view that first and second person (1/2P) plural pronouns are also associative plurals (that is, the meaning of *we* is really ‘the speaker and others’ rather than ‘multiple speakers’). I present novel evidence from both morphosyntax and semantics that 1/2P plural pronouns are not associative plurals. Moreover, I show that some languages in fact possess two kinds of 1/2P plural pronoun: one regular, which has a ‘speaker/hearer and others’ interpretation and a ‘multiple speakers/hearers’ interpretation, and one associative, which uses dedicated associative plural morphology and means ‘speaker/hearer and their associates’.

Chapter 5 discusses two related heterogeneous plurals, simulative plurals (nominal expressions meaning ‘X and things like X’) and approximative plurals (nominal expressions meaning ‘approximately X’). I reveal that there is a typological generalization regarding these plurals too: if the regular plural is used as the simulative or approximative plural, the relevant language lacks free-standing definite articles. I also discuss a novel implicational hierarchy regarding the expression of the associative plural and the simulative plural in languages that have both, and argue that this hierarchy arises as a result of the organization of morphological features associated with regular and heterogeneous plurality.

Cross-linguistic Variation in the Expression of Heterogeneous Plural Marking and its Theoretical Consequences

Beccy Lewis

M.A., University of Connecticut, 2024

M.Phil., University of Cambridge, 2018

B.A., The University of York, 2017

A Dissertation

Submitted in Partial Fulfillment of the

Requirements for the Degree of

Doctor of Philosophy

at the

University of Connecticut

2025

Copyright by

Beccy Lewis

2025

APPROVAL PAGE

Doctor of Philosophy Dissertation

Cross-linguistic Variation in the Expression of Heterogeneous Plural Marking and its Theoretical Consequences

Presented by

Beccy Lewis, B.A., M.Phil., M.A.

Major Advisor

Željko Bošković

Associate Advisor

Adrian Stegovec

Associate Advisor

Vicki Carstens

University of Connecticut

2025

ACKNOWLEDGMENTS

In March of 2019 I sat opposite Željko as a prospective UConn student silently adamant that I would work on some quirk of English, while he gave a spiel about this thing called *generative typology*. Exactly one year later, a global pandemic forced us to replace one isolated bubble (academia) for another, even more isolated bubble (our one-bedroom apartment) while we watched an apocalypse movie play out on the news. What I'm saying is: things never go as planned.

If you've never done one, you might think of a PhD as a very isolating endeavor. But, even in the throes of lockdown and thousands of miles from home, I was far from alone at UConn. I feel enormously privileged to have had Željko Bošković as my major advisor. His enthusiasm for linguistics knows no bounds. Thank you for approaching every half-baked idea or badly written paper of mine with genuine interest and expertise. Thank you for pushing me beyond where I'd probably like to have been pushed. Thank you for reading everything I ever sent you and for every illegible comment. Thank you for always having your students' backs – your support has been immeasurable. Thank you for patiently pointing out every time I misspelled heterogeneous and ambiguous (all *heterogenouses* and *ambigiousses* are my own). I am also very grateful to have worked with Adrian Stegovec and Vicki Carstens as part of my committee. Adrian's knowledge is genuinely inspiring and I fear it may take me the rest of my career to become as well read as he is. He would link my ideas to things that seemed (to me) totally irrelevant, but he was always onto something. For my first year of knowing Vicki, we only met over Zoom for her Field Methods class. I'll be honest here – I was a bit scared of her. But that feeling evaporated when we finally met in person; Vicki has been nothing but encouraging and enthusiastic about my research. I would also like to thank Andrea Calabrese. Andrea was on the committee for this project when it was my first generals paper, and my overwhelming memory of that time was that he thought all of this was wrong. Andrea's speciality, though, is thinking that you're wrong but still telling you that your work is great. He taught me that those two things aren't mutually exclusive. I am also grateful to many other UConn(-affiliated) faculty for their mentorship, teaching, support and feedback on various projects over the years: Diane Lillo-Martin, Jon Gajewski, Harry van der Hulst, Magda Kaufmann, William Snyder, Ian Roberts, and Mamoru Saito. I wouldn't be on this side of grad school were it not for Tamara Cohen, our inimitable Program Coordinator. Thank you for everything that you do.

Getting to UConn would have been impossible without numerous other people. My first taste of linguistics was at The University of York where I was spoiled by some of

the best teachers our field has to offer: Ann Taylor, Shin-Sook Kim, Norman Yeo, Eytan Zweig, Tamar Keren-Portnoy, Nina Radkevich, Kobin Kendrick, Caitl Light, Nino Grillo. Nino especially was instrumental in showing me what it was like to do research and lit a fire that is still going to this day. At The University of Cambridge I learnt a great deal from David Willis, who supervised my MPhil dissertation. I still think of his advice often: the best thing you can be is right, the second best thing you can be is wrong, and the worst thing you can be is unclear.

This dissertation, and much of my other work, would not exist without all of the people who provided me with linguistic judgments: Giulio Ciferri Muramatsu, Yoshiki Fujiwara, Eva Gan, Xueting Yuan, Paula Fenger, Sharmin Ahmadi, Yusuke Yagi, Ivana Jovović, Adrian Stegovec, Rakey Cole, Ognyan Darinov, Burak Öney, Ivan Derzhanski, Andrija Petrović, András Bárány, Gary Thoms, Neil Myler, Becky Woods, Hande Svegi, and Ousmane Cisse. One inevitability of doing cross-linguistic work is that you can't be an expert on every language you discuss. I've learnt, however, that our field is full of kind and generous people who will share their knowledge with you. I am grateful to Camille Simon and Bernard Comrie for discussions regarding portions of this work.

I feel very lucky to have done my first year at UConn with Penelope Daniel, Robin Jenkins, Linghui (Eva) Gan, Maggie Chui Yi Lee and Ari Goertzel. Penelope and Robin are everything you could want in cohort-mates: smart as hell, funny, grounded, kind, generous. Some of my very best memories of Connecticut revolve around you. For all of the coffees, dinners, lunches, drinks, reading groups and office chats I'd like to thank Tarcisio Dias, Xueting Yuan, Giulio Ciferri Muramatsu, Si Kai Lee, Sarah Asinari, Christos Christopoulos, Ivana Jovović, Adrian Stegovec, Mingjiang Chen, Sharmin Ahmadi, Qiushi Chen, Eva Gan, Nic Schrum, Zixi Liu, Walter Shaw, Paula Fenger, Shangyan Pan, Pravaal Yadav, Yusuke Yagi and Marley Beaver. Paula Fenger was in her last year at UConn when I was in my first. We would see each other in Morphology Reading Group but – due to the large gaps between our offices and our respective progress in the program – I don't remember that we spoke very much. Then the world shut down, Paula retreated to a large house in the woods to write her dissertation, and – bizarrely – we texted non-stop. Paula moved to Leipzig, and still we talked. Despite the lack of time spent in the same room (or on the same continent), Paula is now a friend I love dearly. From afar she's shown me how to do a PhD, how to relax, how to be a good friend, how to not be too serious and how to get my man to do what I want. Our brains are stupid, but that's OK.

I feel very lucky to have had family both near and far during my time at UConn. I thank my parents, Alison and Karl, and sister, Hannah, for all of their support, and for their visits to the US when the world finally allowed it. I thank my husband's family – especially Becky, Kris, Reilly and Ashton. Their home has always been open to us for holidays and celebrations, or no-reason-at-all visits. Finally, I really can't express how much Brian, my husband, means to me. He is my best friend, my lunch-maker, my road-tripper, my dessert-bringer, my everything, my peach. I love you.

Contents

1	Introduction	1
1.1	Main findings of the dissertation	9
1.2	Structure of the dissertation	11
2	A typological survey of associative plurals	13
2.1	Introduction	13
2.2	A typological survey of the <i>plural pattern</i>	19
2.3	The language sample	21
2.3.1	The baseline case	23
2.3.2	Regular plural + extra morpheme	26
2.3.3	Regular plural + alternative stem	30
2.3.4	Summary	32
2.4	The typology of <i>plural pattern</i> languages	32
2.4.1	Is the plural pattern predictable?	33
2.5	A novel typological generalization	37
2.5.1	Defining definite articles	38
2.5.2	99 articleless languages	42
2.5.3	Definite article languages	57
2.6	Summary	78
	Appendix 2.A A note on excluded languages	79
3	A new analysis of the <i>plural pattern</i>	90
3.1	The <i>plural pattern</i> as head-movement of Num	93
3.1.1	Motivating head movement	95
3.1.2	A note on the <i>non-plural pattern</i>	114
3.1.3	Summary	116
3.2	Deducing the typological generalization	116
3.2.1	The DP/NP-language distinction	117
3.2.2	The (absence of the) <i>plural pattern</i> in DP languages	126

3.2.3	The <i>plural pattern</i> in articleless languages	129
3.2.4	The <i>plural pattern</i> in affixal article languages	131
3.2.5	Summary	139
3.3	Additional evidence in favor of a movement-based approach	139
3.3.1	Associative plurals and associative duals: one or both?	140
3.3.2	Associative plurals with numerals	144
3.3.3	A restriction against plural possessor agreement	152
3.3.4	Summary	162
4	Associative plurals and 1/2P plural pronouns	164
4.1	Overview	164
4.2	The unified approach: 1/2P plural pronouns are associative plurals	166
4.3	Against a unified analysis: 1/2P plural pronouns are not associative plurals .	172
4.3.1	Number marking on 1/2P pronouns	172
4.3.2	Semantic restrictions on associative plurals	180
4.3.3	Summary and looking ahead	185
4.4	Associative plural pronouns	187
4.4.1	Summary	192
4.5	The associative effect	192
4.5.1	Deriving the associative effect in 1/2P plural pronouns	194
4.6	Summing up	202
5	A broader look at heterogeneous plurals: similitive and approximative plural- ity	204
5.1	Similitive and approximative plurals	205
5.1.1	Heterogeneous plurals and definite articles	209
5.1.2	Implicational relationships	215
5.1.3	Accounting for the implicational hierarchy	228
5.1.4	Accounting for the absence of (free-standing) definite articles in heterogeneous plurals	238
5.1.5	Summary	242
	Bibliography	244

Chapter 1

Introduction

This dissertation is a cross-linguistic investigation into the morphological expression of *heterogeneous* number, which I show is underlyingly syntactic in nature and typologically constrained by independent properties of a languages’ nominal system. “Heterogeneous plural” is an umbrella term for three constructions—*associative plural*, *similative plural* and *approximative plural*—that differ from regular plurals in a crucial way. The associative plural, illustrated with Malayalam in (1), combines with a proper name, *Raaman*, to pick out Ram and people who are associated with Ram (usually his family/friends). The associative plural marker is the word *okke*, meaning ‘all’ (Asher and Kumari [1997](#)).

- | | | |
|-----|----------------------|--|
| (1) | Raaman okke | Malayalam |
| | Ram APL | |
| | ‘Ram and associates’ | (Daniel and Moravcsik 2013) |

The similative plural, shown in (2) for Kuuk Thaayorre, is attached to the noun *pormpr*, ‘house’ to pick out houses and similar things ((2) comes from a conversation about what will be destroyed by a cyclone, so e.g., stores, cars, infrastructure, *etc*). The similative plural marker is =*yuk*, a clitic meaning ‘thing/stuff’ (Gaby [2018](#)).

- (2) pormpr =**yuk**
 houses =SPL
 ‘houses and things’

Kuuk Thaayorre
 (Gaby 2018, p. 151)

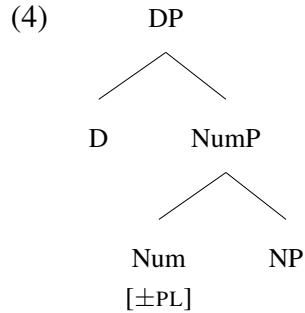
Lastly, the approximative plural, shown in (3) for Cavineña, attaches to nouns of quantity, distance, location or time to give an approximation of amount or duration. The approximative plural marker is the dedicated approximative clitic =*dyane*.

- (3) cuatro kilometro =**dyane**
 four kilometers =XPL
 ‘about four kilometers’

Cavineña
 (Guillaume 2008, p. 535)

Regular plurals, on the other hand, take a nominal stem ‘X’ and produce a plural set whose members are all tokens of ‘X’. That is, the regular plural on a noun *chair* produces a plural set (*chairs*) whose members all have the property of being a chair. This makes the regular plural an *additive* operation that produces *homogeneous* plural sets. The heterogeneous plurals, in contrast, take a nominal stem ‘X’ and produce a plural set whose members are *not* all tokens of ‘X’. In an associative plural, members of the plural set are associated in some way with the nominal stem. In a simulative plural, members of the plural set are similar to the nominal stem in some way. The approximative plural approximates over the quantity specified by the nominal stem to include other, nearby, values.

The syntax and semantics of regular number marking is by now quite well-understood. Since the influential work of Ritter (1992) it has been assumed that there is a universal functional projection in the nominal domain, NumP, the head of which realizes nominal number morphology.



Moreover, there is a clear semantics of plural marking that simply “adds up” tokens of the nominal stem.¹ Chierchia (1998a) argues that the domain of discourse has the shape in (5), which consists of singularities (or atoms) and the plural sets made from those singularities. A singular NP corresponds to one of these atoms. A plural NP has the semantics in (6). A is the set of atoms (see (6b)). ‘*’ is an operator that picks out the set of all sums of elements for any subset X of the domain in (5) (see (6c)). When * is applied to the set of atoms A, the output is a set that consists of all sums of elements of A (see (6d)). A plural feature on Num⁰ thus takes the set in (6d) and removes from it all of the atoms. The result is a set that only consists of pluralities (6e), and this is the additive interpretation of plural.

(5) {a, b, c, d, ...}

{a, b, c} {a, b, d} {b, c, d} {a, c, d}

{a, b} {a, c} {a, d} {b, c} {b, d} {c, d}

a b c d

¹I am specifically referring here to cases where a simple count noun ‘X’ takes plural marking to produce a plurality ‘X+X+X...’. I am not concerned here with other semantically plural constructions like *and*-coordination or collective nouns like *team* that are grammatically singular and which have received various semantic analyses of their own.

(6) *Chierchia's (1998) plural semantics*

- a. $PL(A) = *A - A$
- b. $A = \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$
- c. $*X$ is the closure of X under U (U = the sum of all the elements in X , i.e., the set of all sums of elements of X)
- d. $*A = \{\text{dog}_1\}, \{\text{dog}_2\}, \{\text{dog}_3\}, \{\text{dog}_1, \text{dog}_2\}, \{\text{dog}_1, \text{dog}_3\}, \{\text{dog}_2, \text{dog}_3\}, \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$
- e. $\text{dog}_{PL} = \{\text{dog}_1, \text{dog}_2\}, \{\text{dog}_1, \text{dog}_3\}, \{\text{dog}_2, \text{dog}_3\}, \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$

Thus, any time a noun is marked for plural we might assume that (i) plural morphology is in Num^0 and (ii) it has the semantics in (6).²

A number of independent lines of research, however, show that plural number morphology does not always carry the prototypical additive semantics we'd expect it to. For example, in languages with honorific marking it is plural that is used (among other things) for honorification. French is a well-known example of this: the second person plural pronoun *vous* can be used for polite address toward a *singular* addressee. The plural-marked pronoun in (7) thus lacks plural semantics.

- (7) Avez **vous** le livre?
 have.PRES.2PL 2PL the book?
 'Do you (SG/HON) have the book?'

It is also possible for a plural-marked noun to introduce new meaning on top of additive plurality. For example, in languages like Chinese and Japanese, a bare noun (i.e., a noun with no plural marking) can be interpreted as definite, indefinite, singular or plural (8a). As

²There are languages that lack plural morphology and have, instead, a classifier system. These receive a different analysis in Chierchia (1998b).

expected, attaching the plural marker to the noun makes it unambiguously plural but it also strongly favors a definite interpretation (8b).

- (8) a. John-ga hon-o yon-da
 John-NOM book-ACC read-PST
 ‘John read a book/the book/some books/the books’
- b. John-wa gakusei-**tachi/-ra**-ni at-ta
 John-TOP student-PL/-PL-DAT meet-PST
 ‘John met **the** students.’ (Kurafuji 2004)

The meaning of plural marking may also differ depending on its morphological expression. For example, Kramer (2009, 2016) shows that Amharic nouns can be pluralized using the regular plural suffix *-(w)otf* (9) or via a number of irregular pluralization strategies, including a different suffix (9a), different vowelization (9b) or partial reduplication, among other things.

- (9) bet -**otf**
 house -PL
 ‘houses’
- (10) a. amaät -**at**
 year -PL
 ‘years’
- b. kănfär → kănafir
 lip → lips

Interestingly, while both regular and irregular plural marking may produce an ordinary (i.e., additive) plural noun, Kramer (2016) observes that irregular plural marking may have additional, idiosyncratic interpretations. For example, the noun *näfs* ‘soul’ has an additive plural interpretation when it is suffixed with the regular and irregular plural, but an additional idiosyncratic meaning (‘small insects’) only when it is irregularly pluralized (cf. (11a–b)). The two plurals conform to the traditional inflectional vs derivational morphology distinction—the irregular plural has selectional restrictions on the stem and can convert

certain adjectives to nouns, both typical attributes of derivational morphology. This raises an interesting question regarding whether (and how) the distinction should be reflected in the structure. As we will see below, Kramer argues that it should.

- (11) a. *näfs -otʃtʃ*
 soul -PL
 ‘souls’
- b. *näfs -at*
 soul -PL
 ‘souls, small insects’

Plural semantics may also change depending on the type of noun it is attached to. For instance, Alexiadou (2011) shows that mass nouns in Greek may take plural marking without losing their mass noun semantics (12). This is in contrast to languages like English, where plural marking on mass nouns coerces the noun into having a count reading. Thus, in (13b), *waters* can only be understood as a count noun with an implicit classifier (e.g., cups). Nonetheless, the meaning of plural marking on mass nouns is different: Alexiadou argues that the meaning of plural features on mass nouns is ‘a great amount of [mass noun]’ (see also Kouneli 2019).

- (12) *nero / nera*
 water / water.PL
 ‘water’ / ‘a lot of water’
- (Alexiadou 2011, p. 35)

- (13) a. Water is on the table.
- b. Waters are on the table.

Relevant here is that regular plural morphology can also be used to generate the *heterogeneous* plurals discussed above. In Japanese, the regular plural suffix *-tachi* (14a) is also used as the associative plural (14b). Thus, (14b) means Mika and those associated with her (family/friends).

- (14) a. gakusei **-tachi**
 student -PL
 ‘the students’
- b. Mika **-tachi**
 Mika -PL
 ‘Mika and her friends/family’ (Nakanishi and Ritter 2008, p. 1)

In Eastern Dan, the regular plural suffix *-dhün* (15a) is also used as the simulative plural (15b). Thus, (15b) picks out a butterfly and things like that (e.g., other (non-butterfly) insects).

- (15) a. bɛ̃zɛ̃ **-dhün**
 nephew -PL
 ‘nephews’
- b. dhɛ̃kpāadhɛ̃ **-dhün**
 butterfly -PL
 ‘a butterfly and other beings’ (Vydrin 2022, p. 40)

In Buwal, the regular plural suffix *=égē* (16a) is also used as the approximative plural (16b). Thus, in (16b), the plural marker indicates that the amount ‘twenty two’ is imprecise or approximate.

- (16) a. mēndzēvēk **=égē**
 remedy =PL
 ‘remedies’
- b. rāgbák á xā gbák **=égē**
 twenty PREP over two =PL
 ‘around twenty two’ (Viljoen 2013, pp. 273, 275)

What is responsible for these non-prototypical uses of nominal plural morphology? Some authors argue that it is that Num⁰ is not the only head capable of carrying plural number features in the nominal domain. Taking Chierchia’s (1998) semantics of number to apply to number features on the Num head, the idea is that number features on *other* heads

will have slightly different semantics. In particular, Alexiadou (2011) argues that plural morphology on mass nouns in Greek is the spell-out of plural features on the nominalizing head *n* and, similarly, Kramer (2016) argues that plural features on *n* spell-out irregular plural morphology in Amharic.

Others have argued that it is the unmarked semantics of plural number that allows it to be “re-used” with new meanings. Plural is said to be *semantically unmarked* because it can entail both singular and plural meanings (e.g. Sauerland et al. 2005). That is, if I ask how many *children* you have, you can felicitously answer: *one*. Wang (2023), for example, argues that plural marking is co-opted for honorification because plural is semantically unmarked, permitting both singular and plural meanings. The gist of the proposal is that being vague is more polite than being specific (Brown and Levinson 1987), and since plural permits a singular interpretation without specifying the referent directly, it is recruited for honorification.

This dissertation is a typological investigation into those languages that recruit regular plural morphology for one or more of the heterogeneous plurals (associative, similitive, approximative) and aims to uncover what is responsible for the co-opting of regular plural for this function. That is, when regular plural is recruited for heterogeneous plurality does this involve a [+plural] feature merged on a different head? And if so, which head is the locus of this [+plural] feature? Alternatively, heterogeneous plurality could involve [+plural] on Num, but the semantics of plural (see (6)) can be coerced into having a non-additive (i.e., heterogeneous) reading. I will argue that when regular plural morphology has a heterogeneous plural meaning, we are in fact dealing with a [+plural] feature on a different head. However, whereas others have argued that the [+plural] feature on a non-Num head is base-generated there, I propose that the [+plural] feature in heterogeneous plurality gets on the new head via movement of Num. I show that syntactic constraints

on movement (e.g. locality, intervention) in fact constrain this seemingly morphological phenomenon, indicating that the plural morphology in the heterogeneous plurals is derived via movement.

1.1 Main findings of the dissertation

This dissertation examines the morphological expression of heterogeneous plurality in a broad range of languages (specifically, 109 languages from 33 language families, 4 creoles/pidgins and 2 isolates) and establishes a number of novel typological generalizations regarding languages that use regular plural morphology to express a heterogeneous plural.

I will first establish a novel cross-linguistic generalization regarding languages that use the regular plural as the *associative* plural. Which languages are able to recruit the regular plural for this function was previously thought to be arbitrary. However, I show that it is not by establishing a generalization that uncovers a principled split regarding the types of languages that allow the regular plural to be re-used as an associative plural. In particular, I show that there is a property that characterizes these languages (e.g., Japanese (see above) and Turkish, see (17)) which concerns an independent phenomenon, namely definite articles. Specifically, languages that use the regular plural as (or as part of) the associative plural either lack definite articles or have affixal definite articles. In other words, something like (17) never shows up in languages with free-standing definite articles, like English.

- (17) Ahmet-**ler** çık-tı
Ahmet-PL leave-PST
‘Ahmet and his friends/family left’ (adapted from Görgülü 2011, p. 74)

Furthermore, I offer a deduction of this generalization that ties the expression of asso-

ciative plurality to the amount of functional structure in the nominal domain. I also show that head-movement is crucially involved in the derivation of examples like (17). This is shown to have consequences for a number of theoretical mechanisms including Agree, the nature of head-movement (whether it takes place in syntax or PF), intervention effects, and the structure of the nominal domain.

I will also show that the head movement that I argue takes place in (17) enables us to resolve a long-standing puzzle regarding the co-occurrence of the associative plural and plural possessor agreement in Turkish (18). That is, it is not possible to have both plural possessor agreement (*-mız/-nız*) and the associative plural *-lar* suffixed to a noun in Turkish. If the possessor agreement is singular, both possessor agreement and the associative plural can be suffixed to the noun (19).

- (18) *(bizim) abla-**mız**-ler / *(sizin) abla-**nız**-ler
 (1PL-GEN) elder.sister-1PL.POSS-PL / (2PL-GEN) elder.sister-2PL.POSS-PL
 ‘{**our** / **your.PL**} elder sister and her friends’

- (19) (benim) abla-**m**-ler / (senin) abla-**n**-ler
 (1SG-GEN) elder.sister-1SG.POSS-PL / (2SG-GEN) elder.sister-2SG.POSS-PL
 ‘{**my** / **your**} elder sister and her friends’

I account for this peculiar restriction as a locality constraint on head-movement, which cannot proceed across a [+plural] feature on the Poss head. In other words, there is empirical evidence that using the regular plural as the associative plural is syntactic in nature, as it shows known syntactic locality constraints.

Additionally, I will present a novel implicational hierarchy regarding the expression of heterogeneous number, in particular the expression of associative and similative plurality in a single language. I show that if a language uses the regular plural as the similative plural, it also uses the regular plural as the associative plural. In other words, there is no language that uses dedicated morphology for the associative plural and regular plural morphology

for the similitive plural. I further show that the implicational hierarchy falls out from the organization of morphological features associated with regular and heterogeneous plurality.

1.2 Structure of the dissertation

The dissertation is structured as follows. Chapter 2 presents the results of a broad typological survey of languages that use the regular plural as the associative plural. This survey consists of 109 languages from 33 language families, 4 creoles/pidgins and 2 isolates. It is the largest sample of languages that use the regular plural as the associative plural collected to date. I then show that the morphological expression of associative plurality in these languages is in fact predictable, and establish a novel typological generalization that languages that use the regular plural as the associative plural lack free-standing definite articles (they either lack definite articles or have affixal definite articles).

In Chapter 3, I offer a deduction of the new generalization using Bošković's (2008, 2012) DP/NP-language distinction and the extension of that two-way split in Talić (2017), which treats languages with affixal definite articles differently from languages with free-standing definite articles. I claim that using the regular plural as the associative plural involves head movement, and that the presence of additional functional structure in the nominal domain blocks this movement from occurring. I also discuss a number of empirical puzzles regarding associative number marking—e.g., the presence of associative plurals vs associative duals, the restriction against associative plural morphology co-occurring with plural possessor agreement (discussed above)—and show that the syntactic analysis of heterogeneous plurals proposed in this work offers a straightforward solution to all these issues.

Chapter 4 discusses the claim that 1/2P plural pronouns are universally associative plu-

rals. It has been observed since at least Boas (1911) that 1/2P pronouns do not pluralize like other nouns—their interpretation is not the additive semantics of plural discussed above (i.e., I+I+I..., you+you+you...). Instead, 1/2P plural pronouns appear to have associative plural semantics, in that they (saliently) pick out the speaker/hearer *and others*. This has led a number of authors to argue that 1/2P pronouns actually combine with the associative plural universally (e.g. Cysouw 2003; Ritter and Wiltschko 2019). I show that 1/2P plural pronouns are not the same as true associative plurals, and that their non-additive semantics should be derived differently. I present morphological and semantic evidence to this effect, and discuss a number of languages with dedicated associative plural pronouns (suggesting that the regular 1/2P plural pronoun is not an associative plural). The associative plural is a unique structural configuration that is only found in a subset of languages.

Chapter 5 discusses the morphological expression of the other heterogeneous plurals: simulative plurals and approximative plurals. First, I present a typological survey of languages that use the regular plural as the simulative or approximative plural. These heterogeneous plurals are comparatively under-researched, and thus this chapter presents the largest sample collected to date for simulative plurals and the only one collected for approximative plurals. On the basis of these surveys, I establish a second typological generalization that languages that use the regular plural as the simulative or approximative plural also lack free-standing definite articles. I also investigate whether there is an implicational hierarchy regarding the morphological expression of all three types of heterogeneous plural. That is, if the regular plural is used for one heterogeneous plural, must it also be used for another? I show that there is in fact such an implicational hierarchy for associative and simulative plurals (but not for approximative plurals). I argue that this hierarchy falls out from the organization of morphological features associated with regular and heterogeneous plurality. That is, I propose a morphological account of the hierarchy in question.

Chapter 2

A typological survey of associative plurals

2.1 Introduction

In this chapter, I present the results of a large typological investigation into the morphological expression of the *associative* plural. As noted in Chapter 1, the associative plural combines with an individual-denoting noun ‘X’ (e.g., a proper name, kinship term) to generate the meaning ‘X and X’s associate(s)’. The associates in these plurals most often pick out family/friends. Illustrative examples of the morphological expression of the associative plural cross-linguistically are given in (20). The associative plural in Malayalam (20a) is the word *okke*, meaning ‘all’ (Asher and Kumari 1997). The associative plural in Nias (20b) is a prefix *ira-*, which also has a collective function (Brown 2001). In Belep (20c), the associative plural is a suffix *-ma*, which may derive from the Proto-Oceanic form *MA.4 (*and/with*) (Mauri and Sansó 2021).

- (20) a. Raaman **okke**
 Ram APL
 ‘Ram and associates’ (Daniel and Moravcsik 2013)
- b. **ira-** Basiwa
 APL- Basiwa
 ‘Basiwa and his family’ (Brown 2001, p. 151)
- c. Teâ Polo **-ma**
 Teâ Polo -APL
 ‘Teâ Polo and his people’ (McCracken 2012, p. 248)

Interestingly, the associative plural may also be identical to the regular (additive) plural. This is the case in Japanese (20d)—the suffix *-tachi* is used as both additive and associative plural.

- (21) Mika **-tachi** / gakusei **-tachi**
 Mika -APL / student -PL
 ‘Mika and her friends/family’ / ‘the students’
 (Nakanishi and Ritter 2008, p. 1)

In languages with dual as well as plural number, associative *duals* are also attested. For example, in Cavineña (22), the regular dual clitic =*ekatse* has an associative dual function on kin terms. I will thus sometimes refer to the notion of associative *number* when discussing languages with both associative plurals and associative duals. Mostly, I will also use ‘associative plural’ as an umbrella term throughout.

- (22) e -kwa =ke =**ekatse** / juye =**ekatse**
 3 -mother =3 =DU / ox -DU
 ‘his mother and associated person’ / ‘two oxen’
 (Guillaume 2008, pp. 482, 655)

The morphological expression of associative plural can thus be split into two broad types. In some languages, associative plurality is expressed using a dedicated associative

plural morpheme. In others, regular plural morphology is used as the associative plural. I refer to the latter languages (i.e., Japanese, Cavineña) as *plural pattern* languages. I refer to languages that use dedicated associative plural morphology (Malayalam, Nias, Belep) as *non-plural pattern* languages (because their associative plural does not utilize regular plural morphology).

Regardless of the expression of the associative plural, all of the plurals in (20)–(22) share certain properties that distinguish them from regular (additive) plurality. First, associative plurals are *heterogeneous* while regular plurals are *homogeneous*. That is, regular plurals are *additive* because they take a predicative NP (e.g., *dog*) and produce a plural set that contains only members of the extension of that predicate (i.e., *dog + dog + dog*). Associative plurals, on the other hand, produce a plural set whose members are *not* all described by the predicate. That is, in (20a) *Raaman okke* refers to a group that includes Ram, but who do not all share the property of being called Ram.¹ Another property of the associative plural is that it is restricted to combining with animate nouns (Corbett 2000). To illustrate, consider the examples in (23) from Turkish. Turkish is a *plural pattern* language—the regular plural suffix *-lar* (shown in (23a)) is also used as the associative plural (23b). However, while the plural in (23b)—with an animate noun—is ambiguous between regular and associative plural, the example in (23a)—with an inanimate noun—is unambiguously additive (‘more than one house’). (23a) does not have an associative plural reading ‘the house and

¹Note that the heterogeneity of associative plurals is different from English (ia), where the regular plural *-s* is on a family name. Here the referents of the plural noun do all share the property of having the last name ‘Smith’ (or at least abstractly *being* a Smith), making this homogeneous. Note, too that the associative plural attaches to *given* names; attaching the plural *-s* to a given name in English does not lead to an associative plural interpretation (ib).

- (i) a. The Smiths came over for dinner.
- b. I saw Emmas and then went to a movie.
- ≠ I saw Emma and her family/friends and then went to a movie.

things associated with it’.

- (23) a. ev **-ler**
house -PL
‘houses’ (*the house and things associated with it)
- b. Ahmet **-ler**
Ahmet -PL
‘multiple Ahmets’ / ‘Ahmet and his friends/family’

Associative plurals are thus sensitive to the animacy hierarchy in (24) (adapted from Corbett 2000; Moravcsik 2003, 2020). The hierarchy in (24) represents the order of preference (from left to right) among nouns that the associative plural combines with. That is, an associative plural is most likely to combine with a proper name, followed by a kin term, then a title, then other definite human nouns.²

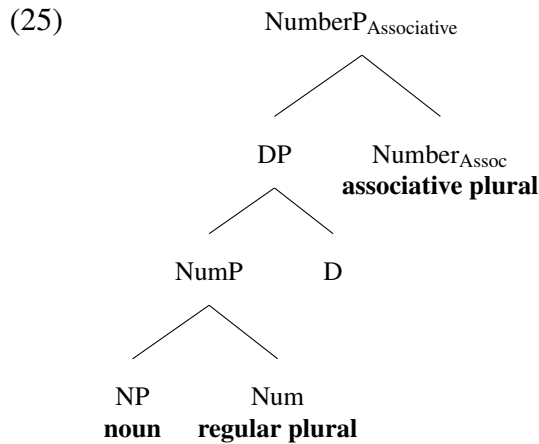
- (24) proper noun > kin noun > title > (other) definite human noun > (non-human animate noun)

Lastly, the noun with which the associative plural combines must be definite (Besten 1996; Daniel 2020). Proper names and possessed kin terms—with which the associative plural most frequently combines—are inherently definite. Other human nouns, however, could in principle be definite or indefinite with the associative plural. Yet, Daniel (2020) notes that we do not find indefinite associatives like *a man-APL*, ‘a man and his family’. This is in contrast to additive plurality, which does not have a definite requirement in all languages.

Turning now to analyses of the associative plural, various approaches have been put forward for individual languages but one analysis in particular is adopted by a number of

²Whether non-human animate nouns can combine with the associative plural is controversial. Daniel (2020) provides an example from Mishar Tatar, where the associative plural *-lar* attaches to the name of a cow.

authors. This analysis proposes that the associative plural heads a functional category high in the nominal domain. For example, Ueda and Haraguchi (2008) argue that the Japanese associative plural *-tachi* is base-generated in D^0 , while Nakanishi and Tomioka (2004) argue that it is realized in the head of a GroupP above DP. Similarly, Görgülü (2011) argues for a functional projection GroupP above DP for the associative plural *-lar* in Turkish, and Dékány (2021) argues for an AssPIP (‘Associative Plural Phrase’) above DP for the Hungarian associative plural *-ék*. Li (1999) argues for Mandarin Chinese that the associative plural *-men* is in D^0 , Biswas (2014) argues for Bengali that the associative plural *-ra* is in a functional projection above DP and Forbes (2013) argues that the Gitskan associative plural *dip* adjoins to DP.³ Drawing on these analyses for individual languages, Cinque (2018) argues for the universal structure in (25) for associative plural constructions. The associative plural heads a functional projection $\text{NumberP}_{\text{Associative}}$ above DP. The regular plural heads NumP (Ritter 1992).

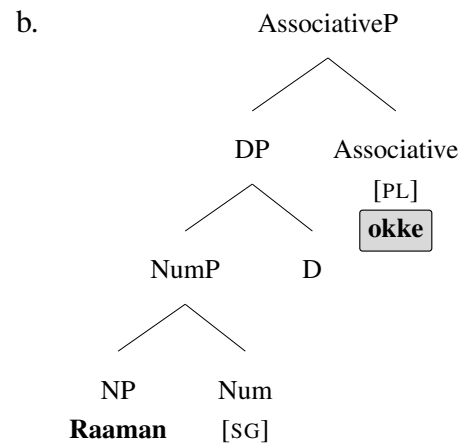


What these approaches all share is the intuition that the associative plural is high in the nominal domain relative to the regular plural, and specifically above DP. These approaches

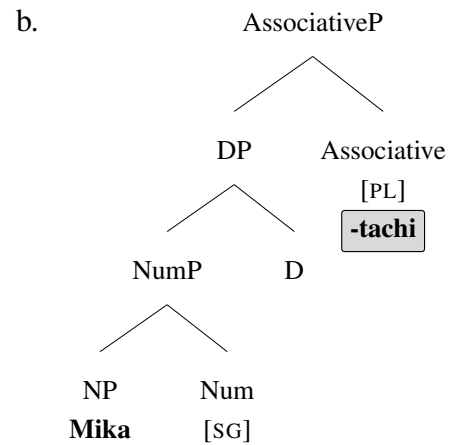
³Not all authors adopt the view that the associative plural occupies a distinct functional projection above DP, though this is by far the most common approach. For some alternative analyses, see den Dikken 2023; Kratzer 2009; Vassilieva 2005.

also all share the assumption that the associative plural is *base-generated* in the head of this functional projection. Thus, using Cinque’s analysis as an illustration, the associative plural construction in Malayalam (26a) has the structure in (26b) in which the associative plural *okke* heads $\text{NumberP}_{\text{Associative}}$. The associative plural construction in Japanese (27a) has the same structure, shown in (27b): The associative plural *-tachi* heads $\text{NumberP}_{\text{Associative}}$.

- (26) a. Raaman **okke**
 Ram APL
 ‘Ram and associates’
 (Daniel and Moravcsik 2013)



- (27) a. Mika **-tachi**
 Mika -APL
 ‘Mika and her friends/family’
 (Nakanishi and Ritter 2008)



In other words, *plural pattern* languages—languages where the regular plural is used as the associative plural—are derivationally identical to *non-plural pattern* languages. That

there is syncretism between regular and associative plural morphology in certain languages has been taken to be incidental. The aim of this chapter is to show that this assumption is wrong. In the following sections I present the findings of a broad cross-linguistic survey of *plural pattern* languages and show that they are not an arbitrary group; the *plural pattern* in fact systematically follows from an independent property of the relevant languages.

2.2 A typological survey of the *plural pattern*

This section presents the results of a large typological survey of *plural pattern* languages. The language sample used for this survey, presented in the next section, consists of 109 languages spanning 33 language families, 4 creoles/pidgins and 2 isolates. It is worth noting that associative plurality is often not reported in the descriptive grammars which traditionally inform large typological studies. That is, building a well-balanced typological survey from scratch—by choosing a genetically diverse, representative sample of languages to study, and then consulting grammars to discern whether a particular phenomenon is present or not—will not yield many results for the phenomenon of associative plurals. It is simply not described.

Thus, when looking for *plural pattern* languages to include in the current sample, two things were important: (i) looking at languages that are genetically related to languages already in the sample and (ii) looking at language families that were not previously represented in typological descriptions of associative plurality (e.g., Daniel and Moravcsik 2013; Mauri and Sansó 2021).⁴ (i) allows for comparison of closely related languages. Thus, while many languages that are genetically related to *plural pattern* languages also

⁴The current language sample was also checked against these existing surveys which are typologically well-balanced studies (Daniel and Moravcsik’s (2013) survey, for example, consists of 65 languages in the WALS 100-language sample, a presumably balanced sample of the world’s languages.)

show the *plural pattern*, some languages that are genetically related diverge in their expression of the associative plural: some show the *plural pattern*, while others use dedicated associative plural morphology. As we will see in Section 2.4.1, we can then look at what additional factors are different between such languages, and make hypotheses as to *why* these related languages differ in their associative plural expression. (ii) diversifies the typological picture, giving us a more complete image of associative plurals cross-linguistically. As we will see, while for most *plural pattern* languages the associative plural is identical to the regular plural, other *plural pattern* languages use the regular plural as only part of the associative plural. Yet other languages use the regular plural form of possessive or genitive nouns as the associative plural. An analysis of *plural pattern* languages cross-linguistically should account for the various sub-types, and the more diverse the language sample the more sub-types we are likely to encounter. Additionally, the more genetically *unrelated* languages in the sample, the more certainly we can say that the *plural pattern* is not constrained by genetic relations but is in fact a cross-linguistically widespread phenomenon. Nonetheless, as we will see, a widespread phenomenon may be constrained by *something else*.

The next section presents the language sample. Given that *plural pattern* associative plurals are found in such a wide variety of languages, Section 2.4 considers whether the distribution of the *plural pattern* is best chalked up to random chance, or whether another factor may be responsible for it. A superficial examination of the languages in the sample reveals that most of them lack definite articles. A detailed investigation reveals a robust new typological generalization: *plural pattern* languages lack free-standing definite articles. I establish this generalization in Section 2.5.

2.3 The language sample

The current language sample consists of 109 languages spanning 33 language families, 3 creoles/pidgins and 2 isolates. The languages are given in (28).

(28) *List of plural pattern languages and their language families:*

1 Turkic:

- 1 Bashkir
- 2 Chuvash
- 3 Crimean Tatar
- 4 Gagauz
- 5 Karachay-Balkar
- 6 Kazakh
- 7 Kumyk
- 8 Mishar Tatar
- 9 Tofa
- 10 Turkish
- 11 Uyghur
- 12 Uzbek

2 Tungusic:

- 13 Evenki
- 14 Manchu
- 15 Nanai
- 16 Udihe

3 Mongolic

- 17 Khalkha

4 East Caucasian

- 18 Aghul
- 19 Archi
- 20 Bagvalal

- 21 Lak

- 22 Lezgian

- 23 Rutul

5 Niger Congo

- 24 Akan

- 25 Dagbani

- 26 (Eastern) Dan

- 27 Ewe

- 28 Fulfulde

- 29 Luganda

- 30 Luvale

- 31 Sango

- 32 Sesotho

- 33 Tswana

- 34 Yoruba

- 35 Zulu

6 Afro-Asiatic

- 36 Margi

7 Tacanan

- 37 Araona

- 38 Cavineña

- 39 Ese-Ejja

8 Chukotko-Kamchatkan

40 Chukchi	64 Iheya
41 Alutor	65 Japanese
9 Nadahup	66 Kin
42 Hup	67 Shiiba
10 Sepik	68 Yoron
43 Awtuw	16 Dogon
44 Manambu	69 Tommo So
11 Mande	17 Salishan
45 Bambara	70 Kalispel
46 Kpelle	18 Uralic
47 Mandinka	71 Hungarian
12 Sino-Tibetan	72 Khanty
48 Belhare	73 Komi-Permyak
49 Burmese	74 Komi-Zyrian
50 Chintang	75 Mari (Hill)
51 Mandarin Chinese	76 Moksha
52 Chantyal	77 Nganasan
53 Hayu	78 Udmurt
54 Lahu	19 Tucanoan
55 Lepcha	79 Desano
56 Limbu	80 Tucano
57 Magar	20 Kadu
58 Newar (Dolkha)	81 Krongo
59 Newar (Kathmandu)	21 Central Sudanic
60 (Eastern) Tamang	82 Lugbara
13 Utu-Aztecan	22 Indo-European
61 Comanche	83 Bengali
14 Cariban	84 Bulgarian
62 Apalai	85 Nepali
63 Hixkaryana	86 Ossetic
15 Japonic	87 Polish

88 Serbo-Croatian	99 Gooniyandi
89 Sinhala	30 Na-Dene
90 Slovenian	100 Koyukon
23 Pama-Nyungan	31 Eskimo-Aleut
91 Ngiyambaa	101 Central Alaskan Yupik
24 Arawakan	32 Austronesian
92 Baniwa	102 Mangap-Mbula
93 Piapoco	33 Austro-Asiatic
94 Tariana	103 Kharia
25 Northwest Caucasian	creoles/pidgins
95 Kabardian	104 Berbice Dutch Creole
26 Tupian	105 Mauritian Creole
96 Urubu-Kaapor	106 Reunion Creole
27 Songhay	107 Tok Pisin
97 Koyraboro Senni	isolates
28 Arawan	108 Ainu
98 Kulina	109 Nivkh
29 Bunuban	

In the following sections we will examine the various ways in which a *plural pattern* associative plural manifests in these languages.

2.3.1 The baseline case

The overwhelming majority of *plural pattern* languages are of the kind presented at the start of this chapter: exactly the same morpheme that expresses regular plural also expresses associative plural. This is what I call a ‘baseline’ *plural pattern* associative plural. Baseline *plural pattern* languages include Turkish (and in fact all of the Turkic languages in the sample) (29), Eastern Dan (and the other Niger Congo languages) (30), Sinhala (Indo-

European) (31) Tok Pisin (a creole spoken in Papua New Guinea) (32) and Nivkh (an isolate spoken in Russia) (33).

(29) *Turkish* [Turkic]

- a. Ahmet-**ler**
Ahmet-PL
'Ahmet and his friends/family' (adapted from Görgülü 2011, p. 74)
- b. Arka sıra-**lar**-da öğrenci-**ler** otur-acak
back row-PL-LOC student-PL sit-FUT
'Students will sit in the back rows' (Göskel and Kerslake 2005, p. 334)

(30) *Eastern Dan* [Niger Congo]

- a. dhǎn-**dhũn**
mother.in.law-PL
'mother in law and another (= her daughter)'
- b. béezǎ-**dhũn**
nephew-PL
'newphews' (Vydrin 2020, p. 40)

(31) *Sinhala* [Indo-European]⁵

- a. Gunapaala-**lā**
Gunapaala-PL
'Gunapala and his family' (Daniel and Moravcsik 2013)
- b. taata-**lā**
father-PL
'fathers' (Garland 2006, p. 4)

(32) *Tok Pisin* [creole]

- a. pater ɔl
priest PL
'the priest and his flock'
- b. ɔl pater
PL priest
'priests'

(Mülhäuser 1981, p. 43)

⁵The relevant plural suffix in Sinhala (-*lā*) consists of a long vowel *ā* (Mallikadevi 2023). Daniel and Moravcsik (2013) transcribe this suffix as *-laa* and Garland (2006) transcribes it as *-la*. For clarity, I have adapted both transcriptions and written the suffix as *-lā*.

(33) *Nivkh* [isolate]

- a. Xevgun-**gu**
Xevgun-PL
'Xevgun together with other people'
- b. if p^h-lele-**gu**-doχ k^hinju-ʃ
3SG REFL-relative-PL-DAT be.nice-IND
'He respects his relatives.'

(Gruzdeva 2022, p. 393)

Note that a language shows the *plural pattern* even if the regular plural that is co-opted for the associative plural is only found on certain kinds of nouns. For example, the associative plural in Burmese is *-tó/-dó*. This is a regular plural marker that is found on pronouns but rarely on common nouns, which use *-twe/-dwe* (34).⁶ Similarly, the associative plural markers in Tariana (Tucanoan) and Desano (Arawakan) are identical to regular plural markers used on kin terms, some with very limited use. In Tariana, the associative plural *-sini* is used with “a few terms” to form the regular plural, including “female ego’s brother’s son” and “father’s younger brother” (Aikhenvald 2003, p. 168). Similarly, the Desano associative plural marker *-sã* expresses the regular plural but only on female relatives (Aikhenvald 2014, p. 51).

- (34) a. Sú=gá Hla Hla-**dó** (=go) twé-géh-deh
Suu=NOM Hla Hla-PL =ACC meet-PAST-NFUT
'Suu met Hla Hla and her associates'

(New 2021, p. 167)

- b. nga-**tó** / min:-**tó**
1P-PL / 2P-PL
'we/us' / 'you guys'

(Soe 1999, pp. 58–9)

⁶The pronominal/associative plural marker *-tó/-dó* does have limited use on common nouns. Thus, (i) has an ambiguous reading.

- (i) Sú=gá s'aya-**dó** (=go) twé-géh-deh
Suu=NOM teacher-PL =ACC meet-PAST-NFUT
'Suu met teachers.' or
'Suu met a teacher and their associate(s).'

(New 2021, p. 169)

- c. Sú=gá s'aya-**dwe** (=go) twé-géh-deh
 Suu=NOM teacher-PL =ACC meet-PAST-NFUT
 'Suu met teachers'

(New 2021, p. 167)

Of the 109 languages in the sample, 99 of them have a baseline *plural pattern* associative plural. Using the regular plural as the associative plural is thus the most common way in which the *plural pattern* manifests in languages cross-linguistically. This fact will be important in the analysis of *plural pattern* languages proposed in Chapter 3. However, this is not the only way a *plural pattern* associative plural manifests. The next section turns to these non-baseline cases.

2.3.2 Regular plural + extra morpheme

A smaller number of languages in the sample express the associative plural using morphology that is not identical to regular plural morphology, but which *contains* the regular plural. For example, the associative plural marker in Central Alaskan Yup'ik is *-nkut* (35a). Central Alaskan Yup'ik also has an associative dual, using the morpheme *-nkuk* (35b). The regular plural and dual markers in Central Alaskan Yup'ik are *-t* and *-k* respectively (36). Thus, one way of analyzing the Central Alaskan Yup'ik data is to decompose the associative marker into the regular plural (or dual), preceded by some dedicated associative morphology (this is the approach adopted by e.g. Cinque 2018; Corbett 2000; Corbett and Mithun 1996; Dékány 2021; Miyaoka 2012)

- | | | |
|------|---|--|
| (35) | a. Cuna-nku- t
Chuna-APL-PL
'Chuna and his family/friends' | b. una-nku- k
Chuna-APL-DU
'Chuna and his friend' |
|------|---|--|

- (36) a. qaya-**t**
 kayak-PL
 ‘kayaks (3 or more)’
 b. qaya-**k**
 kayak-DU
 ‘two kayaks’

(Corbett 2000, pp. 108–9)

Miyaoka (2012), for example, argues that the *-ku-* in the dedicated associative morphology of Central Alaskan Yupi’k likely corresponds to the non-singular expander found on non-singular personal pronouns and nominal demonstratives. This morpheme appears between the pronoun or demonstrative’s stem and any non-singular (plural or dual) number morphology (37).

- (37) u-**ku**-t / wang-**ku**-ta
 this-EX-PL / 1SG-EX-PL
 ‘these’ / ‘we’

Five more languages in the language sample contain regular plural and/or dual morphology plus distinct associative morphology: Ossetic, Mandinka, Khanty, Hup and Hungarian. Erschler (2016) describes the associative plural in Ossetic as involving a dedicated associative plural suffix *-i/-e* that is obligatorily followed by the plural marker. This is shown in (38a).

- (38) *Ossetic* [Indo-European]

- | | | |
|--|---|------------------------------|
| <p>a. šošlan-i/e-te
 Soslan-APL-PL
 ‘Soslan and others’</p> | <p>b. bæles-te
 tree-PL
 ‘trees’</p> | <p>(Erschler 2016, p. 7)</p> |
|--|---|------------------------------|

In Mandinka, the associative plural marker consists of the regular plural marker *-lu* preceded by *-ño* (39a). The morpheme *-ño* also functions as a reciprocal marker, ‘each other’ (own fieldwork).

(39) *Mandinka* [Mande]

- | | |
|--|--|
| a. Báakari-ño- lu
Báakari-each.other-PL
‘Bakari and those with him’ | b. beri- lu
stone-PL
‘stones’ |
|--|--|

(Daniel and Moravcsik 2013)

(Creissels 2013, p. 14)

Khanty has both associative plurals and associative duals. The regular dual and plural markers (-*yen* and -(*ə*)*t*, respectively) preceded by -*s* forms the associative marker. This is shown in (40a). Filchenko (2007) glosses -*s*- as the collective suffix.⁷

(40) *Khanty* [Uralic]

- | |
|---|
| a. jəɣ-s-ä ɣən / ə-s-ä t
father-COLL-DU / mother-COLL-PL
‘A father and his son’ / ‘A woman with her family’ |
|---|

(Filchenko 2007, pp. 79–80)

- | |
|---|
| b. qoj- ɣən / qoj-ä t
man-DU / man-PL
‘two men’ / ‘men’ |
|---|

(Filchenko 2007, p. 76)

The associative plural in Hup is expressed with the regular plural marker -*d’əh* preceded by the morpheme -*ăn* (41a). Epps (2008) suggests that the -*ăn* found in the associative plural marker is identical to the object (or directional) marker -*ăn*.

(41) *Hup* [Nadahup]

- | | |
|---|--|
| a. ʔána-ăn- d’əh
Ana-ăn-PL
‘Ana and they (her children)’ | b. yăʔambǒʔ= d’əh
dog=PL
‘dogs’ |
|---|--|

(Epps 2008, pp. 205–6)

⁷Filchenko (2007, p. 80) glosses -*s*- as collective in the associative dual, but segments the associative plural as *əs-at*, ‘mother-PL’, i.e., as the plural suffix attaching directly to the noun stem. However, the associative plural is segmented as *ə-s-at*, ‘mother-Coll-PL’ later on (Filchenko 2007, p. 105).

Finally, the Hungarian associative plural is expressed using the regular plural *-k* preceded by *-é*. This is shown in (42a). The regular plural in Hungarian may be preceded by a linking vowel (e.g. */o/* in (42b)) that varies depending on the final vowel of the noun stem. Some authors argue that the *-é* in the associative plural is dedicated associative morphology (e.g. Dékány 2021) while others suggest that it is related to the possessive anaphor *-é* (Abondolo 1985; Moravcsik 2003).⁸

(42) *Hungarian* [Uralic]

a. János-**é-k**
 John-*é*-PL
 ‘John and his associates’

b. barát-**ok**
 friend-PL
 ‘friends’

(Corbett 2000, p. 102)

(Dékány 2021, p. 225)

In each of these languages, the regular plural (or dual) makes up part of the associative plural. The supplementary morphology may have other uses in the language (e.g. a reciprocal in Mandinka, a collective suffix in Khanty) or be unique to the associative plural construction (as in Ossetic, and perhaps Hungarian). Note, too, that the ordering of regular number morphology and its supplementary associative morphology is always the same: the regular plural/dual always follows the associative marking. Taking seriously the appearance of regular number morphology in these associative plurals means treating them as minimally different from the baseline *plural pattern* languages. That is, these languages are a sub-type of *plural pattern* language that differ from the baseline case only in overtly exposing associative morphology. We will see that these kinds of languages are going to play an important role in the development of the analysis of *plural pattern* languages later on.

⁸The possessive anaphor is a morpheme that appears on a possessor when the possessed noun is silent.

2.3.3 Regular plural + alternative stem

The last kind of *plural pattern* language is found in the South Slavic languages (Bulgarian, Slovenian and Serbo-Croatian) and Lezgian (East Caucasian). In these languages, the associative plural stem is not a simple noun—like *priest* or *father* or *John*—as has been the case with all of the examples so far. Rather, the associative plural stem is possessive (in the case of South Slavic) or genitive (in Lezgian). Consider first an associative plural in Lezgian, given in (43). The nominal stem, ‘mother’ appears with a reduced form of the genitive marking *di-n*, in which the ending *-in* is dropped.⁹ The stem is pluralized with the substantivized plural *-bur*. The plural form of a genitive noun ‘X’ (with no overt possessum) literally translates to ‘those of X’ (44).

- (43) Dide-d-**bur** bazar.di-z fe-na
mother-GEN-SBST.PL market-DAT go-AOR
‘Mother and those with her went to the market.’ (Haspelmath 1993, p. 79)

- (44) Exirni **xür.ü-n-bur**.u kac tam.u-z čukur-na.
finally village-GEN-SBST.PL(ERG) cat forest-DAT chase-AOR
Finally the villagers (lit. those of the village) chased the cat into the forest.

(Haspelmath 1993, p. 344)

Similarly, in the South Slavic languages associative plurals are identical to plural possessives with a null possessum. Illustrative examples from Bulgarian, Slovenian and Serbo-Croatian are given in (45)–(47). The associative plural stem is possessive, indicated by *-ov*.¹⁰ Plural marking on the possessive is indicated by *-i*. The prototypical interpretation of a plural possessive ‘X’ is ‘those of X’, as shown by the translations in (i). However, this form also has an associative plural interpretation, as shown by the translations in (ii).

⁹Genitive marking in Lezgian involves an ergative (or oblique) form of the stem, the default suffix for which is *-di*. The suffix *-n* is genitive case (Haspelmath 1993).

¹⁰Possessive marking is determined by properties of the stem and is not always *-ov*.

(45) *Bulgarian*

Pešovi oti-do-ha na ribo-lov.

Peter.POSS.PL.M go-PAST-3PL to fish-hunt

i. ‘Those of Peter (e.g., his family) went fishing.’ = possessive

ii. ‘Peter and his friends went fishing.’ = associative

(46) *Slovenian*

Lankotovi so šli na ribo-lov.

Lanko.POSS.PL.NOM.M aux.3PL went.M.PL on fish-hunt

i. ‘Those of Lanko (e.g., his family) went fishing.’ = possessive

ii. ‘Lanko and his family/friends went fishing.’ = associative

(47) *Serbo-Croatian*

Markovi su išli na ribo-lov.

Marko.POSS.PL.NOM.M aux.3PL went.M.PL on fish-hunt.

i. ‘Those of Marko (e.g., his family) went fishing.’ = possessive

ii. ‘Marko and his family/friends went fishing.’ = associative

The meaning difference between a genuine plural possessive and an associative plural is subtle. In the latter, the named referent must be included in the group described (‘Peter *and* his family/friends’) while in the former the named referent may or may not be included in the group (‘Peter’s family/friends, but not Peter’) (see e.g. Daniel 2004; Vassilieva 2005). What is important here is that the examples in (43) and (45)–(47) are the *regular plural* of possessive (or genitive) forms of the noun. Thus, these too are minimally different from the baseline *plural pattern* associative plural. In the baseline case, the regular plural of a simple noun is recruited for associative plural meaning. In these languages, the regular plural of the possessive or genitive form of a noun is recruited for associative plural meaning. Thus, these languages are another sub-type of *plural pattern* language, but this time it is the stem—not the associative plural marker—that takes a slightly different shape. As before,

we should take seriously the appearance of regular number morphology in these associative plurals and try to capture this expression of associative plurality in our analysis of *plural pattern* languages.

2.3.4 Summary

These are thus the three kinds of *plural pattern* language in the language sample: (i) the associative plural is identical to the regular plural, (ii) the associative plural is made up of the regular plural and some invariant supplementary morphology, (iii) the associative plural is identical to the regular plural of a possessive/genitive noun. The next section takes a closer look at the typology of the language sample and at what might constrain the availability of a *plural pattern* associative plural cross-linguistically.

2.4 The typology of *plural pattern* languages

So far it has been established that the *plural pattern* is a cross-linguistically productive way in which languages can generate an associative plural. Regarding the languages in Daniel and Moravcsik's (2013) survey that have an associative plural of any kind, half are characterized as using the regular plural as the associative plural. The current language sample, consisting of 109 languages, replicates the finding that a *plural pattern* associative plural is a cross-linguistically common phenomenon.¹¹ Moreover, *plural pattern* associative plurals do not appear to be genetically constrained: the sample consists of many unrelated languages that otherwise look very different on the surface, but nevertheless share the property of using the regular plural as (or as part of) the associative plural. We could chalk this

¹¹See Appendix for discussion regarding a number of languages that have been characterized as using the regular plural as the associative plural by Daniel and Moravcsik (2013), but which I exclude from the current sample for reasons discussed in the appendix.

up to coincidence. As discussed in Chapter 1, associative plurals come in two broad flavors: (i) the regular plural has a secondary function as (or as part of) the associative plural, or (ii) the associative plural is something else (it does not contain the regular plural). What flavor a language has is a 50/50 chance (recall that Daniel and Moravcsik’s (2013) survey does show a roughly 50/50 split), and the languages in the current sample happen to fall in group (i).

A more interesting hypothesis is that the languages in (28) all show the *plural pattern* because of some additional factor of their grammar. That is, perhaps the *plural pattern* is cross-linguistically constrained, and not random chance. I will show next that this is in fact the case.

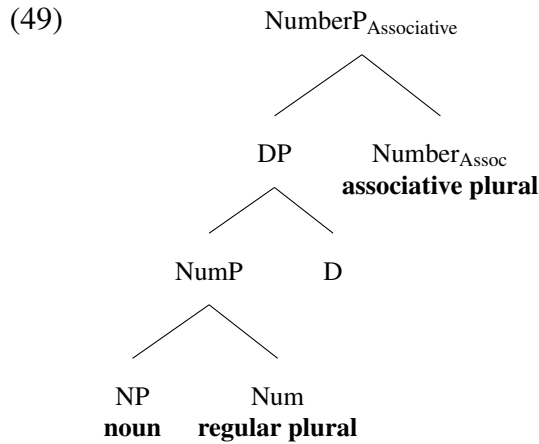
2.4.1 Is the plural pattern predictable?

Some of the earliest analyses of associative plurals suggest that DP structure and associative plurality are closely correlated. Besten (1996), for example, argues that associative plurals in Afrikaans (a non-plural pattern language, see (48a)) involve a DP (i.e., definite) stem followed by a third person plural pronoun (48b).

- (48) a. Pa -**hulle** / my kinder -s -**hulle**
 Dad -them / my children -PL -them
 ‘Dad and his folks’ / ‘my children and others’
- b. [DP₁ DP₂ Pron._{plur}]

Moravcsik (2003) defined associative plural stems as “definite human” individuals and a number of early analyses derived this definiteness requirement by positing that associative plural marking attaches to a DP stem, as in (49) (Biswas 2014; Cinque 2018; Dékány 2021;

Forbes 2013; Görgülü 2011; Hirose 2004; Nakanishi and Ritter 2008).¹²



One piece of compelling evidence that the associative plural marker attaches above DP comes from the relative order of the associative plural and definite articles. Cinque (2018) notes that in Muna (which has a unique particle *ndo* for the associative plural), both the definite article and the associative plural are prenominal but the associative plural precedes the article (50a). Assuming that the linear position of morphemes reflects underlying syntactic structure—that is to say that morphemes farther from the root noun are structurally higher (Baker’s (1985) Mirror Principle)—the article (in D^0) must be in a projection below the associative plural (50b).

- (50) a. no -hamba **ndo** **Wa** Marangkululi
 3sR -chase APL ART Marangkululi
 ‘He chased Marangkululi and her friends.’ (Berg 1989, p. 108)
- b. [AssociativeP Associative **ndo** [DP D **Wa** [NP Marangkululi]]]

Evidence from some of the most well-studied *plural pattern* languages also suggests that the associative plural is in a structurally high position. In Turkish, for example, it has long been observed that the associative plural occupies a higher syntactic position than

¹²Similarly, others have argued that the associative marker is realized in D^0 (e.g. Li 1999; Ueda and Haraguchi 2008).

the regular plural. This can be seen in associative plurals of kin terms which are inflected for possessor agreement, illustrated by (144). Turkish is a head-final language and the associative plural *-lar* follows possessor agreement (thus it is structurally higher than possessor agreement according to the Mirror Principle) (144a). Conversely, the regular plural *-lar* precedes possessor agreement (indicating a lower syntactic position) (144b) (see Görgülü 2011; Göskel and Kerslake 2005; Ileri 2023; Lewis 2000).

(51) *Turkish* [Turkic]

a. Teyze -m **-ler** çık -tı
 aunt -1 SG -PL leave -PAST
 ‘My aunt and her family/friends left.’

b. Teyze **-ler** -im çık -tı
 aunt -PL -1 SG leave -PAST
 ‘My aunts left.’

(Görgülü 2011, p. 74)

That the plural marker has two distinct positions relative to possessor agreement—depending on whether it is regular or associative—has also been observed in Ainu (52) and Hungarian (53). Note that possessed nouns in Hungarian take a possessive suffix *-(j)a/-(j)e* that should not be confused with possessive agreement. Additionally, when plural marking is adjacent to the possessive suffix the *-i* allomorph is used.

(52) *Ainu* [isolate]

a. ku -yup **-utar** -i
 1 SG -elder.brother -PL -POSS
 ‘my elder brothers’

b. ku -yup -i **-utar**
 1 SG -elder.brother -POSS -PL
 ‘my elder brother and others’

(Cinque 2018, p. 497)

(53) *Hungarian* [Uralic]

- a. a te börrönd -je -i -d
the you.SG suitcase -POSS -PL -2SG.POSS
'your(SG) suitcases'
- b. a barát -od -é -k
the friend -2SG.POSS -é -PL
'your friend and his/her associates'

(Dékány 2021, pp. 174, 223)

The current picture thus implies that the associative plural is high in the nominal domain. Languages like Muna provide morphological evidence that the associative plural marker attaches above the DP layer. More pertinent for the current study, which aims to investigate whether *plural pattern* languages are cross-linguistically constrained, *plural pattern* languages like Turkish, Ainu and Hungarian show that the height of plural marking in the nominal domain is crucial in determining whether it is interpreted additively or associatively. When the plural marker is in a low position—preceding possessor agreement and below the merge position of DP—it is obligatorily interpreted as additive. When it is in a high position—following possessor agreement and above the merge position of DP—it is obligatorily interpreted as associative. Given this, the height of associative plural marking suggests that the DP layer is directly implicated in the possibility of the *plural pattern*.

As it turns out, even a superficial examination of the *plural pattern* languages reveals at least a strong tendency for *plural pattern* languages to lack definite articles. Consider first the survey of *plural pattern* languages in Daniel and Moravcsik (2013). Of the 104 languages they characterize as using the regular plural as the associative plural, only 17 are characterized as having a definite article by Dryer (2013).¹³ This distribution of definite articles in *plural pattern* languages is also reflected in the current language sample: out

¹³A language is characterized as having a definite article if it is said to have a definite affix or a definite word distinct from the demonstrative.

of 109 languages, just 10 are characterized as having a definite article by Dryer (2013). Cross-referencing these two phenomena—the *plural pattern* in the associative plural and definite articles—provides a snapshot of the typological picture that suggests that the *plural pattern* is extremely rare in languages that have definite articles. However, information about article status is absent from Dryer (2013) for almost half of the languages (54 in Daniel and Moravcsik’s (2013) survey, 60 in the current sample), meaning we cannot draw any robust conclusions just yet. This is the goal of the next section. I examine all of the *plural pattern* languages in the current sample to determine whether or not they have definite articles. I show that the initial snapshot is confirmed: 99 *plural pattern* languages in the current sample lack definite articles (90.82%). Moreover, of the 10 *plural pattern* languages with definite articles, all are affixal/clitic. This strongly suggests that not only is the presence/absence of definite articles a key factor in the expression of associative plurality, so is the type of definite article.

2.5 A novel typological generalization

This section establishes a novel typological generalization about associative plurals that show the *plural pattern*, given in (54):

- (54) GENERALIZATION: If a language uses the regular plural as (or as part of) the associative plural (the *plural pattern*), it lacks free-standing definite articles (i.e., it either lacks definite articles or has affixal/clitic definite articles).

The generalization is based on a broad typological survey of 109 *plural pattern* languages, the largest sample collected to date. I begin by outlining the definition of “definite article” that I will be using throughout (Section 2.5.1). I then turn to the articleless lan-

guages in the sample, which make up the majority of *plural pattern* languages (Section 2.5.2). Section 2.5.3 deals with the languages in the sample that have a definite article. I show that in all of these languages the definite article is affixal/clitic.

2.5.1 Defining definite articles

Before establishing the typological generalization above it is necessary to lay out precisely what I mean by definite article. The definition of a definite article that I will use throughout this section, from Bošković (2016), is given in (55).

- (55) DEFINITION: A *definite article* (i) is a unique element, in that it occurs once per NP *and* is distinct from demonstratives, (ii) has roughly the meaning of the iota operator, and (iii) is obligatory in NPs with definite interpretation.

The uniqueness requirement in (55i) serves two functions. First, it captures the fact that definite articles are thought to be realized in the head of a functional projection (D). If an expression indicating definiteness can appear more than once per NP, it is not a head and thus not a definite article. Second, it is well-established by now that definite articles in many of the world's languages developed from demonstratives, and that demonstratives and definite articles have different properties (e.g., Dryer (2013), Greenberg (1978), and Himmelmann (2001)). Most notably, demonstratives have a (spatial) deictic function—they point to a perceivable object and encode some information about the location of the object relative to the discourse. Definite articles do not have this property. Thus, in both of the examples in (56), with a demonstrative, the book must be visible. Additionally, (56a) conveys that the book is close to the speaker and (56b) that it is some distance from the speaker. In contrast, in (57), with the definite article, the book need not be visible and no information about the location of the book is encoded.

(56) a. *This book* is fascinating.

b. *That book* is boring.

(57) *The book* I bought yesterday is fascinating.

(55ii)—definite articles have the meaning of the iota operator—comes from Chierchia (1998b). *Iota* is a type-shifting operator that takes a variable of type $\langle e, t \rangle$ (a function from individuals to truth values) and yields an individual-denoting expression of type e . Additionally, Chierchia argues that if a language has a lexical item with a particular semantic function, covert type-shifting is blocked. Thus, because the definite article has the function of the iota operator, covert type-shifting from type $\langle e, t \rangle$ to type e is disallowed in languages with definite articles. This is property (55iii)—the definite article is obligatory in NPs with definite interpretation. In contrast, in languages that lack definite articles the iota-operator can apply covertly, resulting in bare NPs with a definite interpretation. This is illustrated in (58) with Serbo-Croatian.

(58) Mačke razbiše prozor

stone broke window

‘**The** stone broke **the** window.’

Bošković (2016, p. 2)

This will be important in the following discussion. If an alleged “definite article” is not required for definite interpretation, then it is not really a definite article. Note that, following Becker (2021), the absence of the definite article must be *unsystematic* in order for a language to count as an articleless language. If the absence of the article is predictable (based on e.g., morphosyntactic or semantic constraints), a bare NP with a definite interpretation is not necessarily evidence that the language lacks definite articles. For example, the absence of the definite article with proper names in English—and many other languages—does not mean that English lacks a definite article. The absence of the definite article with

these nouns can be explained by the fact that proper nouns are inherently definite, and thus the article is not required for semantic reasons.¹⁴ Thus, when I discuss bare NPs with a definite interpretation in the following sections this is based on an examination of the overall distribution of articles in the language, not isolated examples.

In addition to these properties, it is a well-known distinction that some definite articles are free-standing while some are bound (affixal or clitic). In English, the definite article *the* is free-standing (59a) while in Danish the definite article *en/et* is bound (specifically, it is a suffix) (59b).

- (59) a. **the** house
 b. hus **-et**
 house -DEF
 ‘the house’

What is important for the typological generalization is that all of the languages that show the *plural pattern* in their associative plural and have a definite article have a *bound* (affixal or clitic) definite article.¹⁵ As supporting evidence for the affixal status of the definite article in these languages, I show in Section 2.5.3 that word-internal phonotactic constraints and sandhi rules apply to [noun + definite article] sequences.

Having defined what it means to have a definite article, the next section establishes the typological generalization in Section 2.5, repeated below.

- (60) GENERALIZATION: If a language uses the regular plural as (or as part of) the associative plural (the *plural pattern*), it lacks free-standing definite articles (it either lacks definite articles or has affixal/clitic definite articles).

¹⁴Not required is not the same as not allowed. Some languages, like French, do allow the definite article with proper names.

¹⁵As the generalization in (54) states, for our purposes it does not matter whether the definite article is an affix or a clitic.

That is, all of the languages in the current language sample either lack definite articles or have affixal/clitic definite articles. All of the languages without definite articles are given in (61). The affixal/clitic article languages are given in (62).

(61) *Languages without a definite article:*

[**Turkic**] Bashkir, Chuvash, Crimean Tatar, Gagauz, Karachay-Balkar, Kazakh, Kumyk, Mishar Tatar, Tofa, Turkish, Uyghur, Uzbek, [**Tungusic**] Evenki, Manchu, Nanai, Udihe, [**Mongolic**] Khalkha, [**East Caucasian**] Aghul, Archi, Bagvalal, Lak, Lezgian, Rutul, [**Niger Congo**] Akan, Dagbani, Eastern Dan, Ewe, Fulfulde, Luganda, Luvale, Sango, Sesotho, Tswana, Yoruba, Zulu, [**Tacanan**] Araona, Cavineña, Ese-Ejja, [**Chukotko-Kamchatkan**] Chukchi, Alutor, [**Nadahup**] Hup [**Sepik**] Awtuw, Manambu, [**Sino-Tibetan**] Belhare, Burmese, Chintang, Mandarin Chinese, Chantyal, Hayu, Lahu, Lepcha, Magar, Newar (Dolkha), Newar (Kathmandu), Eastern Tamang, [**Utu-Aztecan**] Comanche, [**Cariban**] Apalai, Hixkaryana, [**Japonic**] Iheya, Japanese, Kin, Shiiba, Yoron, [**Salishan**] Kalispel, [**Uralic**] Khanty, Komi-Permyak, Komi-Zyrian, Mari (Hill), Moksha, Nganasan, Udmurt, [**Tucanoan**] Desano, Tucano, [**Kadu**] Krongo, [**Central Sudanic**] Lugbara, [**Indo-European**] Bengali, Nepali, Ossetic, Polish, Serbo-Croatian, Slovenian, [**Pama-Nyungan**] Ngiyambaa, [**Arawakan**] Baniwa, Piapoco, Tariana, [**Northwest Caucasian**] Kabardian, [**Tupian**] Urubu-Kaapor, [**Arawan**] Kulina, [**Bunuban**] Gooniyandi, [**Na-Dene**] Koyuokon, [**Eskimo-Aleut**] Central Alaskan Yupik, [**Austronesian**] Mangap-Mbula, [**Austro-Asiatic**] Kharia, [**creoles/pidgins**] Berbice Dutch Creole, Mauritian Creole, Reunion Creole, Tok Pisin, [**isolates**] Ainu, Nivkh.

(62) *Languages with an affixal/clitic definite article:*

[**Afro-Asiatic**] Margi, [**Mande**] Bambara, Kpelle, Mandinka, [**Sino-Tibetan**] Limbu,

[Dogon] Tommo So, **[Uralic]** Hungarian, **[Indo-European]** Bulgarian, Sinhala,
[Songhay] Koyraboro Senni.

2.5.2 99 articleless languages

I start with the languages and language families in the sample that are more or less uncontroversially articleless. This is followed by a discussion of a number of language families in which at least one language has been claimed to have a definite article. I show that these languages are, in fact, articleless. Finally, I discuss three language families—Indo-European, Austronesian and Sepik—in which it is possible to compare the expression of associative plurality in languages with and without definite articles. We will see that in related languages that differ with respect to their article status, all of the languages with free-standing definite articles have a *non-plural pattern* associative plural. These languages provide very strong empirical evidence that the absence of definite articles is a pre-requisite for being a *plural pattern* language.

Turkic, Mongolic, and Tungusic languages

All Turkic languages lack definite articles (Johanson 1998). As for the Tungusic languages, definiteness in Evenki, Manchu and Udihe may be marked on direct objects by the accusative case suffix, but there are no articles.¹⁶ Nouns are not marked for definiteness in Nanai, with bare NPs receiving both definite and indefinite interpretations (63).

- (63) **morin** -Ø tutuini / njoani **morin** -jjaji tuxəni Nanai
horse -NOM run / he horse -ABL fall
'A horse is running.' / 'He fell from **the** horse.' (Ko and Yurn 2011, pp. 30, 33)

¹⁶See Nedjalkov (1997, pp. 83, 191) for Evenki, Zikmundová and Wa (2024, p. 471) for Manchu, and Perekhval'skaya and Janhunen (2024, p. 337) for Udihe.

Khalkha, the only Mongolic language in the sample, does not have a definite article (Dryer 2013; Poppe 1954).

TABLE 2.1: Absence of articles in Turkic, Tungusic and Mongolic

Family	Languages	References
TURKIC	Bashkir, Chuvash, Crimean Tatar, Gagauz, Karakhay-Balkar, Kazakh, Kumyk, Mishar Tatar, Tofa, Turkish, Uyghur, Uzbek	Johanson (1998, p. 38)
TUNGUSIC	Evenki Manchu Nanai Udihe	Nedjalkov (1997, passim) Gorelova (2002, passim) Ko and Yurn (2011, passim) Nikoleva and Tolskaya (2001)
MONGOLIC	Khalkha	Dryer (2013), Poppe (1954)
Total: 17		

East Caucasian, Japonic

The sample also consists of 6 East Caucasian languages and 5 Japonic languages. According to Daniel and Lander (2011), all Caucasian languages, with the exception of Armenian and Abkhaz, lack definite articles. In Table 2.2 I have provided additional references for the absence of definite articles in individual East Caucasian languages.

TABLE 2.2: Absence of articles in East Caucasian

Family	Languages	References
EAST CAU-CASIAN	Aghul	Becker (2021, p. 23)
	Archi	Dryer (2013)
	Bagvalal	Fedden and Corbett (2017, passim)
	Lak	Friedman (2020, p. 216)
	Lezgian	Haspelmath (1993, passim)
	Rutul	Polina (2019, p. 7)
Total: 6		

Japonic languages also lack definite articles (Table 2.3).

TABLE 2.3: Absence of articles in Japonic

Family	Languages	References
JAPONIC	Iheya	Carlino (2022, passim)
	Japanese	Hinds (2003, passim)
	Kin	Tamamoto (2022, passim)
	Shiiba	Shimoji and Hirosawa (2022, passim)
	Yoron-Ryukyuan	Kibe et al. (2018, passim)
Total: 5		

Sino-Tibetan languages

All of the Sino-Tibetan languages, with the exception of Limbu (to be discussed in Section 2.5.3), lack definite articles (see Table 2.4). Dryer (2013) categorizes Lepcha as having a definite word distinct from the demonstrative. This is based on Mainwaring (1876, p. 23), who describes two definite articles: *re* and *mu*.

- (64) pa-no **re** / kát **mu**
king the / one the
‘the king / the one’

Plaisier (2006), on the other hand, says that *mu* is not a definite article but an individuating suffix, and that the definite article *re* is suffixal. However, Plaisier also notes that bare NPs (NPs without *-re*) receive a definite interpretation (65a) and that it can occur twice in a single NP (65b). Since definite articles are unique, and their presence is required for definite interpretation in languages that have them, I conclude that Lepcha does not have a definite article.¹⁷

- (65) a. ?óng -re -m **kajú** -nun -do tsúk ma
 boy -DEF DAT **dog** -ABL -self bite AST
 ‘The dog bit the boy.’
- b. maró -**re** -**re** -m
 man -DEF -DEF DAT
 ‘to that very man’

(Plaisier 2006, pp. 64–5)

TABLE 2.4: Absence of articles in Sino-Tibetan

Family	Languages	References
SINO-TIBETAN	Belhare	Bickel (2017, passim)
	Burmese	Wheatley (2017, p. 817)
	Chintang	Michailovsky (2017b, p. 651)
	Mandarin Chinese	Wiedenhof (2015)
	Chantyal	Noonan and Hildebrandt (2017, passim)
	Hayu	Michailovsky (2017a, passim)
	Lahu	Matisoff (1973, p. 65)
	Lepcha	Plaisier (2006, pp. 64–5)
	Magar	Gunrow-Harsta (2011, passim)
	Newar (Dolkha)	Genetti (2017, passim), LaPolla (2017, p. 46)
	Newar (Kathmandu)	Hargreaves (2017, p. 461) LaPolla (2017, p. 46)
	Tamang	Dryer (2013)
Total: 12		

¹⁷If it turns out that Lepcha is better characterized as an affixal article language, this will not affect the typological generalization in Section 2.5.

Niger-Congo languages

There are 12 Niger-Congo languages in the language sample. Dryer (2013) categorizes two of them—Sango and Fulfulde (Adamawa)—as having a definite word distinct from the demonstrative. However, on closer examination these languages do not have articles. The article status of Fulfulde is attributed to Arnott (1970, pp. 21, 138). Arnott specifies three “referentials”—*din*, *nden* and *ɲgel* (which one is used depends on the class of the noun)—that he says often correspond to the definite article in English (see (66a)). However, there are numerous examples of definite NPs without one of these referential markers in Fulfulde (see (66b)), a property only found in articleless languages. Taylor (1921, p. 21) notes this too, stating that there are no articles in Adamawa.

- (66) a. na'i boḏeeji **din**
cattle red the
'the two red cattle' (Arnott 1970, p. 21)

- b. na'i jām / deptere / laamḏo
cattle well / book / chief.CL.C
'**the** cattle are well' / '**the** book' / '**the** chief'
(Arnott 1970, pp. 31, 149, 162)

Thus, whatever these “referentials” in Fulfulde Adamawa are, they do not have the distribution of a definite article.

Sango is classified as having a definite article *ní*, and Samarin (1963, p. 38) calls it a restrictive or anaphoric determiner. However, *ní* is not required for definite interpretation (67). Once again, whatever the determiner *n'i* is in Sango, it does not have the distribution of a definite article.

- (67) mbi tô **susu** tɔ -ngo
1SG fry fish fry -NOM
'I fried **the fish**.' (Pasch 1996, p. 224)

The rest of the Niger-Congo languages in the sample lack definite articles (Table 2.5).

TABLE 2.5: Absence of articles in Niger-Congo

Family	Languages	References
NIGER- CONGO	Akan	Welmers (1946, p. 47)
	Dagbani	Olawsky (2003)
	(Eastern) Dan	Vydrin (2020, p. 459)
	Ewe	Ameka (1991, passim), Dryer (2013)
	Fulfulde (Adamawa)	Taylor (1921, p. 21)
	Luganda	Baertlein and Ssekitto (2014, p. 2)
	Luvala	Horton (1949, p. 187), Dryer (2013)
	Sango	Pasch (1996, passim)
	Sesotho	Mojapelo (2007, p. 72)
	Tswana	Creissels (2022, passim)
	Yoruba	Bamgbose (1966, passim)
	Zulu	Dryer (2013)
Total: 12		

Uralic

There are 8 Uralic languages in the sample. Hungarian has a definite article and will be discussed in Section 2.3.3. Of the remaining 7, it has been observed for 6 of these languages—Khanty, Komi (Permyak and Zyrian), Mari, Nganasan, and Udmurt—that possessive marking functions like a definiteness marker (Gerland 2014). (68) illustrates this in Nganasan, Komi and Mari. The noun is suffixed with a third person singular possessive marker, but the interpretation is *non-possessive* definite.

- (68) a. Məu -δu Nganasan
earth -3SG
‘**the** earth’
- b. vɛr -as Komi
forest -INE.3SG
‘in **the** forest’

c. awa -**se**
 mother -3SG
 ‘**the** mother’

Mari

(Gerland 2014, pp. 271, 276)

However, Gerland (2014) argues that the possessive suffixes in Uralic are not definite articles based on three properties of grammaticalized definite articles in Indo-European: (a) morphological reduction/phonetic erosion, (b) semantic bleaching/extension, (c) obligatoriness. Thus, a typical Indo-European definite article is often a morphologically reduced form of a demonstrative (property (a)) that has lost its deictic function (property (b)) and is required for definite interpretation (property (c), see also (55iii)). The possessive suffixes in Uralic, in contrast, are not morphologically reduced, have retained their possessive function (i.e., they have not undergone semantic bleaching), and are not obligatory for definite interpretation, as shown by Komi in (69).

(69) pop lokt -i.s
 priest come -PRT.3SG
 ‘**The** priest arrived.’

(Gerland 2014, p. 281)

Following Gerland (2014), I thus do not treat the possessive suffixes as definite articles in Uralic.¹⁸ This is also in line with Hajdú (1983), who argues that (barring Hungarian) Uralic languages lack definite articles.

¹⁸See also Becker 2021, pp. 109–114.

TABLE 2.6: Absence of articles in Uralic

Family	Languages	References
URALIC	Khanty, Mari (Hill), Nganasan, Udmurt, Komi-Permyak, Komi-Zyrian Moksha	Gerland (2014), Becker (2021), Hajdú (1983) Hajdú (1983), Pleshak (2021)
Total: 7		

Arawakan

3 Arakwakan languages are in the sample—Baniwa, Piapoco and Tariana. Baniwa lacks definite articles; third person pronouns may be used to indicate definiteness (70a) but this marking is not obligatory for a definite interpretation on almost all of the Baniwa nouns in Aikhenvald (1995). Some examples are given in (70b).

- (70) a. **h̥ia** ita
 he canoe
 ‘the canoe’(Aikhenvald 1995, pp. 163, 170)
- b. pimi / mari / hekwapi / pedze / itfida
 colibri (a small bird) / heron / day / vulture / turtle
 ‘**the** colibri’ / ‘**the** heron’ / ‘**the** day’ / ‘**the** vulture’ / ‘**the** turtle’

Piapoco also lacks definite articles. According to Klumpp and Burquest (1983), Piapoco nouns can be modified by demonstratives or possessive pronouns (71a), but bare NPs also have a definite interpretation (71b).

- (71) a. **yái** Felife
 3M.DEM Phillip
 ‘that Phillip’
- b. capii
 house
 ‘**the** house’(Klumpp and Burquest 1983, pp. 389, 392)

According to Dryer (2013), Tariana has a definite word distinct from the demonstrative, citing Aikhenvald (2003). Aikhenvald mentions a specifier article, *diha*, which is described as marking “specific and/or definite referents” (72a). However, the specifier article can also occur with indefinite NPs (72b), suggesting *diha* is not a definite article. Confirming this view, Aikhenvald (2010, p. 21) states that Tariana lacks definite articles.

- (72) a. **diha** yawi
 ART jaguar
 ‘the jaguar’
- b. **diha** hema hipa -ma -pe nu -ka diha hinipu -nuku
 ART tapir leg.INDEF -CL:PAIR -PL 1SG -see ART road -TOP.NON.A/S
 ‘I see **a tapir**’s footprints on the road.’

Bunuban

The only Bunuban language in the sample is Gooniyandi, which Dryer (2013) categorizes as having a definite word distinct from the demonstrative. This is based on McGregor (1990, p. 144), who lists two “endophoric determiners” *niyaji* ‘this’ and *niyi* ‘that’. However, these determiners are not required for a definite interpretation of the NP, suggesting that they are not definite articles.

- (73) waya thiddgirli -windi
 wire straight -it.got
 ‘**The** wire got straight.’ (McGregor 1990, p. 154)

Northwest Caucasian

The language sample contains one Northwest Caucasian language: Kabardian. Dryer (2013) categorizes Kabardian as having a definite affix. However, according to Colarusso (1992), Kabardian distinguishes between referential and non-referential rather than definite and indefinite. Furthermore, referential marking is not a unique definite suffix but case

marking. Thus, the case ending in (74a) marks the noun as referential, but it can be definite or indefinite. Various case endings have a definite (or referential) function in Kabardian (74b). These are not definite articles.

- (74) a. $\lambda'\text{ə}$ -**r**
 man -ABS
 'the man / a man / some man' (Colarusso 1992, p. 56)
- b. pśāśa -**m**
 girl -ERG
 'the girl' (Matasović 2010, p. 23)

Indo-European

There are 8 Indo-European (IE) languages in the sample. With the exception of Bulgarian and Sinhala (to be discussed in Section 2.3.3), all of them lack definite articles.

Family	Languages	References
INDO- EUROPEAN	Bengali	Thompson (2012, p. 50)
	Nepali	Riccardi (2003, p. 605), Poudel (2017, p. 75)
	Ossetic	Erschler (2019, p. 880)
	Polish	Sadowska (2012, p. 209), Dryer (2013)
	Serbo-Croatian	Jasenovic (1983, p. 21), Dryer (2013)
	Slovenian	Marušič and Žaucer (2006)
Total: 6		

It is worth dwelling on the fact that not only is the *plural pattern* extremely rare in Indo-European (given the number of IE languages and the extent of their description), but so are associative plurals more generally. Associative plurals are almost entirely absent from Germanic languages (found only in Afrikaans, Icelandic and Norwegian), and totally absent from Romance languages, which overwhelmingly have free-standing definite articles. That such a well-attested phenomenon is a rarity in a subset of IE languages with free-standing

definite articles—but is found in related articleless and affixal article languages—strongly suggests that associative plurality is constrained, in part, by a language’s article status.

Even more importantly for the typological generalization is that Afrikaans—the only IE language that I am aware of with a free-standing definite article and an associative plural—shows the non-plural pattern.¹⁹ The associative plural in Afrikaans is the third person plural pronoun *hulle* (75).

- (75) Pa **-hulle** / my kinder -s **-hulle**
 Dad -them / my children -PL -them
 ‘Dad and his folks’ / ‘my children and others’

That related languages which differ in their article status also systematically differ in their expression of associative plurality is particularly strong evidence that the absence of definite articles is a pre-requisite for being a *plural pattern* language. Austronesian, which I turn to immediately below, is a particularly striking example of this.

Austronesian

Mangap-Mbula is the only Austronesian language in the sample, and it too lacks a definite article: there is no (non-demonstrative) definite marking on any nouns (76).

- (76) Puulu kembei ta zɔŋ
 moon like SPEC sun
 ‘**The** moon is like **the** sun.’ (Bugenhagen 1995, p. 152)

Even more so than Indo-European, a comparison of associative plurals in Austronesian provides compelling evidence for the typological generalization. I am aware of 7

¹⁹Daniel and Moravcsik (2013) list German and Frisian as having a unique periphrastic associative plural, and both have free-standing definite articles. However, I put these languages aside because their “associative plural” takes the form ‘X and them’ just like English “John and them”. I treat these as simple coordination structures which are distinct from grammaticalized associative plurals.

Austronesian languages with an associative plural that also have a free-standing definite article: Hawaiian, Dupaninan Agta, Malagasy, Manadonese (or Manado Malay), Tuvaluan, Rapanui and Chamorro.²⁰ What is striking is that all of these languages have a *non-plural pattern* associative plural. For example, Malagasy nouns are not inflected for number. A plural demonstrative before an unmarked noun indicates plurality. Before a proper name, the plural demonstrative has an associative function. In Chamorro, the associative plural is a prenominal particle *ha*, which is preceded by the definite article *si*. The regular plural is *siha*. Although the two are homophonous, they are compositionally distinct (for a more detailed discussion of these languages, see the Appendix).

(77) *Malagasy* [Austronesian]

a. **ireo** zaza
DEM.PL child
'the children'

b. **ry** Ratsimba
DEM.PL Ratsimba
'the Ratsimba family'

(Paul 2009, p. 10)

(Paul 2018, p. 329)

(78) a. **si ha** Pedro
DEF APL Peter
'Peter and friends/companions'

b. **chalan / chalan siha**
road / road PL
'road' / 'roads'

(Safford 1903, pp. 298–304)

The associative plural in Hawaiian is *mā* (79a)²¹ and the regular plural is *mau* (79b).

²⁰For the article status of Hawaiian, see Elbert and Pukui (1979, p. 42); for Dupaninan Agta, see Becker (2021, p. 365); for Malagasy, see Bennett (1986, p. 41); for Manadonese, see Shiohara and Jukes (2018, p. 116); for Tuvaluan, see Besnier (2000, p. 367); for Rapanui, see Becker (2021, p. 278); for Chamorro, see Safford (1903, p. 298) (the article status of Chamorro is actually disputed (see Bošković 2016, fn. 10)).

²¹Mauri and Sansó (2021) suggest that this particle can be traced back to Proto-Oceanic *MA.4, 'and/with'.

(79) a. Lakou **mā**
 them APL
 ‘they and the others’

b. kēia **mau** wāhine
 DEM PL woman
 ‘these women’

(Elbert and Pukui 1979, pp. 129, 163)

The Manadonese associative plural is *dong*, the third person plural pronoun. In Tuvaluan, the associative plural is *saa*, a noun meaning ‘group’. Lastly, the associative plural in Rapanui is *kua*, which Feu (1996) glosses as ‘persons group’.

(80) a. Anis **dong**
 Anis 3PL
 ‘Anis and associates’

Manadonese

(Daniel and Moravcsik 2013)

b. **saa** Sinaa
 APL Sina
 ‘Sina and her group’

Tuvaluan

(Besnier 2000, p. 364)

c. **kuá** Rui
 APL Rui
 ‘Rui and the others’

Rapanui

(Feu 1996, p. 94)

The expression of associative plurality in Austronesian thus provides strong evidence that the *plural pattern* is constrained by a language’s article status.

Sepik

Another language family that confirms the typological generalization is Sepik. Awtuw and Manambu are the only Sepik languages in the sample. Manambu lacks definite articles (Aikhenvald 2008, p. 507). According to Dryer (2013), citing Feldmen (1986, p. 116), Awtuw has a free-standing definite article. However, what Feldmen describes as determiners in Awtuw are kinship terms (81a), pronouns (81b), quantifiers (81c) and possessors (81d).

- (81) a. **eywe** Wiltiw
 ancestor Wiltiw
 ‘Grandpa Wiltiw’
- b. **tey** tale
 3.F.SG woman
 ‘the woman’
- c. **kokot** æwre
 all house
 ‘every house’
- d. **nom -ke** yaw
 1PL -POSS pig
 ‘our pig’

Of these, only the third person pronoun in (81b) could plausibly be a definite article (Elbourne 2001). However, the third person pronoun is not required for a definite interpretation of NPs in Awtuw, suggesting that they are not definite articles.

- (82) wiytape-ke / æye / uy / yiyte
 river-LOC / food / hole / fence
 ‘at **the** river’ / ‘**the** food’ / ‘**the** hole’ / ‘**the** gate’

(Feldman 1986, pp. 16, 44, 55, 71)

Thus, both Awtuw and Manambu lack definite articles and they both show the *plural pattern* in their associative plural. Another Sepik language with an associative plural is Alamblak. Unlike Awtuw and Manambu, Alamblak does have a free-standing definite article (Bruce 1984). Crucially, it also shows the *non-plural pattern* in its associative plural — the associative plural marker is the third person plural pronoun and the regular plural (on third person nouns) is *-m* (83).

(83) a. Yoni **rēm**
 Yoni 3PL
 ‘Yoni and his associates’

(Mauri and Sansó 2017, p. 2)

b. yima-**m**
 person-PL
 ‘people’

(Bruce 1984, p. 96)

Once again, the article status of these related languages directly implicates the availability of the *plural pattern* in the associative plural.

Remaining languages

So far I have covered 73 of the 99 articleless languages in the sample, focusing on particularly well-represented language families and individual languages that required some additional discussion. The remaining languages in the sample—small numbers of languages from a single language family, creoles/pidgins and isolates—are given in Table 2.8.

TABLE 2.8: Absence of articles in the remaining languages

Family	Languages	References
TACANAN	Araona Cavineña Ese-Ejja	Dryer (2013) Guillaume (2008, p. 453) Vuillermet (2012, pp. 4, 298)
CHUKOTKO-KAMCHATKAN	Chuckchi Alutor	Dunn (1999, passim) Nagayama (2016, passim)
NADAHUP	Hup	Epps (2008, passim)
UTU-AZTECAN	Comanche	Charney (1993, passim)
CARIBAN	Apalai Hixkaryana	Koehn and Koehn (1986, passim) Derbyshire (1979, passim)
SALISHAN	Kalispel	Dryer (2013)
TUCANOAN	Desano Tucano	Lima Silva (2012, pp. 175–6) Dryer (2013)
KADU	Krongo	Reh (1983, passim)

CENTRAL SUDANIC	Lugbara	Crazzolara (1960 , p. 17)
PAMA- NYUNGAN	Ngiyambaa	Dryer (2013)
TUPIAN	Urubu-Kaapor	Dryer (2013)
ARAWAN	Kulina	Dienst (2014 , passim)
NA-DENE	Koyuokon	Thompson (1989 , passim)
ESKIMO- ALEUT	Central Alaskan Yupik	Reed (1977 , p. 40)
AUSTRO- ASIATIC	Kharia	Peterson (2014 , passim)
CREOLES/ PIDGINS	Berbice Dutch Creole Mauritian Creole Reunion Creole Tok Pisin	Kouwenberg (2007 , p. 447) Syea (2007 , p. 3) Albers (2020 , passim) Verhaar (1995 , p. 160)
ISOLATES	Ainu Nivkh	Bugaeva (2022), Masayoshi (1990 , passim) Gruzdeva (2020 , p. 14)
Total: 26		

2.5.3 Definite article languages

I turn now to the languages in the sample that have definite articles. These are:

- (84) Bambara, Bulgarian, Sinhala, Tommo So, Hungarian, Koyraboro Senni, Limbu, Margi, Kpelle, Mandinka.

In the following sections, I will discuss the article status of these languages. Where the article status is unclear from the literature, I show that the definite article in these languages behaves like a bound (affixal or clitic) element.

Koyraboro Senni, Limbu, Sinhala, Margi

The definite articles in Koyraboro Senni, Limbu, Sinhala and Margi are all suffixes. The definite article in Koyraboro Senni is *-oo/-aa* (DEF.SG) or *-ey* (DEF.PL). This is shown in (85a–b).

- (85) a. woy **-oo**
 woman -DEF.SG
 ‘the woman’
- b. woy **-ey**
 woman -DEF.PL
 ‘the women’
- (Heath 1999, p. 128)

Vowel contraction rules indicate that the definite article in Koyraboro Senni is suffixal. Koyraboro Senni has what Heath (1999) calls Phrasal VV-contraction, whereby a stem-final vowel is deleted when followed by a vowel-initial word. That is, Phrasal VV-contraction applies across words. This is shown in (86a), which involves a pronominal possessor and a possessed noun. However, Phrasal VV-contraction does not apply if the stem-final vowel is high and the following word starts with a low vowel. Instead, the stem-final high vowel is replaced by its corresponding semi-vowel (/w/ for /u, uu/ and /y/ for /i, ii/). This is shown in (86b).

- (86) a. **ŋga iz-oo** → /ŋgi:zo/
 3SGF child-DEF.SG
 ‘her child’
- b. a **buu a** ra → /abwa:ra/
 3SG.S die 3SG LOC
 ‘He died of it.’

Koyraboro Senni also has a Suffixal VV-contraction rule, whereby a stem-final short vowel is deleted when followed by a vowel-initial suffix. This is shown in (87a), in which

the stem *deke*, ‘put up’ becomes *dek* after the causative suffix *-andi* is attached. This rule is word-internal, and unlike Phrasal VV-contraction it applies even if the stem-final vowel is high and the suffix-initial vowel is low, as shown in (87b). Crucially, the definite articles in Koyraboro Senni behave like suffixes when it comes to vowel contraction: a stem ending in a high vowel followed by the definite article *-oo* or *-ey* (both of which start with low vowels) is deleted; it is not replaced by a semi-vowel. Thus, the definite articles behave like suffixes.

- (87) a. *deke* → *dek* -*andi*
 put up put up -CAUS
 ‘cause to mount’

- b. *deeši* → *deesh* -*andi*
 fly fly -CAUS
 ‘cause to fly’

(Heath 1999, p. 40)

- c. *hasu* → *has* -*oo*
 tree tree -DEF.SG
 ‘the tree’

(Heath 1999, p. 67)

The definite article in Limbu is *-ʔin* (88a).²² That the definite article is bound (specifically, a suffix) is first evidenced by the fact that Limbu syllable structure only allows /ʔ/ as an onset word-internally (Driem 1987, p. 16).²³ The definite article also participates in glottal-nasal assimilation (88b). According to Hildebrandt (2007), this assimilation occurs across morpheme (not word) boundaries. Other suggestive evidence that the definite article is not free-standing in Limbu is that it triggers a restricted kind of vowel harmony. When

²²The definite article *-ʔin* is used in definite absolutive case contexts. Following Driem (1987) I treat this suffix as an article, and not a case ending, because it forms part of other case suffixes in definite contexts, e.g., indefinite ergative *-le*, definite ergative *-ʔille* (i.e., *-ʔin* plus *-le*). Despite this, Driem glosses the definite article as absolutive (ABS). I have altered the glossing to DEF for clarity.

²³In addition to syllable structure, the other word-hood test discussed by Driem is hiatus. Hiatus inserts a glottal stop before a vowel word-initially, and intervocally word-internally. Since the definite article begins with a consonant it will never participate in hiatus, and thus this test is uninformative.

attached to any derivational suffix ending in *-a*, the definite article fronts the *-a* to *-ε* (and is itself reduced to *-n*) (88c). It also triggers fronting of /e/ in the genitive suffix to /ε/ (88d).

- (88) a. phak -**ʔin**
 pig -DEF
 ‘the pig’
- b. waʔdhin -**nin** cεʔ!
 egg -DEF eat
 ‘Eat the egg!’
- c. yumlakpε -**n** haɲd -ε -i mε -haɲs ε -n -ni?
 acār -DEF be.hot -PT -Q NEG -be.hot -PT -NEG -Q
 ‘Is the acār spicy hot or isn’t it?’
- d. kəŋ kheŋ məna -rε -**n**
 this that man -GEN -DEF
 ‘This is that man’s.’
- (Driem 1987, pp. 36–7, 108)

The definite (singular) article in Sinhala is *-a/-ə/-i* (89a).²⁴ We can tell that this definite article is affixal because it also participates in word-internal phonological processes. For example, when a vowel-final stem is followed by a vowel-initial suffix in Sinhala, a semi-vowel is inserted; *-y-* if the final vowel is *i*, *e*, or *æ*, *-w-* if the final vowel is *u*, *o*, or *a* (Chandralal 2010, p. 91). This is illustrated in (89b) using two nouns inflected with the nominative singular case suffix *-a*. Definite NPs that end in one of these vowels also insert the relevant semi-vowel (89c), suggesting that they are suffixal.

- (89) a. gaɲg -ə
 river -DEF
 ‘the river’
- b. ball -**a** / gowi -**y** -**a**
 dog -NOM / farmer -SV -NOM
 ‘dog’ / ‘farmer’
- (Garland 2006, pp. 7, 22)

²⁴Which suffix is used is determined by animacy, with inanimate nouns only using *-ə*.

- c. baaldi / baaldi -y -ə
 bucket / bucket -SV -DEF.SG
 ‘bucket’ / ‘the bucket’

(Sultan 2006, p. 132)

The definite article in Margi is *-ári*. Showing that this morpheme is a suffix, rather than free-standing element, requires a short diversion into the rules of Margi phonology. When a sequence of vowels arises in Margi, a phonological process of Vowel Elision deletes the first vowel.²⁵ This is illustrated in (90). Vowel Elision applies at the juncture of a stem and its suffix (90a), across words (90b), and between a noun and the definite article (90c). Thus, this rule alone cannot tell us if the definite article is suffixal or not.

- (90) a. là + árí → làrí
 dig -DEF
 ‘to dig’
- b. ómckà óskwàr → ómcká skwàr
 soup pot
 ‘soup pot’
- c. fà ári → fǎrí
 farm DEF
 ‘the farm’

(Tranel 1993, pp. 139, 141–2)

However, Vowel Elision also creates stranded tones. In (90a), the stem ends in a low tone vowel (/à/) and the verbal suffix is vowel-initial, carrying high tone (/á/). Vowel Elision deletes the first vowel, stranding the low tone (91a). Tranel (1993) argues that a tone sandhi rule, L-association, rescues the stranded low tone. L-Association attaches a stranded low tone to a high tone vowel on its right, and concurrently delinks the high tone from this vowel. As a result, the initial vowel of the suffix bears low tone (91b).²⁶

²⁵There is a peculiarity here which is that in sequences [/a/ + /ə/] it is /a/ that remains after Vowel Elision, despite it being the first vowel in the sequence. Tranel (1993, pp. 145–6) offers an explanation for this that maintains V1 deletion.

²⁶As for why the vowel that surfaces is /a/, see footnote 25.

- (91) a. Vowel Elision: $l\grave{a} + \acute{o}r\acute{i} \rightarrow l\grave{ } + \acute{o}r\acute{i}$
 b. L-association: $l\grave{ } + \acute{o}r\acute{i} \rightarrow l + \grave{a}r\acute{i}$

Something slightly different happens when Vowel Elision applies between a noun and the definite article. As with (90a), the noun ends in a low tone vowel (/à/) and the definite article is vowel-initial, carrying a high tone (/á/). Vowel Elision deletes the first vowel—stranding the low tone (92a)—and L-association applies (92b). This time, however, delinking of the high tone during L-association strands the high tone. In (91a), the verbal suffix has two high tones—one on /ó/ and one on /í/. Tranel (1993) argues that this is actually multiple attachment of a single tone, and thus when the high tone is delinked it does not become stranded (it is still attached to /í/). The tonal melody of the definite article is HL—a (singly attached) high tone on /á/ and a low tone on /i/. Thus, when L-association applies, the singly attached high tone is stranded after delinking. Tranel argues that another tone sandhi rule, Rising-Tone Formation, applies to rescue the stranded high tone. This rule relinks the high tone, creating a rising tone (92c).

- (92) a. Vowel Elision: $f\grave{a} \acute{a}r\grave{i} \rightarrow f\grave{ } + \acute{a}r\grave{i}$
 b. L-association: $f\grave{ } + \acute{a}r\grave{i} \rightarrow f + \acute{ } \grave{a}r\acute{i}$
 c. Rising-Tone Formation: $f + \acute{ } \grave{a}r\acute{i} \rightarrow f\check{a}r\acute{i}$

Importantly, Tranel shows that the tone sandhi rule L-association does not apply across words. To see this, consider (90b), repeated below as (93a). The first word in (93a) is vowel-final and this vowel carries a low tone; the second word is vowel-initial and carries a high tone. Vowel Elision deletes the first vowel in the sequence, stranding the low tone (93b).

- (93) a. ómckà óskwàr → ómcká skwàr
 soup pot
 'soup pot'
- b. Vowel Elision: ómckà óskwàr → ómck` óskwàr

Tranel shows that when low tones are stranded in word-final position they do not migrate to the following word. For example, when the word-final low tone vowel ì in (94a) is deleted (via Vowel Apocope), shown in (94b), the tonal melody of the following word is not affected (i.e., /yé/ does not receive low tone).

- (94) a. àfól -érì yé
 dance -DER 1SG.SUBJ
 'I danced'
- b. Vowel Apocope: àfólér yé (Tranel 1993, p. 118)

Likewise, Vowel Elision and the subsequent low tone stranding in (93b) does not affect the tonal melody of the following word. As (93a) shows, the high tone on the overt vowel is maintained. If L-association had applied, this vowel should carry the stranded low tone. Instead, the low tone remained stranded and is unrealized. Tranel concludes, based on these data, that L-association is a word-internal tone sandhi rule. Crucially, since L-association must have applied for the rising tone to surface on the definite noun in (92c), the definite article must be part of a [stem + suffix] domain rather than a [word + word] domain.

Tommo So

The definite article in Tommo so is a clitic =gɛ. Both Plungian (1995) and McPherson (2013) describe this article as a clitic. Evidence for this comes from the fact that words (more specifically, stems) in Tommo So have underlying tone (either high (H) or low (L)) and are minimally bimoraic (McPherson 2013, p. 74). First, McPherson shows that the def-

inite article carries neither H or L tone, but is underspecified for tone. One piece of evidence she provides is that the article appears to carry H tone, sometimes L tone and sometimes neither. When the article is surrounded by syllables of the same tonal value (i.e., H=*ge* H, L=*ge* L) the pitch contour remains level and the article appears to carry H or L tone, respectively. Second, the definite article consists of a single mora while words in Tommo So must be at least bimoraic. In the case of monosyllabic words, bimoraicity is achieved either via a long vowel (CVV) (95a) or a sonorant coda (CVC) (95b). Monomoraic equivalents of the words in (95) are not found (e.g., **kú*).²⁷ The definite article is monomoraic, consisting only of a short vowel (CV) (95c). This is typical of clitics in Tommo So (McPherson 2013, p. 36). Finally, the definite article does not attach to the noun stem but to the end of the NP, as further indication of its clitic status (95c). Recall that the distinction between affixal and clitic articles is irrelevant for the typological generalization established in this chapter. What is important is that all languages that show the *plural pattern* in their associative plural either lack definite articles or have bound definite articles.

- (95) a. kúú
head
- b. gé̃m
black
- c. Jàndùlù gé̃m =gɛ
donkey black =DEF
'the black donkey'

(McPherson 2013, p. 133)

²⁷The exception to this is three verbal stems *gé* 'say', *yé* 'see', and *jé* 'take'.

Bulgarian

The definite article in Bulgarian is *-a*, *-to*, *-te*, or *-ta* depending on the gender, number and phonological shape of the noun (96).

- (96) knìga **-ta** / vìnò **-to** / učíteli **-te**
book -DEF.F / wine -DEF.N / teacher -DEF.PL
'the book' / 'the wine' / 'the teachers'

It is well-established in the literature that the Bulgarian article is bound, though there is a debate as to whether it is a suffix or a clitic (see e.g. Elson 1976; Franks 1998, 2001; Franks and King 2000; Harizanov 2014; Haspelmath 2023). Whether it is a suffix or a clitic does not matter for our typological generalization. Evidence of its bound status comes from stress shift and word-final devoicing (Harizanov 2014). Stress placement in Bulgarian can be affected by other morphemes within the same word. Crucially, the definite article attracts stem-initial stress, shown in (97), suggesting that we are dealing with a [stem + affix] structure here. Additionally, word-final devoicing is bled by suffixes, illustrated in (98b) with the plural suffix. The definite article also bleeds word-final devoicing (98c).

- (97) *hùbost* 'beauty' — *hubost-tà* 'the beauty'

- (98) a. *mǎž* → [mǎš] 'man'
b. *mǎž* + e_{PL} → [mǎže] 'men'
c. *mǎž* + a_{DEF} → [mǎža] 'the man' (Harizanov 2014, p. 122)

Kpelle, Bambara, Mandinka

According to Dwyer (1985), the definite article in Kpelle is *n̄-* (a prefix bearing low tone) (99a). Thach and Dwyer (1981) describe the article differently, saying that it is a floating low tone that is sometimes accompanied by *n*. When the low tone is placed on a syllable,

it triggers consonant mutation (99b). Dwyer (1985) analyzes cases like (99b) not as a floating low tone, but as the low tone *n*- prefix assimilating with the stem-initial consonant, such that the prefix is obscured. Thus, (99b) has undergone the phonological processes in (99c)—voicing spreads to the stem-initial consonant (/p/ → /b/) and gemination obscures the nasal altogether. The phonological processes that *n* triggers on the noun suggest that it is affixal.²⁸

- (99) a. wúlu / **ṇ**- wúlii
 tree / DEF- tree
 ‘tree’ / ‘the tree’
- b. péle / **b**éleí
 house / the house
- c. *n*-prefixation: **ṇ**- péle
 Voicing assimilation: **ṇ**- **b**éle
 Gemination: **ḃ**- **b**éle

In Bambara, the definite article is a floating low tone. According to Dwyer (1985) this suffix may be cognate with the nasal low tone prefixes in other Southwestern Mande languages (like Kpelle). The low tone of the definite article raises the tone of a preceding low tone noun (100a). I treat tone as an autosegmental feature, assigned to a segment that may correspond to a morpheme (root, affix) or word. Thus, in the case of the Kpelle definite article, low tone is assigned to the prefix *n*-. In Bambara, the low tone is assigned to a covert (i.e., phonologically deficient) segment. Because this element is prosodically deficient, the tone has to find an adjacent (overt) segment to attach to (100b). Thus, floating tones are often thought of as a kind of affixation (Bye and Svenonius 2012; Roberts 1994;

²⁸The article may also be a circumfix, sometimes appearing with a *-i* suffix as well (e.g. Thach and Dwyer 1981; Winkler 1997).

Wolf 2006), hence I treat the article in Bambara as an affix.²⁹

- (100) a. mùsò tè / mùsó tè
 woman NEG / woman NEG
 ‘It’s not a woman.’ / ‘It’s not **the** woman.’ (Bird et al. 1977, p. 22)
- b. mùsò + ` → mùsó tè

The definite article in Mandinka is -ò, and it adds a final low tone to its host (101a). The article varies in form depending on the phonological shape of the noun. After short vowels (*a, e, i, o, u*) and the long vowel *oo*, the article is lengthened to -*oo*. After the long vowel /*aa*/ the article is -*aa*. After the long vowels /*ee*, *ii*/ and /*uu*/ a semivowel is inserted (101b). Creissels (2013) notes that the way the article interacts with its host is typical of affixes in Mandinka. Despite this, Creissels calls the Mandinka definite article an enclitic because it behaves like one in its distribution—it attaches to the entire NP rather than the nominal stem (101b). Recall, however, that the affix/clitic distinction does not matter for our purposes.

- (101) a. sǔŋ / sùŋ -**ôo**
 thief / thief -DEF
 ‘thief’ / ‘the thief’
- b. sàtée ~ sàtée -**wòo** / jí ~ jí -**yo** / súu ~ sú -**wóo**
 village ~ village -DEF / water ~ water -DEF / house ~ house -DEF
- c. dǐndǐŋ -**o** / dǐndǐŋ màlùbál -**òo**
 child -DEF / child shameless -DEF
 ‘the child’ / ‘the shameless child’

It is worth noting that of the three Southwestern Mande languages discussed here, Bambara and Mandinka could also be characterized as articleless. The articles in these

²⁹Note that if the definite article in Kpelle is in fact a floating low tone (rather than *ŋ-*), it will also be affixal for the same reasons.

languages did, at one stage, systematically encode definiteness. However, Creissels (2013) notes that in Mandinka, the article only contrasts with its absence in negative clauses (cf. (102a-b)), interrogative clauses and NPs containing a numeral (see also Lewis (2021)). Outside of these contexts, an *o*-marked noun is ambiguous between definite and indefinite (102c). Similarly, Creissels (2007, fn:1) notes that the tonal modifications that distinguish definite from indefinite nouns in Bambara are only meaningfully contrastive in negative and interrogative contexts. These “definite articles” may then be becoming (or have already grammaticalized into) noun class markers whose only function is to identify an expression as nominal Greenberg (1977).

- (102) a. η má η mùs -ôo je
 1SG CPL.NEG woman -DEF see
 ‘I did not see **the** woman.’
- b. η má η mùsú je
 1SG CPL.NEG woman see
 ‘I did not see any woman.’
- c. η á mùs -ôo je
 1SG.CPL woman -DEF see
 ‘I saw the/a woman.’

(Creissels 2013, p. 13)

Hungarian

The last language to discuss is Hungarian, which is often said to have a free-standing definite article *a/az*, the latter being used before words beginning with a vowel (103).

- (103) a. **a** ház
 the house
- b. **az** esemény
 the event

(Rounds 2009, p. 76)

This makes Hungarian a conceivable counter-example to the typological generalization

that *plural pattern* languages lack free-standing definite articles. Nonetheless, following recent work by Oda (2022) and Bošković (2023), the goal of this section is to show that the definite article in Hungarian in fact behaves like a bound element and should be considered affixal (specifically, a prefix).

So far the bound status of definite articles has been determined based on phonological properties. One phonological argument for the prefixal nature of the Hungarian definite article given in MacWhinney (1976) concerns the morphophonemic alternation it shows depending on the shape of the root noun. That is, when the noun is vowel-initial, the definite article is *az*; when the noun is consonant-initial, the definite article is *a*. This alternation is typical of affixes. A potential counter-argument to the claim that the definite article in Hungarian is affixal is that it does not participate in some word-internal phonological processes, like Vowel Harmony (VH). Consider first (104a), in which the stem *kez* contains a front, unrounded vowel /e/ and vowel harmony ensures that the suffixes also contain front vowels. When the stem contains a back vowel, like /a/ (104b), the vowels in the suffixes harmonize in backness. In [word + word] compounds, vowel harmony does not apply (the vowels' [$\pm$ FRONT] values are mismatched) (104c). Similarly, the definite article does not harmonize with the vowels in the noun in (104d). This could thus be taken as phonological evidence that the Hungarian definite article does not form a word with its noun. However, this is not an argument against the affixal nature of the definite article. Vogel (1989) shows that vowel harmony fails to apply to prefixes in general in Hungarian.³⁰ Thus, the stem in (104e) contains back vowels (/a/ and /o/) but the prefix *fel-*, 'up' has only a front vowel (/e/). The definite article—if it is affixal—is a prefix, and thus behaves like other prefixes in this respect.

³⁰There are in fact more general differences of this sort between prefixes and suffixes; see Bobaljik and Wurmbrand (2001), Bogolmolets (2020), Booij and Rubach (1984), Fenger (2020), and Hyman (2008).

- (104) a. *kez -em -ben*
 hand -1SG.POSS -LOC
 ‘in my hand’
- b. *ház -am -ban*
 house -1SG.POSS -LOC
 ‘in my house’
- c. *halál büntetés*
 death punishment
 ‘death penalty’
- d. *az esemény*
 the event
- e. *fel- darabol*
 up- cut
 ‘cut up’

(Vogel 1989, p. 335)

The status of a definite article as affixal or free-standing can be tested directly by looking at phonological processes that are known to apply at the word-level. When phonological tests are not conclusive—e.g., the Hungarian definite article could be analyzed as a prefix or a free-standing element in the domain of Vowel Harmony—we can look to other domains for indirect evidence of affixhood. In particular, the PF status of the definite article (as an affix or a free-standing element) may in fact correlate with syntactic behavior. To see this, consider first a generalization established by Bošković (2005, 2012b) that only languages without definite articles allow Left Branch Extraction of adjectives (see also Corver 1992; Uriagereka 1988 and Section 3.2.1). Thus, extraction of an adjective is disallowed in English (a language with definite articles) (105), but allowed in Serbo-Croatian (a language without definite articles) (106a). LBE is also disallowed in Bulgarian, an affixal article language (106b). That is, the affixal article language Bulgarian patterns with the non-affixal article language English.

(105) *Expensive_i he saw [t_i cars]

(106) a. Skupa_i je vidio [t_i kola] Serbo-Croatian (Bošković 2012b, p. 1)
expensive is seen car

b. *Novata_i prodade Petko [t_i kola] Bulgarian (Bošković 2005, p. 3)
new-DEF sold Petko car

However, it has recently been shown that languages with affixal definite articles may also pattern like articleless languages. For example, Reuland (2011) and Despić (2011, 2015) observe that languages that have possessive reflexive pronouns either lack definite articles or have affixal definite articles. English, a language with a non-affixal definite article, does not have a reflexive possessive (107a). In contrast, Serbo-Croatian (an articleless language) and Danish (an affixal article language), have them (cf. (107b-c)).³¹

(107) a. *She loves **herself's** pets. (She loves **her** pets.)

b. Svaki dečak_i je video **svog**_i oca. Serbo-Croatian
every boy is seen self's father
'Every boy_i saw his_i father.'

c. John_i læste **sin**_i artikel. Danish
John read self's article
'John_i read his_i article' (Despić 2015, p. 209)

In fact, Talić (2017), Oda (2022), and Bošković (2023) show that there are a number of other cases where languages with affixal articles do not pattern with languages with non-affixal articles but with languages without articles. With regard to a number of these cases, Hungarian behaves like an affixal article language too. For example, regarding possessives, Hungarian the reflexive possessives *maga*, 'oneself' and *önmaga* 'themselves'

³¹For a more detailed discussion, including a deduction of these syntactic and semantic differences between articleless, affixal article and free-standing article languages, see Chapter 3.2.1.

(108a-b). Recall from above that reflexive possessives are only found in languages without definite articles or with affixal definite articles. Hungarian (along with a number of other affixal article languages) thus patterns with articleless languages when it comes to reflexive possessive pronouns.

- (108) a. A fiúk felfedzt -ék a **maguk** határ -a -i -t
 the boys discovered -3PL the themselves limit -POSS -PL -ACC
 ‘The boys discovered their (own) limits’
- b. A fiúk felfedzt -ék **önmaguk** határ -a -i -t
 the boys discovered -3PL themselves limit -POSS -PL -ACC
 ‘The boys discovered their (own) limits’

(Oda 2022, adapted from Rákosi 2020)

Another relevant generalization concerns indeterminate pronoun systems. Indeterminate pronouns are pronouns that have both indefinite and interrogative functions (Haspelmath 1997) and can be divided into two types (Watanabe 2004).³² In one type of indeterminate pronoun system, found in Chinese, the interrogative and indefinite pronouns share the same form (109). In the other type of indeterminate system, found in Japanese, the interrogative pronoun has an indefinite function when a particle is added to it (110). Oda (2022) calls the two types *bare indeterminate pronouns* and *compositional indeterminate pronouns* respectively.

(109) *Bare indeterminate pronouns*

- a. Tā bǎ **shénme** shū diū le?
 she ACC what book throw PFV
 ‘What books did she throw away?’
- b. Tā bǎ **shénme** shū diū le.
 she ACC what book throw PFV
 ‘She threw away a certain book.’

(Haspelmath 1997, p. 206)

³²Indeterminate pronouns may also have other functions, such as NPI, universal quantification and negative concord. These will not be relevant here and so I restrict the discussion to interrogative and indefinite forms.

(110) *Compositional indeterminate pronouns*

- a. Kare -wa watashi -ga **nani** -ga sukida to omotteiru no?
he -TOP I -NOM what -NOM like C think Q
'What does he think I like?'
- b. Kare -wa watashi -ga **nani -ka** -ga sukida to omotteiru.
he -TOP I -NOM what -KA -NOM like C think
'He thinks I like something.' (Oda 2022, p. 67)

Interestingly, Watanabe (2004) observes that languages with compositional indeterminate pronouns are found in languages without definite articles (he discusses Japanese, Russian and Latin). Latin provides diachronic evidence of this generalization: the Romance languages that descended from Latin (e.g., French) developed overt definite articles *and* lost their indeterminate pronoun system (cf. Tables 2.9 and 2.10).

TABLE 2.9: Indeterminate pronoun system in Latin

Latin	interrogative	indefinite
person	quis	ali-quis
thing	quid	ali-quid
place	ubi	ali-ubi
time	quando	ali-quando

TABLE 2.10: No indeterminate pronoun system in French

French	interrogative	indefinite
person	qui	quelqu'un
thing	quoi/que	quelque chose
place	où	quelque part
time	quand	(en quelque temps)

(Watanabe 2004, p. 394)

Following Watanabe's (2004) observation, Oda (2022) conducts a large cross-linguistic survey of indeterminate pronoun systems and finds that, of the 80 languages with productive compositional indeterminate pronouns, 68 lack definite articles and 11 have affixal definite articles, establishing the generalization that only languages without definite articles or with affixal definite articles have productive indeterminate pronouns. The remaining language is Hungarian. In Hungarian, whose indeterminate pronoun system is illustrated in Table 2.11, the sole exception to the generalization that languages that have productive compositional indeterminate pronouns either have affixal definite articles or lack definite articles — a generalization that holds for 80 typologically diverse languages?

TABLE 2.11: Indeterminate pronoun system in Hungarian

Hungarian	interrogative	indefinite
person	ki	vala-ki
thing	mi	vala-mi
place	hol	vala-hol
time	mikor	vala-mikor

(Watanabe 2004, p. 402)

Oda argues that it is not. Rather, the fact that Hungarian has compositional indeterminate pronouns is evidence that its article is prefixal.

A related piece of evidence comes from Bošković (2023) who, based on a typologically diverse group of 18 multiple *wh*-fronting (MWF) languages, shows that MWF languages all have a particular type of indeterminate pronoun system and, importantly, either lack definite articles or have affixal definite articles. MWF is allowed in Hungarian (111).

- (111) Kiváncsi vagyok hogy **ki** **mit** vett
I.wonder that who what.ACC bought
'I wonder who bought what.'
(Dikken 2003, p. 90)

Hungarian also fails to behave like a non-affixal definite article language with respect to adjunct extraction. Adjunct extraction is disallowed in English (202b). Stjepanović (1998) observes, however, that adjunct extraction is permitted in Serbo-Croatian and Russian, both of which lack definite articles (202a). Bošković (2012b), who also looks at Slovenian, Czech, Polish, Ukrainian, Hindi, Bangla, Angika and Magahi (which all lack articles) and Spanish, Icelandic, Dutch, German, French, Arabic and Basque (which all have definite articles), establishes the generalization that adjunct extraction is only allowed in languages without definite articles.

- (112) *From which city_i did John want to meet [professors t_i]?

- (113) Iz kojeg grada_i je Petar sreo [djevojke t_i]? Serbo-Croatian
from which city is Peter met girls

(Bošković 2012b, p. 3)

However, Dubinsky and Tasseva-Kurktchieva (2014) show that in Bulgarian, an affixal article language, adjunct extraction is allowed so long as the definite article is dropped

(114). For detailed discussion as to why the presence/absence of the article matters here, see Chapter 3.2. What is important now is that adjunct extraction is allowed in both articleless and affixal article languages, but never in non-affixal article languages.

- (114) a. *Ot koj universitet_i sreštna -ha nyakolko -to studenti t_i?
 from which university met -they several -the students
 ‘From which university did they meet several students?’
- b. Ot koj universitet_i sreštna -ha nyakolko studenti t_i?
 from which university met -they several students
 ‘From which university did they meet several students?’

(Dubinsky and Tasseva-Kurktchieva 2014)

Importantly, adjunct extraction is also possible in Hungarian. In the examples in (115), the adjunct *from which cities* has been extracted from an embedded NP. Note, too, that similarly to the Bulgarian judgments from Dubinsky and Tasseva-Kurktchieva (2014), the speaker finds the example with the definite article degraded.

- (115) a. ?Melyik városokból_i akart János [a professzorokkal t_i] találkozni?
 which cities.out.of want János the professors met
 ‘From which cities does John want to meet the professors?’
- b. Melyik városokból_i akart János [professzorokkal t_i] találkozni?
 which cities.out.of want János professors met
 ‘From which cities does John want to meet professors?’ (András Bárány, p.c)

Finally, Talić (2017) shows that affixal article languages may omit the article in weak definites. Weak definites are nominals that, in languages like English, appear with the definite article but do not carry the uniqueness presupposition typically attributed to definite nominals (Carlson et al. 2006; Schwarz 2014). For example, the definite NP in the English example (116a) does not refer to a unique hospital, but could in fact refer to various different hospitals. The set of nouns that can occur as weak definites varies cross-linguistically,

but typical examples include *the radio* (116b) and *the store* (116b). Talić shows that affixal article languages, like Icelandic, Bulgarian and Romanian can omit the definite article in a weak definite context. Thus, in Bulgarian the definite article is omitted in an NP like *the radio* with a weak definite interpretation (117). Omitting the definite article is not possible in English, a language with a free-standing definite article.

- (116) a. John and Mary are in **the hospital**. (John is at Manchester Memorial Hospital
and Mary is at Massachusetts General Hospital)
- b. John went to **the store**.
- c. Mary is listening to **the radio**.
- (117) (Toj) slusha radio. Bulgarian
(he) listens radio.
'He is listening to **the radio**.'

Crucially, Hungarian also omits the definite article in weak definites. This is shown in (118). The ability to drop the definite article when it is semantically superfluous—as in weak definites—is a property only of affixal article languages. This is particularly strong evidence that the definite article in Hungarian is a prefix.

- (118) a. Bolt -ba ment.
store -into go.PST.3SG
'He went to **the store**'
- b. Rádió -t hallgat.
radio -ACC listen.3SG
'He listens to **the radio**.'
- (András Bárány, p.c)

Hungarian thus behaves like an affixal article language for five typological generalizations: the presence of reflexive possessives, the presence of compositional indeterminate pronouns, MWF, the permissibility of adjunct extraction and omitting the article in weak

definites. We can add to this list the typological generalization that I have established in this chapter—*plural pattern* languages either lack definite articles or have affixal definite articles. To treat Hungarian as a non-affixal definite article language would require us to also treat it as the lone exception to all of these typological generalizations. On the other hand, treating Hungarian as an affixal article language means that it behaves exactly as we expect it to regarding the generalizations just discussed. This is, in my view, a better conceptualization of the facts.³³

2.6 Summary

This chapter has presented the largest cross-linguistic survey of languages with a *plural pattern* associative plural to date. These are associative plurals that use the regular plural as (or as part of) the associative plural. The survey consists of 109 languages from 33 language families, 4 creoles/pidgins and 2 isolates. Furthermore, I have shown that the expression of associative plurality—as *plural pattern* or *non-plural pattern*—correlates with another property of the nominal domain: the presence vs absence of free-standing definite articles. This led to the novel typological generalization in (119).

- (119) GENERALIZATION: If a language uses the regular plural as (or as part of) the associative plural (the *plural pattern*), it lacks free-standing definite articles (it either lacks definite articles or has affixal/clitic definite articles).

In the next chapter, I offer a new analysis of associative plurals that accounts for this generalization.

³³Note that the associative plural in Hungarian is a kind that appears to consist of regular plural morphology (-*k*) and dedicated associative plural morphology -*é*-. Thus, another option—if the reader is not convinced that Hungarian is an affixal article language—is to say that Hungarian is not a *plural pattern* language. That is, the -*k* of the associative plural is not the regular plural, but rather -*ék* is a monomorphemic dedicated associative plural. See Dékány (2021, p. 228) for some evidence in favor of this view.

2.A A note on excluded languages

Although the current sample includes the majority of languages also characterized as *plural pattern* languages by Daniel and Moravcsik (2013), 25 languages that Daniel and Moravcsik list as showing the *plural pattern* have been excluded. These are:

- (120) Bandi, Chamorro, Dani, Garo, Gbeya Bossangoa, Jamaican Creole, Kanuri, Khakas, Korean, Koryak, Kulung, Loko, Loma, Makah, Malagasy, Malayalam, Mupun, Ngan’gityemerri, Nias, Nobiin, Punjabi, Papiamentu, Pulaar, Takelma, Wichí.

This section discusses the reasons for their exclusion, which fall into four groups: (1) the language does not use regular nominal plural morphology as the associative plural (i.e., it is not a *plural pattern* language), (2) the language does not inflect for number on its nouns (and therefore is also not a *plural pattern* language), (3) the language does not have an associative plural, and (4) I could not verify the language as showing the *plural pattern* in its associative plural. I start with group (1).

Group (1): Regular plural $\neq$ associative plural

9 of the languages fall into group (1).

- (121) Bandi, Chamorro, Wichí, Korean, Garo, Loma, Malayalam, Kanuri, Makah.

Bandi has an associative plural using the clitic =*ni* (122a). Although Daniel and Moravcsik (2013) categorize =*ni* as the regular plural, according to Grossman (1992) the regular plural clitics are =*a* and =*tii*, and =*ni* has an “accompaniment” function. Outside of the associative plural, it is only found on examples like (122b), ‘X who is *with* Y’.

- (122) a. Kpehe =**ni**
 Kpehe =APL
 ‘Kpehe and his companions’ (Daniel and Moravcsik 2013)
- b. Offanko =**ni** ngaa Váani ti yé -ngo taa lí
 Offanko =ACCOMP REL Vanni they PROG -REM.PST PROG go
 ‘Offanko and Vanni were going’ (Grossman 1992, p. 31)

The Chamorro associative plural is a pre-nominal particle *ha* (123a). The definite article *si* is also used before proper names and titles. The regular plural is the word *siha* (123b). Thus, the Chamorro regular and associative plural look identical on the surface, but they are compositionally distinct. The regular plural is a non-segmentable word, and the associative plural is *ha* which happens to be preceded by an article, *si*.

- (123) a. si **ha** Pedro
 DEF APL Peter
 ‘Peter and friends/companions’
- b. chalan / chalan **siha**
 road / road PL
 ‘road’ / ‘roads’ (Safford 1903, pp. 298–304)

Wichí has an associative plural using the suffix *-lhayis* (124a). According to Terraza and Baito (2014, p. 214), *-lhayis* is a collective suffix, not a plural suffix (124b). The plural suffixes are *-j*, *-Vs*, *-tsel*, *-t* and *-haj* (124c).

- (124) a. Loida **-lhayis**
 Loida -APL
 ‘Loida and her husband’ (Daniel and Moravcsik 2013)
- b. late sej toj nech’e iche siwel **lhayis**
 CONJ CAUS.CONJ newly have white.man group
 ‘because the whites were barely there’ (Nercesian 2011, p. 453)
- c. halo **-j**
 tree -PL
 ‘trees’ (Terraza and Baito 2014, p. 213)

In Korean, two suffixes are said to be used to form the associative plural: *-tul* and *-ney*. *-ney* is a dedicated associative plural suffix and *-tul* is the regular plural (125a–b). However, the associative plural use of *-tul* is disputed. Kim (2005) says that *-tul* does not have an associative plural function (125c), and the Korean speakers I consulted did not accept *-tul* as an associative plural (they did accept *-ney*).³⁴ Thus, while Korean could in principle be included in the sample, I chose to exclude it because I could not verify that the regular plural has this function with native speakers.

- (125) a. Chelswu **-ney** -ka ikyessta
Chelswu -APL -NOM won
'Chelswu's team won' (Kim 2005, p. 51)

- b. haksayng **-tul** / Minca **-tul**
student -PL / Minca -PL
'students' / 'Minca and her friends'

(Kim (2005, p. 49), Daniel and Moravcsik (2013))

- c. Edison **-tul**
Edison -PL
'(many) Edisons' / *'Edison and his associates' (Kim 2005, p. 50)

According to Daniel and Moravcsik (2013), citing Burling (1961), the associative plural in Garo is *-maj* (126a), which is the same as the regular plural according to their classification.³⁵ I was unable to consult the original source, but Burling (2003) lists the associative plural as *-ong* (126b). He lists the regular plural suffixes as *-rang*, *-dang* and *-drang* (126c). Based on the information in Burling (2003), I excluded Garo from the sample.

- (126) a. najan **-ma'ŋ** -ni nok
Nangan -APL -POSS house
'the house of Nangan and his family' Daniel and Moravcsik 2013

³⁴See also Kim and Melchin (2018).

³⁵Daniel and Moravcsik (2013), again citing Burling (1961), say that Garo has a nominal plural clitic but do not provide an example.

- b. Ma -gip -a **-ong**
 Mother -APL
 ‘Mother and her people’

- c. nok **-drang**
 house -PL
 ‘houses’

(Burling 2003, pp. 179–80)

The regular plural and associative plural in the remaining languages—Loma, Malayalam, Kanuri and Makah—are given in Table 2.12. The regular plural in Loma is *-a* or *-iti*, and the associative plural is *-ni*.³⁶ In Malayalam, the regular plural is *-kaḷ* or *-maar*, depending on the gender of the noun and phonological properties of the stem. The associative plural is *okke*, ‘all’. The Kanuri associative plural is *-so*, which is the collective marker; the regular plural marker is *-wa*.³⁷ Finally, the regular plural in Makah has many forms, shown in Table 2.12, but none of them are the same as the associative plural, which is the clitic *=aṭ*.

TABLE 2.12: Regular and associative plural in Loma, Malayalam, Kanuri and Makah

Language	PL	APL	References
Loma	<i>-a</i> , <i>-(i)ti</i>	<i>-ni</i>	Wilhoit 1999, pp. 27, 32
Malayalam	<i>-kaḷ</i> , <i>-maar</i>	<i>okke</i>	Asher and Kumari (1997, pp. 125, 249)
Kanuri	<i>-wa</i>	<i>-so</i>	Hutchinson (1981, pp. 43, 208)
Makah	<i>-minḥ</i> , <i>-badaḥ</i> , <i>-aqa</i> , - <i>yu</i> , <i>-iḥ</i> , <i>-iḥ</i> , <i>-t-</i> , <i>-Vy</i> , <i>-ṭ-</i> , <i>-ṭa-</i>	<i>=aṭ</i>	Davidson (2002, pp. 206–9, 317)

³⁶Note that this is just like Bandi, a related Western Mande language.

³⁷Daniel and Moravcsik (2013) give (ia) as a regular plural, ‘mats’, using *-s* [sic]. However, the example in the original source is (ib), where the gloss is not ‘mats’, but ‘mats and all’. This is probably a simulative plural; Hutchinson (1981) also gives *fṛr -sò*, horse-COLL ‘horses etc’.

- (i) a. bə̀jì -s
 mat -PL
 ‘mats’

- b. bə̀jì -so
 mat -COLL
 ‘mats and all’ (Hutchinson 1981)

Group (2): Plural is not inflectional marking on the noun

6 languages fall into group (2).

(127) Malagasy, Mupun, Nias, Ngan'gityemerri, Jamaican Creole, Papiamentu.

Malagasy and Mupun nouns do not have inflectional plural marking on the noun. In Malagasy, the only way to indicate the plurality of a noun is by using a plural demonstrative or a quantifier (128a) (Paul 2009, 2015, 2018). A plural demonstrative before a proper name generates an associative plural (128b).

(128) *Malagasy* [Austronesian]

a. **ireo** zaza
DEM.PL child
'the children' (Paul 2009, p. 10)

b. **ry** Ratsimba
DEM.PL Ratsimba
'the Ratsimba family' (Paul 2018, p. 329)

Similarly, plurality in Mupun is expressed via the third person plural/demonstrative pronoun *mo* (129a). Using *mo* with a proper name generates an associative plural (129b).

(129) a. saar **mo**
hand 3PL
'hands'

b. James **mo**
James 3PL
'James and company' (Frajzyngier 1997, p. 201)

Although the mechanism for marking regular and associative plurality is the same in these languages (a plural demonstrative), neither involve inflectional number marking on the noun and they are therefore excluded from the sample. Note, too, that (128b) does not

have an additive plural interpretation (e.g., more than one person called Ratsimba). That reading is only allowed when the name is bare.

Turning to Nias, Nias nouns are not marked for number (Brown 2001, p. 109). The associative plural is a prefix, *ira-* (130a). This is a collective prefix that appears on a small number of human nouns (see (130b)) which is perhaps why Daniel and Moravcsik (2013) categorized Nias as a language with a *plural pattern*. However, even if *ira-* is analyzed as contributing plural meaning in (130b) (rather than collective), Brown (2001) argues that these forms have been lexicalized because they no longer participate in regular phonological processes. I therefore do not treat *ira-* as a plural marker in Nias, which means that Nias is not characterized as a *plural pattern* language.

(130) *Nias* [Austronesian]

- a. **ira-** Gusti
COLL- Gusti
'Gusti and her family/friends'
- b. **ira-** ina
COLL- mother
'women (lit. mothers)'

(Brown 2001, pp. 111–2)

Ngan'gityemerri nouns are also not marked for number. Nouns in Ngan'gityemerri are modified by one of 15 class prefixes. For example, the human nouns in (131) are prefixed with the class markers *wur-* (female) and *wa-* (male). However, the class marker is neutral with respect to number: (131) is amenable to a singular or plural interpretation.

(131) *Ngan'gityemerri* [Southern Daly]

- wur-** / **wa-** nugumang
CL.F- / CL.M- orphan
'female/male orphan'

One noun class marker, *awa-*, is prefixed to human groups and is glossed as 'mob'

(132a). This is also the associative plural (132b). However, *awa-* is not the regular plural—it does not take a singular noun and pluralize it.

(132) *Ngan'gityemerri* [Southern Daly]

- a. **awa-** kinyi / **awa-** purrpurrk
mob -this / mob- little'uns
'this mob' / 'a mob of kids'
- b. **awa-** Tyemeny
mob -Tyemeny
'Tyemeny and her kids/friends/family' (Brown 2001, p. 297)

Finally, in Papiamentu and Jamaican Creole, regular and associative plural marking is identical to the third person plural pronoun (133a–b).

- (133) a. **Marianan** Papiamentu
Maria.3PL
'Maria and her folks' (Besten 1996, p. 14)
- b. **Miss Waaka dem** Jamaican Creole
Miss Waaka 3PL
'Miss Waaka and the others' (Patrick 2013, p. 232)

However, the plural marking in Papiamentu does not seem to be *inflectional* plural marking. Wiltschko (2008) presents a number of diagnostics that can be used to determine whether plural marking is inflectional (where 'inflectional' means that the plural morphology is merged as a syntactic (functional) head (Num⁰)). For example, inflectional plural marking is restricted to a single category (nouns). Yet in Papiamentu, the plural marker *nan* freely attaches to nouns, demonstratives, adjectives, adverbs, participles and already plural personal pronouns (Castillo 2022). The example in (134a) shows *nan* attached to a participle. Second, bare plurals can function as arguments if the marking is inflectional (e.g., English *Children are intelligent*). Wiltschko shows that in languages where plural

marking is not inflectional (e.g. Halkomelem) bare plurals cannot function as arguments.

Papiamentu bare plurals cannot function as arguments either (134b).

(134) *Papiamentu* [creole]

- a. e supermercado paki koko raspa -**nan** kaba
DEF supermarket bag coconut shaved.PTCP -PL PFV
'the supermarket packets of already shaved coconut' (Castillo 2022, p. 21)
- b. *muchu -**nan** ta inteligente
child -PL are intelligent
'children are intelligent' (Kester and Schmitt 2007, p. 114)

The Jamaican Creole plural marker *dem* also fails to behave like an inflectional plural marker. That *dem* is not an inflectional plural is most clear in examples like (135), which involve a plural suffix *-s* co-occurring with *dem*. This suffix resembles the inflectional plural marking from English (the superstrate language). Importantly, since both number markers may co-occur, they cannot both be inflectional plural marking.

(135) *Jamaican Creole*

- some a di helper -s -**dem** in our area
some of DEF helper -PL -PL in our area
'some of the helpers in our area' (Patrick 2013, p. 224)

Group (3): No associative plural

Daniel and Moravcsik (2013) list Takelma as having an associative plural that is identical to the regular plural. Their example involves the dual marker *-dīl* on the noun 'coyote', and it is glossed as 'coyote and another' (136).

(136) *Takelma* [isolate]

- sgi'si -**dīl**
coyote -DU
'coyote and another' (Daniel and Moravcsik 2013)

However, Sapir (1922) actually translates (136) as ‘two coyotes (lit. coyote-and-another [coyote])’ (137), and says that the Takelma dual picks out two entities “of the same kind”. I take this to mean that (137) is the regular dual (that is, it is homogeneous). The Takelma dual can, when two referring expressions are adjacent, be suffixed to one of them as a coordinator, as in (137b). I do not take this to be an associative dual either because all members of the duality are named (the coyote and the mother); associative number names a single referent and leaves the others implicit. Moreover, I have not been able to find any other sources claiming that Takelma has an associative dual (or plural), and thus I exclude it from the sample.

- (137) a. sg’isi -dīl
 coyote -DU
 ‘two coyotes (lit. coyote and another [coyote])’
- b. sgisi ni’xa -dīl
 coyote mother -DU
 ‘coyote and his mother’
- (Sapir 1922, p. 249)

Group (4): Unable to verify the *plural pattern*

The last group of languages that I exclude from the survey are those I could not independently verify as having a *plural pattern*. These are:

- (138) Dani, Gbeya Bossangoa, Khakas, Koryak, Kulung, Loko, Nobiin, Pulaar, Punjabi.

According to Daniel and Moravcsik (2013) the associative plural in Dani is *-lak*. I was able to verify this in Bromley (1972), but could not verify what the regular plural is in Dani. I did not find any examples of nouns suffixed with *-lak* with a non-associative interpretation.

For Khakas, Nobiin, Pulaar and Punjabi, I was unable to verify the associative plural given in Daniel and Moravcsik (2013), but could verify at least one distinct regular number

marker. For example, Daniel and Moravcsik give *-nu* as the associative plural in Khakas. I could not find independent evidence of this, and the only plural marker I have found is *-lar* (or perhaps *-(C)ar*) (139).

- (139) pula **-lar** / urok **-tar** Khakas [Turkic]
 child -PL / lesson -PL
 ‘children / lessons’ (Maltseva and Sultrekova 2023, pp. 246, 249)

The Nobiin associative plural, according to Daniel and Moravcsik, is *-kuu* (or perhaps *-Cuu*). I could not verify this, and the only regular plural suffix I have encountered is *-(r)i* (140).

- (140) noog **-ri** / burw **-i** Nobiin [Eastern Sudanic]
 house -PL / girl -PL
 ‘houses / girls’ (Alamin 2014, pp. 9, 11)

Daniel and Moravcsik (2013) give *-‘en* as the associative plural in Pulaar. I could not verify that *-‘en* is used as an associative or regular plural. I have only found *-be* being used as a regular plural on human nouns in Pulaar (141).³⁸

- (141) wor **-be** Pulaar [Niger-Congo]
 man -PL
 ‘men’ (Ba 2013, p. 8)

Lastly, Daniel and Moravcsik say that *vaam* is the associative plural in Punjabi. I could not verify that *vaam* had an associative usage, or that it is used as a regular plural marker. According to Bhatia (1993), number is fusional with case in Punjabi (142).

³⁸It is worth noting that Pulaar is related to Fulfulde (Adamawa)—they are both Niger-Congo languages in the genus Peul-Serer—and Fulfulde does use *-‘en* as the regular and associative plural.

(142) Punjabi [Indo-European]

- | | |
|------------|-------------|
| a. -e/Ø/āā | DIRECT.PL |
| b. -iāā/āā | OBLIQUE.PL |
| c. -io/-io | VOCATIVE.PL |

The associative plural in Loko is said to be *-ni*. Although I could not independently verify that *-ni* is an associative plural in Loko, Loko is related to Bandi and Loma (Western Mande), which both have an associative plural *-ni/=ni*. While I was able to verify the regular plural in Bandi and Loma as distinct—and thus characterize them as not having the *plural pattern*—I could not check what the regular plural is in Loko, hence it falls into Group (4).

Finally, I was unable to verify any information regarding the associative or regular plurals in the last three languages—Gbeya Bossangoa, Koryak and Kulung.³⁹

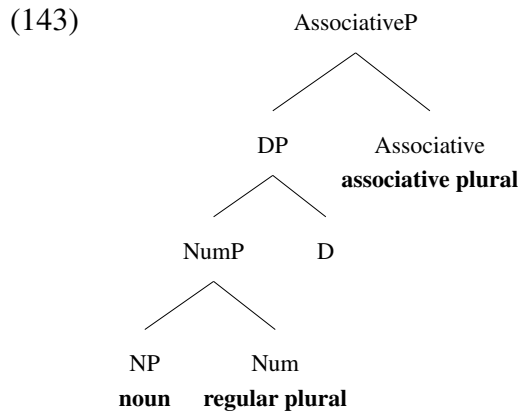
³⁹Note that none of these languages would pose a problem for the typological generalization that languages that show the *plural pattern* in their associative plural lack free-standing definite articles. According to Dryer (2013) Gbeya Bossangoa has a definite affix. Koryak lacks definite articles (Abramovitz 2021, p. 159), as does Kulung (Michailovsky 2017b, p. 651). Thus, if it turns out that these languages in fact show the *plural pattern*, they will not be a problem for the generalization (they would in fact confirm it).

Chapter 3

A new analysis of the *plural pattern*

It was noted in Chapter 2 that a number of authors have adopted essentially the same syntactic analysis for associative plurals in a variety of different languages. The dominant view is that associative plural morphology is realized in the head of a functional projection above DP. This kind of analysis is adopted by Hirose (2004) and Nakanishi and Ritter (2008) for Japanese, Görgülü (2011) for Turkish, Forbes (2013) for Gitksan, Biswas (2014) for Bengali, Dékány (2021) for Hungarian and Cinque (2018) who, building on the analyses of the authors cited above, argues for the general structure in (143) for a variety of languages. I will call the functional projection whose head realizes the associative plural AssociativeP (143).¹

¹I use head-final tree structures for illustrational purposes because, in the majority of the languages in the sample, the associative plural is a suffix. If we used head-initial structures instead, the suffixal nature of the associative plural morpheme would be derived via post-syntactic reordering (e.g., *loweing*), or with roll-up movement of the NP to the specifier of AssociativeP (cf. Cinque 2018).



This general structure is motivated by a number of empirical facts regarding associative plurals cross-linguistically. Firstly, it captures the fact that associative plural morphology attaches to individual-denoting stems (i.e., expressions of type *e*) (Nakanishi and Tomioka 2004; Smith 2020). Assuming that an object of type *e* corresponds to the highest projection in the noun phrase, Associative⁰ must combine with this projection.² Secondly, the associative plural occupies the head of a distinct functional projection, which explains why associative and regular plural morphology appear in different positions relative to other morphemes. Recall that in Turkish, Hungarian and Ainu, the regular and associative plural are realized on opposite sides of possessor agreement: the associative plural follows possessor agreement (144a) while the regular plural precedes it (144b).

- (144) a. Teyze -m **-ler** çık -tı
 aunt -1SG -PL leave -PAST
 ‘My aunt and her family/friends left.’
- b. Teyze **-ler** -im çık -tı
 aunt -PL -1SG leave -PAST
 ‘My aunts left.’

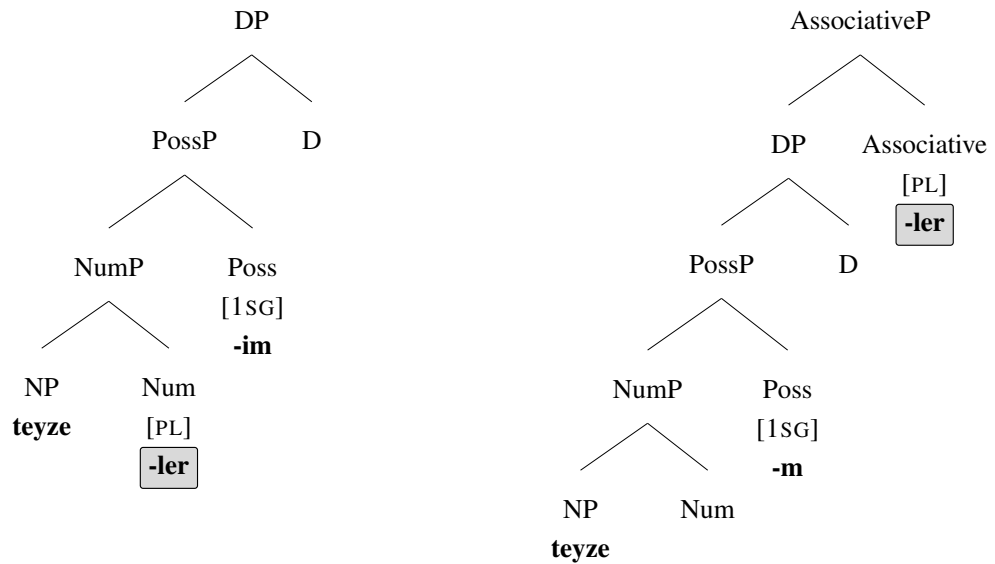
(Görgülü 2011, p. 74)

The structure in (143) captures this nicely. According to the Mirror Principle (Baker

²This projection can be DP, but in later sections I will adopt the view that languages without definite articles lack DP (Bošković 2005, 2008a). In this case, a DP-less noun becomes type *e* via covert type-shifting (Chierchia 1998b). See Section 3.2.3.

1985) the surface order of morphemes reflects underlying syntactic structure. The regular plural, which is in Num⁰, is closer to the root noun than possessor agreement, which /is in Poss⁰ (145a). In contrast, the associative plural, which is in Associative⁰, is further from the root noun than possessor agreement in Poss⁰ (145b).

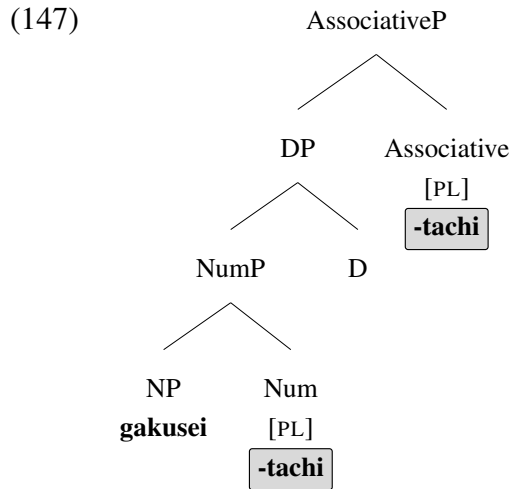
- (145) a. *Regular plural* ([Num > Poss]) b. *Associative plural* ([Poss > Num])



A third piece of empirical evidence for a distinct functional projection for the associative plural comes from languages that allow the regular plural and the associative plural to co-occur. Japanese is a language like this (146). The *-tachi* closest to the noun is the regular plural (‘more than one student’) and the *-tachi* farther from the noun is the associative plural (‘and their associates’). This ordering of double *-tachi*s also reinforces the claim that the associative plural is structurally higher than the regular plural (147).³

³The availability of such ‘double’ plural marking has been explored for Japanese, Hungarian, Turkish and Mandarin Chinese. Although, according to Ueda and Haraguchi (2008), it is possible for the regular plural *-tachi* and the associative plural *-tachi* to co-occur, Cinque (2018, 498:fn8) notes that not all speakers accept (146). The Japanese speakers that I asked also found (146) degraded (but not unacceptable). It is not possible for regular plural *-lar* and associative plural *-lar* to co-occur in Turkish, nor for regular plural *-men* and

- (146) *gakusei -tachi -tachi*
 student -PL -(A)PL
 ‘the students and their associates’ (Ueda and Haraguchi 2008)



The general structure in (143) thus accounts for a range of empirical phenomena that we want to preserve. On its own, however, (143) is inadequate in light of the typological generalization established in Chapter 2. That is, if all associative plural morphology is base-generated in the head of AssociativeP, why does the *plural pattern* never show up in languages with free-standing definite articles? The goal of this chapter is to develop a new analysis of *plural pattern* associative plurals that retains the empirical coverage of the previous approaches while simultaneously accounting for the novel typological generalization.

3.1 The *plural pattern* as head-movement of Num

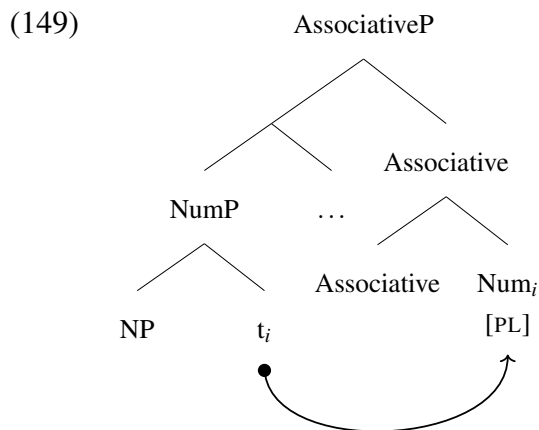
Until now it has been tacitly assumed in the literature that the *plural pattern* is a case of syncretism between two separate categories—plural number and associativity (Corbett

associative plural *-men* to co-occur in Mandarin Chinese (Ueda and Haraguchi 2008). Double plural marking is acceptable in Hungarian only if the focal referent is a possessed noun (e.g. *a-barát-a-i-d-ék*, ‘your friends and their associates’ (Dékány 2021, p. 223)). I leave these peculiar facts aside for now.

2000). The problem with maintaining this view is that it cannot account for the syntactic typological generalization established in Chapter 2, given again below. (148) implies that the *plural pattern* is not a morphological phenomenon but a syntactic one.

(148) GENERALIZATION: If a language uses the regular plural as (or as part of) the associative plural (the *plural pattern*), it lacks free-standing definite articles (it either lacks definite articles or has affixal/clitic definite articles).

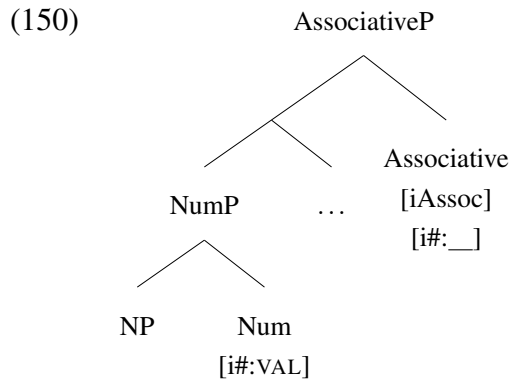
In this regard, I propose that the *plural pattern* actually involves movement of the functional head Num to Associative⁰.



I will show that the associative plural is forced to show the *non-plural pattern* when head movement of Num is blocked. I adopt the view that languages with free-standing definite articles and languages without free-standing definite articles differ in the amount of functional structure in the nominal domain (Section 3.2), and argue that this structural difference blocks the requisite movement.

3.1.1 Motivating head movement

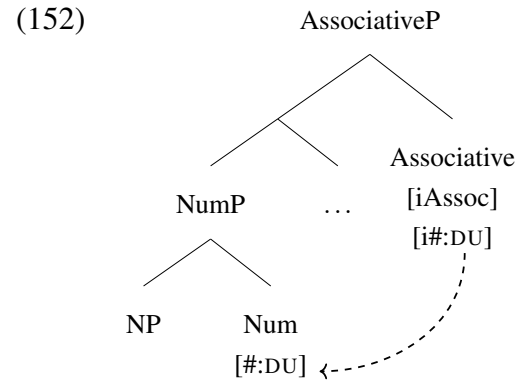
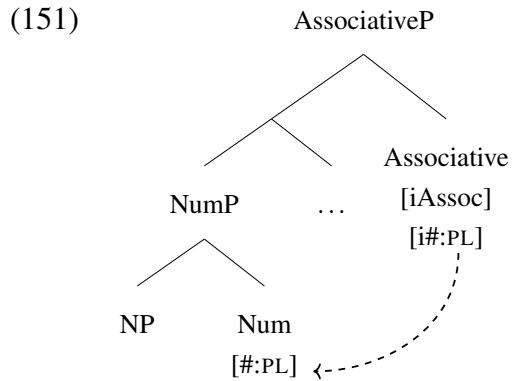
I propose that *plural pattern* associative plurals have the structure in (150). Broadly speaking, (150) is very similar to the structure in (143)—there is a dedicated functional projection where associative plural morphology is realized, above DP—allowing us to retain all of the empirical insights discussed in the introduction to this chapter. What is new is an unvalued, interpretable number feature [i#:__] on the Associative head.



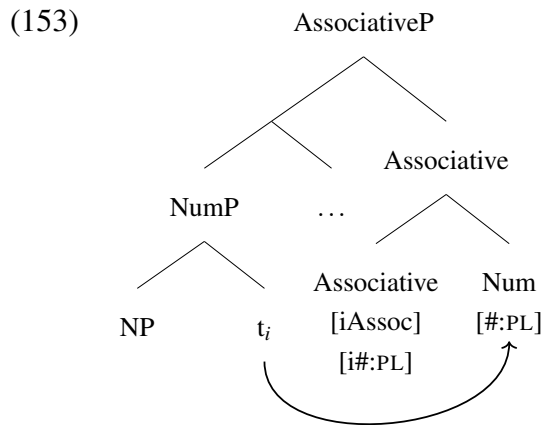
In languages that show the *plural pattern* in their associative plural, I claim that [i#:__] is valued by a matching, valued number feature on Num⁰. Any value on Num⁰ is theoretically capable of valuing [i#:__]. Thus, plural pattern associative *plurals* involve [i#:__] being valued by a Num head bearing the value [PL] (151); *plural pattern* associative *duals* involve [i#:__] being valued by a Num head bearing the value [DU] (152).⁴ I assume that singular number is the absence of number features (Nevins 2011).⁵

⁴As we will see later on, some languages have associative plurals but not associative duals, or associative duals but not associative plurals. An analysis that involves feature valuation in fact captures these asymmetries in a way that the base-generation analysis cannot.

⁵A theory of number that treats number feature values as bivalent (i.e., [±SG], [±AUG], (Harbour 2011)) rather than privative (i.e., [PL], [DU]) is compatible with this analysis. I use the privative features [PL] and [DU] for expository purposes only.



I propose that as a result of Agree between [i#:__] and a matching number feature on Num, the Num head incorporates into Associative⁰ in narrow syntax.



The above analysis assumes that head movement is syntactic, and there are well-known arguments to this effect (see e.g., Roberts 2010). For example, Roberts (2010) shows that head movement cannot be an exclusively PF-domain operation because head movement can have semantic effects, meaning that LF must be able to “see” that movement has taken place. Consider the examples in (154), which show that negative polarity items (NPIs), like *any*, must be asymmetrically c-commanded by their licenser (a negative element). In (154b–c), the licenser is embedded (in a CP subject and a ConjP, respectively) and thus not c-commanding the NPI.

- (154) a. *Anyone didn't see John. (John didn't see anyone.)
 b. *That nobody trusts him bothers anyone.
 c. *Few professors and any students were at the party.

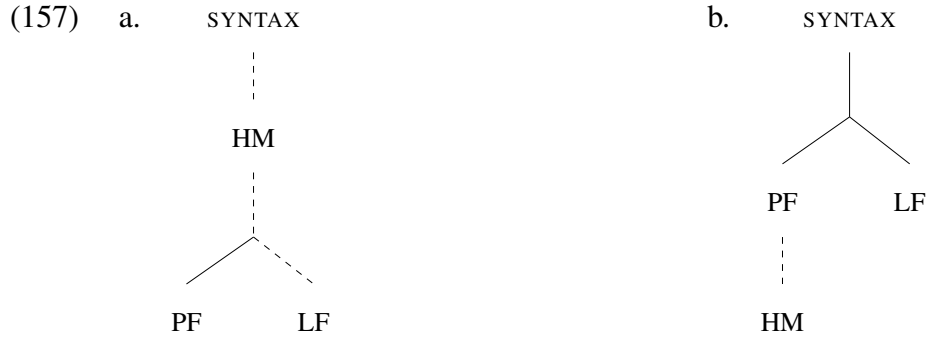
Importantly, the right licensing condition can be created by moving the licensor to a c-commanding position. This is shown in (155a–b) with a raising construction.

- (155) a. *After the meeting, it seemed to anybody that nobody was satisfied with the outcome.
 b. After the meeting, nobody_i seemed to anybody t_i to be satisfied with the outcome.

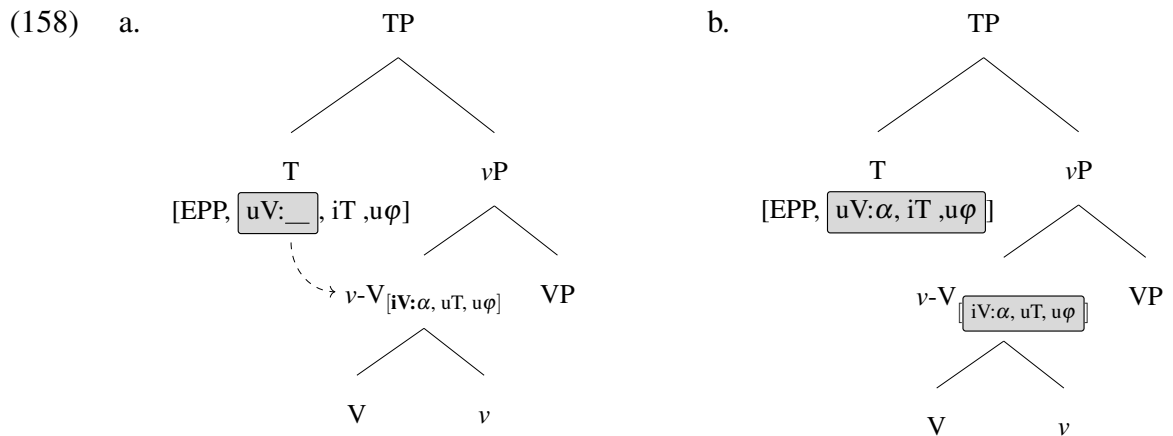
Crucially, head movement can also create the right licensing condition for NPIs. (156a) shows that when the NPI (*anybody*) is not c-commanded its licensor ((*does*)*n't*), the result is ungrammatical. (156b) involves an embedded *wh*-question – which does not involve T-C (head) movement – and the NPI is also not licensed. Crucially, however, subject-auxiliary inversion—which is standardly assumed to involve T-to-C (head) movement of the auxiliary—does license the NPI (156c).

- (156) a. *Anybody doesn't like John.
 b. *They succeeded in finding out which one of them anybody didn't like.
 c. Which one of them doesn't anybody like?

Given the standard Y-model organization of the grammar, in order for LF to “see” the head movement in (156c) it must have happened in narrow syntax (157a) rather than at PF (157b).

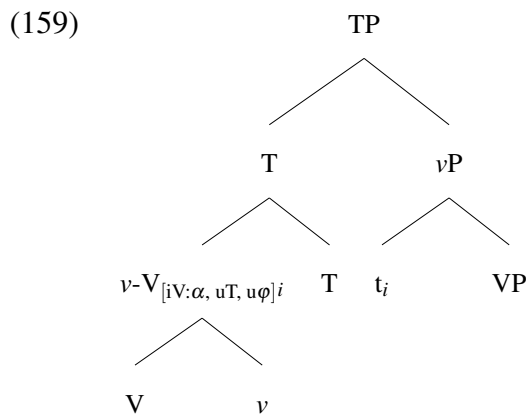


Roberts (2010) proposes that (syntactic) head movement of X^0 to Y^0 occurs following Agree between X^0 and Y^0 , and iff X^0 is a defective goal (a goal whose formal features are a proper subset of the probe's formal features). To illustrate, consider (158), which shows ν -V to T movement under Roberts' analysis. ν -V bears the features $[iV]$, $[uT]$ and $[u\phi]$, and T bears the features $[uV]$, $[iT]$, $[u\phi]$ and an EPP feature. The $[uV]$ feature on T probes for a goal, finding $[iV]$ on ν (158a). The features on ν -V are copied into the probe.



What is important here is that copying the $[iV:\alpha]$ feature onto the probe exhausts the feature content of ν -V. This is because all of the features that are on ν -V are now also represented on T (a matching valued V feature, a T feature, and uninterpretable phi features).

This causes a well-known problem for linearization: the two copies in (158b) cannot be ordered by the LCA (Kayne 1994) because the same element cannot precede and follow itself. One of the copies must be deleted. Standardly, the head of the chain survives and all lower copies are deleted (Bošković and Nunes 2007). The effect of Agree between a probe and a defective goal, followed by deletion of lower copies, is that the goal has incorporated into the probe. This is shown in (159).



Returning now to head movement in the *plural pattern*, Num⁰ is a defective goal. The probe (Associative⁰) bears a categorial feature [iAssoc] and an interpretable, unvalued number feature [i#:__]. Num⁰ bears a valued (categorial) number feature [i#:VAL]. Thus, Num⁰'s formal features are a proper subset of Associative⁰'s formal features.

At its core the *plural pattern* involves different interpretations of plural morphology in different positions. We saw this most clearly in languages like Turkish when plural marking co-occurs with possessor agreement: when plural morphology precedes possessor agreement it is interpreted additively; when it follows possessor agreement it is interpreted associatively. In other words, plural morphology has different semantics when it is higher in the nominal domain. This is another case of head movement in narrow syntax having semantic effects. In its base position, Num⁰ combines with a predicative NP (type $\langle e, t \rangle$).

The function of Num^0 is to divide the extension of the predicate into atoms (singularities), making the noun countable (160a-b) (Borer 2005). A singular NP picks out one atom in the extension of the predicate (160). A plural NP has the semantics in (161) (Chierchia 1998a). A is the set of atoms (161b). ‘*’ is an operator whose definition is in (161c). When * is applied to the set of atoms A, the output is a set that consists of all sums of elements of A (161d). A plural feature on Num^0 thus takes the set in (161b) and removes from it all of the atoms. The result is a set that only consists of pluralities (161e). This is the additive interpretation of plural.

(160) a. $[\text{NumP} [\text{NP } \text{dog}_{\langle e,t \rangle}] \#]$

b. $\text{dog} = \{\text{dog}_1, \text{dog}_2, \text{dog}_3, \text{dog}_4, \text{dog}_5 \dots \text{dog}_n\}$

c. $\text{a/the dog} = \{\text{dog}_{37}\}$

(161) *Chierchia’s (1998) plural semantics*

a. $\text{PL}(A) = *A - A$

b. $A = \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$

c. *X is the closure of X under U (U = the sum of all the elements in X, i.e., the set of all sums of elements of X)

d. $*A = \{\text{dog}_1\}, \{\text{dog}_2\}, \{\text{dog}_3\}, \{\text{dog}_1, \text{dog}_2\}, \{\text{dog}_1, \text{dog}_3\}, \{\text{dog}_2, \text{dog}_3\}, \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$

e. $\text{dog}_{\text{PL}} = \{\text{dog}_1, \text{dog}_2\}, \{\text{dog}_1, \text{dog}_3\}, \{\text{dog}_2, \text{dog}_3\}, \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$

However, once Num^0 has incorporated into Associative^0 the plural feature no longer combines with a predicative NP. Rather, plural features combine (as part of a complex head) with an individual-denoting object of type *e*. Thus, the additive interpretation of the plural is unavailable after incorporation. That additive plurality is incompatible with

objects of type e is explored in Daniel (2020), who argues that an associative plural (e.g., *Ahmet-ler*, ‘Ahmet and his family’ in Turkish) is just the interpretation of the additive plural combined with a unique individual. When plural Num^0 combines with a predicative NP, the predicate is divided into atoms of the same type which can be combined to yield plural sets (162). When plural Num^0 combines with an individual-denoting argument of type e , however, the individual cannot be divided into atoms and is thus incompatible with the semantics in (160).⁶ Daniel (2020) argues that, instead, the meaning of plural number is coerced such that the plural set yielded from a unique individual must minimally contain that individual (163).

- (162) a. $\text{dog} = \{\text{dog}_1, \text{dog}_2, \text{dog}_3, \text{dog}_4, \text{dog}_5 \dots \text{dog}_n\}$
 b. $\text{dog}_{\text{PL}} = \{\text{dog}_1, \text{dog}_2\}, \{\text{dog}_1, \text{dog}_3\}, \{\text{dog}_2, \text{dog}_3\}, \{\text{dog}_1, \text{dog}_2, \text{dog}_3\}$
- (163) a. $\text{Ahmet} = \{\text{Ahmet}\}$
 b. $\text{Ahmet}_{\text{PL}} = \{\text{Ahmet}, \text{Faruk}, \text{Ali}\}$

Following Daniel (2020), I propose that (163b) is the result of combining plural number with an argument of type e , which is what happens when plural Num^0 has incorporated into Associative⁰. However, Daniel’s (2020) analysis does not derive the fact that the members of plural set that contains *Ahmet* in (163b) stand in an associative relation with Ahmet. Thus, I also assume that Associative⁰ has its own semantics. I adopt a slightly modified version of Tomioka’s (2021) semantic analysis for Associative⁰, shown in (164c).⁷ Here,

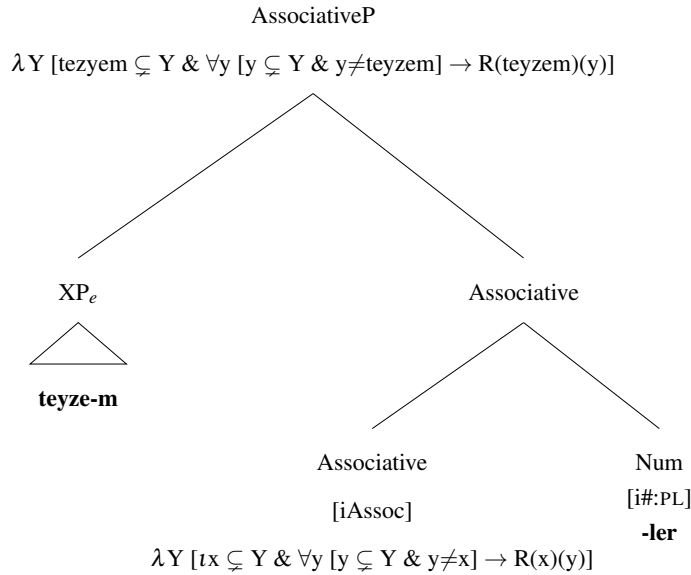
⁶So-called unique individuals – or the focal referents of associative plurals – are only incompatible with the semantics in (160) when they are uniquely referring. Thus, if a proper name like ‘Ahmet’ is construed as a predicate, it is compatible with ordinary additive semantics (‘multiple people with the name Ahmet’).

⁷Tomioka’s (2021) original semantic analysis is given in (i). Here, Associative⁰ combines with an object of type e and yields a maximum plural entity that contains it. In the modified version outlined above, it is the plural feature that yields a plural entity and the [iAssoc] feature combines with this plural entity.

(i) $\llbracket \text{PL}_{\text{Associative}} \rrbracket = \lambda x. \iota Y [x \subsetneq Y \ \& \ \forall y [y \subsetneq Y \ \& \ y \neq x] \rightarrow R(x)(y)]$

Associative⁰ takes a maximal plural entity Y, and says that if there is a unique individual x in Y, and for all y , y is also in Y (and where y is not x), then y stands in a relation R with x . This R relation is one of contextually salient association.

- (164) a. tezye -m -ler
 aunt -1SG.POSS -PL
 ‘my aunt and her friends’
- b. [AssociativeP [... [PossP [NumP [NP **tezye**_{<e,t>}] **t_i**] -m] ...]_e Associative + #_i -ler]]
- c. [iAssoc] = $\lambda Y [tx \subsetneq Y \ \& \ \forall y [y \subsetneq Y \ \& \ y \neq x] \rightarrow R(x)(y)]$
- d.



The plural feature first combines with the individual-denoting argument of type e to yield a plural set consisting of the named referent and others. The associative feature then takes this plural set and establishes a contextually salient association between the named referent and the ‘others’ in the plural set. Note that incorporation is the *only* way for plural morphology to carry associative semantics on this analysis. If [i#:__] is valued by the matching number feature on Num⁰, but Num⁰ does not incorporate, then it combines with a predicative NP in its base position and has only ordinary, additive semantics. Agree,

without movement, is not enough to derive a *plural pattern* associative plural.⁸

⁸Two additional implications of the proposed approach to *plural pattern* associative plurals are worth noting. First, head movement of Num⁰ “relocates” the semantic import of Num⁰ from its base position to its derived position. That is, in order to derive a plural set consisting of the focal referent plus ‘others’ the plural feature must be interpreted “upstairs”. There are cases where head movement does not affect interpretation in this manner – e.g. head movement V-T does not change where V is interpreted (it is always interpreted “downstairs”). This difference could be due to the kind of feature on the probe that triggers movement—an uninterpretable [uV:___] feature on T in the case of V-T (Roberts 2010) and an *interpretable* [i#:___] feature on Associative⁰ in the present case. In the former, only the lower copy is visible at LF and the verb is interpreted as though it hadn’t moved. In the latter, both the lower and higher copies are visible at LF. Assuming that either one of these features can be interpreted at LF, interpretation upstairs is predicted to be possible. This leads us to the second implication for the proposed approach: we may expect that Num⁰ could reconstruct – that is, pronounced “upstairs”, in Associative⁰, but interpreted “downstairs”, in its base-position. Reconstruction of Num⁰, schematized in (i), would derive an additive plural interpretation of the focal referent, Ahmet. Associative⁰ would combine with the (additively) pluralized focal referent and establish a contextually salient association between all of the Ahmets.

- (i) [AssociativeP [NumP [NP [N *Ahmet*]] [Num [i#:PL]]] [Associative [Associative + Num *-ler*]]

It is not clear if reconstruction of Num⁰ is possible. That is, the relevant association between all of the Ahmets may be that they share the same name, making it indistinguishable from the regular (additive) plural. Moreover, recall that the associative plural *-lar* in Turkish appears to the right of possessor agreement (*abla-m-ler* ‘my elder sister and her friends’). Interestingly, *ablamler* also has an additive interpretation (‘my elder sisters’) according to my consultants. This interpretation could be derived via reconstruction of the plural feature, or it could be a special interpretation of the associative plural (i.e., my elder sister’s associates also happen to be my elder sisters).

Suppose for the sake of argument that reconstruction of Num⁰ is not possible. Why should this be? The answer might again be due to the kind of probe that is triggering movement – an unvalued [#] feature. Klingvall (2018) shows that in Swedish the subject of an adjectival tough construction cannot reconstruct when there is ϕ -agreement between the subject and the predicate (iia-b) (CMN stands for common gender, which is a non-neuter gender in the singular). When there is no ϕ -agreement, reconstruction is possible (iic).

- (ii) a. [Få personer]₁ var svår-a för Johan att prata med ___₁
 few people were difficult-PL for Johan to talk to
 ‘Few people were difficult for Johan to talk to.’ *few > difficult, *difficult > few*
- b. [En bok]₁ var svår för Anna att ge bort ___₁
 a book was difficult.CMN for Anna to give away
 ‘One (particular) book was difficult for Anna to give away.’ *one > difficult, *difficult > few*
- c. Var [ett enda plagg]₁ svår-t för Johan att köpa ___₁
 was one single piece of clothing difficult-NEUT for Johan to buy
 ‘Was one single piece of clothing such that it was difficult for Johan to buy?’ or ‘Was it difficult for Johan to buy just one piece of clothing?’ *one > difficult, difficult > one*
 (Klingvall 2018)

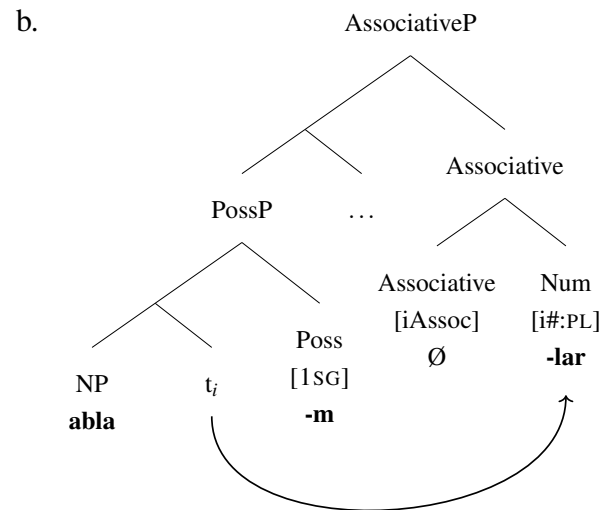
Klingvall argues that the neuter marking on the adjective in (iic) is a default form that arises in the absence of ϕ -agreement, and proposes that ϕ -agreement in Swedish blocks reconstruction. This is language specific — ϕ -agreement does not block reconstruction in English: *Every student has not passed the exam (not>every)*. Nevertheless, reconstruction of Num⁰ may be similarly blocked as a result of the movement-inducing feature being a ϕ -feature.

In the following sections, I show that head movement of Num⁰ captures the full range of *plural pattern* languages presented in Chapter 2. This includes (i) the baseline languages (regular and associative plural are identical), (ii) languages where the regular plural appears alongside dedicated associative plural morphology, and (iii) languages where the *plural pattern* involves using the regular plural of an alternative stem.

Head movement & the baseline case

The most common type of *plural pattern* language in the sample is one in which the associative plural is identical to the regular plural. In this pattern, Num⁰ has incorporated into Associative⁰ under Agree and the lower copy is deleted.⁹ Additionally, the categorial feature on Associative is zero. This is illustrated in (165) with Turkish.

- (165) a. *abla -m -lar*
 elder.sister -1SG.POSS -PL
 ‘my elder sister and her friends’



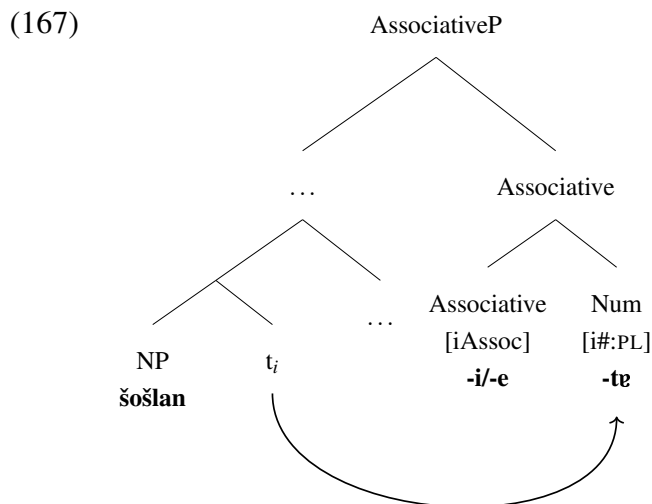
⁹Recall that the plural feature on the probe is a copy of the plural feature on Num⁰, hence they are the *same* feature. The lower copy of the feature is deleted, and the higher one is pronounced. That is, plural morphology is pronounced in a higher syntactic position because Num⁰'s plural feature has been copied onto a higher head. To show that this is really (the plural feature on) Num⁰ being pronounced in a higher syntactic position, I illustrate this in the tree structures as incorporation of the Num head.

Head movement & regular plural + extra morpheme

A handful of *plural pattern* languages in the sample—Ossetic, Mandinka, Khanty, Hup and Hungarian—involve regular plural morphology plus some dedicated associative plural morphology. For example, the associative plural in Ossetic consists of the regular plural suffix *-tə* plus dedicated associative morphology *-i/-e-*.

- (166) šošlan **-i** **-tə** / šošlan **-e** **-tə**
 Soslan -APL -PL / Soslan -APL -PL
 ‘Soslan and others’ (Erschler 2016, p. 7)

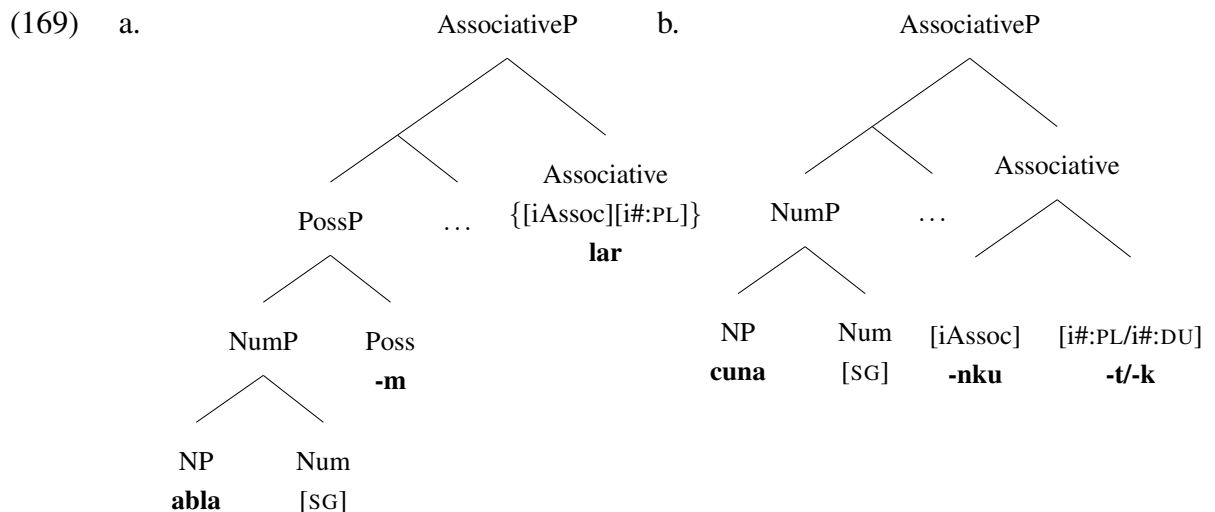
These languages are actually a crucial piece of evidence for the claim that the *plural pattern* involves head movement of Num⁰. After Num⁰ has incorporated into Associative⁰, the result is a complex head that consists of both Num, bearing [i#:PL] and Associative, bearing [iAssoc]. It is therefore expected that at least some *plural pattern* associative plurals will be morphologically complex, overtly expounding both features. This is illustrated in (167) for Ossetic.



Dékány (2021) offers a similar analysis for the associative plural (and dual) in Cen-

tral Alaskan Yup'ik (168). She proposes that the Associative head is made up of multiple features corresponding to the two semantic components of associative number. Thus, one feature is responsible for the non-atomic (i.e., non-singular) meaning, and another is responsible for the non-homogeneous meaning. Most languages spell-out the entire feature bundle with a single morpheme (e.g., Turkish, see (169a)), but some (like Central Alaskan Yup'ik) reflect this underlying complexity with bimorphemic associative plural morphology (169b)

- (168) cuna -nku -t / cuna -nku -k
 Chuna -APL -PL / Chuna -APL DU
 'Chuna and his family/friends' / 'Chuna and his friend' (Corbett 2000, p. 108)



Where I depart from Dékány is that the feature responsible for non-atomicity is not an inherent feature of Associative⁰. Rather, this feature gets on the Associative head via head movement of Num⁰.

Head movement & regular plural + alternative stem

The last set of *plural pattern* languages are those that involve the ordinary plural of a possessive or genitive noun. The South Slavic languages and Lezgian fall into this group.

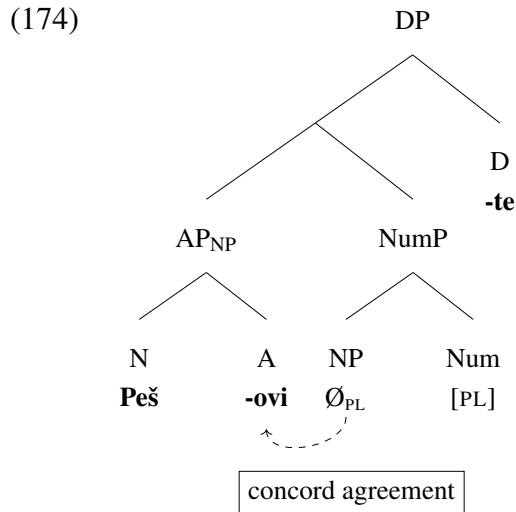
- (170) Dide-d-**bur** bazar.di-z fe-na Lezgian
mother-GEN-SBST.PL market-DAT go-AOR
'Mother and those with her went to the market.'
(Haspelmath 1993, p. 79)
- (171) Peš**ovi** oti-do-ha na ribo-lov. Bulgarian
Peter.POSS.PL.M go-PAST-3PL to fish-hunt
'Peter and his friends went fishing.'

As we saw in Chapter 2, these associative plurals are identical on the surface to headless possessives (possessive constructions in which the possessum is null). Headless possessives can involve ellipsis (see (173a)), in which case the referent of the null possessum is recovered via the antecedent, or not (see (172) and (173b)), in which case the referent is determined by extra-linguistic contextual information.

- (172) Exirni xür.ü-n-**bur**.u kac tam.u-z čukur-na.
finally village-GEN-SBST.PL(ERG) cat forest-DAT chase-AOR
Finally the villagers (lit. those of the village) chased the cat into the forest.
- (Lezgian; Haspelmath 1993, p. 344)
- (173) a. negov-i-te roditel-i b'axa po-bogati ot Irin-**ini**-te
his-PL-DEF.PL parent-PL they.were CMPR-rich from Irina-POSS.PL.F-DEF.PL
'His **parents** were richer than Irina's <parents>.'
- b. Peš-**ovi**-te zaminali na more
Pete-POSS.PL.M-DEF.PL they.departed to sea.ACC
'Pete's family went to the seaside.'
- (Bulgarian; Daniel 2004)

I assume that the possessors in (172) and (173) are denominal adjectives (they involve adjectivizing morphology on a nominal stem; see e.g., Corbett 1987 for Slavic and Haspelmath 1993 for Lezgian) that agrees in number with the covert noun. I also assume that the

headless possessive adjectives are substantive (they are semantically nouns; see Bošković 2012a; Daniel 2004; Haspelmath 1993). The structure of the headless possessive in (173b) is given in (174). The possessive adjective (AP_{NP}) modifies a plural null noun, with which the adjectivizing morphology in A shows concord agreement.



Daniel (2004) argues that associative plurals in these languages—which he calls possessive-like associative plurals—are derived from the headless possessive in (174). According to Daniel, the path to a possessive-like associative plural is (175) (illustrated with Bulgarian). A language begins with an adnominal possessive (175a), in which the possessive adjective modifies an overt noun. The possessive then acquires a headless (substantivized) usage that picks out a contextually determined noun (175b). Lastly, the headless possessive in (175b) undergoes a “shift” (Daniel’s (2004) term) to the associative plural (175c).

(175) a. *Adnominal possessive*

[NP [AP Pešovi] kotoratsi]
Pete.POSS.PL.M cat.PL
‘Pete’s cats’

b. *Headless possessive*

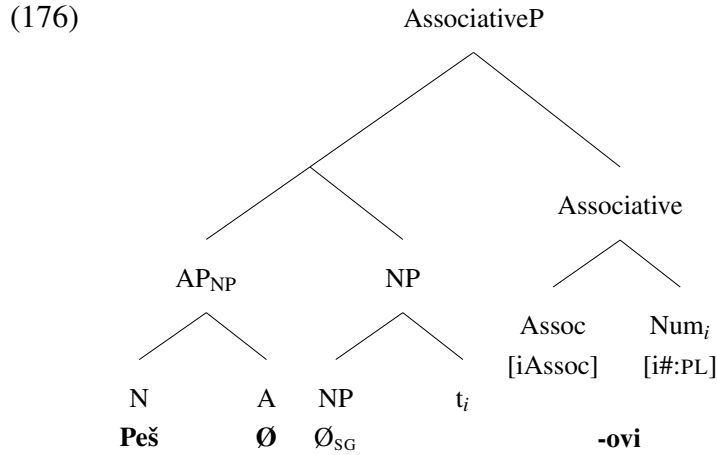
[_{DP} [_{NP} [_{AP} Pešovi]_{NP} [_{NP} Ø_{PL}]]] D -te] zaminali na more
Pete.POSS.PL.M -DEF.PL they.departed to sea.ACC
‘Pete’s family went to the seaside.’

c. *Associative plural*

[_{APL} Pešovi] oti-do-ha na ribo-lov.
Peter.POSS.PL.M go-PAST-3PL to fish-hunt
‘Peter and his friends went fishing.’

I propose that this “shift” between (175b) and (175c) (from headless possessive to associative plural) is the development of AssociativeP, and that after this development Num⁰ incorporates into Associative⁰. Recall that one consequence of head movement of Num⁰ is that it no longer combines with a predicative NP, eliminating the additive interpretation of plural morphology. In the headless possessive, the possessive adjective shows concord agreement with the null noun—there is a (pragmatically salient) X (e.g., family/friends) in a possessive relation to Y (e.g., Peter). If the null noun is no longer interpreted as plural (because Num⁰ has incorporated into Associative⁰) then there is no longer a plurality X that Y could be in a possessive relation to. In other words, the genuine possessive reading is lost.

I propose that incorporation of Num⁰ into Associative⁰—and the subsequent loss of the genuine possessive reading—results in reanalysis of the adjectivizing/possessive morphology as the exponent of the (complex) Associative head. That is, the adjectivizing morphology is reanalyzed as associative plural morphology. The adjectivizing head is null. The reanalyzed structure for the Bulgarian associative plural is given in (176).



Reanalysis of plural possessive morphology in headless possessives as associative plural morphology involves a process of grammaticalization whereby adjectival structure loses morphological exponence. The result of this is that only nominal material is overtly exponed in the AP in (176). One expected consequence of this is further reanalysis of the substantivized adjective itself as a simple NP. The associative plural may then reflect that change by being realized as basic nominal plural inflection.

Interestingly, Daniel (2004) notes that Lezgian also uses the ordinary plural as the associative plural, and that this is in fact the primary way associative plurals are formed. The possessive-like associative plural is more likely to be interpreted as a genuine possessive. There may then be an additional reanalysis process happening in Lezgian in which the substantivized stem of the possessive-like associative plural is being reanalyzed as a simple nominal, which is reflected in the associative plural morphology changing from the substantivized plural *d-bur* to the ordinary plural *-ar*. The entire grammaticalization path is shown in (177). Lezgian begins with a genuine headless possessive in which a substantivized adjective modifies a silent null noun (177a). A possessive-like associative plural arises following the development of AssociativeP, head movement of Num⁰ to Associative⁰ and reanalysis of the genitive plural morphology as associative plural morphology (177b).

Eventually, the substantivized adjective is reanalyzed as a simple NP, and associative plural morphology is the ordinary plural (177c).¹⁰

(177) Grammaticalization of associative plurals in Lezgian:

a. *Headless possessive*

Exirni [NP [AP xür.ün**bur**]NP [NP Ø_{PL}]] kac tam.u-z čukur-na
 finally village.GEN.SBST.PL cat forest-DAT chase-AOR
 ‘Finally those of the village chased the cat into the forest.’

b. *Possessive-like associative plural*

[AssociativeP [NP [AP dide]NP [NP Ø_{SG}]] Assoc + Num **-d-bur**
 mother -GEN-SBST.PL
 ‘Mother and those with her’

c. *Reanalysis of AP_{NP} as NP (baseline associative plural)*

[AssociativeP [NP dide] Assoc + Num **-jar**
 mother -PL
 ‘Mother and those with her’

Hungarian provides additional evidence that associative plurals can be derived from headless possessives, and that this involves various stages of grammaticalization. Recall that the associative plural in (standard) Hungarian is *-ék* (178a). Interestingly, *-é* is also suffixed to possessors when the possesessum is null (i.e., it marks headless possessives). This is shown in (178b). Note that after possessive endings, the plural is *-i* rather than *-k*.

(178) a. János **-é -k** elbújtak
 John -é -PL hid.PST.3PL
 ‘John and them hid’

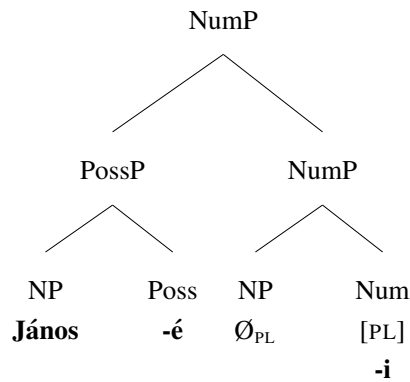
b. János **-é -i** elbújtak
 John -POSS -PL hid.PST.3PL
 ‘Those of John hid’

(Dékány 2021, p. 229)

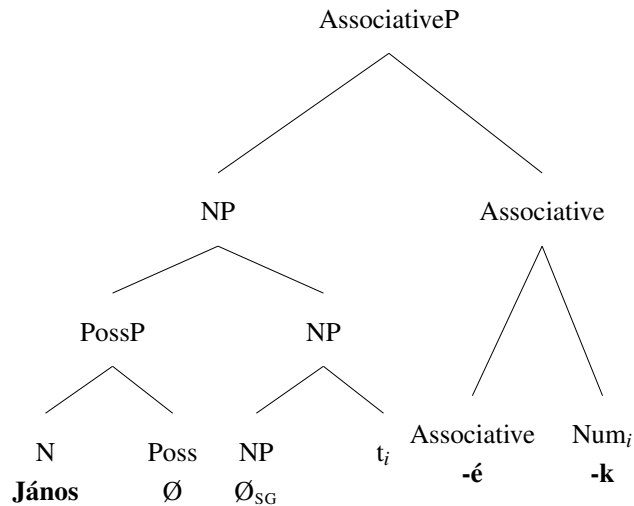
¹⁰The example in (177c) has been constructed based on the information in Daniel (2004) about Lezgian associative plurals and Haspelmath (1993) about Lezgian plural morphology (although Daniel says that the regular plural is used as the associative plural in Lezgian, he does not provide an example). I have not been able to check (177c) with a native speaker.

If (178b) had an associative plural function (that is, if *-éi* was reanalyzed as associative plural morphology after Num⁰ incorporated into Associative⁰), it would be like the South Slavic languages and Lezgian. Instead, it seems that *-é-* (which I treat as heading a PossP following É. Kiss (2002); see (179a)) has been reanalyzed as dedicated associative morphology alone (179b). This in turn led to a change in plural morphology from *-i* to the ordinary (non-possessive) plural *-k*.

(179) a.



b.



Crucially, however, in the Transdanubian dialects Örsef, Hetés and Southwest Göcsej, the associative plural is the ordinary nominal plural *-k* (180). One way of understanding the variation between Standard Hungarian and the Transdanubian dialects is that the latter

have undergone an additional process of reanalysis that Standard Hungarian has not. In the dialectal varieties, the possessor has been reanalyzed as a simple NP.¹¹ This is shown in (181).

- (180) Kovaács Pistá **-k** -ot nēm híták meg
 Kovács Pistá -PL -ACC not invite.PST.DEF.OBJ.3PL PRT
 ‘Kovács Pistá and his family were not invited.’ (Dékány 2021, p. 232)

- (181) *Grammaticalization of associative plurals in Hungarian:*

a. *Headless possessive*

[NumP [PossP [NP János] -é] [NumP [NP Ø_{PL}] -i]] elbújtak
 János -é -PL hid.PST.3PL
 ‘Those of John hid.’

b. *Head movement + reanalysis of é*

[AssociativeP [PossP [NP János] [NP Ø_{SG}]] Assoc **-é** + Num **-k**
 János -é -PL
 ‘János and his family/friends’

c. *Reanalysis of PossP as NP* (Transdanubian dialects only)

[AssociativeP [NP János] Assoc + Num **-k**
 János -PL
 ‘János and his family/friends’

To sum up, associative plurals with a possessive or genitive stem (as opposed to a simple nominal stem) are also amenable to a head-movement analysis of the *plural pattern*. Head movement of Num⁰ leads to the genuine possessive reading being lost and plural possessive morphology being reanalyzed as associative plural morphology.

¹¹Although I am treating the possessor in Hungarian as a PossP and the possessors in Slavic and Lezgian as APs, this has no affect on the reanalysis argument. In both Hungarian and Slavic/Lezgian, the possessive/possessive adjective morphology is reanalyzed as associative plural morphology. This leads to the PossP and the AP losing overt possessive and adjectival morphology respectively, and eventually being reanalyzed (in the case of Lezgian and the Transdanubian dialects of Hungarian) as a simple NP.

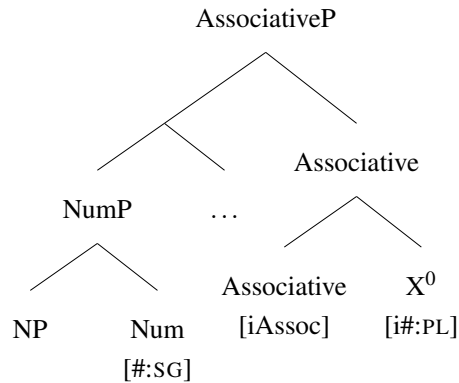
3.1.2 A note on the *non-plural pattern*

Recall that the typological generalization regarding the *plural pattern* was a one-way generalization. That is, all languages that have a *plural pattern* associative plural either lack definite articles or have affixal definite articles, but not all articleless and affixal definite article languages have a *plural pattern* associative plural. Some articleless and affixal article languages use dedicated associative plural morphology. Yagua is a language like this. It does not have a definite article (Dryer 2013) and its associative plural is an accompaniment suffix *-ve* (182a). The regular plural is marked by the suffixes *-ri* and *-mijy* (182b).

- (182) a. Manúngu **-ve**
Manúngu -APL
'Manungu and company' (Payne 1997, p. 99)
- b. deera **-mijy**
child -PL
'children' (Payne 2007, p. 455)

I propose that languages that have a *non-plural pattern* associative plural lack [i#:__] on Associative⁰ (183). This is just like Roberts's (2010) analysis of *v-V-to-T* movement: languages with *v-V-to-T* movement have a [uV: __] feature on T; languages without *v-V-to-T* movement lack it. Thus, the only feature on the Associative head in these languages is [iAssoc]. Recall, however, that [iAssoc] alone cannot generate the right semantics for an associative plural. [iAssoc] combines with a plural entity consisting of the focal referent and others, and establishes a contextually salient association between the named referent and the others in the plural set. Thus, in *non-plural pattern* languages it is necessary for something bearing *inherent* plural features to first be adjoined to Associative⁰.

(183)



This requirement straightforwardly explains why the associative plural in *non-plural pattern* languages is itself plural. Mauri and Sansó (2021) find that, in languages in which the regular plural is not used as (or as part of) the associative plural, associative plural markers are third (and sometimes second) person plural pronouns, plural demonstratives, plural possessives, words denoting ‘group’, or ‘family’, collective markers, universal quantifiers and conjunctions like ‘and/with’ and ‘also’. These are all syntactic objects that clearly have an interpretable plural feature.¹² The plural feature on these syntactic objects serves to yield a plural set consisting of the named referent and others. The associative feature can then take this plural set and establish a contextually salient association between the named referent and the others. The upshot of this is that *plural pattern* and *non-plural pattern* associative plurals differ only minimally — either the required [i#:PL] feature is *internally* merged (in *plural pattern* associative plural languages) or *externally* merged (in *non-plural pattern* associative plural languages).

¹²One kind of associative plural marker that does not seem to meet this criteria is found in Kuuk Thaayorre and Central Asmat, which use morphemes that may derive from ‘low down by’ and ‘close by’ respectively as their associative plural. However, Mauri and Sansó (2021) categorize these diachronic sources as ‘unlikely’ and ‘uncertain’ respectively.

3.1.3 Summary

I have established in this section that head movement can derive all three kinds of *plural pattern* languages (baseline, regular plural + extra morpheme, regular plural + alternative stem). In the next section I return to the typological generalization, established in Chapter 2, that *plural pattern* languages lack free-standing definite articles. I show that an analysis of *plural pattern* languages that involves head movement of Num⁰ allows us to deduce the typological generalization in (184).

- (184) GENERALIZATION: If a language uses the regular plural as (or as part of) the associative plural (the *plural pattern*), it lacks free-standing definite articles (it either lacks definite articles or has affixal/clitic definite articles).

3.2 Deducing the typological generalization

I propose that the *plural pattern* is restricted to languages without free-standing definite articles because head movement of Num⁰ is blocked in precisely these languages. Understanding why a free-standing definite article would have a blocking effect on movement requires a proposal by Bošković (2005, 2008b, 2012b) about languages with and without definite articles (185), and an observation by Talić (2017) about affixal article languages (186).

- (185) *The DP/NP-language distinction:*

Languages with definite articles project DP above NP (“DP languages”), but languages without definite articles do not (“NP languages”).

- (186) *Functional structure in affixal article languages:*

Affixal article languages behave like NP languages when the article is dropped.

Bošković argues, based on a number of cross-linguistic generalizations, that there is a structural difference in the nominal domain between languages with and without definite articles (DP vs no DP). More recently, however, Talić has refined this two-way split regarding nominal functional structure. Specifically, she shows that affixal article languages behave as a third type, patterning with both non-affixal article languages and articleless languages.

In the following sections I present just some of the empirical observations motivating the DP/NP-language distinction proposed by Bošković (2008b, 2012b) and the treatment of affixal article languages in Talić 2017. I then show how the DP/NP approach can be used to deduce the generalization regarding *plural pattern* languages established in this work.

3.2.1 The DP/NP-language distinction

Bošković (2008b, 2012b) argues for a fundamental distinction between languages with definite articles and languages without definite articles. Specifically, he argues that the absence of definite articles cannot be a PF phenomenon (i.e., a covert *the*), but that languages without articles lack DP in their nominal projection.¹³ This distinction is based on a number of cross-linguistic generalizations about the syntactic and semantic behavior of languages with and without definite articles. Some of these generalizations are given in (187). Bošković has uncovered many more cross-linguistic generalizations like this, all of which target some syntactic or semantic property (e.g., radical pro-drop, island-sensitivity, superiority effects in multiple *wh*-fronting, adjunct extraction, etc; see Bošković (2012b, 2016, 2023)).

¹³For similar analyses regarding individual languages without articles, see e.g., Baker 1996; Corver 1992; Despić 2011; Jiang 2012; Marelj 2011; Runić 2014; Zanon 2015; Zlatić 1997.

- (187) a. Only languages without articles may allow Left Branch Extraction of adjectival elements out of a nominal phrase.
 b. Only languages with articles allow the majority reading of MOST.
 c. Only languages without articles may allow (Japanese-style) scrambling.
 d. Only languages with articles may allow clitic doubling.
 e. Second position clitics are found only in languages without articles.
 f. Languages without articles disallow negative raising (i.e. strict clause-mate NPI licensing under negative raising from finite (i.e. indicative) clauses).

To illustrate, consider the generalization in (187a) that only languages without articles may allow Left Branch Extraction (LBE). As we saw in Chapter 2, it is impossible to extract an adjective from a noun phrase in English, a language that has free-standing definite articles (188). It is also impossible to extract an adjective in Bulgarian, a language with affixal definite articles (189a).¹⁴ In contrast, LBE is allowed in Serbo-Croatian, a language without definite articles (189b).

(188) *Expensive_i he saw [t_i cars]

(189) a. *Novata_i prodade Petko [t_i kola] Bulgarian (Bošković 2005, p. 3)
 new-DEF sold Petko car

b. Skupa_i je vidio [t_i kola] Serbo-Croatian (Bošković 2012b, p. 1)
 expensive is seen car

Bošković (2012b) gives Serbo-Croatian, Russian, Polish, Czech, Ukrainian, Slovenian, Latin, Mohawk, Southern Tiwa and Gunwinjguan languages as allowing adjectival LBE, and all of these languages lack definite articles. Particularly strong evidence that it is the article status of these languages that allows LBE comes from Finnish. Franks (2007) observes that colloquial Finnish, which has developed a definite article, disallows LBE (190a). Literary Finnish, on the other hand, lacks a definite article and allows LBE (190b).

¹⁴Recall that LBE is allowed in Bulgarian if the article is dropped. We will see why the absence of the article matters here in Section 3.2.1.

- (190) a. ?*Punaisen osten (sen) auton (spoken Finnish)
red.ACC buy.PST.1SG (the.ACC) car.ACC
- b. Punaisen ostin [t_i auton] (literary Finnish, poetic style)
red.ACC buy.PST.1SG car.ACC
‘I bought the red car’

Similar evidence is found when looking at Ancient Greek, which has undergone a change from an articleless language (Homeric Greek) to a definite article language (Koine Greek). Taylor (1990) observes that split wh-phrases (which involve extraction of just the wh-word) and extraction out of NP (which Bošković (2012b) notes often involves LBE) are found significantly more frequently in the Homeric period (68% for split wh-phrases, 25% for NP extraction) than in the Koine period (15% for split wh-phrases, 0% NP extraction). This finding is particularly significant for (187a): as Ancient Greek gained a definite article, it lost LBE. Lastly consider Latin, which lacked definite articles and allowed LBE. Crucially, Latin differs from Modern Romance languages which have definite articles and disallow LBE.

Consider, too, an observation by Despić (2019) regarding the anaphoric use of bare nouns. Despić shows that bare count nouns in articleless languages can be used anaphorically (he discusses Turkish, Serbo-Croatian, Chinese, Japanese and Hindi). This is illustrated in (191) with Japanese. The bare noun *roojin-wa*, ‘old.man-TOP’ picks out *Yahachi* in the antecedent clause. In contrast, English (a language with free-standing definite articles) must use the definite article in this context (192).

- (191) Yahachi -o miru -to, roojin -wa damatte unazuita
Yahachi -ACC see -when old.man -TOP silently nodded
‘When I saw Yahachi, the old man silently nodded.’

- (192) Yesterday I read *Crime and Punishment*. I really liked *(the) book.

Interestingly, bare *mass* nouns in kind-denoting contexts in articleless languages cannot be used anaphorically (this is true for all of the languages listed above). This is illustrated in (193) with Turkish—the bare noun *meyve*, ‘fruit’ cannot refer to *üzüm*, ‘grapes’ in the antecedent clause. As with singular count nouns, mass nouns in English can be used anaphorically when they are accompanied by the definite article. This is shown in (194).

(193) *Ömrüm boyunca üzü yetiştirdim. Meyve herşeyim oldu.
 my life throughout grape produce.1.PR. Fruit my everything became.
 ‘I have been producing grape my whole life. This fruit is everything to me.’

(194) We have been growing grapes for generations – and you know, we have made millions on *(the) fruit.

If the only difference between English and the articleless languages is that the article is covert in the latter—and if the covert article is responsible for the anaphoric reading of the bare count noun in (191) (as the overt article is in (192))—why does this covert article not produce an anaphoric reading of the bare mass noun in (193)? This cross-linguistic variation shows that the absence of definite articles is not phonological in nature.

Variations in functional structure (i.e., DP vs no DP), on the other hand, can be used to deduce these differences. To illustrate, consider again the ban on LBE of adjectives (which Bošković (2013) argues are adjoined to NP) in English (195) compared to the possibility of such LBE in articleless languages like Serbo-Croatian (196).

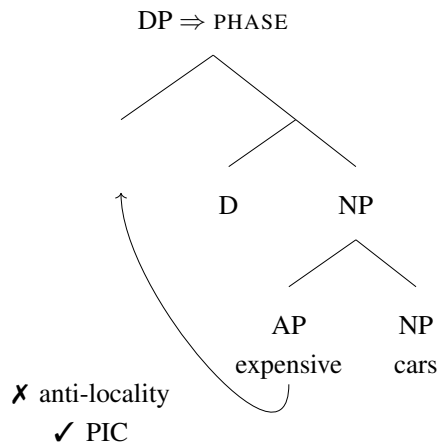
(195) *Expensive_i he saw [_i cars]. English

(196) Skupa_i je vidio [_i kola].
 expensive is seen car
 Serbo-Croatian(Bošković 2012b)

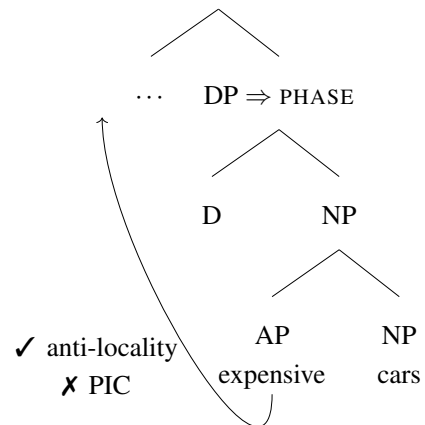
To account for this, Bošković (2005) adopts two independent conditions on movement: Chomsky’s (2000) Phase Impenetrability Condition—only the phase head and its specifier

(the phase edge) are able to undergo further movement outside the phase—and the anti-locality condition—movement must cross at least one phase (Abels 2003; Bošković 1994; Grohmann 2000, among others). Additionally, Bošković (2014) argues that the highest phrase in the extended projection of a lexical head (e.g., N) is a phase. This means that in languages with definite articles, which project DP, DP is a phase. In languages without definite articles, which lack DP, NP is a phase. Thus, to extract an NP-adjoined adjective out of DP in English the adjective must first move to SpecDP to satisfy the PIC because DP is a phase. However, this movement violates the anti-locality condition (197a). In order for movement to obey anti-locality, it must violate the PIC by skipping SpecDP (197b). There is thus no permissible way for languages with definite articles—DP languages—to permit LBE.

(197) a. *Anti-locality violation*



b. *PIC violation*

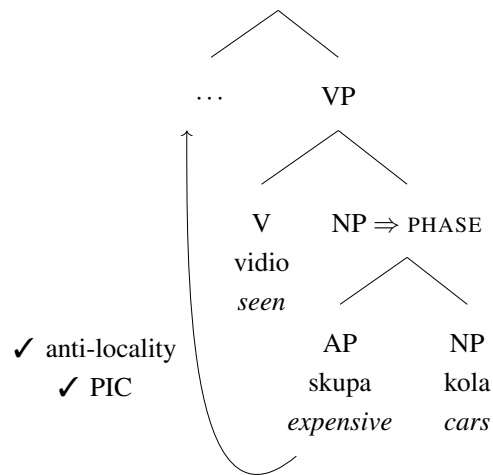


In an articleless language like Serbo-Croatian, there is no DP and NP is a phase.¹⁵ An

¹⁵See Bošković (2013, 2014) for evidence for the phasehood of NP in Serbo-Croatian.

NP-adjoined adjective can thus move out of NP without violating the PIC (it is already at the phase edge) or anti-locality (the movement step crosses VP). This is shown in (198).

(198)



Thus, the presence/absence of DP affects whether certain locality-based operations, like movement, are possible because DP is a phase (for deductions of other generalizations, see Bošković 2008b, 2012b, 2016, 2018, 2023).

Crucially, the DP/NP-language distinction only makes a two-way language cut—a language either has definite articles and is a “DP language” or it lacks definite articles and is an “NP language”. However, it has been observed that languages with affixal definite articles sometimes behave—in their syntax and semantics—like articleless languages.¹⁶ Recall, for example, that Reuland (2011) and Despić (2011, 2015) observe that there are no reflexive possessives in languages with free-standing definite articles (e.g., English) (199a). However, languages without definite articles (e.g., Serbo-Croatian) and languages with affixal definite articles (e.g., Danish) have them (199b–c). In other words, these affixal definite article languages behave like articleless (NP) languages, rather than non-affixal definite

¹⁶Some of these observations are also discussed in Chapter 2.5.3.

article (DP) languages.

- (199) a. *She loves **herself's** pets. (She loves **her** pets.)
- b. Svaki dečak_i je video **svog**_i oca. Serbo-Croatian
 every boy is seen self's father
 'Every boy_i saw his_i father.'
- c. John_i læste **sin**_i artikel. Danish
 John read self's article
 'John_i read his_i article' (Despić 2015, p. 209)

Pancheva and Tomaszewicz (2012) also show that affixal article languages sometimes behave like articleless languages. In particular, they show that the interpretation of superlatives differs depending on the article status of the language. Superlatives have in principle three readings depending on the comparison class they are evaluated against, illustrated in (200): (a) the absolute reading, where the comparison class is *the best albums by U2*, (b) the relative reading with *internal* focus (RIF), where the comparison class is *by U2* (it is *internal* to the superlative), and (c) the relative reading with *external* focus (REF), where the comparison class is *John*.

- (200) John has the best albums by U2.
- a. 'John has the best albums U2 has ever made.' Absolute
- b. 'John has better albums by U2 than by any other band.' RIF
- c. 'John has better albums by U2 than anyone else does.' REF

In languages with free-standing definite articles (e.g., English) the Relative Internal Focus (RIF) reading is unavailable (201a) but in articleless languages (e.g., Polish) the RIF reading of superlatives is available (201b). Crucially, Bulgarian (an affixal article language) can pattern with articleless languages in allowing the RIF reading (201c). However, this is only so if the article is dropped; when the article is present, the RIF reading is

unavailable (201d). If the semantic difference between English and Polish is due to the presence/absence of DP (see Shen 2014), there must also be no DP in (201c).

(201) a. *English*

John has the best albums by U2.

≠ John has better albums by U2 than by any other band.

✗ RIF

b. *Polish*

Ivan ma naj-lepsze albumy U2.

Ivan has SPRL-better album.ACC.PL U2

= Ivan has better albums by U2 than by any other band.

✓ RIF

c. *Bulgarian (definite article)*

Ivan ima naj-dobri-te albumi ot U2.

Ivan has SPRL-good-the albums by U2.

≠ Ivan has better albums by U2 than by any other band.

✗ RIF

d. *Bulgarian (no definite article)*

Ivan ima naj-dobri albumi ot U2.

Ivan has SPRL-good albums by U2.

= Ivan has better albums by U2 than by any other band.

✓ RIF

Similarly, Dubinsky and Tasseva-Kurktchieva (2014) show that adjunct extraction—which is allowed in articleless languages (202a) but not languages with free-standing definite articles (202b)—is allowed in Bulgarian in the absence of the definite article (202c). When the article is present, adjunct extraction is impossible (202d). Bošković (2008b) deduces the ban on adjunct extraction the same way that he deduces the ban on LBE, discussed above (see (197))—the presence of DP leads to a violation of either the PIC or the anti-locality condition. Thus, if adjunct extraction is possible in Bulgarian when the article is absent, DP must be absent in (202c) as well.

- (202) a. Iz kojeg grada_i je Petar sreo [djevojke t_i]? Serbo-Croatian
 from which city is Peter met girls
 (Bošković 2005, p. 10)
- b. *From which city_i did John want to meet [the professors t_i]?
- c. Ot koj universitet_i srešna -ha nyakolko studenti t_i? Bulgarian
 from which university met -they several students
 ‘From which university did they meet several students?’
- d. *Ot koj universitet_i srešna -ha nyakolko -to studenti t_i?
 from which university met -they several -the students
 ‘From which university did they meet several students?’
- (Dubinsky and Tasseva-Kurkchieva 2014)

Finally, Talić (2017) also shows that affixal definite article languages behave like articleless languages. She shows that articleless languages allow LBE of an intensifier adverb out of an Adjective Phrase (AP) (203a).¹⁷ Languages with free-standing definite articles do not allow this movement (203b). Crucially, Talić observes that Bulgarian, Icelandic and Romanian—all of which have affixal definite articles—also allow adverb LBE. This is shown in (203c) with Icelandic. She proposes the typological generalization in (204).¹⁸

- (203) a. Strašno_i je bila t_i unmora Serbo-Croatian
 terribly is been tired
 ‘She was terribly tired.’

¹⁷She discusses Serbo-Croatian, Polish, Russian and Slovenian.

¹⁸Adverb extraction involves movement out of AP, not NP/DP. To account for the generalization in (204), Talić (2017) proposes the Structural Parallelism hypothesis in (i). Talić argues that APs have a functional layer XP_{AP} above AP that corresponds to the functional layer DP above NP in the nominal domain. Thus, the hypothesis in (i) says that if a language lacks DP, it may lack XP_{AP} as well. XP_{AP} blocks movement out of the adjectival domain for the same reason that DP blocks movement out of the nominal domain (PIC/anti-locality violations).

(i) *Structural Parallelism:*

- a. If a language always requires functional structure within TNP (DP), it also always requires functional structure in TAP (XP_{AP}).
- b. If a language allows a bare NP, it also allows a bare AP.

b. *Terribly_i, she was t_i tired.

c. Rosalega_i er hún t_i falleg
 extremely is she beautiful
 ‘She is extremely beautiful.’

Icelandic

(Talić 2017, pp. 10, 23)

(204) *Adverb Extraction Generalization (Predicative TAPs):*

Languages with non-affixal articles disallow Adv-extraction out of predicative TAPs, but languages without articles and languages with affixal articles may allow it.

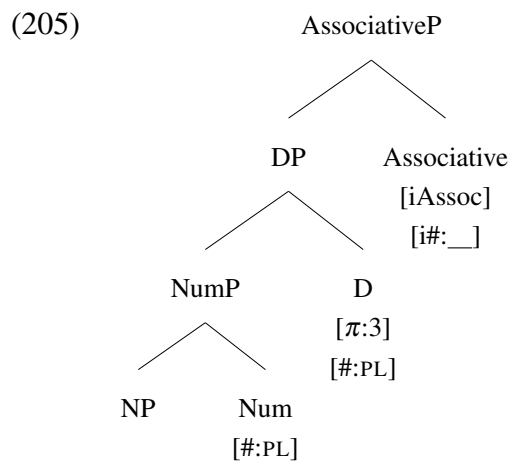
Talić (2017) argues that this behavior of affixal definite article languages indicates that the two-way cut proposed by Bošković (2008b, 2012b) should actually be a three-way distinction. That is, languages with free-standing definite articles *always* project DP, articleless languages *never* project DP, and affixal article languages do not project DP *when the article is absent* (see also Oda 2022). This is why affixal article languages sometimes behave like a “DP language” and sometimes like an “NP language”, and why we find cross-linguistic generalizations in which articleless and affixal definite article languages cluster together to the exclusion of languages with free-standing definite articles. The typological generalization established in this work is another example of this clustering: articleless languages and languages with affixal definite articles may have *plural pattern* associative plurals, but languages with free-standing definite articles never do.

In the following sections I show how the three-way DP/NP-language distinction (i.e., DP vs no DP vs sometimes DP), coupled with head movement of Num⁰, deduces the typological generalization.

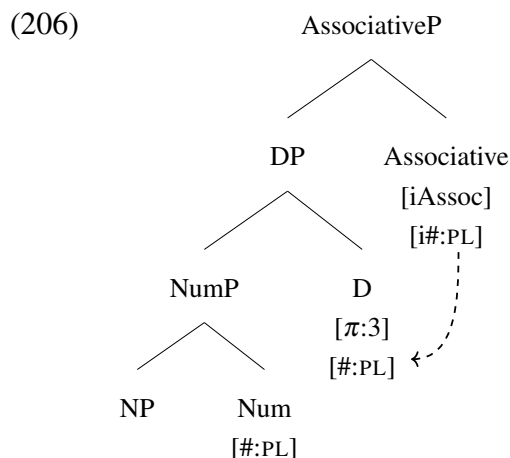
3.2.2 The (absence of the) *plural pattern* in DP languages

I have argued that the *plural pattern* arises when the Associative head bears an interpretable, unvalued [#] feature that is valued by a matching valued [i#] feature on Num⁰. Agree be-

tween Associative⁰ and Num⁰ triggers incorporation of Num⁰ into Associative⁰ because the latter is a defective goal. If [i#:__] is absent from Associative⁰ in a particular language, that language will show the *non-plural pattern*. Given this, consider the basic structure of an associative plural in a DP language in (205). Since we are trying to derive the (unattested) *plural pattern*, Associative⁰ bears a [i#:__] feature. Num⁰ bears a valued number feature. Following Danon (2011), I also assume that D bears a complete set of valued ϕ -features (for ease of exposition, only number and person are shown).



Now consider what happens when [i#:__] on Associative⁰ probes in search of a matching number feature. The first matching number feature it finds is on D⁰, which values the [i#:__] feature on Associative⁰ (206). There is no incorporation of Num⁰ (because Num⁰ has not established an Agree relationship with the probe) nor of D⁰ (because D⁰ is not a defective goal).



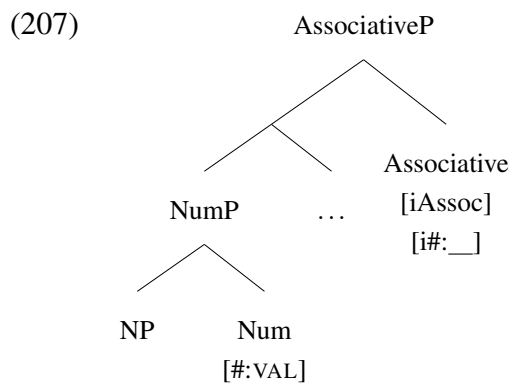
In other words, the presence of D^0 between Associative^0 and Num^0 is an intervenor for movement: D^0 is an element of the same structural type as Num^0 (a head) that c-commands Num^0 and bears the feature that Associative^0 attracts ([#]). It is not possible for the probe to establish an Agree relation with Num^0 across D^0 because D^0 causes an intervention effect. Because incorporation of Num^0 is blocked, the *plural pattern* is ruled out (recall that the only way to interpret plural morphology associatively is if it incorporates). The only other option is for the feature bundle on Associative^0 to be expounded with distinct associative plural morphology.

Thus, DP languages are forced to show the *non-plural pattern* in their associative plural regardless of the featural make-up of Associative^0 . When $[\text{i\#}:__]$ is absent, the *non-plural pattern* arises. When $[\text{i\#}:__]$ is present, the presence of D^0 prevents incorporation of Num^0 , resulting in the *non-plural pattern* again.¹⁹

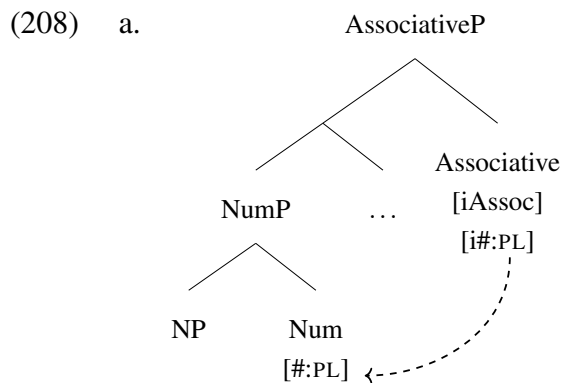
¹⁹For the sake of completeness, consider what would happen if D^0 possessed a $[\text{i\#}:__]$ feature but Num^0 lacked number features (i.e., Num^0 is singular). In this derivation, $[\text{i\#}:__]$ would fail to be valued by anything in its probing domain. Thus, the only way a $[\text{i\#:PL}]$ feature can be introduced into this derivation is if it is *externally merged*, leading once more to the *non-plural pattern*.

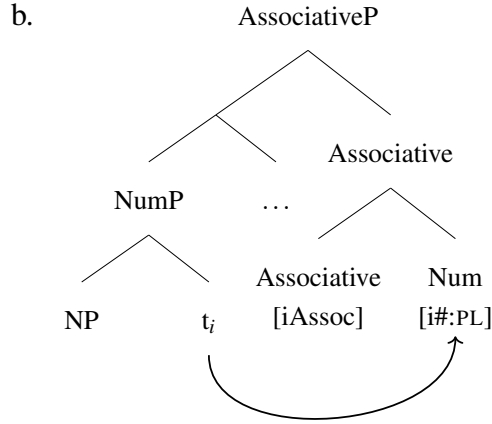
3.2.3 The plural pattern in articleless languages

Now consider the basic structure of an associative plural in an NP language (207). Associative⁰ bears an interpretable, unvalued number feature ([i#:__]) and Num⁰ bears a matching valued number feature. Since we are dealing with an articleless language, DP is not projected between NumP and AssociativeP.



[i#:__] probes in search of a matching number feature. The closest matching feature it finds is the [#] feature on Num⁰ (208a). Agree between the probe and Num⁰ triggers incorporation of Num⁰ into the Associative head. As part of the complex Associative⁰ head, plural morphology now carries associative semantics (208b).





Note that the lack of DP does not affect the semantics of associative plurals discussed in Section 3.1.1 (that is, that Num⁰, from its incorporated position in the Associative head, combines with an individual-denoting argument of type e). In articleless languages, arguments of type e are generated via a covert type-shifting operation *iota* at LF (Chierchia 1998b; Dayal 2017; Partee 1987). This type-shifting takes a predicative NP (of type $\langle e, t \rangle$) and produces an argument of type e. In order for Associative⁰ to combine with the noun, this type-shifting must occur before Associative⁰ is merged to avoid a type mismatch. Thus, in an articleless language (e.g., Shiiba), an associative plural has the LF structure in (209b): the proper name starts off as type $\langle e, t \rangle$,²⁰ *iota* type-shifts the noun to type e and Associative⁰ takes this type e object as its argument.

- (209) a. Taroo -domo
 Taro -PL
 ‘Taroo and his associates’ (Shimoji and Hirosawa 2022, p.300)

b. [AssociativeP [_λP [NP **Taro**]_{⟨e,t⟩} [_λι]_e [Assoc + Num **-domo**]]

Thus, the *plural pattern* is available in articleless languages because the closest goal

²⁰See e.g., Jambrović (2022) and Matushansky (2005) for arguments that proper names start off as predicates.

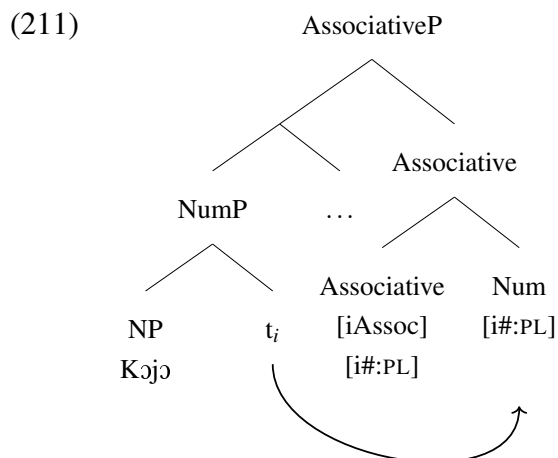
for [i#:___] is the matching valued feature on Num⁰, leading to incorporation of Num⁰ and a new semantics of plural morphology.

3.2.4 The *plural pattern* in affixal article languages

So far I have argued that D⁰ blocks incorporation of Num⁰ (i.e., the *plural pattern*) in DP languages, and its absence allows incorporation of Num⁰ in NP languages. In this section, I show that affixal article languages (which, recall, lack DP when the article is absent but project DP when the article is present) provide compelling evidence that it is indeed the presence of D⁰ that blocks the *plural pattern* in DP languages. However, we will also see that some affixal article languages allow a *plural pattern* associative plural even when D⁰ is projected. Following proposals by Oda (2022) and Bošković (2023), I argue that some languages head-adjoin D⁰ without projecting DP and this makes them less DP-like than other definite article languages.

Recall the 10 languages in the sample with affixal definite articles: Bambara, Bulgarian, Sinhala, Tommo So, Hungarian, Koyraboro Senni, Limbu, Margi, Kpelle, and Mandinka. Given the three-way DP/NP-language distinction proposed by Talić (2017) (that is, that DP is absent in affixal article languages when the definite article is dropped) we expect an affixal article language to be able to show the *plural pattern* so long as the article is absent (i.e., so long as there is no DP). This is what we find: (210) shows an associative plural attached to a proper noun in Tommo So. Since the stem (*Kɔjɔ*) is not affixed with the definite article (=gɛ), DP is absent and Num⁰ can incorporate into Associative⁰ (211).

- (210) Kɔjɔ =mbe
 Kodio =PL
 ‘Kodio and his family’ (Plungian 1995, p. 11)



Proper nouns and kinship terms are the most common stems for the associative plural and—because they are inherently definite—they are usually incompatible with the definite article. Since the definite article is generally absent in the same contexts that give rise to associative plurals, we expect the *plural pattern* to be found in affixal article languages since it is precisely in these contexts that DP is missing. Compelling evidence that it is the absence of DP that allows the associative plural interpretation of (210) comes from Bulgarian. Recall that Bulgarian associative plurals are identical to plural possessive adjectives (212a). Crucially, these adjectives may take the definite affix, even with proper names and some kinship terms (212b). However, while (212b) allows a genuine possessive reading, it does not have an associative plural interpretation.

- (212) a. **Pešovi** oti-do-ha na ribo-lov.
 Peter.POSS.PL.M go-PAST-3PL to fish-hunt
 i. ‘Peter and his friends went fishing.’ = associative
 ii. ‘Those of Peter (e.g., his family) went fishing.’ = possessive
- b. **Pešovi-te** oti-do-ha na ribo-lov.
 Peter.POSS.PL-DEF go-PST-3PL to fish-hunt

- a. ‘Those belonging to Peter [his friends/family] went fishing’ POSS
 b. # ‘Peter and his friends went fishing.’ # APL

(Daniel 2004, I. Derzhanski p.c.)

In other words, the definite affix in (212) blocks the associative plural interpretation. This is expected if DP plays a crucial role in (dis)allowing the *plural pattern*: in articleless languages the absence of DP makes the *plural pattern* possible; in affixal article languages, the *plural pattern* is possible when DP is absent (i.e., (210) and (212a)) but not when it is present (212b).

However, a definite affix does not always block the *plural pattern* from arising. In Hungarian, the definite article can co-occur with an associative plural. This is shown in (213).

- (213) a. **az** -igazgató **-ék**
 the -director -APL
 ‘the director and his company’
 b. **a** -barát -a -i -d **-ék**
 the -friend -POSS -PL -2SG.POSS -APL
 ‘your friends and their associates’ (adapted from Dékány 2011, p. 3)

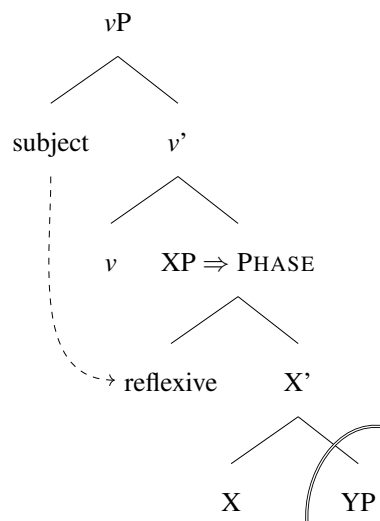
Interestingly, this is not the only case where Hungarian behaves like an articleless language (that is, an NP language) even though the article is present. Recall from Chapter 2 that Hungarian has reflexive possessive pronouns (214a), which Reuland (2011) and Despić (2011, 2013) observe are only found in languages without definite articles and languages with affixal definite articles (cf. (214b–d)).

- (214) a. A fiúk felfedzt **-ék önmaguk** határ -a -i -t
 the boys discovered -3PL themselves limit -POSS -PL -ACC
 ‘The boys discovered their (own) limits’

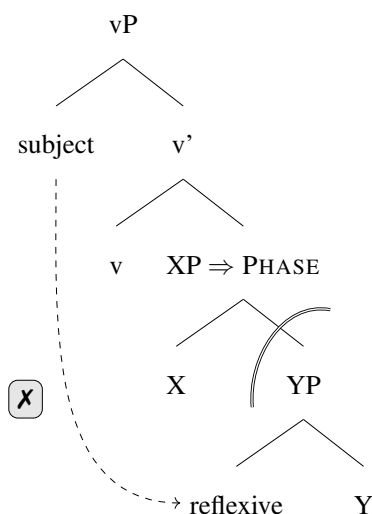
- b. Svaki dečak_i je video **svog**_i oca. Serbo-Croatian (no article)
 every boy is seen self's father
 'Every boy_i saw his_i father.'
- c. John_i læste **sin**_i artikel. Danish (affixal article)
 John read self's article
 'John_i read his_i article' (Despić 2015, p. 209)
- d. *She loves **herself's** pets. (She loves **her** pets.)

The basic idea in Despić 2011 is that reflexive anaphors need to be bound in their binding domain, where binding domains correlate with phasal spell-out domains. Thus, an anaphor must have a phase-mate antecedent, but can be bound outside of its phase if it is located at the phase edge. A reflexive anaphor at the nominal phase edge can thus be bound by the subject in Spec,νP because the reflexive and the subject are in the same binding domain (cf. (215a-b)).

(215) a. Bound reflexive at phase edge: ✓



b. Free reflexive inside spell-out domain: ✗



The possessive pronouns are argued to be in SpecPossP.²¹ This means that in DP languages, reflexive anaphors are not at the phase edge—Spec,DP—and are therefore unable to be bound by the subject (a Binding Condition A violation (216)).

(216) [_{vP} *subject_i* v [_{DP} D [_{PossP} **himself_i* Poss [_{NP}]]]]

In articleless languages, there is no DP phase above PossP. Thus, a reflexive anaphor in SpecPossP is at the phase edge and can be bound by the subject in Spec,_{vP} (217).

(217) [_{vP} *subject_i* v [_{PossP} *himself_i* Poss [_{NP}]]]]

Now consider the Hungarian examples in (218). Recall that Hungarian has a pronominal possessive pronoun *ő*, ‘his’ and the reflexive possessives *maga*, ‘oneself’ and *ön maga* ‘themselves’. When the pronominal possessor is used, it must appear with the definite article (218a). This is expected if DP is projected. DP is a phase between the pronoun in SpecPossP and the subject in Spec_{vP}, preventing the pronoun from being bound as required by Condition B. Moreover, when the reflexive possessor *ön maga* is used, it *cannot*

²¹Possessives may also be in Poss⁰, since the head of the phase is also at the phase edge. Oda (2022) argues that simple possessives are in Poss⁰ in Hungarian.

appear with the definite article (218b). This is expected if DP is absent—only then can the reflexive be bound by the subject in SpecvP.

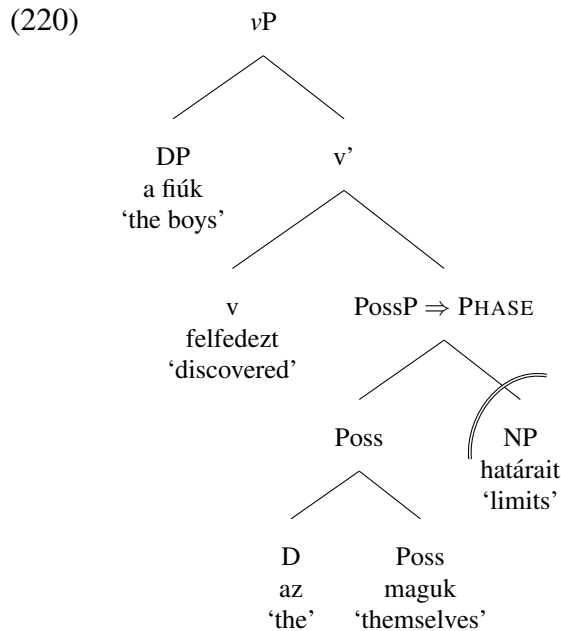
- (218) A fiúk felfedezték ...
the boys discovered-3PL ...
- a. **(az) ő* (kis) határ-a-i-k-at.
the he (little) limit-POSS-PL-3PL-ACC
‘The boys discovered their (little) limits.’
- b. *(??az) önmaguk* határ-a-i-t
the themselves limit-POSS-PL-ACC
‘The boys discovered their (own) limits.’ (Rákosi 2017, pp. 24–5)

However, when the simple reflexive anaphor *maguk* is used, it too must appear with the definite article. That is, the reflexive anaphor in (219) is bound by the subject even in the presence of the definite article.

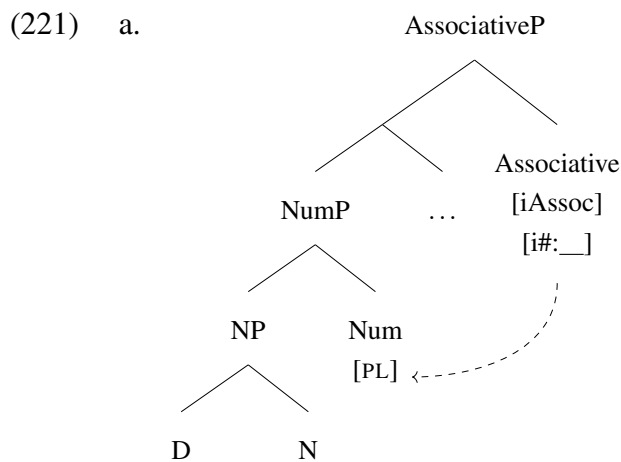
- (219) A fiúk felfedezték **(a) maguk* határ-a-i-t.
the boys discovered-3PL the themselves limit-POSS-PL-ACC
‘The boys discovered their (own) limits’

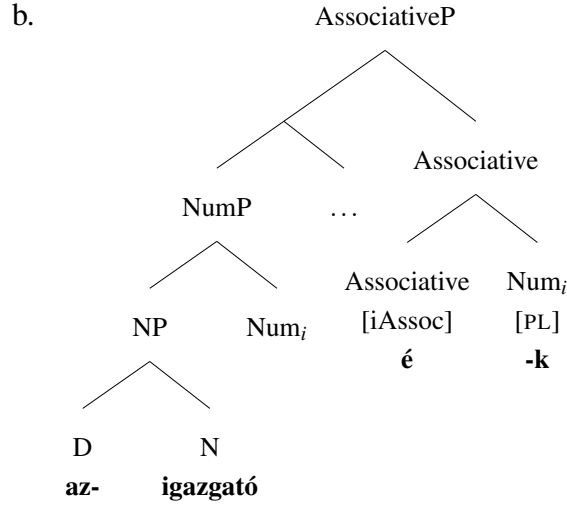
Oda (2022) explains this by proposing that definite articles can be base-generated adjoined to a nominal head without projecting DP (see also Bošković (2023)). Oda proposes that D is base-generated adjoined to Poss⁰ in (219), making PossP the nominal phase as the highest projection in the extended projection of N. The simple reflexive possessive *maguk* (which Oda argues is in Poss⁰) is then at the phase edge in (219) and can be bound by the subject in SpecvP.²² (220) illustrates this.

²²Note that there is a parallelism here between minimal (i.e. heads) and maximal (i.e. phrases) projections. It is standardly assumed that adjoined phrases can be base-generated adjoined or move to undergo phrasal adjunction. Likewise, heads should be able to either achieve head-adjunction by movement or they can be base-generated adjoined.



Following Oda, I claim that D^0 is also head-adjoined in the associative plural. In particular, I argue that the Hungarian affixal article in D^0 is adjoined to N^0 in the associative plural (this is on par with how Bošković analyzes it in Bošković (2023)). This is shown in (221). Crucially, since the definite article is adjoined to a nominal head below Num^0 , it does not intervene for head movement. This is shown in (221b).





Oda’s claim, that articles can adjoin to a nominal head rather than project DP, essentially extends Bošković’s (2012) DP/NP typology and Talić’s (2017) three-way split to introduce a ‘DP/NP-scale’ – some languages are DP languages at one end (e.g. English), others are NP languages at the other (e.g. Serbo-Croatian), but many languages fall at various points along a spectrum of more or less DP-like. Hungarian is one of these languages, and thus we sometimes find it behaving like DP is not projected even when the article is present (e.g., with the simple reflexive possessive *maguk*). The associative plural is another context in which Hungarian behaves less DP-like than other (affixal) definite article languages, e.g., Bulgarian. This difference can be captured if the presence of the definite article in Bulgarian entails the presence of DP, while the definite article in Hungarian can be head-adjoined without projecting DP.²³

²³This claim—that Bulgarian is an affixal article language that projects DP while Hungarian has a head-adjoined D—departs in some respects from Oda (2022). Oda argues that in some cases free-standing definite articles can be head-adjoined, while the current proposal allows the base-generation adjunction option only for affixal definite articles. This is in fact in line with the arguments in Bošković (2023). However, the current proposal also departs from Bošković in some respects. Bošković argues that the affixal article is adjoined to N⁰ in all languages that have multiple wh-fronting (MWF), which he shows all either lack articles or have affixal articles. Bulgarian is an affixal article language with MWF, and thus D⁰ is base-generated adjoined to N⁰ in Bulgarian on his analysis. However, on the present analysis, D⁰ projects DP in Bulgarian.

3.2.5 Summary

We have now established why the *plural pattern* is never found in languages with free-standing definite articles. I adopted a DP/NP-language distinction—based on the cross-linguistic generalizations in Bošković (2012b) *et seq* and Talić (2017)—that languages with free-standing definite articles always project DP, articleless languages never project DP and affixal article languages only project DP when the article is present. I showed that the presence of D^0 causes an intervention effect in the *plural pattern* that prevents Num^0 from incorporating into Associative^0 . Compelling evidence for this analysis of the *plural pattern* came from Bulgarian, an affixal article language that disallows the associative plural only when the article is present. Moreover, following Oda’s (2022) observation that Hungarian is a less DP-like language—such that it behaves like an NP language even when the definite article is present—I argued that in Hungarian associative plurals D^0 is head-adjoined to N^0 .

In the next section, I present three additional pieces of empirical evidence for a head-movement analysis of the *plural pattern*.

3.3 Additional evidence in favor of a movement-based approach

Among the literature on associative plurals, there are few works that take a generative typological approach. Researchers are usually concerned with associative plurals in a specific language or language family, and often treat their findings as idiosyncratic properties of the particular language/language family they are looking at. The approach taken in this work, on the other hand, views associative plurality (and in particular the *plural pattern*) as a syntactic phenomenon (head movement) that should show similar properties across all

languages. Moreover, the properties should follow from the head movement analysis.

In this regard, this section discusses three properties of *plural pattern* associative plurals that have been described for various languages (sometimes just one or two, sometimes more). I show that these properties can be explained if the *plural pattern* involves head movement of Num⁰.

3.3.1 Associative plurals and associative duals: one or both?

We saw in Chapter 2 that in languages with both plural and dual number, there are both associative plurals and associative duals. For example, both Central Alaskan Yup'ik and Khanty nouns inflect for plural and dual (222) and they both have associative plurals and associative duals that contain the same morphology (223).

- (222) a. qaya-**k** / qaya-**t** Central Alaskan Yup'ik
 kayak-DU / kayak-PL
 'two kayaks' / 'kayaks (3 or more)' (Corbett 2000, p. 108)
- b. qoj-**ɣən** / qoj-**ət**
 Khanty man-DU / man-PL
 'two men' / 'men' (Filchenko 2007, p. 76)
- (223) a. cuna-nku-**k** / Cuna-nku-**t** Central Alaskan Yup'ik
 Chuna-APL-DU / Chuna-APL-PL
 'Chuna and his friend' / 'Chuna and his family/friends'
 (Corbett 2000, p. 109)
- b. jəɣ-s-**äɣən** / ə-s-**ät** Khanty
 father-COLL-DU / mother-COLL-PL
 'A father and his son' / 'A woman with her family'
 (Filchenko 2007, pp. 79–80)

Interestingly, however, not all *plural pattern* languages with both plural and dual number have plural and dual associatives. Charney (1993), for example, says that the dual suffix

has an associative usage in Comanche (224). No mention is made, however, of the plural suffix (*-nii*) having this function.

- (224) Francine-**nikwih**-tsa
 Francine-DU-TOP
 ‘Francine and [her husband] Phil’

(Charney 1993, p.52)

Similarly, Guillaume (2008) states that an associative function is attested for the plural and dual suffixes in Cavineña. However, the examples Guillaume gives for the “associative plural” (225) do not look like typical associatives—most notably, the plural suffix is attached to an inanimate noun. Rather, these examples could be similitive plurals, which denote ‘X and things similar to X’ (see Chapter 5).

- (225) karetu -**kwana**
 cart -PL
 ‘cart and everything’

The dual suffix in Cavineña, on the other hand, is more clearly associative. It attaches to kin terms and picks out the named referent and one other associated person.

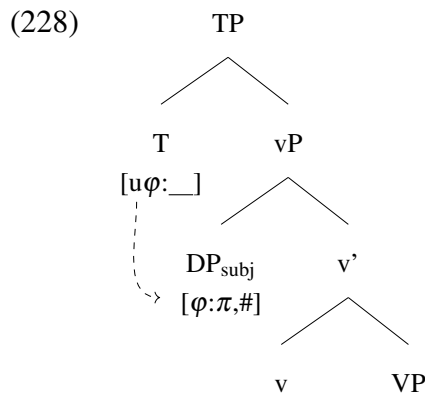
- (226) e -kwa =ke =**ekatse** / juye =**ekatse**
 3 -mother =3 =DU / ox -DU
 ‘his mother and associated person’ / ‘two oxen’

(Guillaume 2008, pp. 482, 655)

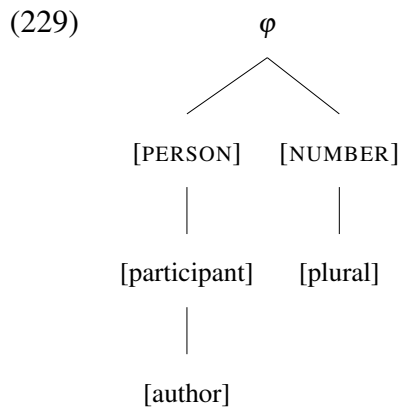
Cavineña may then have associative duals, but not associative plurals. Finally, it has been explicitly noted that Slovenian—which, recall, has an associative plural that is identical to the plural possessive (227a)—does not have associative duals (Vassilieva 2005). That is, Slovenian has a dual possessive that can be adnominal (227b) or headless (227c), but which cannot be used with an associative interpretation.

- (227) a. Lankot -ov **-i** so šli na ribo-lov.
 Lanko -POSS -PL.NOM.M aux.3PL went.M.PL on fish-hunt
 i. 'Lanko and his family/friends went fishing.' = associative
 ii. 'Those of Lanko (e.g., his family) went fishing.' = possessive
- b. Lankot -ov **-a** prijatelja
 Lanko -POSS -NOM.DU friends.DU
 'Lanko's (two) friends'
- c. Lankot -ov **-a** ta prišla
 Lanko -POSS -NOM.DU aux.3.DU arrive.M.DU
 'Lanko's two (e.g., children)' = possessive
 *'Lanko and his associate' $\neq$ associative

If associative number is dependent on regular number in the way that I have proposed (at least in the *plural pattern*) then we can explain why some languages have associative duals but not associative plurals, and why Slovenian has an associative plural but not an associative dual. The explanation involves Preminger's (2011) notion of *relativized probing*. To illustrate, consider first a straightforward example from English. In English, the verb shows person and number agreement with a DP subject. This is because T^0 bears a ϕ -probe that Agrees with and copies the ϕ -features of the closest DP, and the closest DP is the subject (228).



According to Preminger, English T^0 is relativized to the root of the feature geometry in (229) (i.e., ϕ). As a result, any DP bearing ϕ -features is a viable goal, and any ϕ -features on that DP are copied onto the probe. However, Preminger argues that probes may be relativized to a subset of the features in (229). Thus, he argues that in Kichean, the ϕ -probe on T^0 is relativized to [participant] (i.e., 1/2P). This is because verbs in the Agent-Focus construction²⁴ show agreement with a 1/2P argument, regardless of whether the argument is the Agent or the Patient (230). In other words, ϕ -agreement is tracking 1/2P features specifically. The probe will skip a DP that lacks a [participant] feature because it is not a viable goal.



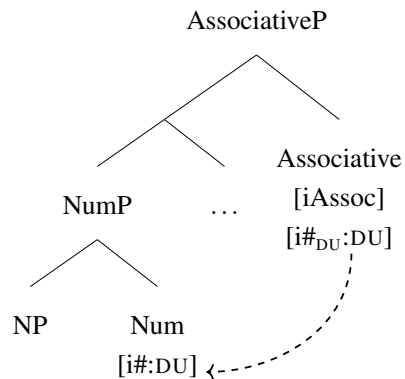
- (230) a. ja rat x-at/*Ø-axa-n ri achin
 FOC you.(SG) PRFV-2SG/*3SG.ABS-hear-AF the man
 ‘It was you that heard the man.’
- b. ja ri achin x-at/*Ø-axa-n rat
 FOC the man PRFV-2SG/*3SG.ABS-hear-AF you.(SG)
 ‘It was the man that heard you.’

Just as a $[\pi]$ probe may be relativized to a particular person value, I claim that the $[\#]$ probe on Associative⁰ may be relativized to a *particular* number value, [plural] or [dual].

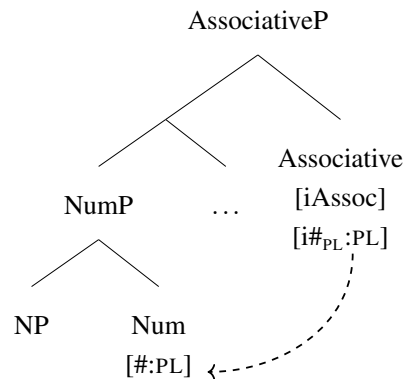
²⁴As the name suggests, the Agent-Focus construction involves focusing the Agent argument and suffixing an agent focus marker to the verb.

In Comanche, the probe on Associative⁰ is relativized to [dual], and thus only a Num⁰ bearing the feature [i#:DU] is a viable goal and capable of incorporating into Associative⁰ (231). In Slovenian, the probe is relativized to [plural] and only a Num⁰ bearing the feature [i#:PL] is a viable goal (232).

(231) *Relativized probing in Comanche*



(232) *Relativized probing in Slovenian*



To sum up, an Agree-based approach to the *plural pattern* explains the cross-linguistic variation in permissible associative number values. Some languages that have both dual and plural number have both associative plurals and associative duals, while others have one or the other but not both. Note, too, that the only attested associative number values are *subsets* of regular number values. This strongly implies that associative number is dependent on regular number in the way that I have proposed (see also Corbett 2000).

3.3.2 Associative plurals with numerals

A second puzzle regarding associative plurals is that they cannot be preceded by a numeral. This is the case in at least Mandarin Chinese, Japanese, Turkish, Hungarian and the South Slavic languages. (233) illustrates this with Hungarian.

- (233) a két igazgatóó -(*ék)
 the two director -APL
 ‘the two directors and their company’

In Mandarin, Turkish and Hungarian, the regular plural is also incompatible with numerals. Thus, as the example in (234) shows for Hungarian, when preceded by a numeral the noun must also remain unmarked for regular number.

- (234) a két igazgatóó -(*k)
 the two director -PL
 ‘the two directors’

One may think, then, given the analysis of the *plural pattern* that I have presented in this chapter, that numerals are incompatible with the associative plural *because* they are incompatible with the regular plural. That is, the presence of a numeral forces Num⁰ (for whatever reason) to be unrealized. Since the *plural pattern* is just the realization of Num⁰ in a different syntactic position, the restriction against realizing both elements holds here too. This would be a plausible solution if it weren’t for the South Slavic languages. These, too, disallow the associative plural to be preceded by a numeral but they *allow* numerals to precede the regular plural (possessive). This is shown in (235) with Serbo-Croatian (see Vassilieva (2005, pp. 35, 40) for this restriction in Bulgarian and Slovenian).

- | | |
|--|--|
| (235) a. tri Markova brata
three Marko brother.GEN.SG
‘three of Marko’s brothers’ | b. tri Markova
three Marko
*‘Marko & his associates, all three’ |
|--|--|

It thus seems that there is an independent restriction on numerals preceding an associative plural. I propose that this restriction arises due to conflicting requirements on Num⁰. Following Borer (2005), I assume that (in its base position) Num⁰ is responsible for divid-

ing the extension of the predicate NP, making the noun countable.²⁵ To illustrate, consider the English noun *apple* at the point at which it is a predicative NP. The NP combines with Num⁰ (-s) at LF to create an infinite number of divisions in the extension of *apple*. This divided mass can then be used in its bare form (*apples*) or combine with a counter (e.g., a numeral or a quantifier) to pick out a fixed number of the divisions (e.g., *three apples*). Crucially, counters cannot combine with undivided nouns, i.e., nouns that have not combined with Num⁰. Thus, Borer (2005) argues that mass nouns (e.g., I like *apple*) involve the absence of dividing structure (i.e., Num⁰).

With this in mind, consider again the ungrammatical example in (233), repeated below as (236a), in which a cardinal numeral precedes an associative plural in Hungarian. The LF structure of this ill-formed associative plural is given in (236b). Crucially, because Num⁰ has incorporated into Associative⁰, it does not combine directly with the noun at LF. As a result, the NP (*igazgato*) in the scope of the cardinal is uncountable.

- (236) a. a két igazgatóó -(***ék**)
 the two director -APL
 ‘the two directors and their company’
- b. [AssociativeP [NumP [#P **két**] [NumP [NP **igazgato**]-COUNT *t_i*] Assoc + Num_{*i*} -**ék**]]

Thus, (236b) reveals a conflict: cardinal numerals must combine with a count noun, which requires Num⁰ to combine directly with the noun (it must be in its base position). At the same time, the *plural pattern* requires Num⁰ to incorporate into Associative⁰, above the numeral. Since Num⁰ cannot be interpreted in both positions simultaneously, associative plurals preceded by cardinal numerals are ungrammatical.

Two pieces of evidence support this view. First, cardinal numerals (more specifically

²⁵Borer (2005) calls this head Div⁰, but the precise label is not important. What’s important is that it is the head that spells out plural morphology that divides the predicate, and I call that head Num⁰.

numeral-classifiers) can *follow* an associative plural in Mandarin (237a), but a numeral-classifier after a regular plural is still ungrammatical (237b). This implies that the restriction against numerals preceding associative plurals is independent from the restriction against numerals preceding regular plurals (if the restriction was the same, we would expect it to disappear when the numeral is postnominal for both plural types).

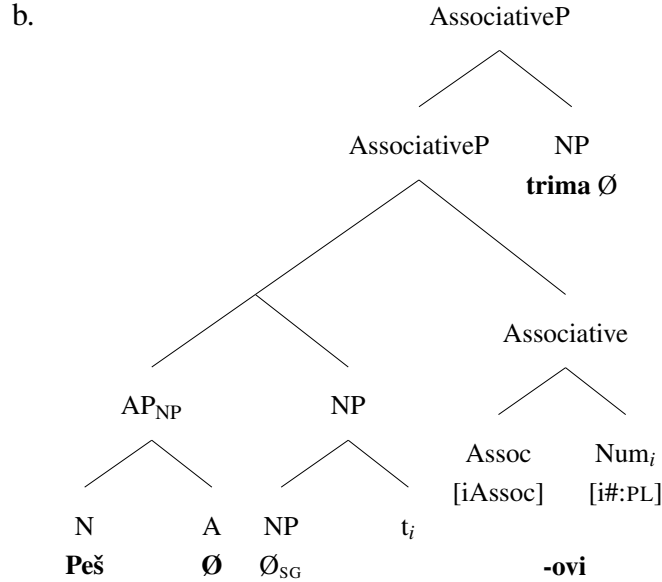
- (237) a. wo qing {XiaoQiang **-men** / xiaozhang -men} san -ge chifan
 I invite XiaoQiang -PL / principal -PL three -Cl eat
 ‘I invited {XiaoQiang/the principal and others} for a meal.’
- b. *wo qing {pengyou **-men** / xiaozhang -men} san -ge chifan
 I invite friend -PL / principal -PL three -Cl eat
 ‘I invited {three friends / three principals} for a meal.’

Following Li (1999), Jiang (2017) argues that the postnominal numeral-classifier is an *appositive modifier*. This is because a noun denoting ‘person’ is allowed after the numeral-classifier (238), suggesting that there are two N slots in the *noun > cardinal* order.

- (238) wo qing xiaozhang -men san -ge (**ren**) chifan
 I invite principal -PL three -Cl (person) eat
 ‘I invited the principal and others for a meal.’

Additional evidence that the *noun > cardinal* order involves two N slots comes from South Slavic (specifically Bulgarian and Slovenian), which also allows a cardinal numeral to follow the associative plural (239). Following Jiang, I propose that in the *noun > cardinal* order in South Slavic, the cardinal is an appositive modifier, i.e., it projects its own noun phrase. This is shown in (239b) for Bulgarian; the numeral is contained inside a separate nominal domain that is adjoined to AssociativeP.

- (239) a. Peš -ov -i **trima** (B) / Lankot -ov -i **tri**je (S)
 Peter -POSS -PL three / Lanko -POSS -NOM.PL three
 ‘{Peter / Lanko} and his associates, all three.’

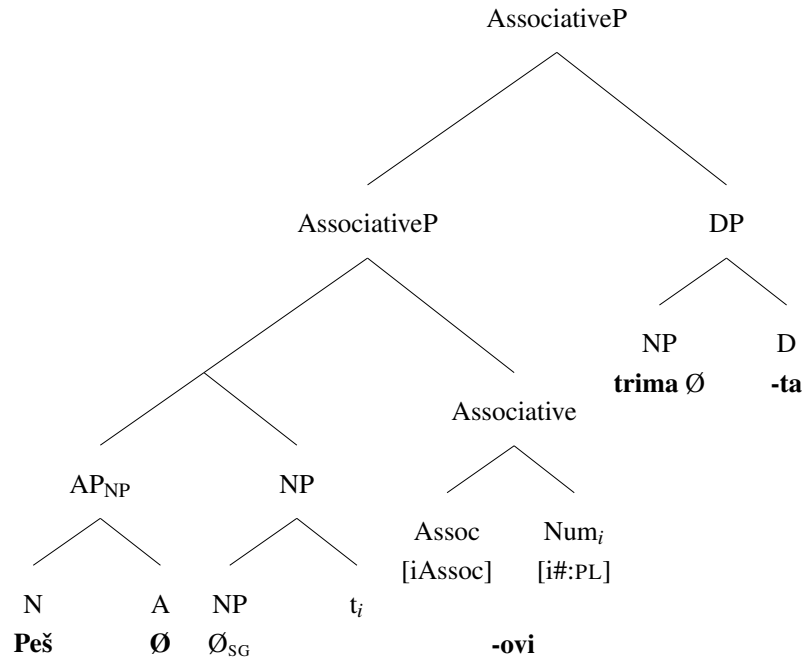


That we are dealing with two noun phrases here is evidenced by the fact that the definite suffix can surface on the post-nominal cardinal in Bulgarian (240).

- (240) Pešovi (i) trima **-ta**
 Peter.POSS.PL (all) three -DEF
 ‘Peter and his associates, (all) three’

We saw earlier that the definite suffix is unacceptable on a Bulgarian associative plural, and it was argued that this is because the φ -features on D^0 block Num^0 from incorporating into Associative^0 . If the *noun* > *cardinal* order in (240) involved a single DP, Num^0 must have illicitly moved over D^0 to generate the associative plural interpretation. This problem does not arise if (240) involves an appositive DP adjoined to a DP-less AssociativeP (241). In this configuration, Num^0 does not cross D^0 when it incorporates into Associative^0 .

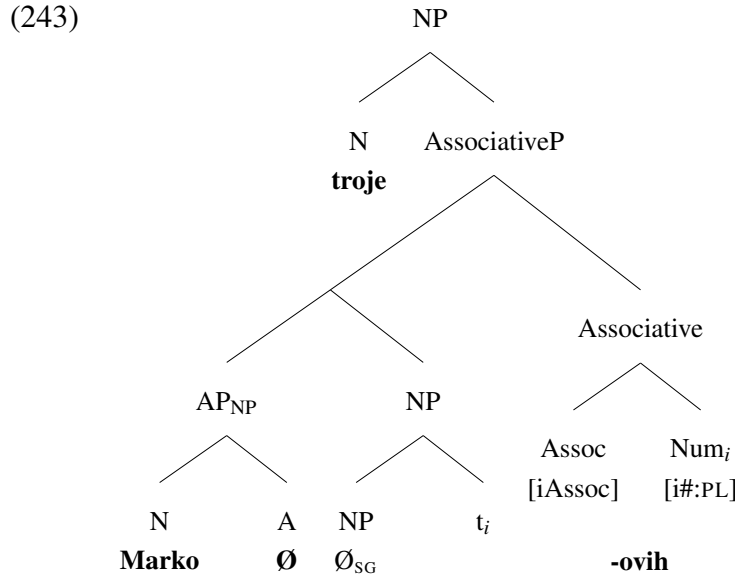
(241)



Second, *collective* numerals can precede associative plurals in Slovenian and Serbo-Croatian.

- (242) **troje** Lankot -ov -ih (S) / **troje** Marko -ov -ih (SC)
 three.COLL Lanko -POSS -GEN.PL / three.COLL Marko -POSS -GEN.PL
 ‘{Lanko / Marko} and his family/friends, all three’

I propose that the collective numerals in (242) head their own NP that takes the associative plural as its complement (243). Under this view, Num⁰ remains in the scope of the numeral even when it has incorporated into Associative⁰—that is, the collective numeral takes as its complement a countable nominal expression.



Two empirical observations support this analysis. The first, only observable in SC, concerns the case and number that the collective numeral assigns to the noun it modifies.²⁶ While the cardinals 2–4 assign genitive singular (244a), collective numerals and clear nominal numerals assign genitive plural (244b).²⁷ In other words, collective numerals pattern with clear nominal numerals in the case and number that they assign the noun.

- (244) a. tri studenta
 three student.GEN.SG
 ‘three students’
- b. {troje / trojica} studenata
 {three.COLL / three.NOUN} student.GEN.PL
 ‘three students’

Second, demonstratives show concord agreement with the numeral when it is nominal, and may do so when it is collective, but must agree with the noun when it is cardinal.

²⁶In Slovenian, the cardinals 2 and 3 are adjectival and do not affect the inflection on the noun; the cardinal numerals above 4, collective numerals and nominal numerals all assign genitive plural, and thus no distinction between the three can be drawn (see Stegovec 2022).

²⁷Not all numerals that assign genitive plural are nominal numerals—cardinal numerals above 5 also assign it. Another diagnostic for nominal numerals is that they decline for case (numerals 5 and above do not decline for case in SC).

(245a)–(245c) illustrates this. In (245a) the demonstrative modifies (and thus φ -Agrees) with the head noun *dog*. In contrast, demonstratives preceding clear nominal numerals modify (and thus φ -Agree with) the numeral (245b). Crucially, the demonstrative in (245c) can φ -Agree with the collective numeral—not the noun *psov*, ‘dog’—suggesting that the collective numeral is itself a noun and it is with this noun that the demonstrative agrees.²⁸

- (245) a. **teh** pet **psov**
 those.GEN.PL five dogs.GEN.PL
 ‘those five dogs’
- b. **ta** **peterica** psov
 those.NOM.SG.(F) five.NOM.SG.(F) dogs.GEN.PL
 ‘those five dogs’
- c. % **to** **petero** psov
 those.NOM.SG.(N) five.COLL dogs.GEN.PL
 ‘those five dogs’
- (Adrian Stegovec, p.c)

Thus, if collective numerals in Slovenian and SC are nouns that take the associative construction as their complement, as illustrated in (243), Num⁰ is in the scope of the numeral even when it has incorporated into Associative⁰. This is crucial for the requirement that numerals combine with countable entities.

To sum up, I have argued that the *cardinal* > *noun* order is ungrammatical with associative plurals because of conflicting requirements on Num⁰: cardinal numerals must combine with a count noun, which requires Num⁰ to be in its base position. Simultaneously, the *plural pattern* requires Num⁰ to incorporate into Associative⁰, above the numeral, where it combines with an individual-denoting entity of type e. Since Num⁰ cannot be interpreted in both positions simultaneously, associative plurals preceded by cardinal numerals are ungrammatical. This issue does not arise in the *noun* > *cardinal* order because the cardinal

²⁸(245c) is marginal (indicated by %), and may not be accepted by all speakers.

is an appositive modifier, nor does it arise in Slovenian and Serbo-Croatian when the preceding numeral is collective because collective numerals are nouns that take AssociativeP as their complement.

3.3.3 A restriction against plural possessor agreement

A final puzzle regarding the *plural pattern* to be discussed in this chapter was noted by Bartos (2000). Bartos observes that in Hungarian the co-occurrence of plural possessor agreement and the associative plural is ungrammatical. Consider first the acceptable examples in (246a), with singular possessor agreement. The kin term *novér* ‘sister’ is suffixed with first, second or third person singular possessor agreement (*-em/-ed/-é*, respectively) followed by the associative plural *-ék*. Now consider the examples in (246b), in which the singular possessor agreement has been replaced with *plural* possessor agreement. These examples are ungrammatical across the person paradigm.

- (246) a. *nővér-em-ék* / *nővér-ed-ék* / *nővér-é-ék*
 sister-1SG.POSS-APL / sister-2SG.POSS-APL / sister-3SG.POSS-APL
 ‘{my / your / his/her} sister and her friends’
- b. **nővér-ünk-ék* / **nővér-etek-ék* / **nővér-ük-ék*
 sister-1PL.POSS-APL / sister-2PL.POSS-APL / sister-3PL.POSS-APL
 ‘{our / your.PL / their} sister and her friends’

I add to Bartos’ observation that the same restriction against plural possessor agreement and associative plural morphology arises in Turkish. This is shown in (247).²⁹

²⁹Turkish associative plurals are degraded with third person possessor agreement regardless of number (i) which is why I have not included the singular/plural distinction in the third person for Turkish. For discussion of this restriction, see Ileri (2023) and Kunduracı (2013).

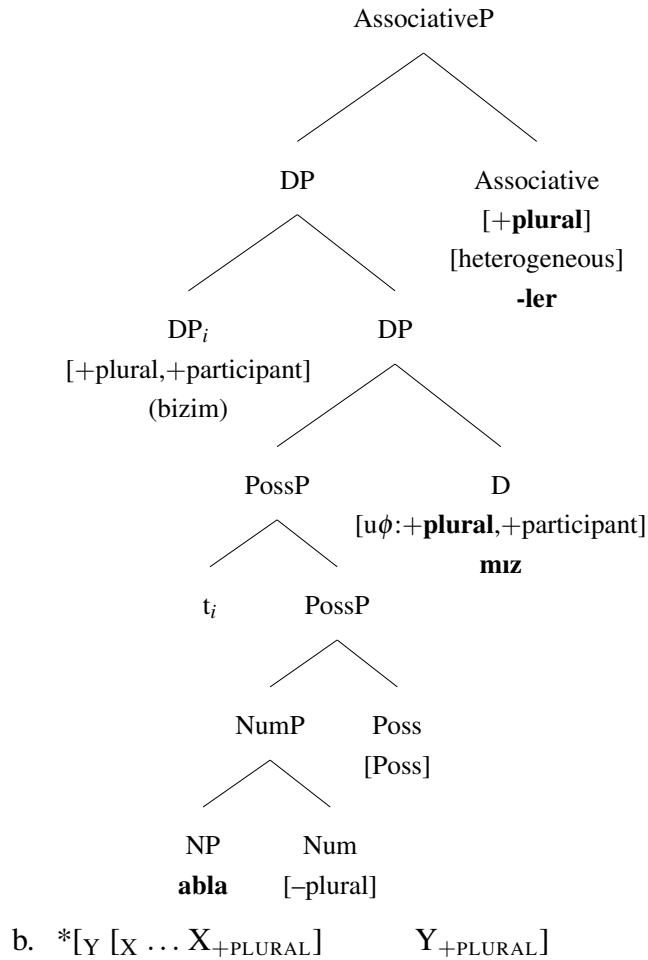
- (i) a. **abla-si-lar* b. **abla-lari-lar*
 elder.sister-3SG.POSS-PL elder.sister-3PL.POSS-PL
 ‘his/her elder sister and her friends’ ‘their elder sister and her friends’

(247a) shows that it is possible to have first or second person singular possessor agreement on a kin term (*abla*, ‘elder sister’) followed by the associative plural *-lar*. Replacing singular possessor agreement for plural, however, is ungrammatical (247b).

- (247) a. (benim) *abla-m-lar* / (senin) *abla-n-lar*
 (1SG-GEN) elder.sister-1SG.POSS-PL / (2SG-GEN) elder.sister-2SG.POSS-PL
 ‘{**my** / **your**} elder sister and her friends”
- b. *(bizim) *abla-miz-ler* / *(sizin) *abla-niz-ler*
 (1PL-GEN) elder.sister-1PL.POSS-PL / (2PL-GEN) elder.sister-2PL.POSS-PL
 ‘{**our** / **your.PL**} elder sister and her friends’

Only recently have authors tried to provide a solution to this puzzle. Ileri (2023) argues that the ungrammatical *-miz/-niz + ler* combination in Turkish is a morphophonological restriction against adjacent [+plural] features. He assumes the structure in (248a) for an associative plural with a possessor argument in Turkish. The possessor (whether it is overt or covert) is generated in SpecPossP but moves to SpecDP where it values [u ϕ :__] on D. The Associative head hosts the features [+heterogeneous] and [+plural] (following Dékány (2021)). Thus, when the possessor is plural, two adjacent nodes (D and Associative) host a [+plural] feature. Ileri argues that Turkish has a rule (given in (248b)) against the co-occurrence of two adjacent plural features in a single morphological word.

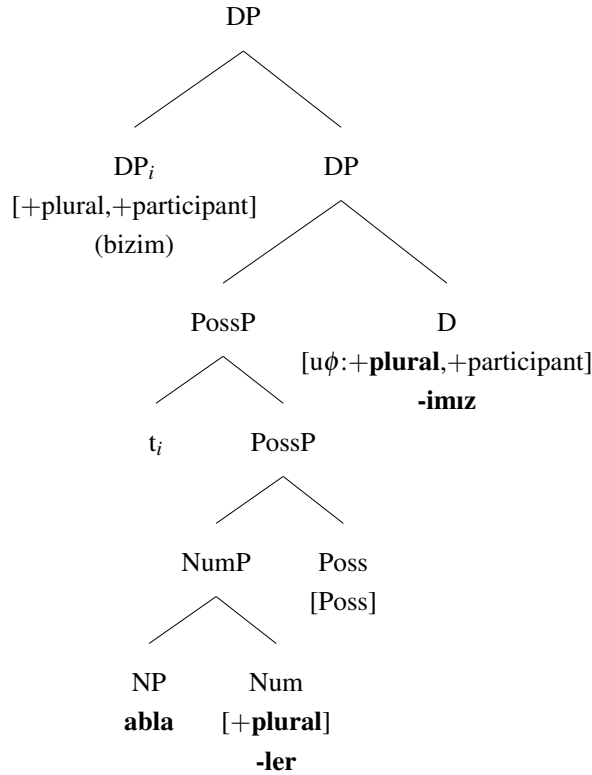
- (248) a. Structure for **(bizim) ablamızlar* ('our sister and her friends')



The rule in (248b) is not falsified by the grammatical (249a), which involves two string-adjacent plural morphemes (additive plural and plural possessive agreement). This is because the additive plural feature is in Num⁰ and the plural possessor features are in D⁰, separated by Poss⁰ (249b).

- (249) a. (bizim) abla -ler -ımız
1PL-GEN elder.sister -PL -1PL.POSS
'Our elder sisters'

b. Structure of *ablalerimiz* ('our sisters')



The crux of Ileri's (2023) analysis is that possessor agreement features are in D^0 , which makes them structurally adjacent to the associative (but not the additive) plural. One problem with this account, however, is that we have seen compelling evidence that articleless languages (like Turkish) lack DP. Additionally, if the **-mız/-nız+ -ler* combination is due to structurally adjacent plural features, this should ideally also explain why plural possessor agreement and the associative plural cannot co-occur in Hungarian either (i.e., **-ünk/-etek/-ük+ -ék*). However, nouns in Hungarian can be inflected for both definiteness and possessor agreement simultaneously (250). Assuming the definite article is in D^0 , possessor agreement cannot also be in D^0 —and, in fact, it is standardly assumed that possessor agreement in Hungarian is in Poss^0 (e.g., Dékány 2011; É. Kiss 2002; Szabolcsi 1994). Thus, Ileri's (2023) analysis of the restriction in Turkish does not extend to Hungarian.

- (250) ez a -ház -am
 this the -house -POSS.1SG
 ‘this house of mine’

(adapted from Dékány 2021, p. 111)

A different approach is taken by den Dikken (2023), who suggests that the restriction against *-ünk/-etek/-ük+ -ék* in Hungarian is semantic. This is based on the fact that not only is it ungrammatical for the possessor to be plural, but that it’s ungrammatical for the noun itself to be plural too (251). He speculates that there is a semantic restriction against plural elements being (part of) the focal referent.³⁰

- (251) *a -fiú -k -ék
 the -boy -PL -APL
 ‘the boys and their associates’

But this is problematic. First, there is no restriction against plural focal referents in Hungarian (as den Dikken in fact notes). If the focal referent is possessed, it can be plural (252). Recall that the plural suffix is *-i* (rather than *-k*) when adjacent to possessive morphology.

- (252) a -baráta -i -d -ék
 the -friend -PL -2SG -APL
 ‘your friends and their associates’

Moreover, Turkish shows us that the problem with plural possessors co-occurring with the associative plural is the plural possessor *agreement*, not the (semantics of the) plural possessor itself. Certain kin terms in Turkish have two variants: one that takes possessor agreement (253a) and one that does not (253b). Ileri (2023) shows that both of these kin terms can be used as the stem of an associative plural with a singular possessor (254). (254a) takes singular possessor agreement while (254b) does not.

³⁰In den Dikken’s (2023) terminology, the restriction is against the subject (or the subject’s possessor) of a silent group-denoting NP being plural.

- (253) a. benim baba -**m**
 1SG-GEN father -1SG.POSS
 ‘my father’
- b. benim peder -(***im**)
 1SG-GEN father -(1SG.POSS)
 ‘my father’
- (254) a. benim baba -**m** -lar
 1SG-GEN father -1SG.POSS -PL
 ‘my father and his associates’
- b. benim peder -ler
 1SG-GEN father -PL
 ‘my father and his associates’

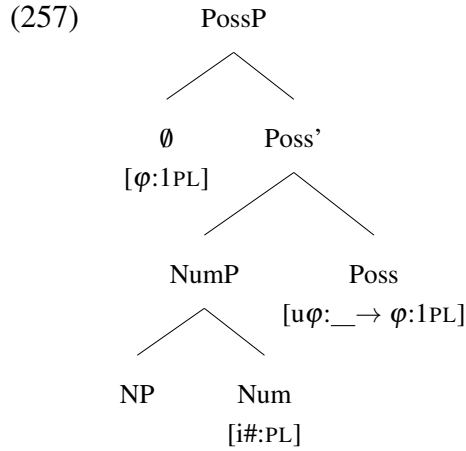
Interestingly, while *baba* is ill-formed as the stem of an associative plural with a plural possessor as expected (255a), *peder* is not (255b). Ileri (2023) thus concludes that the plural possessor alone is not the reason why examples like (255a) are ungrammatical. Plural possessor *agreement* is needed for the associative plural to be ungrammatical.

- (255) a. *bizim baba -mız -lar
 we-GEN father -1PL.POSS -PL
 Intended: ‘Our father and his associates’
- b. bizim peder -ler
 we-GEN father -PL
 ‘Our father and his associates’

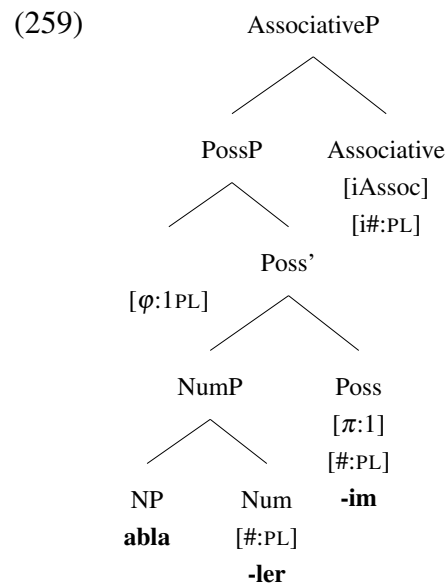
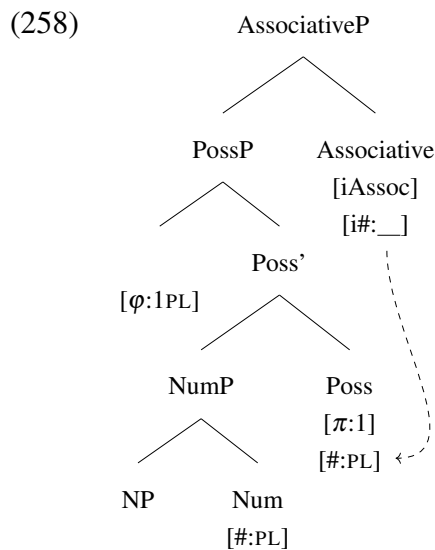
As additional evidence that it is possessor *agreement* that matters here, consider the associative plural in (256) from Japanese. Japanese does not have possessor agreement, and it is possible for a plurally possessed kinship term (e.g., *boku-tachi no oneechan*, ‘our elder sister’) to serve as the focal referent of an associative plural. Thus, both Turkish and Japanese indicate that there is no general semantic restriction on plurally possessed focal referents in associative plurals.

- (256) a. **boku-tachi** no oneechan-tachi Japanese
 1 P.PL POSS elder.sister-PL
 ‘**our** elder sister and her friends’
- b. **kimi-tachi** no oneechan-tachi
 2 P.PL POSS elder.sister-PL
 ‘**your.PL** elder sister and her friends’ (Yoshiki Fujiwara p.c)

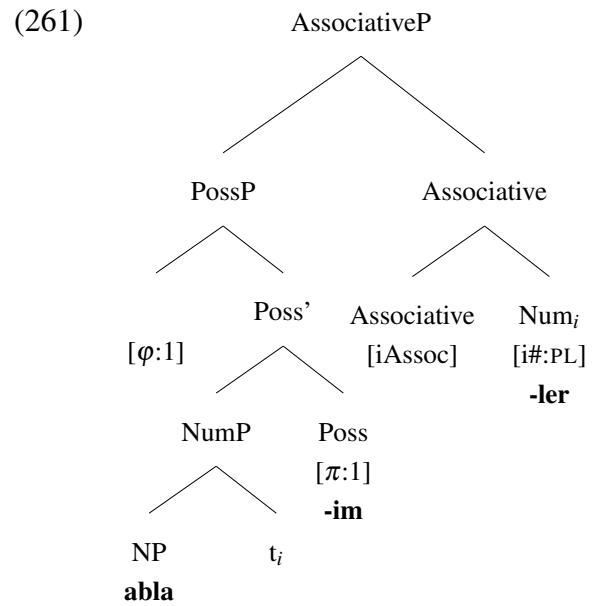
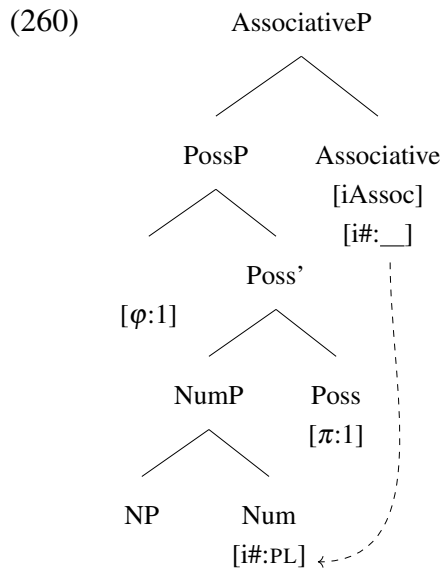
I will now show that a head movement analysis of the *plural pattern* can account for the ban on the sequence [-POSS.AGR(PL) + -APL]. Recall that under the current approach to the *plural pattern* Associative⁰ establishes an Agree relation with Num⁰ in order to value an interpretable number feature [i#:__]. Because of the featural composition of both Associative⁰ and Num⁰—that is, because Num⁰’s features are a proper subset of Associative⁰’s features—Num⁰ incorporates into Associative⁰ (Roberts 2010). Now consider how the derivation of an associative plural with plural possessor agreement in Turkish or Hungarian proceeds under this approach (257). I assume the possessor (whether it is overt or a non-overt *pro*) is base-generated in the specifier of PossP and the morphological reflex of possessor agreement is realized on the Poss head (e.g. Dékány 2011; É. Kiss 2002; Szabolcsi 1994). In (257) the ϕ -features of the possessor are first person plural and the uninterpretable ϕ -features on the Poss head are valued under Spec-head agreement. Num⁰ bears a plural [#] feature.



Next, Associative⁰ is merged and [i#:#] probes in search of a goal. However, the first appropriate goal that it finds is [i#:PL] on Poss⁰ (259). Just as we saw earlier with respect to DP languages, [i#:#] is valued but Num⁰ does not incorporate into Associative⁰ because it has not established an Agree relationship with the probe. The presence of a plural number feature on Poss⁰ thus causes an intervention effect. Failure of Num⁰ to incorporate into Associative⁰ means that (i) the morpheme order [-POSS.AGR(PL) + -APL] is underivable, and (ii) plural morphology has only an additive interpretation in its base-position.



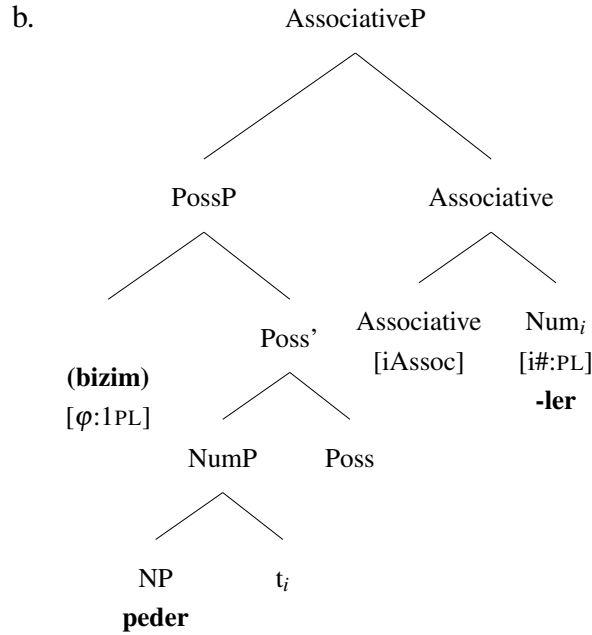
Now consider the well-formed cases of associative plurals with singular possessor agreement. Recall that I adopt the view that singular number is the absence of a [#] feature. That is, the Poss head does not possess a [#] ϕ -feature, only a [π] ϕ -feature, when the possessor is singular. Crucially, this means that Poss⁰ does not intervene between Associative⁰ and Num⁰ and thus the closest viable goal for [i#:__] is the valued number feature on Num⁰. (260) illustrates this for a Turkish associative plural with first person singular possessor agreement. As a result of Agree, Num⁰ incorporates into Associative⁰ and is pronounced (and interpreted) in this position (261).



Finally, the current movement-based approach predicts that there will be no intervention effect in *plural pattern* languages when there is no possessor agreement morphology, as is the case with Turkish nouns like *peder* (262a) and Japanese. If there is no possessor agreement, there are no [#] ϕ -features on the Poss head that intervene between Associative⁰ and Num⁰. Num⁰ is the closest head capable of valuing [i#:__] on Associative⁰ and it moves

to Associative⁰ as a reflex of this Agree relation (262b).

- (262) a. (bizim) peder -ler
 (we-GEN) father -PL
 ‘Our father and his associates’



That plural possessor agreement is ungrammatical with the associative marker in certain *plural pattern* languages has long been an unresolved puzzle in the literature. The current approach—in which *plural pattern* associative plurals involve movement of Num⁰ to Associative⁰—offers a straightforward explanation of the facts: plural possessor agreement, which involves [#] ϕ -features on a Poss head between Associative⁰ and Num⁰, causes an intervention effect that prevents Num⁰ from incorporating into Associative⁰. No intervention effect arises with singular possessor agreement since the Poss head lacks a [#] feature in this context, nor in languages that lack possessor agreement. Interestingly, the current approach predicts that all *plural pattern* languages that have possessor agreement should disallow plural possessor agreement with the associative plural. Whether this prediction is borne out requires further research but, should it be confirmed, it will be strong

evidence for a movement-based analysis of *plural pattern* languages.

3.3.4 Summary

In this chapter I have presented a new analysis of the *plural pattern* associative plural. The core of the analysis is that, in languages that show the *plural pattern* in their associative plural, the Associative head bears an interpretable, unvalued number feature. $[i\#:_]$ is valued by the matching number feature on Num^0 and, as a result of this Agree relation, Num^0 incorporates into Associative^0 . A head-movement analysis was initially motivated by the novel typological generalization established in Chapter 2: *plural pattern* languages either lack definite articles or have affixal definite articles.

In this chapter, I argued that the presence of D^0 between Num^0 and Associative^0 in languages with free-standing definite articles blocks head-movement of Num^0 and, thus, the *plural pattern*. Additionally, I showed that the proposed analysis allows for affixal article languages to have a *plural pattern* associative plural so long as D^0 is not projected between Num^0 and Associative^0 , which I argued is a theoretical possibility in such languages. The prediction was shown to be borne out in Bulgarian, which loses the associative plural interpretation of the plural possessive when the definite article is present. I also showed that a movement-based analysis is able to explain a number of independent properties of *plural pattern* associative plurals, which have not received an explanation in the previous literature. This included the observation that some languages, while they have both plural and dual additive number have only associative plurals or associative duals, but not both. Under the current proposal, languages such as these are predicted to exist given relativized probing (the probe on Associative^0 can be relativized to $[i\#_{\text{PL}}:_]$ or $[i\#_{\text{DU}}:_]$). The observation that cardinal numerals cannot precede an associative plural in a variety of languages was shown

to follow from the proposed analysis; in particular, it is a result of conflicting requirements on Num⁰. The role of Num⁰ in its base position is to divide the extension of a predicate into countable entities which can then be combined with a counter (a numeral or quantifier). On the current proposal, Num⁰ is not in its base position in *plural pattern* associative plurals, but instead must be interpreted in its incorporated position in Associative⁰. Since Num⁰ cannot be interpreted in both positions simultaneously, associative plurals preceded by cardinal numerals are ungrammatical. Finally, it was shown that the head-movement analysis of *plural pattern* languages resolves a long-standing puzzle regarding the ungrammaticality of plural possessor agreement co-occurring with the associative plural. I showed that the puzzle is really about possessor *agreement*, and that the plural φ -features on the Poss head block incorporation of Num⁰ into Associative⁰.

Chapter 4

Associative plurals and 1/2P plural pronouns

4.1 Overview

So far I have been assuming that associative plurals exist only in a subset of languages, namely those that use the regular plural as (or as part of) the associative plural or that have dedicated associative plural morphology. Thus, Turkish (263) and Sawila (264) have associative plurals but English does not (English can only express an analogous meaning using a coordinate structure (265)).

(263) *Ahmet-ler cik-ti*
Ahmet-PL leave-PST
'Ahmet and his family left.' (adapted from Görgülü 2011, p. 74)

(264) *Simon nang noo=li=da suwi onu*
Simon AM where=be.DIST=REAL arrive PART
'Where did Simon with his friends come from?' (Kratochvíl 2014, p. 139)

(265) *John and his family* left.

However, a number of authors posit that associative plurals are universal. In particular, they argue that first and second person (1/2P) plural pronouns, which have available an inherently associative meaning, are associative plurals in all languages (e.g. Cysouw 2003; Dékány 2021; Ghomeshi and Massam 2020; Moravcsik 2003; Ritter and Wiltschko 2018; Vassilieva 2005). That is, a first person plural pronoun like *we* has as one of its interpretations ‘the speaker and others’ and a second person plural pronoun can refer to ‘the addressee and others’ (Jespersen 1924). This has been called the ‘associative effect’ of 1/2P plural pronouns (Ackema and Neeleman 2018).

This chapter begins by discussing the 1/2P-plural-pronoun-as-associative-plural analysis (henceforth the *unified* analysis), presenting the data that has motivated these authors to treat 1/2P plural pronouns as associative plurals (Section 4.2). However, following Ackema and Neeleman (2018), I will argue that 1/2P plural pronouns are not associative plurals. Evidence for this comes from (i) regular, rather than associative, number marking on 1/2P pronouns (Section 4.3.1) and (ii) semantic constraints on associative plurals that do not extend to 1/2P plural pronouns (Section 4.3.2). After establishing that the unified analysis cannot be maintained, I present novel empirical evidence in Section 4.4 that 1/2P plural pronouns are not associative plurals: a number of languages, in addition to having regular plural pronouns, may also build a distinct associative plural on their 1/2P singular pronouns. This is predicted to be impossible under a unified analysis—if 1/2P plural pronouns are associative plurals, a distinct associative plural pronoun should not exist. Having shown that 1/2P plural pronouns are not associative plurals, Section 4.5 considers what, instead, might be responsible for their associative effect.

4.2 The unified approach: 1/2P plural pronouns are associative plurals

The observation that there is a semantic parallelism between 1/2P plural pronouns and associative plurals is an old one (Boas 1911; Jespersen 1924; Lyons 1968; for more recent discussion, see e.g. Cysouw 2003; Kratzer 2009; Moravcsik 2003; Wechsler 2010). These works note that like associative plurals, but unlike regular plurals, 1/2P plural pronouns (on at least one reading) are heterogeneous: the referents of the plurality are not all tokens of the same type. That is, the first person plural pronoun *we* typically refers to the speaker plus other *non-speakers*. It may have an additional additive interpretation (‘multiple speakers’)—known as ‘chorus *we*’—but this is not its most salient reading (in fact some authors argue it does not exist at all (e.g. Boas 1911)). Likewise, the second person plural pronoun *you* often refers to one addressee plus other *non-addressees*. On the other hand, regular plurals never have a heterogeneous reading: the referents of *dogs* must all have the property of being a dog. Given this semantic parallelism, a number of authors have argued that 1/2P pronouns obligatorily combine with the associative plural rather than the regular plural (Bobaljik 2008; Cysouw 2003; Dékány 2021; Ghomeshi and Massam 2020; Moravcsik 2003; Ritter and Wiltschko 2019; Vassilieva 2005). Cysouw (2003), for instance, argues that while 1/2P plural pronouns have both associative and additive interpretations, only the associative has been grammaticalized in human language; the ‘multiple speakers’ reading is only a (marked) semantic category. Going one step further, Dékány (2021) argues that 1/2P plural pronouns combine with the associative plural but that, even in chorus *we*, each speaker is actually referring to him/herself and others—the others just happen to also be speakers. Thus, there is no such thing as a true additive first person

plural pronoun. One piece of evidence often cited for the claim that 1/2P pronouns are obligatorily inflected with the associative plural is the generalization that no language has a distinct 1/2P plural pronoun used solely for the ‘multiple speakers’ or ‘multiple addressees’ reading. These are two of Bobaljik’s (2008) *person universals*, given in (266).

(266) U1: No language distinguishes [1+1] from [1+3].

U2: No language distinguishes [2+2] from [2+3]. (Bobaljik 2008, p. 206)

The idea is that if 1/2P pronouns could combine with the regular plural in addition to the associative plural, we would expect at least some languages to morphologize the ‘multiple speakers’ and ‘multiple addressees’ interpretations with a distinct form. Since this is not attested, 1/2P plural pronouns must be obligatorily inflected with the associative plural.

In addition to the semantic parallelism between 1/2P plural pronouns and associative plurals, some languages also show a morphological parallelism: the associative plural marker is also the pronominal plural marker. In Tamang, for example, associative plurals are formed by suffixing *-ni* to an appropriate nominal stem (e.g., a proper name, kinship term) (267a). The same suffix is the plural marker on 1/2P pronouns (267b), but not common nouns (267c).¹

(267) *Tamang* [Sino-Tibetan]

a. lap **-ni**
 father -APL
 ‘father and his party’ (Moravcsik 2003, p. 493)

b. ŋa **-ni** / e: **-ni**
 1 -PL / 2 -PL
 ‘we’ / ‘you (PL)’ (Lee 2011, p. 37)

¹Moravcsik (2003) gives the second person plural pronoun in Tamang as *zai-ni*. I have used the form given in Lee (2011). What is important is that both authors agree that the plural suffix on the pronoun is *-ni*.

- c. kitap **-kade**
book -PL
'books'

(Lee 2011, p. 31)

Similarly, Burmese associative plurals are formed by attaching *-tó/dó* to the nominal stem (268a) and the same suffix functions as a plural marker on 1/2P pronouns (268b) but not common nouns (268c).²

(268) *Burmese* [Sino-Tibetan]d

- a. Hla Hla **-dó**
Hla Hla -APL
'Hla Hla and her associates'

(New 2021, p. 167)

- b. nga **-tó** / min: **-tó**
1P -PL / 2P -PL
'we' / 'you guys'

(Soe 1999, pp. 58–9)

- c. s'aya **-dwe**
teacher -PL
'teachers'

(New 2021, p. 167)

These data are taken as morphological evidence by Moravcsik (2003) and Dékány (2021) that 1/2P pronouns combine with the associative plural rather than the regular plural. In languages like Burmese and Tamang, 1/2P plural pronouns are made up of the singular pronoun plus associative plural morphology; it follows that *all* 1/2P pronouns combine with the associative plural but that this is obscured by the fact that pronominal number is generally suppletive.

²Similar, though less compelling, evidence comes from languages like Japanese, where the associative plural is formed by suffixing *-tachi* to an appropriate stem (i). Japanese pronouns are also suffixed with *-tachi* to create their plural form (ii), but so are common nouns (iii). In languages that use the same marker for all three kinds of plurality (associative, pronominal, additive), whether 1/2P pronouns combine with the associative or regular plural is obscured by homonymy.

- | | | |
|--|--|--|
| (i) Taroo -tachi
Taroo -PL
'Taroo and his associates' | (ii) watasi -tachi
I -PL
'we' | (iii) gakusei -tachi
student -PL
'students' |
|--|--|--|

Thus, semantics and morphology seem to suggest that 1/2P plural pronouns are actually associative plurals (ones with a pronominal—instead of a nominal—focal referent). What this means in practice is that 1/2P plural pronouns and associative plurals are syntactically identical—whatever structure is adopted for one is also adopted for the other. One implementation of this idea comes from Vassilieva (2005). Vassilieva’s (2005) analysis of associative plurals is different from the one that I have been arguing for in this work. On her analysis, associative plural meanings arise because the focal referent moves to Spec,DP where it modifies a silent NP with group reference. Plural features on Num ensure that the silent group is interpreted as plural, while the high position of the focal referent, outside the scope of Num, means that it is interpreted as singular.

(269b) illustrates this for an associative plural from the creole Tok Pisin. The focal referent is *pater*, ‘priest’, occupying Spec,DP. The (regular) plural marker is *ɔl*, which occupies its usual position in Num and pluralizes the silent group NP.

- (269) a. *pater ɔl*
 priest PL
 ‘the priest and his congregation’
- b. [DP **pater** [D’ D [NumberP Num_[+PL] *ɔl* [NP $\emptyset$]]]

For Vassilieva, the relationship between the focal referent and the silent NP is one of predication, similar to the relationship between the two noun phrases in possessive and demonstrative constructions. That is, ‘this book’ predicates of ‘book’ that it is *near* the speaker; ‘John’s book’ predicates of ‘John’ that it *belongs* to John; in associative plurals it is predicated of the (silent) group NP that it is *represented by* the focal referent. Thus, the meaning of (269b) is ‘the group represented by the priest’.³

³To derive the fact that *pater ɔl* ‘the priest and his congregation’ includes the priest, Vassilieva argues that the focal referent starts out inside a PP headed by ‘with’ in the complement of the silent group NP.

Vassilieva argues that 1/2P plural pronouns have an associative effect because the structure of 1/2P plural pronouns is identical to (269b) (with the addition of GenderP). Like associative plurals, 1/2P plural pronouns have a focal referent (*speaker* or *hearer*) that is inherently associated with a group. Vassilieva's structure of the English first person plural pronoun is shown in (270). *Speaker* is in Spec,DP from where it modifies a (silent) plural NP. The entire structure in (270) is spelled out by the monomorphemic *we*.

(270) [DP **speaker** [D' D [NumberP Num_[+PL] [GenderP Gender_[+HUM] [NP Ø]]]] = **we**

A slightly different unification approach is taken by Dékány (2021). Her syntactic analysis is on par with what I have argued for in this work: the structure of an associative plural consists of a functional projection high in the nominal domain, the head of which realizes associative plural morphology. Thus, on her analysis, the relevant unifying property of associative plurals and 1/2P plural pronouns is the presence of an Associative Plural projection (AplP). The (Hungarian) associative plural in (271a) has the structure in (271b), in which the associative plural marker *-ék* is realized in the head of AplP. The semantic contribution of API is that the referent of the DP it attaches to ('the director') is part of a (non-homogeneous) group.

(271) a. az-igazgato-ék
 DEF-director-APL
 'the director and his/her associates'

b. [AplP [DP D **az-** [NumP Num_[+SG] [NP **igazgato**]]] Apl **-ék**]

1/2P plural pronouns have the same underlying structure, shown in (272). The features [+PROX] and [+PARTICIPANT]] on Deix specify the referent of the DP as first person. Num is specified for singular. Plural semantics comes from the API head which, as mentioned above, says that the referent of the DP that it attaches to is part of a non-

homogeneous group. The plural pronoun (e.g., *mi* ('we')) expones the entire structure as a single morpheme.

(272) [AplP [DP D [DeixP Deix_[+PROX,+PART] [NumP Num_[+SG] [NP]]]] Apl] = **mi** ('we')

According to a unified analysis, then, all languages have associative plurals in the pronominal domain because 1/2P pronouns and associative plurals are syntactically identical. This is why 1/2P plural pronouns are heterogeneous and why some languages use the same morpheme for associative plurals and 1/2P plural pronouns. This idea has been challenged, however, most notably by Ackema and Neeleman (2018). They argue that while 1/2P plural pronouns do have an associative interpretation ('speaker/hearer + others'), the associative effect is not due to 1/2P pronouns combining with the associative plural. Instead, they attribute the associative effect to a property of person features. In the following sections I build on two of Ackema and Neeleman's arguments against the view that 1/2P plural pronouns combine with the associative plural. First, Ackema and Neeleman wish to show that when plural marking on 1/2P pronouns is inflectional (rather than suppletive) in languages that have inflectional associative plural marking, it is regular, not associative, plural marking that appears on 1/2P pronouns. Second, Ackema and Neeleman show that there are semantic restrictions on associative plurals that do not extend to 1/2P plural pronouns. The following sections present new empirical evidence that strengthens Ackema and Neeleman's (2018) original claim.

4.3 Against a unified analysis: 1/2P plural pronouns are not associative plurals

4.3.1 Number marking on 1/2P pronouns

Ackema and Neeleman (2018) note that if 1/2P pronouns obligatorily combine with the associative plural, then when pronominal number involves a segmentable inflectional morpheme (as opposed to suppletion) we should always see associative plural morphology—not regular plural morphology—pluralizing pronouns. They discuss two patterns, one from Hungarian and one from Central Alaskan Yup'ik, in which plural marking on the pronoun is inflectional but the plural marker is *not* the associative plural.

The first pattern concerns oblique 1/2P pronouns in Hungarian. Assuming the morphological template [case–person–number] for Hungarian oblique pronouns, these 1/2P pronouns are pluralized with what looks like the regular Hungarian plural marker *-k*, not the associative plural marker *-ék*. That is, the dative 1/2P person plural pronouns in Hungarian can be segmented into dative case (*nek*), first or second person (*-ün*, *-te*) and regular plural (*--k*) (273). Ackema and Neeleman argue that if 1/2P pronouns obligatorily combined with the associative plural, inflectional number marking on the oblique 1/2P pronouns in Hungarian should be associative.

(273) *Hungarian dative 1/2P pronouns* (Ackema and Neeleman 2018, p. 92)

	Singular	Plural
1	nek-em	nek-ün- k /(*ék)
2	nek-ed	nek-te- k /(*ék)

In the second pattern, from Central Alaskan Yup'ik (CAY), the associative number marker is *-nku-t* (associative plural) or *-nku-k* (associative dual). Recall that the associative number marker in CAY is comprised of a dedicated suffix *-nku-* and the regular plural (*-t*) or dual (*-k*) suffix. If 1/2P non-singular pronouns are associative, we might expect to find the dedicated associative suffix (*-nku-*) in the non-singular pronominal forms. The relevant data is shown in (274). CAY pronouns are formed from two roots: first person *waŋ(a)*⁴ and second and reflexive third person *il-* (Miyaoka 2012). At first glance, the non-singular forms of the first person pronoun do seem to contain associative morphology. That is, both the first person plural and dual pronouns contain *-ŋku-*.

(274) *Central Alaskan Yup'ik 1/2P pronouns* (Miyaoka 2012, p. 383)

	Singular	Plural	Dual
1	wii wiŋa	waŋkuta	waŋkuk
2	ilpit	ilpici	ilpitik

However, Ackema and Neeleman argue that the *-ŋku-* in these pronouns is not a segmentable morpheme. Rather, *-ku-* is the non-singular expander that is also used in demonstratives—*u-ku-t*, (this-EX-ABS.PL), ‘these ones’ (Miyaoka 2012, p. 1168)—and *ŋ* is part of the first person root *waŋ(a)*-. Their underlying segmentation of the first person non-singular forms is shown in (275). On this analysis, the associative marker *-nku-* is absent.

(275) *Underlying forms of 1P non-singular pronouns in CAY*

a. *waŋ-ku-ta* = 1PL

⁴The surface forms of the first person singular pronouns are *wii* and *wiŋa* because of a phonological rule that changes /aa/ to /ii/ before *ŋ* (Miyaoka 2012, p. 205). This change is phonologically conditioned in the long form of the pronoun and retained in the short form.

b. *waŋ-ku-k* = 1DU

The non-singular second person forms clearly do not contain associative morphology. Instead, the second person plural is composed of the second person root *iŋ-* and the second person plural absolutive/relative case marker *-pici*. Likewise, the second person dual is composed of the same root and the second person dual absolutive/relative case marker *-pitik*. Thus, first and second person non-singulars in CAY do not contain associative morphology (*-nku-*), a strange state of affairs if 1/2P non-singular pronouns are associatives.

Nevertheless, the evidence presented by Ackema and Neeleman is somewhat unsatisfactory. First, Dékány (2021) shows that the morphological form of the oblique 1/2P plural pronouns in Hungarian is not evidence that 1/2P pronouns take the regular plural. She argues that the oblique pronouns are not based on the template [case–person–number], but rather [droppable (nominative) pronoun–case–(possessive) agreement for person and number]: a full (nominative) plural pronoun has been dropped from the position preceding dative case, and *-ünk* and *-tek* reflect person and number agreement with this dropped pronoun. This is shown in (276). When the data is interpreted this way, the *-k* in the plural form of the dative pronoun is not the regular plural marker in Num⁰ but part of a complex Agr(eement) head with both person and number features represented.

(276) *Hungarian dative 1/2P pronouns*

(Dékány 2021, p. 248)

	Singular	Plural
1	(én)nek-em	(mi)nek-ünk
2	(te)nek-ed	(ti)nek-tek

The Central Alaskan Yup'ik data is more compelling, but not without its issues. First, Ackema and Neeleman (2018, p. 300) themselves concede that the first person non-singular

forms could be derived from the underlying forms in (277), which do contain the associative morpheme *-nku-*. The initial /n/ of *-nku-* is deleted at PF to prevent the sequence /ŋ-n/.

(277) *Alternative underlying forms of 1P non-singular pronouns in CAY*

a. *waŋ-nku-ta* = 1PL

b. *waŋ-nku-k* = 1DU

Second, the second person non-singular pronouns realize case and number simultaneously on a single head, meaning that the absolutive/relative plural and dual suffixes are not segmentable into separate case and number marking. This, in turn, means that the absence of the associative marker is not in and of itself revealing. This is because the surface form of simultaneous exponence does not represent a one-to-one mapping of categories, but rather categories are “fused” (Noyer 1992). This fusion arguably obscures the underlying categories in these second person non-singular pronouns, meaning that second person pronouns combining with associative number in CAY cannot be ruled out.

However, I will now show that there is other compelling evidence from inflectional number marking on 1/2P pronouns which suggests that these pronouns combine with the regular, rather than the associative, plural. The first piece of evidence comes from Desano, a Tucanoan language that has an associative plural marked with *-sã*, a particle meaning ‘also’ and the regular plural marker on female kin terms (Aikhenvald 2014, p. 51). This is shown in (278a). The other regular plural markers in Desano are *-a/-da* (used with animates) (278b) and *-di* (used with inanimates) (278c).

(278) a. *aʔl -sã*

dad -APL

‘dad and the others’

(Mauri and Sansó 2021, p. 20)

- b. yahi / yahi -a
heron / heron -PL.AN
'heron' / 'herons' (Lima Silva 2012, p. 112)
- c. wii / wii -di
house / house -PL.IA
'house' / 'houses' (Lima Silva 2012, p. 120)

The 1/2P pronouns in Desano are given in (279). The first person plural forms are suppletive and thus obscure the type of the underlying number. However, Lima Silva (2012, p. 141) argues that the second person plural can be segmented into the second person singular pronoun *bʉ* and the animate plural suffix *-a*. This pattern is expected if 1/2P pronouns combine with the regular plural, but puzzling if they obligatorily combine with the associative plural (where we would expect to see *-ša*).

(279) *Desano 1/2P personal pronouns*⁵ (Lima Silva 2012, p. 140)

	Singular	Plural
1	yʉʉ	~badi (INCL) / gua (EXCL)
2	~bʉʉ	~bʉa

Similar evidence is found in Buriat (Mongolic). Buriat has an associative plural marked by the suffix *-tan* (280a), which also serves to form collective nouns (Mauri and Sansó 2021, pp. 21–2). According to Poppe (1960), various suffixes mark nominal plurality in Buriat: *-d*, *-uud*, *-nuud* and *-nar* (280b–c). The choice between *-d*, *-uud* or *-nuud* is determined by phonological properties of the nominal stem. The suffix *-nar* is restricted to nouns denoting kinship relations and socially important people ending in vowels; it is also used with loan words that end in palatalized consonants.

⁵A tilde preceding the pronoun indicates that the surface form of the morpheme is nasalized.

- (280) a. Badma -**tan**
 Badma -APL
 ‘Badma and his family’ (Daniel and Moravcsik 2013)
- b. nojo -**d** / ang -**uud** / íetrad -**nuud**
 prince -PL / wild animals -PL / notebook -PL
 ‘princes’ / ‘wild animals’ / ‘notebooks’ (Poppe 1960, p. 34)
- c. axa -**nar**
 older brother -PL
 ‘older brothers’ (Poppe 1960, p. 35)

The 1/2P pronouns in Buriat are given in (281). Like Desano, Buriat shows suppletion in the 1/2P plural forms—*bi* ~ *biden/man*⁶ (1P) and *ši* ~ *ta* (2P)—obscuring the underlying number. However, as (281) shows, the plural forms of these pronouns may also take the regular plural suffix *-nar*.

- (281) *Buriat (nominative) 1/2P pronouns* (Poppe 1960, pp. 50–2)

	Singular	Plural
1	bi	biden / man / biden nar
2	ši	ta / taan nar

There is no meaning difference between *biden* and *bidenar* in the first person plural, but a distinction is made in the second person: *ta* has taken on the role of an honorific plural, used to address a singular addressee politely. To refer to a plurality that includes the addressee, *taanar* is used. In other words, to differentiate the (semantically) singular form *ta* from plural *ta* the pronoun is inflected for number, and the exponent of this number

⁶Historically, *biden* and *man* delineated clusivity: *biden* was the inclusive first person plural and *man* was the exclusive first person plural. This distinction has since been lost.

value is *-nar*. Thus, when number on a 1/2P pronoun is inflectional in Buriat it is the regular plural, not the associative plural, that surfaces.

A version of the regular plural, rather than the associative plural, also surfaces on 1/2P pronouns in Wutun, a vernacular of Northwest Mandarin. The associative plural marker in Wutun is *-mu* (282a).⁷ Lexical nouns are marked for plural using the suffix *-dera*, though some human nouns are marked with *-men* (the Mandarin Chinese plural marker) (282b–c). Wutun also has a paucal marker, *-jhege* (282d).

- (282) a. aba **-mu**
 father -APL
 ‘father and his associates’(Janhunnen et al. 2008, p. 66)
- b. alak / alak **-dera**
 lama / lama -PL
 ‘lama’ / ‘lamas’
- c. niren / niren **-men**
 ‘woman’ / ‘woman’ -PL
 ‘wife/wives’
- d. zhawa / zhawa **-jhege**
 disciple / worker -PAUC
 ‘worker’ / ‘(a couple of) workers’(Sandman 2016, pp. 47–50)

The 1/2P pronouns in Wutun are given in (283). This time, number is clearly inflectional on both first and second person stems, and is identical to the nominal paucal (*-jhege*).⁸ Taking paucal to be a sub-type of regular number (Harbour 2014; Noyer 1992)—not a different category of number—pluralization of the 1/2P pronouns in Wutun involves regular number marking.

⁷According to Janhunnen et al. (2008), *-mu* can attach to human nouns to generate an associative plural only in some idiolects.

⁸Although the pronominal *plural* marker is identical to the nominal *paucal*, the interpretation of the 1/2P pronouns in (283) is not paucal (i.e., *a few, some*), but simply plural (Janhunnen et al. 2008; Sandman 2016).

(283) *Wutun 1/2P pronouns*

(Sandman 2016, p. 69)

	Singular	Plural
1	ngu	ngu- jhege
2	ni	ni- jhege

Finally, it is the regular plural marker, not the associative plural marker, that pluralizes 1/2P pronouns in Amdo Tibetan (Sino-Tibetan). A number of associative plural suffixes have been described for Amdo Tibetan: *-ts^hang*—with allomorphs *-zaŋ* and *-saŋ*—(Simon 2021), *-lasokkwa* and *-tɕenbo* (Ebihara 2010). An associative plural with *-ts^hang* is given in (284a) and another one with *-lasokkwa* in (284b). The regular nominal plural is marked by *-tʃ^hao* or one of the following allomorphs: *-tʃ^ho*, *-zo*, *-so* (285).

- (284) a. ^hlə^mbəmts^heraŋ -**ts^haŋ** -kə k^ha^rtsaŋ rə mə -^fkwa -nəere
L. -APL -ERG yesterday tent NEG -move -GENER.FACT.Q
'Didn't Ləmbəm Tserangs [i.e. his family] move their camp yesterday?'

(Simon 2021, p. 71)

- b. awa -**las^hokkwa**
father -APL
'father and his company'

(Ebihara 2010, p. 53)

- (285) a. ʃaʒə -**tʃ^hao** -kəŋen -i za -jokə
children -PL -ERG listen -CONV stay -PERF.SENS
'The children are listening.'

- b. s^ha -**zo** koɖək je -zoknə
fields -PL distribution LIGHTV -RES.PERF.FACT
'[They] have redistributed the fields [of the villagers].' (Simon 2021, pp. 55–6)

The 1/2P pronouns in Amdo Tibetan are given in (286). Both first and second person stems are inflected with an allomorph of the regular plural marker (*-tʃ^ho* / *-zo*) and the

associative plural marker(s) is absent.⁹

(286) *Amdo Tibetan 1/2P pronouns*

(Ebihara 2010, p. 49)

	Singular	Plural
1	ŋa	ŋə (EXCL) / ə (INCL) {-tʃ ^h o / -zo}
2	tʃ ^h o	tʃ ^h i-tʃ ^h o

Thus, each of these languages—Desano, Buriat, Wutun and Amdo Tibetan—has an associative number marker, and pronominal plurality is (at least sometimes) marked inflectionally on a pronominal stem. If a unified analysis is on the right track—that is to say, if 1/2P pronouns obligatorily combine with the associative plural—it is precisely in these languages that we would expect to see the associative plural pluralizing 1/2P pronouns. Instead, the regular plural marker surfaces. The absence of the associative plural, and the presence of the regular plural, receives a straightforward explanation if 1/2P pronouns combine with the regular plural just like other nominals do.

The morphological evidence is thus in line with the view that 1/2P pronouns inflect for the regular plural, not the associative plural. In the next section, I turn to the semantics of associative plurals and show that they do not carry over to 1/2P plural pronouns.

4.3.2 Semantic restrictions on associative plurals

Besides their heterogeneity, associative plurals have been shown to possess other semantic properties that regular plurals do not. For example, it has been claimed that, at least in

⁹I have adapted the orthography in Ebihara (2010) to match that of Simon (2021) to keep things consistent. What Simon writes as <tʃ^h> Ebihara writes as <tʃ^h>.

some languages, associative plurals are restricted to a collective interpretation while regular plurals allow both collective and distributive readings. For example, Görgülü (2011) shows that this semantic restriction on associative plurals holds in Turkish. Thus, the regular plural in (287) can either describe a situation in which (a) the speaker's brothers went to Ankara together or (b) each brother went at a different time. The associative plural in (288), however, can only describe a situation in which the speaker's brother and his friends went together.

- (287) **Abi-ler-im** **Ankara-ya** **git-ti**
 brother-PL-1SG Ankara-DAT go-PAST
 'My brothers went to Ankara'
- a. My brothers went to Ankara together. (collective)
- b. My brothers went to Ankara at different times. (distributive)
- (288) **Abi-im-ler** **Ankara-ya** **git-ti**
 brother-1SG-PL Ankara-DAT go-PAST
 'My brother and his family / friends / associates went to Ankara'
- a. My brother and his associates went to Ankara together. (collective)
- b. *My brother and his associates went to Ankara at different times. (distributive)
- (Görgülü 2011, p. 74)

Ackema and Neeleman (2018) then show that 1/2P plural pronouns in Turkish behave like the regular plural in (287)—they are compatible with distributive readings.

- (289) **Biz** **farklı** **zaman-lar-da** **gel-di-k**
 we different time-PL-LOC come-PST-1PL
 'We came at different times.'
- (290) **Siz** **farklı** **zaman-lar-da** **var-dı-nız**
 you.PL different time-PL-LOC arrive-PST-2PL
 'You arrived at different times.'

This interpretation of Turkish 1/2P plural pronouns makes sense if they are inflected with the regular plural—regular plurals allow both collective and distributive interpretations.

A second semantic difference between associative plurals and 1/2P plural pronouns is the relationship between the named referent and their associates. Ackema and Neeleman (2018) argue that the relationship between *speaker* or *addressee* and their associates in 1/2P plural pronouns is in no way fixed; associates of 1/2P pronouns are entirely dependent on the utterance context and/or linguistic antecedents. In contrast, the relationship between the named referent and their associates in associative plurals is much more constrained. To see this, consider first the utterance in (291). In this example, “that girl over there” is introduced to the discourse context in the first sentence and contained in the reference of the first person plural pronoun in the second sentence. The same utterance is felicitous in Japanese (292). Thus, so long as an individual is introduced into the discourse context, it is a possible associate of the *speaker* or *hearer*.

(291) Look at [**that girl over there**]_j. [**We**]_{i+j} are wearing the same socks.

(adapted from Ackema and Neeleman 2018)

(292) [**Ano on’nanoko-o**]_j mite. [**Watasi-tachi**]_{i+j} onaji kutushita-o haite-iru
 that girl-ACC look.IMP. I-PL same socks-ACC wear-PROG
 ‘Look at that girl. We are wearing the same socks.’

(Giulio Muramatsu, p.c)

If 1/2P plural pronouns are associative plurals, replacing the pronoun in (292) with an associative plural should be felicitous. (293) shows that this is not the case; just being in the discourse context is not enough to make “that girl” an associate of Taro.

- (293) #[Ano on'nanoko-o]_j mite. [Taroo-tachi]_{i+j} onaji kutushita-o haite-iru
that girl-ACC look.IMP. Taroo-PL same socks-ACC wear-PROG
Intended: 'Look at that girl. Taroo and her are wearing the same socks.'

(Giulio Muramatsu, p.c)

Thus, any individual in the discourse context is a possible associate of the speaker/hearer with 1/2P plural pronouns. In contrast, associates in associative plurals are restricted to a small number of pre-determined relations (e.g., family/friends/colleagues).¹⁰

Additionally, according to Moravcsik (2003), the named referent and their associates in associative plurals are members of a conceptually coherent group; they are not a set of unconnected individuals. Regular plurals, on the other hand, may refer to members of a group or to a set of unconnected individuals. This is illustrated in (294). In (294a), *the boys* could pick out a set {Arthur, Bill, Chris}, where none of the individuals in the set know each other and each member will play on a different team. On this reading *the boys* picks out unconnected individuals.¹¹ In (294b), *the boys* ({Arthur, Bill, Chris}) now make up a single team, and are thus understood as members of a conceptually coherent group.

- (294) a. The boys will be allowed to play soccer after school.
b. The boys won 3–0.

Moravcsik argues that associative plurals always behave like (294b). Thus, she says that an associative plural could not refer to 'X and his enemies', because 'X and his enemies' does not form a coherent group. Testing Moravcsik's (2003) claim on Japanese, this in fact

¹⁰The associates in an associative plural are not entirely pre-determined; they can be influenced by the context. Moravcsik (2003) discusses an example of what she calls "incidental association": a group of school children going on a school trip are divided between two buses, each supervised by a parent; the associative plural of a parent's name can be used to refer to that parent and their bus-load of children. What is important is that a linguistic antecedent is not enough to license an associative plural, but this is routinely how referents of 1/2P plural pronouns are introduced.

¹¹(294a) also has a reading where the set {Arthur, Bill, Charles} form a group. What is important is that a non-group reading is available.

holds. Consider the example in (295), which describes how one group (*Taroo-tachi*, ‘Taro and his associates’) is about to do battle with an enemy group. Using a predicate like *fight*, we can test whether this enemy group can be included in the reference of *Taroo-tachi*, ‘Taro and his associates’. This is because *fight* can take both implicit and explicit arguments. If I say *John and Mary are fighting*, both arguments (*John, Mary*) are expressed, but if I say *John is fighting*, who John is fighting is left implicit. When the explicit argument of *fight* is plural, it can serve as the reference of both arguments: in *the boys are fighting*, it’s possible for *the boys* to refer to John and Bill and the meaning of the sentence is that the boys are fighting each other (i.e., *John is fighting Bill* or vice versa). Thus, if the associative plural in (295) is capable of referring to Taro and his associates *and* Taro’s enemy and his associates, we would expect it to behave like a regular plural noun and mean something like ‘Taro (and his group) and his enemy (and his group) will fight each other’. However, (295) is only felicitous if it refers to Taroo and those fighting on his side, with the other argument remaining implicit.

(295) **Scenario:** During a superhero movie, the hero (Taroo) and his group are about to have a final showdown with the villain and his group.

Mousugu T.-no teki-ga yattekuru. **T.-tachi-wa** sinu-made tatakau-daroo
soon T.-GEN enemy-PL arrive. T.-PL-TOP die-until fight-EVID
‘Taroo’s enemies are coming ...’

- a. = Taroo and his group will fight to the death.’
- b. ≠ Taroo (and his group) and his enemy (and his group) will fight to the death.’

(Giulio Muramatsu, p.c)

If 1/2P plural pronouns are associative plurals, they should also be unable to be used to refer to Taroo *and* his enemy in (295). This is not the case, as shown in (296). The plural pronoun can pick out both arguments simultaneously: ‘me (and my group) and my enemy

(and his group)’.

(296) **Scenario:** During a superhero movie, the hero and his group are about to have a final showdown with the villain and his group. The hero says:

Mousugu teki-ga yattekuru. **Watashi-tachi-wa** sinu-made tatakau-daroo
soon enemy-PL arrive. I-PL-TOP die-until fight-EVID
‘My enemies are coming ...’

a. = Me and my group will fight to the death.’

b. = Me (and my group) and my enemy (and his group) will fight to the death.’

(Giulio Muramatsu, p.c)

Thus, 1/2P plural pronouns can pick out both arguments of a predicate even if they do not form a cohesive group. Associative plurals cannot.

4.3.3 Summary and looking ahead

Building on the arguments in Ackema and Neeleman (2018), I have shown in this section that 1/2P pronouns are inflected with the regular, not the associative, plural. Morphological evidence for this came from regular plural marking on 1/2P pronouns in Desano, Buriat, Wutun and Amdo Tibetan, a pattern which is difficult to account for if 1/2P pronouns obligatorily combine with the associative plural. Semantic differences between associative plurals and 1/2P plural pronouns also indicate that the latter take the regular plural, with 1/2P plural pronouns patterning like regular plurals in allowing distributive readings in Turkish, in being able to update their referents in the linguistic context and in having both group and unconnected individual interpretations.

Ackema and Neeleman point out that if 1/2P pronouns take the regular plural there is a possibility that some languages will have, in addition to regular 1/2P plural pronouns, dedicated associative plural pronouns. That is, a subset of languages may allow a 1/2P pronoun

to combine with the associative plural. The form of this pronoun should be distinct from the regular plural and lack the additive reading. Ackema and Neeleman suggest that Dutch may be one such language.¹² Dutch has a periphrastic associative plural (297a) and, crucially, allows 1/2P singular pronouns in this construction (297b). That is, Dutch 1/2P pronouns have a regular plural form (297c) and a distinct (periphrastic) associative plural form. The regular 1/2P plural pronouns have both additive and associative readings, but the associative plurals involving 1/2P pronouns only have an associative interpretation.

- (297) a. Marie en de haren
 Marie and the hers.PL
 ‘Marie and those associated with her’

¹²They also suggest that Haruai (Piawi) has a regular first person plural pronoun (i) and what Corbett (2000) and Ackema and Neeleman (2018) call an associative plural with a first person pronoun (ii).

- | | |
|---|--|
| (i) An hön pay-n-ŋ-a
we pig hit-FUT-1PL-DEC
‘We will hit the pig.’ | (ii) N dy-n-ŋ-a
I go-FUT-1PL-DEC
‘I and some other(s) will go.’ |
|---|--|

However, Bernard Comrie (personal communication), to whom (ii) is attributed, disagrees with the claim that this is an associative plural. He suggests that Haruai has an inclusory construction consisting of two independent pronouns—PRO₁ and PRO₂—and exhibits subject agreement on the verb (iii); when the inclusory construction is in subject position, as in (iii), agreement is with PRO₁. Notably, PRO₁ itself may also be dropped in the inclusory (iv), leaving only PRO₁ information indexed on the verb. Thus, an inclusory construction like *we, including you* may consist of an overt second person singular pronoun with first person plural agreement on the verb.

- | | |
|--|---|
| (iii) An₁ nagö₂ dy-n-ŋ ₁ -a
we you.SG go-FUT- 1PL -DECL
‘We, including you, will go.’ | (iv) nagö₂ dy-n-ŋ ₁ -a
you.SG go-FUT- 1PL -DECL
‘We, including you, will go.’ |
|--|---|

Comrie suggests that what has been called an associative plural with a first person singular pronoun is actually a special case of the inclusory construction: *we, including me* (v). The first person plural pronoun—PRO₁—has been dropped, but first person plural agreement remains on the verb. The first person singular pronoun—PRO₂—remains overt.

- (v) **N₂** dy-n-ŋ₁-a
 I go-FUT-**1PL**-DEC
 ‘We, including me, will go.’

- b. **ik** en de mijnen / **jij** en de jouwen
I and the mine.PL / **you** and the yours.PL
 ‘Me and those associated with me’ / ‘You and those associated with you’
 ≠ ‘multiple speakers’ / ‘multiple addressees’
- c. wij / jullie
 we / you.PL
 ‘I and others’ / ‘You and others’
 ‘multiple speakers’ / ‘multiple addressees’

But there is a problem with Ackema and Neeleman’s (2018) example: Dutch does not have two distinct forms of a 1/2P plural pronoun—regular and associative. It has a regular plural pronoun and an associative plural construction that contains an uninflected singular form. In the next section, I present novel evidence from three languages that do have distinct regular and associative 1/2P plural pronouns. These languages provide direct evidence that 1/2P pronouns must be capable of combining with both the regular plural and the associative plural. The former generates a regular 1/2P plural pronoun that is ambiguous between an additive and associative reading—akin to English *we*. The latter generates an unambiguously associative pronoun, for which there is no equivalent in English. Given that the unambiguous number marking is associative, the ambiguous number must be regular.

4.4 Associative plural pronouns

Nambu, a Japanese dialect spoken in Tohoku, marks plurality on lexical nouns with the suffix *-ⁿdo* (298). This same suffix is used to pluralize pronouns (299).

- (298) warasa-**ⁿdo**
 child-PL
 ‘children’

(Tanaka 1996, p. 171)

(299) a. *Nambu 1/2P pronouns*

	Singular	Plural
1	wa, ora	wa- ⁿ do, ora- ⁿ do
2	ome, na	ome- ⁿ do

- b. kore ora-ⁿdo minna=de taberu besi
 this 1-PL all-INST eat INFR
 ‘This one, we all are going to eat it.’

(Nakagawa 2022, pp. 198–9)

Nambu has another plural suffix (*-ho*) used to refer to people who belong to the same group (e.g., the same family, village or organization). Notably, *-ho* can also be suffixed to 1/2P pronouns (300).¹³ The pronouns in (300) are glossed as ‘we, our family’ and ‘you, your family’, indicating that the ‘others’ associated with the 1/2P pronoun are in a somewhat fixed relationship with the speaker/addressee (e.g. family members, colleagues). This is typical of associative plurals (see Section 4.3.2), but not true of the 1/2P plural pronouns in (299), in which the ‘others’ are contextually dependent on the discourse.

- (300) **ora-ho**=daba mikan mame=no ue=sa age-te jat-ta=n
 1-GR=TOP mandarin bean=GEN top=DAT offer.to.alter-SEQ give-PST=NMLZ
 da-kedomo **ome-ho**=daba desu
 COP-though 2-GR=TOP how.do
 ‘We (our family) offer a mandarin orange on beans to the Buddhist altar. What do
 you (your family) do?’

(Nakagawa 2022, p. 199)

Nambu 1/2P pronouns thus combine with two different suffixes to generate two different 1/2P plural pronouns—a regular plural pronoun with both additive (‘multiple speakers/hearers’) and associative (‘I/you and others’) interpretations, and an unambiguously

¹³According to Nakagawa (2022, p. 198), *-ho* does not attach to third person pronouns. There are also no examples of *-ho* on a non-pronominal stem in Nakagawa 2022, but it is not clear if this is impossible or just not described.

associative plural pronoun (‘me and my family/you and your family’). Given that the interpretation with a distinct form is associative, the ambiguous 1/2P plural pronouns must arise by combining with the regular plural. If the ambiguous pronoun was inflected for the associative plural (as argued by proponents of a unified analysis of 1/2P plural pronouns and associative plurals), then a 1/2P plural pronoun that disambiguates between the two readings with a distinct form should morphologize the additive reading. However, we do not find the additive reading morphologized in any language.

Another language with two 1/2P plural pronouns is Wutun. As we saw earlier, number is marked on Wutun lexical nouns by the plural suffixes *-dera* and *-men* and the paucal suffix *-jhege* (301).

- (301) a. alak / alak **-dera**
 lama / lama -PL
 ‘lama’ / ‘lamas’
- b. niren / niren **-men**
 ‘woman’ / ‘woman’ -PL
 ‘wife/wives’
- c. zhawa / zhawa **-jhege**
 disciple / worker -PAUC
 ‘worker’ / ‘(a couple of) workers’

(Sandman 2016, pp. 47–50)

1/2P pronouns in Wutun are also pluralized with *-jhege*. The paucal interpretation of *-jhege* is neutralized on pronouns (see footnote 8).

(302) a. *Wutun* 1/2P pronouns

	Singular	Plural
1	ngu	ngu- jhege
2	ni	ni- jhege

- b. ngu-**jhege** hai lai-de kuli zhai-lio
 1-PAUC even come-ATTR time pick-PFV
 ‘And we (each of us individually) even picked (ears of the wheat from the field)
 while coming back (from the school)’ (Sandman 2016, pp. 69, 71)

Finally, recall that *Wutun* has an associative plural marker *-mu*, shown again in (303).

- (303) aba-**mu**
 father-APL
 ‘father and his associates’ (Janhunen et al. 2008, p. 66)

Crucially, *-mu* can also attach to personal pronouns to form associative plural pronouns (Janhunen et al. 2008; Sandman 2012, 2016).¹⁴ These pronouns only have a collective interpretation just like the associative plurals in Turkish (see Section 4.3.2) and are functionally differentiated from ordinary 1/2P plural pronouns in their requirement that the group that is picked out has some internal cohesion. This is again reflected in the original gloss ‘we (our whole family)’.

- (304) nga-**mu** gu qhichai mai-she-ma-li
 1-APL that car buy-RES.AO-RES.PO-SEN-INF
 ‘We (our whole family) bought that car.’ (Sandman 2016, p. 71)

¹⁴Unlike Nambu, *-mu* is said to appear on personal pronouns across all three persons (Janhunen et al. 2008). Sandman (2016) says that *-mu* appears most commonly on first and third person pronouns in her fieldwork, and rarely on second person.

Lastly, distinct associative plural pronouns are also found in Amdo Tibetan. As we saw earlier, plural is marked on lexical nouns by the suffixes *-tʃhao*, *-tʃho*, *-zo* or *-so* (305) and these suffixes also mark plural on 1/2P pronouns (306a).

- (305) ʃajə **-tʃ^hao** / s^ha **-zo**
 child -PL / field -PL
 ‘children’ / ‘fields’ (Simon 2021, pp. 55–6)

- (306) a. *Amdo Tibetan 1/2P pronouns* (Ebihara 2010, p. 49)

	Singular	Plural
1	ŋa	ŋə (EXCL) / ə (INCL) { -zo / -tʃ^ho }
2	tʃ ^h o	tɕ ^h i- tʃ^ho

- b. ŋə-tʃ^hao ndɛ sa-jo ga=gə
 1PL.EXCL rice eat.IPF-NMLZ like=AUX
 ‘We (EXCL) like eating rice.’ (Ebihara 2014, p. 87)

Recall now that one of the associative plural markers in Amdo Tibetan is *-ts^haŋ* (307). According to Simon (2021), this associative plural marker also appears on pronouns (308). It appears on a third person pronoun in (308a) and a first person pronoun in (308b).

- (307) ^hləmbəmts^heraŋ **-ts^haŋ** -kə k^ha^rtsaŋ rə mə -^fkwa -nəere
 L. -APL -ERG yesterday tent NEG -move -GENER.FACT.Q
 ‘Didn’t Ləmbəṃ Tserangs [i.e. his family] move their camp yesterday?’ (Simon 2021, p. 71)

- (308) a. ap-ki ta k^hə-**-ts^haŋ**-kə [...] ⁿɖa ^rdzak-kokə
 father-ERG TOP 3-APL-GEN [...] picture LV-IPV.SENS
 ‘The father takes a picture of his [family].’ (Simon 2021, p. 71)
- b. A: ^rtoŋ-ts^ho ^rji təmo ⁿt^hak-kogə ^rtoŋ-ts^ho ^rji-ra s^həm-^Vdza
 thousand-PL two like.that mill-IPFV.SENS thousand-PL two-COM three-hundred
 ‘We mill two thousand [kilos of barley], like that. Two thousand and three hundred.’

B: $\text{V}_{\text{zə}} \text{V}_{\text{dza}}$ təmo $\text{ᵐt}^{\text{h}}\text{ak-kə}$ ta $\text{ŋə-ts}^{\text{h}}\text{aŋ-a}$ sa-no
 four-hundred like.that mill-IPFV.STAT.SENS disc 1-APL-DAT eat.ICP-NML
 ‘[We mill two thousand and] four hundred, like that. Well, what we eat (in our family).’

(Camille Simon, p.c.)

Once again, the glosses indicate that when the pronoun is suffixed with the associative plural, the referents of the pronoun are restricted to internally cohesive, familial relationships.

4.4.1 Summary

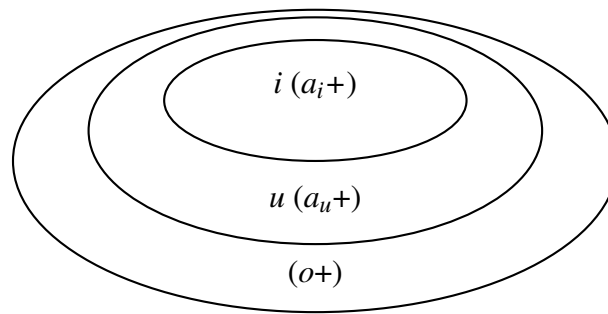
This section has presented new empirical evidence that 1/2P are capable of combining with dedicated associative plural morphology to generate an unambiguously associative plural pronoun. Moreover, because it is the associative plural interpretation that is morphologized with a distinct form, the inherently ambiguous 1/2P plural pronouns that are present in all languages (like English *we*) must be inflected with the regular plural. If regular 1/2P pronouns combined with the associative plural (as argued for by e.g. Cysouw 2003; Dékány 2021; Moravcsik 2003) we would expect it to be an ambiguously additive reading that is morphologized, contrary to what we find. Thus, the existence of dedicated associative plural 1/2P pronouns is additional evidence that ordinary 1/2P pronouns are inflected with the regular plural.

4.5 The associative effect

If the associative effect observed in 1/2P plural pronouns is not due to 1/2P pronouns combining with the associative plural, there must be an alternative explanation. As mentioned at the start of this chapter, Ackema and Neeleman (2018) propose that the associative effect

is a property of the person system. They argue for a person system that consists of three privative features: *i* (for speaker), *u* (for addressee) and *o* (for other). These features are divided into three layers, where the layers corresponding to *i* and *u* may also contain *others*. However, they also argue that the status of *others* in the person domain varies; *others* in the *i* or *u* layers must be *associates* of the participants ($a_{i/u}$), while *others* in the outermost layer are genuine others.

(309) *Organization of the person space* (Ackema and Neeleman 2018, p. 6)



The separation of *others* into ‘associates’ and genuine others is motivated by examples like (310). Ackema and Neeleman describe a situation in which Peter and Ad—who are supporting opposite teams—are at a football game. If Ad returns from the concession stand and asks Peter what happened, Peter might utter (310a) or (310b).

- (310) a. We scored.
b. You scored.

Crucially, the reference of *we* in (310a) cannot be Peter and an arbitrary group of others; it must be Peter and those he is associated with. In this case: Peter, the players of his chosen football team and other supporters of that club. Likewise, *you* in (310b) refers to Ad, the

players of his chosen football team and any other supporters.¹⁵ Thus, according to Ackema and Neeleman, the associative effect results from the requirement that the *others* that are in the same layer as the participant must be associated with the participant, thus placing the cause of the associative effect firmly in the person domain.¹⁶ This is, however, little more than a description of the facts. The larger question is *why* this should be.

4.5.1 Deriving the associative effect in 1/2P plural pronouns

Daniel (2020): Unique referents cannot be additively pluralized

Daniel (2020) offers a very different analysis of associative plurals to the one presented in this work, but which nonetheless implicitly supports Ackema and Neeleman’s claim that the associative effect is a property of the person system. Just as I have been throughout this work, Daniel is concerned specifically with languages that use the regular plural as the associative plural (*plural pattern* languages in this work, ‘non-dedicated APL’ in Daniel 2020). Specifically, Daniel makes the argument that the availability of associative readings of the regular plural is not due the animacy of the stem (as is typically assumed) but its unique reference. The long-held view is that associative plurals are governed by the Animacy Hierarchy (Corbett 2000; Moravcsik 2003); they are most common on proper names and kin terms, also allowed on common human nouns, rarely found on non-human nouns and never on inanimates. Moreover, as this chapter has discussed, 1/2P pronouns—at the top of the Animacy Hierarchy—also have associative interpretations. However, Daniel argues that appealing to the Animacy Hierarchy to explain the interpretation of *plural pattern*

¹⁵Note that being associated with speaker/hearer is not the same as forming a semantically cohesive group with speaker/hearer, which differentiates 1/2P plural pronouns from associative plurals. See Section 4.3.2.

¹⁶Ackema and Neeleman (2018) give other reasons, not relevant to this discussion, for separating *others* into ‘associates’ and genuine others, and for restricting the privative person features to *i*, *u* and *o*. See Ackema and Neeleman (2018, pp. 6–9).

associative plurals is questionable. For starters, while 1/2P plural pronouns have associative interpretations, and proper names and kin terms are frequently found as associative plural stems, third person plural pronouns are generally homogeneous (i.e., additive). That is, given the Animacy Hierarchy in (311), the associative plural appears to freely attach to 1/2P pronouns and proper names/kin terms, but “skip” third person pronouns.

(311) *Associative semantics along the animacy hierarchy “skipping” third person*

1 > 2 > 3 > proper > kin > human > animate > inanimate

✓ ✓ ✗ ✓ ✓ ✓ ✓

Some authors argue that third person plural pronouns do have a heterogeneous use in addition to their ordinary, additive function (Daniel 2020; Vassilieva 2005). The allegedly heterogeneous third person plural pronoun is shown in (312). However, experimental evidence shows that people almost always interpret third person plural pronouns as additive (Maldonado and Gascon 2022).

(312) A: And what became of John?

B: Oh, **they** moved to DC a few years ago. (they = John and family)

(Vassilieva 2005, p. 47)

Another argument given by Daniel against the *plural pattern* being governed by the Animacy Hierarchy is that, independently of associative plurals, there is little evidence that proper names and kin terms form a typological class separate from other human-denoting common nouns (see Daniel 2020, pp. 5–6 for details). That is, typological cut-off points relating to animacy often involve animate vs. inanimate, human vs. non-human or pronoun vs. common noun. They tend not to involve proper names/kin terms vs. other common human nouns.

What Daniel proposes instead is that the relevant factor governing associative plurality (at least in *plural pattern* languages) is unique reference. 1/2P pronouns and human proper names have unique reference and are thus resistant to the regular (additive) interpretation of the plural. According to him, the regular plural is coerced into having an associative interpretation by “activating associative links” that are available to the stem. Daniel presents a number of quite compelling (but anecdotal) pieces of empirical evidence that coercion of the regular plural by a unique referent derives associative interpretations in at least some contexts. For example, he claims that Russian given names are not amenable to associative plural formation, but Russian nicknames (which are lexically unique) are. Other “occasional APLs” include lexicalized forms in Indo-European: e.g., Latin *Castores* (Castor-PL) ‘Castor and Pollux’ or Spanish, *los reyes* (the king-PL) ‘the king and the queen’. The use of the regular plural with associative meaning is not productive in these languages, meaning that the regular plural marker on other proper names or socially important people does *not* produce an associative interpretation. Rather, the regular plural having an associative reading has been grammaticalized on a very small number of nouns. According to Daniel, it is the unique reference of these nouns that allowed such grammaticalization to take place. Other evidence that unique reference may coerce the regular plural into having an associative interpretation comes from the fact that associative-like meanings are not restricted to nominals (something we might expect if associative plurality was strictly governed by the Animacy Hierarchy). Place names, which Daniel argues are unique, can also have an associative interpretation, as in Aghul (313).

- | | | |
|-------|--|-------|
| (313) | quruh- ar
Qurux-PL
‘the villagers of Qurux’ | Aghul |
|-------|--|-------|

Daniel (2020) attempts to analyze all *plural pattern* languages as being derived through

coercion of the semantics of the regular plural. I do not adopt this strong version of Daniel’s claim. One reason for this is that Daniel is forced to speculate as to how kin terms and some common human nouns (e.g., teacher)—which clearly do not resist additive interpretations but are nonetheless frequent associative plural stems—fit into his analysis. He posits that kin terms are “often highly individuated in the discourse” and thus their plural forms can be coerced into having an associative interpretation.¹⁷ That is, when a speaker refers to someone as *father*, the idea is that they are using a descriptive expression to nonetheless refer to a unique individual—there is only one ‘father’ who is the father of the speaker. This idea is more difficult to extend to associative plurals built on nouns like *teacher*, which are less individuated (a person can plausibly have more than one teacher). Moreover, it is unclear how this analysis would deal with examples of associative plurals with plural stems. In Japanese, for example, the plural marker *-tachi* can be suffixed to a stem twice, with the first *-tachi* expressing the regular (additive) plural and the second *-tachi* expressing the associative plural (314). Thus, associative plural *-tachi* combines with the plural noun *gakusei-tachi* ‘the students’.

- (314) *gakusei -tachi -tachi*
 student -PL -PL
 ‘students and their associates’ (Ueda and Haraguchi 2008, p. 237)

The plural noun in (314) is clearly not a unique individual, and yet the second *-tachi* receives an associative interpretation. (314) also shows that not all associative plurals can be explained by resistance to additive pluralization. Daniel’s idea is that the regular plural is coerced into having an associative interpretation because the noun that it combines with resists the regular (additive) reading. Thus, his analysis says that the plural marker in (315)

¹⁷The same argument is given as to why third person plural pronouns allow homogeneous (additive) and associative readings.

is interpreted as associative because *gakusei* resists additive pluralization. But (314) shows that this is not the case; the noun *gakusei* can be additively pluralized.

- (315) *gakusei* -tachi
 student -PL
 ‘the student(s) and their associates’

Nevertheless, Daniel’s idea that some expressions are resistant to additive pluralization can explain the associative effect in 1/2P plural pronouns and, as I will show, the semantic differences between them and associative plurals. I turn to this immediately below.

1/2P pronouns cannot be additively pluralized

Recall the semantic analysis of associative plurals discussed in Chapter 3. There I argued, following the ideas of Daniel (2020) just discussed, that the additive semantics of Num^0 is incompatible with incorporation of Num^0 into Associative^0 because, from this incorporated position, Num^0 combines with an individual-denoting argument of type *e*. The plural set that is yielded from combining the plural with an object of type *e* consists of the focal referent and others (318).

- (316) a. Ahmet = {Ahmet}
 b. Ahmet_{PL} = {Ahmet, Faruk, Ali}

I then argued that the Associative head has its own semantics, adapted from Tomioka (2021), which determines a contextually salient association between the focal referent and others.

- (317) $\llbracket i\text{Assoc} \rrbracket = \lambda Y [\iota x \subsetneq Y \ \& \ \forall y [y \subsetneq Y \ \& \ y \neq x] \rightarrow R(x)(y)]$

I propose that the associative effect in 1/2P plural pronouns is the result of the same semantic coercion of the regular plural that we find in associative plurals. That is, 1/2P pronouns

are individual-denoting arguments of type *e*, just like focal referents in associative plurals, and are thus resistant to additive pluralization. Consequently, regular plural semantics is coerced with 1/2P pronouns such that the plural of a 1/2P pronoun must minimally contain the referent of the 1/2P pronoun.

- (318) a. $I = \{\text{speaker}\}$
 b. $I_{\text{PL}} = \{\text{speaker, other, other}\}$

The syntax of these pronouns does not, however, contain an Associative head whose semantic denotation specifies a contextually salient association between the 1/2P pronoun and the others in the plurality (see (317)). This is why there are differences between 1/2P plural pronouns and associative plurals with respect to their ‘others’, as discussed in Section 4.3.2. An ‘other’ in 1/2P plural pronouns is any individual in the discourse context, whereas an ‘other’ in an associative plural must be a member of a smaller set of pre-determined relations. These pre-determined relations are specified by the semantics of Associative (317), which 1/2P plural pronouns lack. The ‘others’ in 1/2P plural pronouns also do not have to form a conceptually coherent group (e.g., *we* = ‘me and my enemies’) whereas associative plurals do (*Taroo-tachi* $\neq$ ‘*Taroo and his enemies*’). This is also explained by the presence of Associative⁰, with the semantic denotation in (317), in associative plurals and the absence of this element in 1/2P plural pronouns.

There is thus a separation along the Animacy Hierarchy when it comes to ‘associative’ semantics: in the pronominal domain, the semantic type of 1/2P pronouns triggers semantic coercion of the regular plural to yield a plural set that consists of the speaker/hearer and others. In the nominal domain, the semantic type of the focal referent triggers semantic coercion of the regular plural to yield a plural set that consists of the focal referent and others *and* there is a distinct functional projection (AssociativeP) that serves to establish a

contextually salient association between them.

(319) *Separation of associative semantics along the Animacy Hierarchy*

$$\boxed{1 > 2} > 3 \quad / \quad \text{proper} > \text{kin} > \text{human} > \text{animate} > \text{inanimate}$$

semantic coercion semantic coercion *and* functional structure

Importantly, this separation enables us to explain why associative semantics seems to “skip” third person. Third person pronouns, unlike 1/2P pronouns, are not individual-denoting objects of type *e*, but variables (Heim and Kratzer 1998).¹⁸ They are not resistant to additive pluralization, and hence lack a heterogeneous reading when additively pluralized.¹⁹

Moreover, the proposed analyses of the associative effect in 1/2P plural pronouns and of associative plurals in the nominal domain also captures the large amount of overlap in the meaning of 1/2P plural pronouns and associative plurals. Both 1/2P plural pronouns and associative plurals involve a plural number feature combining with an object of type *e*, resulting in a plural set consisting of the speaker/hearer or the focal referent and ‘others’ who are not the speaker/hearer or focal referent.²⁰ Hence, both 1/2P plural pronouns and associative plurals have generally the same meaning — ‘X and others’, where X is either a 1/2P pronoun or a nominal focal referent. As discussed above, the semantic differ-

¹⁸This is not to say that 1/2P pronouns are never variables. They may be used as bound variables in examples like (i) (e.g. Kratzer 2009; Rullman 2004).

- (i) a. Only I got a question that I understood (= nobody else got a question that they understood).
- b. Only you got a question that you understood (= nobody else got a question that they understood).

¹⁹They may receive a heterogeneous interpretation if a dedicated associative plural marker can attach to them, as in Amdo Tibetan.

²⁰An issue for any semantic analysis of 1/2P plural pronouns is that they appear to have both homogeneous and heterogeneous interpretations (i.e., *the speaker/hearer and others* and *multiple speakers/addressees*). Under the analysis proposed here, the default interpretation of 1/2P plural pronouns is heterogeneous. Following Dékány (2021), I assume that the homogeneous interpretations of these pronouns arise when the ‘others’ in the plural set happen to also be speakers or hearers in the utterance context.

ences between them come from the presence of Associative⁰, which contributes semantic information about the ‘others’, in associative plurals and which is absent in 1/2P plural pronouns.

Note, however, that nothing in the proposed analysis says that 1/2P pronouns *can’t* combine with Associative⁰, only that ordinary 1/2P plural pronouns (like English *we*) do not. This means that there can be languages in which the focal referent of an associative plural is *pronominal*. This is in fact what we saw in Section 4.4: Nambu, Wutun and Amdo Tibetan all have regular 1/2P plural pronouns (with their associative flavor derived via semantic coercion only), as well as 1/2P pronouns that are inflected with dedicated associative plural marking. The relevant data is illustrated with Wutun in (320).

- (320) nga-**mu** gu qhichai mai-she-ma-li
 1-APL that car buy-RES.AO-RES.PO-SEN-INF
 ‘We (our whole family) bought that car.’ (Sandman 2016, p. 71)

It is quite natural to assume that the associative plural morphology on these 1/2P pronouns is the realization of an Associative head (just as it is in the nominal domain). This is morphological evidence for the presence of Associative⁰ with 1/2P pronouns. The gloss in (320) also suggests that the meaning of these 1/2P pronouns has changed. The ‘others’ associated with the speaker are now restricted to kinship relations. This is precisely what the semantic contribution of Associative⁰ yields — a contextually salient association between the focal referent (now a first person pronoun) and the others.

The proposed analysis of the interpretation of ordinary 1/2P plural pronouns (relying on semantic coercion) and associative plurals (relying on semantic coercion + functional structure) may also explain why the latter is cross-linguistically rare for 1/2P pronouns. I am only aware of Nambu, Wutun and Amdo Tibetan as languages with dedicated associa-

tive plural pronouns.²¹ If it is possible for 1/2P pronouns to combine with Associative⁰, why don't we have overt evidence of this happening in more languages? The suggestion is that the semantic coercion of the plural that happens with 1/2P plural pronouns yields something close enough to the meaning of a true associative plural that encoding a very similar meaning with additional functional structure is not appealing from the point of view of economy.

4.6 Summing up

This goal of this chapter was to show that ordinary 1/2P plural pronouns are not true associative plurals. Building on the evidence provided by Ackema and Neeleman (2018) for this view, I presented new evidence that inflectional number marking on 1/2P pronouns is, in at least some contexts, the same as regular plural marking rather than associative plural marking (in languages that have both). I also showed that there are semantic differences regarding who the 'others' in associative plurals and 1/2P plural pronouns are. Only in the former, 'others' are restricted to a small set of predetermined relations. I argued that this is the result of the semantics of Associative⁰ combining with the focal referent. 1/2P plural pronouns lack the Associative head, hence lack its semantic import. The claim that ordi-

²¹According to Fried (2010), Bonan (Mongolic) may have associative plural 1/2P pronouns. Bonan pronouns make a collective/non-collective distinction, with non-collective pronouns appearing with the plural clitic =*la* (i) and collective pronouns lacking it (ii).

- (i) atɕaŋ **mangɕalada** ts^hamhtoɕ=gə ɔɕ-tɕo
3SG 1PL.INCL.LOC gift=SG.INDEF give-IMPF.OBJ
'He gave us a gift.' (allows for the reading, 'He gave us each a gift.')
- (ii) ɕoŋɔ **manda** da ts^ha wi-saŋ
originally 1PL.INCL.COLL.LOC also land COP.SUBJ-POS
'Originally, we (the whole village) also owned this land.'

Fried (2010, p. 118) notes that "the absence of the enclitic =*la* has less to do with the referents acting together as a whole and more to do with indicating that the referents comprise an intimately connected group (such as a clan or family)". The pronoun in (ii) may then be an associative plural pronoun.

nary 1/2P plural pronouns (like English *we*) are not true associative plurals also predicts that 1/2P pronouns should be able to combine with associative plural morphology to generate a dedicated associative plural pronoun. I showed that such pronouns are in fact attested – in Nambu, Wutun and Amdo Tibetan.

In the next chapter I discuss some of the implications of the analysis of *plural pattern* associative plural marking developed in this thesis. This includes investigating whether there are any cross-linguistic generalizations to be made about other kinds of heterogeneous plurals, namely simulative and approximative plurals. We will see that the restriction of the *plural pattern* in heterogeneous plurals to languages without free-standing definite articles may in fact extend beyond associatives, and that there is an interesting implicational hierarchy holding for two of the heterogeneous plurals (simulative and associative).

Chapter 5

A broader look at heterogeneous plurals: similative and approximative plurality

So far it has been established that cross-linguistic variation regarding associative plurals is not arbitrary, but constrained by external factors. In particular, we have seen that using the regular plural as (or as part of) the associative plural (the *plural pattern*) is restricted to languages without free-standing definite articles. No such constraint holds if associative plural marking is distinct from the regular plural. I have offered an analysis of the *plural pattern* that aimed to provide an explanation for this typological generalization. The analysis that I proposed tied the multifunctionality of nominal plurality to differences in nominal structure (namely, the presence or absence of DP) and in doing so provided a principled account of what was before considered an arbitrary difference regarding whether a language uses the regular plural as the associative plural or it uses something else. This work thus adds to a growing body of literature that argues that a variety of syntactic and semantic effects observed in languages without articles (or with affixal articles) have a syntactic explanation.

In this chapter I discuss some implications of the thesis. Specifically, I turn to two other

types of heterogeneous plurals: *similative* and *approximative* plurals. These plurals have not received the same kind of attention as associative plurals in the literature — Mauri and Sansó (2017, 2019, 2021) and Smith (2020) are the two cross-linguistic studies of similative plurals; no cross-linguistic study of approximative plurals has been conducted. Thus, the first goal of this chapter is to assemble a broad typological survey of similative and approximative plurals. The survey presented in Section 5.1 is the largest so far for similative plurals, and the only one collected for approximative plurals. The second goal of this chapter is to investigate whether there are any typological generalizations regarding similative and approximative plurals. We will see that there are in fact tendencies that emerge regarding the current sample of languages. I offer an initial account of the generalizations established in this chapter in Sections 5.1.3 and 5.1.4.

5.1 Similative and approximative plurals

The discussion so far has been a typological investigation into *associative* plurals, a type of heterogeneous plural that picks out a focal referent ‘X’ and people associated with X. The focal referent (or stem) is generally restricted to animate, human nouns; associative plural marking may be dedicated associative plural morphology (321a), the same as the regular plural (321b), or a combination of the two (321c).

- (321) a. **en** yá Kotoko
 3PL mother
 ‘mother and her people’ (Allison 2012, p. 107)
- b. Ahmet **-ler** Turkish
 Ahmet -PL
 ‘Ahmet and his friends/family’ (Göskel and Kerslake 2005, p. 151)

- c. cuna -**nku -t** Central Alaskan Yup'ik
 chuna -APL -PL
 'Chuna and his family/friends' (Corbett 2000, p. 108)

Two related phenomena, which this chapter will deal with, are *similative* and *approximative* plurals. Similative plurals (abbreviated as SPL) are also heterogeneous plurals, but they abstract over the referent of the nominal stem to pick out 'X and things like X' (Mauri and Sansó 2017, 2019, 2021; Smith 2020). The nominal stem does not have to be animate. Like associative plurals, similative plural marking may be dedicated similative plural morphology (322a) or identical to the regular plural (322b).¹

- (322) a. wuuc =**yuk** Kuuk Thaayorre
 dance =SPL
 'dances and things' (Gaby 2018, p. 127)
- b. phuŋ -**ce** Chintang
 flower -PL
 'flowers and so' (Paudyal 2013)

Approximative plurals (abbreviated as XPL) are less well described in the literature, both for individual languages and cross-linguistically. The approximative plural takes a nominal stem 'X' and approximates or widens its meaning. They are typically found on temporal nouns (days, temporal adverbs), times, distances, locations and ages. Like associative and similative plurals, approximative plural marking may be dedicated approximative plural morphology (323a) or identical to the regular plural (323b).²

¹I have not found any examples of similative plurals that combine dedicated similative plural morphology with the regular plural. This may be due to the small sample size.

²One phenomenon, which could be labelled as an approximative plural, has received a lot of attention in the literature: *approximative inversion* in Russian (and other East Slavic languages). Approximative inversion involves inverting the ordinary numeral noun order to get an approximative reading (i). See e.g., Bošković (2006), Franks (1994), and Yandroff and Billings (1998).

- (323) a. cuatro kilometro =**dyane** Cavineña
 four kilometers =XPL
 ‘about four kilometers’ (Guillaume 2008, p. 535)
- b. rāgbák á xā gbák =**égē** Buwal
 twenty PREP over two =PL
 ‘around twenty two’ (Viljoen 2013, p. 275)

Table 5.1 lists all of the languages I know of so far that have a simulative or approximative plural, and whether the language shows the *plural pattern* or the *non-plural pattern*. There are a total of 48 languages in the survey with a simulative plural. This is an extension of Mauri and Sansó (2021)—who are only concerned with *non-plural pattern* languages—and is thus the largest sample of simulative plural languages collected to date. There are a total of 8 languages in the survey with an approximative plural.

-
- (i) a. pjat' **knig**
 five book.GEN.PL
 ‘five books’

- b. **knig** pjat'
 book.GEN.PL five
approximately five books’
 (Yadroff and Billings 1998)

TABLE 5.1: Similitive and approximative plural languages

Similitive (SPL)			Approximative (XPL)	
<i>plural pattern</i>	<i>non-plural pattern</i>		<i>plural pattern</i>	<i>non-plural pattern</i>
Cavineña	Arapaho	Makalero	Buwal	Cavineña
Chintang	Bargam	Manambu	Crimean Tatar	Meithei
Tommo So	Belep	Martuthunira	Kharia	
Eastern Dan	Bengali	Nama	Mishar Tatar	
Kharia	Cupeño	Nankina	Tagalog	
Nepali	Diu	Norf'k	Turkish	
Shiiba	Galo	Nungon		
	Goemai	Persian		
	Gooniyandi	Punjabi		
	Hindi	Sandawe		
	Hinuq	Shughni		
	Iatmul	Skou		
	Ingush	Tamil		
	Japanese	Tshangla		
	Karbi	Turkish		
	Kayah Li	Udi		
	Koasti	Vietnamese		
	Kuche	Wardaman		
	Kuuk Thaayorre	Yongning Na		
	Lao	Yukarè		
	Ma Manda			
7	41		6	2

In this section, I investigate the *plural pattern* in similitive and approximative plurals and show that there are typological generalizations that emerge here too. However, as Table 5.1 shows, the language sample for similitive and approximative plurals is much smaller than that for associative plurals, especially regarding the *plural pattern*. Thus, the generalizations presented in the following sections should be considered tentative.

First, I show that similitive and approximative plurals that show the *plural pattern* also lack (free-standing) definite articles. Second, I show that there is an implicational relationship regarding the realization of the *plural pattern* in heterogeneous plurals: if the

similative plural shows the *plural pattern*, the associative plural does too. In other words, there is no language that has dedicated associative plural morphology and that also uses the regular plural as the similative plural. This implicational relationship does not hold for the approximative plural (with either similative or associative plural). I offer an account of these generalizations in Sections 5.1.3 and 5.1.4.

5.1.1 Heterogeneous plurals and definite articles

This thesis has established that when regular plural morphology is used as (or as part of) the associative plural, the relevant language lacks free-standing definite articles. Given that this generalization holds for one heterogeneous plural, it is worth considering whether or not it would hold for other heterogeneous plurals, too. In this section, I show that it does. All of the similative and approximative plurals that show the *plural pattern* in Table 5.1, repeated below, in fact lack free-standing definite articles.

TABLE 5.2: Similative and approximative plural pattern languages

Similative (SPL)	Approximative (XPL)
Cavineña	Buwal
Chintang	Crimean Tatar
Tommo So	Kharia
Eastern Dan	Mishar Tatar
Kharia	Tagalog
Nepali	Turkish
Shiiba	

Below, I present the relevant data which show that all of these languages show the *plural pattern* in their heterogeneous plural and that, furthermore, they all lack free-standing definite articles.

It should, however, be noted that all of the languages in the sample except for Buwal

and Tagalog also show the *plural pattern* in the associative plural, ensuring that they lack free-standing definite articles. This is perhaps due to a bias in the literature—similative and approximative plurals are more likely to be described for languages already known to have an associative plural, and most associative plurals show the *plural pattern* (Daniel and Moravcsik 2013). We thus may need more languages that have similative and/or approximative plurals but no associative plural to confirm the generalization, which I leave to future work.

Similative and approximative plurals and definite articles

Let's start with the 7 similative plural languages. One of the regular plural markers in Cavineña is =*kwana* (324a).³ This plural clitic is also used as the similative plural (324b).

(324) Cavineña [Tacanan]

- | | | |
|----|--|-----|
| a. | ebakwa = kwana
child =PL
'children' | PL |
| b. | karetu = kwana
cart -PL
'the cart and everything (the oxen, the load, etc)' | SPL |

(Guillaume 2008, pp. 177, 482)

Chintang has a non-singular suffix *-ce* that is also used as the similative plural (325).⁴
Tommo So has a plural clitic =*mbe* that is also used as the similative plural (326).⁵

³The other regular plural marker is the clitic =*ekana*, used with kinship nouns that are possessed by first and second persons.

⁴Paudyal (2013) calls the example in (325b) an associative plural—which Chintang also has. However, the inanimate stem and the gloss are analogous to a similative plural.

⁵Plungian (1995, p. 11) calls the example in (326b) an approximative plural, but describes its meaning as 'X and similar objects' or 'X and something like X'.

(325) *Chintang* [Sino-Tibetan]

- a. cha -**ce**
child -NS
'children'
- b. phuŋ -**ce**
flower -NS
'flowers and so'

(Paudyal 2013, pp. 35–6)

(326) *Tommo So* [Dogon]

- a. ɛnɛ =**mbe**
goat =PL
'goats'
- b. isu =**mbe** nie **mbe**
fish =PL oil =PL
'fish, oil and similar things'

(Plungian 1995, p. 11)

In Eastern Dan, nominal plural is marked by the plural word *dhũn*. The plural word is also used as the simulative plural (327). Kharia has a plural clitic =*ki* that is also used as the simulative plural (328).⁶

(327) *Eastern Dan* [Niger Congo]

- a. Dhán -**dhũn**
child -PL
'children'
- b. dhěŋkpāadhě -**dhũn**
butterfly -PL
'a butterfly and the other beings'

(Vydrin 2022, p. 40)

(328) *Kharia* [Austro-Asiatic]

- a. lebu =**ki**
man =PL
'men, people'
- b. daʔ =**ki**
water =PL
'water or whatever'

(Peterson 2014, pp. 85, 92)

In Nepali, the regular plural is the suffix *-haru*.⁷ This suffix is also used as the simulative plural (329). Finally, Shiiba nouns take the plural suffix *-domo*, which also has a simulative function (330).

⁶Peterson (2014) calls the example in (328b) an approximative plural.

⁷This suffix is used very infrequently on common nouns (Riccardi 2003).

(329) *Nepali* [Indo-European]

- a. choraa **-haru**
son -PL
'sons' (Hari 1973, p. 15)

- b. kitab **-haru**
book -PL
'books and other things'
(Riccardi 2003, p. 606)

(330) *Shiiba* [Japonic]

- a. togi **-domo**
friend -PL
'friends'

- b. sumoo **-domo**
sumo -PL
'Sumo wrestling and suchlike'
(Shimoji and Hirosawa 2022, p. 300)

All of these languages also lack free-standing definite articles. As a reminder (see Chapter 2), a definite article: (1) has the meaning of the iota operator, (2) occurs only once per TNP, (3) obligatorily occurs in a nominal phrase with a definite interpretation (i.e., there are no bare definite NPs), and (4) is distinct from demonstratives. A free-standing definite article is one that has the status of an independent word. Affixal and clitic articles are not free-standing. Cavineña, Chintang, Eastern Dan, Kharia, Nepali and Shiiba lack definite articles.⁸ Tommo So has a clitic article =gε (see Plungian 1995, p. 11, McPherson 2013, p. 133).

Turning now to the approximative plurals, Buwal has a general plural clitic =égē that is also used as an approximative plural on numerals (331).

(331) *Buwal* [Chadic]

- a. mēndzēvēk =égē
remedy =PL
'remedies'
- b. rāgbák á xā gbák =égē
twenty PREP over two =PL
'around twenty two'

(Viljoen 2013, pp. 273, 275)

⁸See Guillaume (2008, p. 61) for Cavineña, Michailovsky (2017b, pp. 647, 651) for Chintang, Vydrin (2020, p. 459) for Eastern Dan, Peterson (2011, p. 142) for Kharia, Poudel (2017, p. 75) and Riccardi (2003, p. 605) for Nepali and Shimoji and Hirosawa (2022, passim) for Shiiba.

Crimean Tatar and Mishar Tatar both use the plural suffix *-lar* as the approximative plural (332). This suffix is used to approximate over time, space and distance. The Kharia plural clitic *=ki* also has an approximative use on temporal nouns like *yesterday*, *tomorrow*, and days of the week (333).

(332) *Tatar* [Turkic]

- a. at **-lar**
horse -PL
'horses'
- b. egerme metr **-lar**
twenty meter -PL
'approximately twenty meters'

(Burbiel 2018, pp. 51–2)

(333) *Kharia* [Austro-Asiatic]

- a. lebu **=ki**
man =PL
'men, people'
- b. sanicar **=ki**
Saturday =PL
'Saturday or so'

(Peterson 2014, pp. 85, 92)

Tagalog nominal plurality is expressed using *mga* (334a). With cardinal numbers or time expressions, *mga* also has an approximative function (334b). Finally, the Turkish plural suffix *-lar* (335a) has an approximative usage on diectic expressions (e.g. here, there), temporal nouns (e.g., months), years and ages (335b).

(334) *Tagalog* [Austronesian]

- a. **mga** ito
PL chair
'chairs'
- b. **mga** sampu
PL ten
'approximately ten'

(Schachter and Otnes 1972,
pp. 111–12)

(335) *Turkish* [Turkic]

- a. ev **-ler**
house -PL
'houses'
- b. 1995 orta **-lar** -in -da
1995 middle -PL NC -LOC
'about the middle of 1995'

(Göskel and Kerslake 2005,
p. 150)

These languages also all lack free-standing definite articles. Buwal has what Viljoen

(2013) calls a “definite determiner”, which is, however, identical to the third person singular possessive pronoun (336a). Furthermore, Viljoen likens the possessive marking in Buwal to that in Uralic, discussed in Chapter 2. Recall that in a number of Uralic languages, third person singular possessive suffixes are interpreted as *non-possessive* definite (336b). Recall also that these possessive markers are not definite articles; their form has not been phonologically or morphologically reduced, they retain their possessive interpretation and they are not obligatory for definite interpretation.

- (336) a. *tekeḏ anta* *Buwal*
 calabash 3SG.POSS
 ‘**the** calabash’ (Viljoen 2013, p. 239)
- b. *Məu -ḏu* *Nganasan*
 earth -3SG
 ‘**the** earth’ (Gerland 2014, p. 271)

These properties also hold of the Buwal determiner. It is identical in form to the possessive pronoun (i.e., non-reduced), it retains its possessive interpretation (337a) and it is not obligatory for definite interpretation (337b). All these indicate that the element in question in Buwal is not a definite article.

- (337) a. *ujek anta a təwse*
 hut 3SG.POSS PREP1 there
 ‘**his** house is there’
- b. *ujek caw*
 hut there
 ‘**the** hut is there’ (Viljoen 2013, pp. 254, 257)

For the article status of Crimean and Mishar Tatar see Burbiel (2018, p. 54) and Johanson (1998, p. 38). For Tagalog see Paul et al. (2015) and Collins (2019). For Turkish, see Göskel and Kerslake (2005, p. 180) and Johanson (1998, p. 38).

Thus, based on the current sample of similitive and approximative *plural pattern* languages a tentative typological generalization emerges: languages that use the regular plural as the similitive or approximative plural lack free-standing definite articles.

- (338) GENERALIZATION: If a language uses the regular plural as the similitive or approximative plural, it lacks free-standing definite articles.

5.1.2 Implicational relationships

We know that languages can use regular plural morphology as the associative, similitive and approximative plural. We have also established that this pattern is cross-linguistically restricted for associative plurals, as well as for similitive and approximative plurals. Another possibility is that the *plural pattern* shows an implicational hierarchy across heterogeneous plurals. That is, if the *plural pattern* is found in one kind of heterogeneous plural it will also be found with another. Note that an implicational hierarchy of this sort would not mean that no language can have just one kind of heterogeneous plural—an implicational hierarchy of this sort would kick in only when a language has more than one type of heterogeneous plural.

An implicational hierarchy with similitive and associative plurals

Focusing on associative and similitive plurals for now, if a language has both of these constructions there are four logically possible pattern combinations: (i) associative plurals and similitive plurals show the *plural pattern*, (ii) associative plurals and similitive plurals use dedicated morphology, (iii) associative plurals show the *plural pattern* and similitive plurals use dedicated morphology, (iv) associative plurals use dedicated morphology and similitive plurals show the *plural pattern*. Importantly, only (i)–(iii) have so far been

attested in my research. There is no language that uses dedicated morphology for the associative plural and regular plural morphology for the simulative plural. I therefore propose the implicational hierarchy in (339):

(339) *Implicational hierarchy on heterogeneous plurals*

If a language shows the *plural pattern* in the simulative plural, it shows the *plural pattern* in the associative plural too (if it has one).

I abbreviate the implicational hierarchy as (340). ‘pp’ stands for *plural pattern*.

(340) $SPL_{pp} \rightarrow APL_{pp}$

We saw in the previous section that seven languages make up the pattern combination in (i): Shiiba, Chintang, Eastern Dan, Nepali, Cavineña, Tommo So and Kharia. The examples in (341) and (342) show the regular plural, the associative plural and the simulative plural—all of which are identical—in Shiiba and Chintang respectively.

(341) *Shiiba* [Japonic]

- a. togi **-domo**
friend -PL
‘friends’
- b. Taro **-domo**
Taro -PL
‘Taro and his associates’
- c. sumoo **-domo**
sumo -PL
‘Sumo wrestling and suchlike’

(Shimoji and Hirosawa 2022, p. 300)

(342) *Chintang* [Sino-Tibetan]

- a. cha **-ce**
child -PL
‘children’
- b. Mankumar **-ce**
Mankumar -PL
‘Mankumar and friends’
- c. phuŋ **-ce**
flower -PL
‘flowers and so’

(Paudyal 2013, pp. 35–6)

There are also a number of languages that use dedicated morphology for both the associative and simulative plural (pattern combination (ii)). For instance, Kuuk Thaayorre uses

a suffix *-mangka* (possibly from ‘low down by’) as the associative plural (343a) and *=yuk* (‘thing/stuff’) as the simulative plural (343b).⁹ Kuuk Thaayorre nouns are not inflected for regular plural (Gaby 2018).

(343) *Kuuk Thaayorre* [Pama-Nyungan]

a. Dan **mangka**
 Dan APL
 ‘Dan and his associates’

b. wuuc **=yuk**
 dance =SPL
 ‘dances and things’

(Gaby 2018, pp. 127, 335)

In Belep, the associative plural is *-ma* (344a), which is homophonous with the linker *ma* that conjoins NPs.¹⁰ The simulative plural is *-mene* (diachronic source unknown) (344b). Belep also lacks inflectional (regular) number marking on nouns (McCracken 2012, p. 246).

(344) *Belep* [Oceanic]

a. Teâ Polo **-ma**
 Teâ Polo -APL
 ‘Teâ Polo and his people’

b. uruânya **-mene**
 hurricane -SPL
 ‘hurricanes and stuff’

(McCracken 2012, pp. 248, 252)

Sometimes the associative plural and the simulative plural are identical to each other, though neither of these are the regular plural. For example, the associative plural and the

⁹The diachronic sources of associative and simulative plurals come from Mauri and Sansó (2021). Mauri and Sansó estimate the likelihood of the proposed diachronic source for each language in their sample. The status of the associative plural *-mangka* as being derived from ‘low down by’ is ‘unlikely’ and of the simulative plural *-yuk* being derived from ‘thing/stuff’ is ‘likely’.

¹⁰This diachronic source is ‘likely’.

simulative plural in Ma Manda is *kadek*, a noun meaning ‘group’ (345b–c).¹¹ Regular non-singular marking, *-ye*, is found on possessed kinship terms (345a).

(345) *Ma Manda* [Papuan]

- a. daamin -nek -ye
ancestor -1NSG.POSS -NSG
‘our ancestors’
- b. Pandi **kadek**
Pandi APL
‘Pandi and his group’
- c. mi **kadek**
water SPL
‘water and such’

(Pennington 2016, pp. 249–50, 577)

In Kayah Li and Goemai, the associative plural and the simulative plural are identical to the second person plural pronoun (see (346a–b) and (347b–c)).¹² Kayah Li lacks regular nominal number marking (Solnit 1997). Productive nominal number marking in Goemai are the prefixes *gòe-* (SG) and *mòe-* (PL) (347a).¹³

(346) *Kayah Li* [Sino-Tibetan]

- a. ʔamɔ sī
Amaw APL
‘Amaw and that group’
- b. thwí kɔ sī
lime-box SPL
‘the lime-box & similar things’

(Solnit 1997, p. 184)

(347) *Goemai* [Afro-Asiatic]

- a. **mòe-** sháár
NMLZ(PL)- friendship
‘friends’
- b. **gwén** Shalyen
APL Shalyen
‘Shalyen and his people’
- c. **gwén** dask’oom
SPL elders.PL
‘those kinds of elders’

(Hellwig 2011, pp. 123, 143)

¹¹This diachronic source is ‘likely’.

¹²This diachronic source is ‘uncertain’.

¹³These prefixes also function as nominalizers.

Pattern combination (iii)—the associative plural shows the *plural pattern* and the similative plural uses dedicated morphology—is found in Bengali, Turkish, Japanese, Gooniyandi and Manambu. For example, Turkish forms similative plurals using *m*-reduplication (348c), while the regular plural and the associative plural are identical. Japanese has dedicated similative plural suffixes *-toka/-tari* (349c), and identical associative and regular plurals.

(348) *Turkish* [Turkic]

- a. ev **-ler**
house -PL
'houses'
- b. Ahmet **-ler**
Ahmet -PL
'Ahmet and his family'
- c. bulut **m-** ulut
cloud M- cloud
'clouds and the like'

(Görgülü 2011, Gillon et al. 2014)

(349) *Japanese* [Japonic]

- a. gakusei **-tachi**
student -PL
'students'
- b. Taroo **-tachi**
Taroo -PL
'Taro and his friends/family'
- c. Taroo **-toka**
Taro -SPL
'Taro and someone like him'

(Ueda and Haraguchi 2008, Smith 2020)

Importantly, no language shows pattern combination (iv): the associative plural uses dedicated morphology and the similitive plural shows the *plural pattern*. There is no cross-linguistic survey that focuses on languages in which the similitive plural is identical to the regular plural (Mauri and Sansó (2017, 2019, 2021) are explicitly concerned with similitive plurals that use dedicated morphology). Instead, we have to look for similitive plurals in languages that are known to use dedicated associative plural morphology, and see which pattern surfaces. To do this, I took 86 languages from Daniel and Moravcsik (2013) that have either a unique affixal or a unique periphrastic associative plural.¹⁴ I added to this 29 languages from Mauri and Sansó (2021)¹⁵ for a total of 115. Of these, I was able to examine 48 languages with dedicated associative plural morphology in detail. These are given in (350).

- (350) Abkhaz, Abui, Afrikaans, Arapesh, Barasano, **Bargam**, **Belep**, Buwal, Central Pomo, Chechen, Dargwa, **Goemai**, Greenlandic West, Hausa, **Iatmul**, **Ingush**, Inuktitut, Kambaata, Karardild, **Kayah Li**, **Kuche**, **Kuuk Thaayorre**, **Lao**, **Ma Manda**, Maltese, Mangarrayi, Mauwake, Meithei, Mian, Muna, Mwotlap, **Nama**, **Nork’k**, **Nungon**, **Persian**, Rapanui, **Sandawe**, Sawila, Sheko, **Shughni**, Slave, Toqabaqita, Tuvaluan, **Wardaman**, Wolof, Yagua, Yidiny, Yukaghir (Kolyma).

For 17 of these languages (those in bold), I was able to find reference to a similitive plural in the literature. We have already seen that in Kuuk Thaayorre (343), Belep (344), Ma Manda (345), Kayah Li (346) and Goemai (347) the similitive plural uses dedicated morphology. Crucially, the remaining 12 also use dedicated similitive plural morphology. For example, Bargam has a dedicated similitive plural word *nagah*, ‘something’ (351a).

¹⁴The total number of languages in these categories in Daniel and Moravcsik (2013) is 95. I did not include Bengali, Desano, Hungarian, Karachay-Balkar, Tamang, Tatar, Central Alaskan Yup’ik, Mandarin or Bulgarian because I categorize them as *plural pattern* languages, as discussed in Chapter 2.

¹⁵This is the total number of associative plurals in Mauri and Sansó’s (2021) survey, minus any duplicates with Daniel and Moravcsik (2013) and not including Bulgarian, Desano, Ese Ejja, Hungarian, Hup, Kabardian, Kharia, Komo-Zyrian, Lezgian, Manambu, Mandarin, Tariana, Tswana and Central Alaskan Yup’ik, which, as noted above, I categorize as *plural pattern*.

The associative plural is a clitic =*nen* (351b).¹⁶ Inalienably possessed nouns are pluralized with one of the plural suffixes *-an*, *-niz* or *-gniz* (351c).

(351) *Bargam* [Trans-New Guinea]

- a. sansan **nagah**
leaves something
'leaves and things'
- b. Anna =**nen**
Anna =APL
'Anna and those with her'
- c. abay **-niz**
brother/sister.in.law -PL
'brothers/sisters in law'

(Hepner 2006, pp. 31, 58, 145)

In Iatmul, the simulative plural is *wudi da* (a plural demonstrative + 'thing') (352a). The associative plural is a suffix *-du*, 'man' (352b). Regular number marking is not found on nouns, only on determiners, modifiers and through verb agreement.

(352) *Iatmul* [Sepik]

- a. glaasi'k **wudi da**
pineapple DEM.PL thing
'pineapples and similar'
- b. Magina **-du**
Magina -APL
'Magina and her mother/family'

(Jendraschek 2012, pp. 133, 149)

Similarly, Ingush has a dedicated simulative plural *yzhazh=a* (353a), made up of the third person plural demonstrative and the coordinating clitic particle =*'a* (Nichols 2011, pp. xi, 179). The associative plural is *-aar* (353b). The nominal plural suffixes are *-zh* and

¹⁶Possibly derived from 'mother'.

-i/-j-, and are determined by the case on the noun (353c).¹⁷

(353) *Ingush* [East Caucasian]

- a. q'amealazh **yzhazh='a**
conversation.PL these=&
'conversations and so forth'
- b. Easiet **-aar**
Easiet -APL
'Easiet and her friends/relatives'
- c. xi **-zh**
river -PL(NOM)
'rivers'

(Nichols 2011, pp. 138, 164–5)

Table 5.3 shows the regular and heterogenous plurals for all 17 languages. All of these languages use dedicated morphology (i.e., not the regular plural) for the similitive plural. This means that there is no language that uses dedicated morphology for the associative plural and regular plural morphology for the similitive plural.

(354) *Implicational hierarchy on the exponence of heterogeneous plurals*

If a language shows the *plural pattern* in the similitive plural, it shows the *plural pattern* in the associative plural too (if it has one).

¹⁷ -i/-j- is used on nouns in the genitive, lative, and comparison case, and for some nouns the nominative. Elsewhere -zh is used (Nichols 2011, p. 138).

TABLE 5.3: Exponence of regular and heterogenous plurals

Language	Regular	Associative	Similitative	References
Bargam	-an, -niz, -gniz	=nen	<i>nagah</i>	Hepner (2006, pp. 31, 58, 145)
Belep	N/A	-ma	-mene	McCracken (2012, pp. 248, 252)
Goemai	mòe-	gwén	gwén	Hellwig (2011, pp. 123, 143)
Iatmul	N/A	-du	wudi da	Jendraschek (2012, pp. 133, 149)
Ingush	-zh / -i / -j-	-aar	yzhazh = 'a	Nichols (2011, pp. 138, 154–5)
Kayah Li	N/A	sī	sī	Solnit (1997, p. 184)
Kuche	bā-	bā-nà	bā-nà	Wilson (2002, p. 161)
Kuuk Thaayorre	N/A	mangka	=yak	Gaby (2018, pp. 127, 355)
Lao	N/A	khaw3	REDUP + ñang3	Enfield (2007, pp. 82, 309)
Ma Manda	-ye	kadek	kadek	Pennington (2016, pp. 249–50, 577)
Nama	-ku(M), -tĩ(F), -(i)ñ	hāa	hāa	Hagman (1977, pp. 22, 29)
Norŋk	N/A	dem	en dem	Mühlhäusler (2020, p. 142)
Nungon	-na-i, -a-i, -i/-n-i	-nit, oemma, gamong	nunon	Steeleman (2014, pp. 196, 199–204, 538)
Persian	-hā	inā	REDUP	Yousef and Torabi (2018, p. 26), Mauri and Sansó (2021, p. 14), Smith (2020, p. 47)
Sandawe	-sò	-x	-x	Steeleman (2011, pp. 61–2, 65–6)
Shughni	-(y)en	wāḏ, wav	REDUP	Parker (2023, pp. 148, 151–2)
Wardaman	-mulu ~ -bulu	-garma ~ -warra	wayana	Merlan (1994, pp. 97–90)

There is one language in (350) that at first looks like a counterexample to the generalization in (354). Kambaata (Cushitic) has dedicated associative plural morphology -'VV (355). The regular plural takes two forms in Kambaata depending on the phonological shape of the stem. Nouns with a simplex stem-final consonant are pluralized via stem-final consonant reduplication (+ the case/gender (C/G) suffixes *á-ta*) (356a). Nouns that end with a consonant cluster are pluralized with the suffix *-aakk* (+ the case/gender suffixes) (356b).

- (355) aachch -e -'ée
 mum -F.ACC -APL
 'Mum and her family/friends' (adapted from Treis 2014, p. 126)

- (356) a. miní / min -n -á -ta
 house / house -REDUP -C/G -C/G
 'house' / 'houses'
- b. lankaanná / lankaann -aakk -á -ta
 paternal.uncle / paternal.uncle -PL -C/G -C/G
 'paternal uncle' / 'paternal uncles' (Treis 2014, p. 115)

Interestingly, Treis (2014) provides an example of regular plural marking with what looks like simulative plural interpretation. This is shown in (357).¹⁸

- (357) buur -aakk -áta
 butter -PL -C/G
 'butter and other things (associated with butter, e.g. butter spices)'

If (357) is a simulative plural using regular plural morphology, it contradicts the implicational hierarchy in (354). This is because the associative plural uses dedicated morphology (see (355)). However, there is reason to think that (357) is not a simulative plural. Treis notes that the 'N + others' interpretation of the regular plural in (357) has so far only been

¹⁸Treis (2014) actually calls the construction in (357) an associative plural, but the gloss is reminiscent of simulative plurality. Note, too, that associative plurals are not attested on inanimate nouns.

attested on mass nouns. This is particularly important because the meaning of plural marking on mass nouns is often different from that on count nouns (e.g. *a great amount of N* in Greek (Alexiadou 2011; see also Kouneli 2019)). In fact, Treis (2014) notes that plural marking on inherently plural nouns (of which mass nouns are a kind) in Kambaataa does come with a change in meaning. For example, plural marking on the inherently plural *lal-ú*, ‘cattle’ individuates: *lal-l-a-sí*, lit. ‘cattles’ refers to multiple individuated cows rather than multiple herds. Plural marking on nouns that may be interpreted as singular or plural can also pick out species or types: *haqq-aakk*, lit. ‘trees’ refers to types/species of trees. Most notably, Treis (2014, 125:fn15) also acknowledges that plural marking on mass nouns in Kambaataa may express ‘types of N’, meaning that (357) may be better translated as *types of butter*. Note, too, that of all of the languages with simulative plurals discussed in this section, none of them are restricted to mass nouns. Given this, and that idiosyncratic interpretations of regular plural marking on mass nouns are found both cross-linguistically and in Kambaataa, it is very likely that (357) is not a simulative plural. The implicational hierarchy in (354) holds.

No implicational hierarchy with approximative plurals

I now turn to approximative plurals. We have seen that there is an implicational hierarchy regarding the morphological expression of simulative and associative plurals. Interestingly, there does not seem to be a similar implicational hierarchy with approximative plurals (XPL). There are four logically possible hierarchies regarding the exponence of the approximative plural with another heterogeneous plural. These are given in (358).

- (358) a. $APL_{pp} \rightarrow XPL_{pp}$
 b. $SPL_{pp} \rightarrow XPL_{pp}$

c. $XPL_{PP} \rightarrow APL_{PP}$

d. $XPL_{PP} \rightarrow SPL_{PP}$

All four potential implicational hierarchies have a counterexample. Cavineña defies (358a-b), which say that the approximative plural will be identical to the regular plural if the associative or simulative plural is too. Cavineña has an associative dual that is identical to the regular dual marker (recall that no examples of an associative plural were found in Cavineña) (cf. (359a–b)). It also has a simulative plural that is identical to the regular plural (359c). Yet, it has a dedicated approximative plural suffix (359d).

(359) $*APL_{PP} \rightarrow XPL_{PP}$ and $*SPL_{PP} \rightarrow XPL_{PP}$

- | | | |
|----|---|-----|
| a. | juye = ekatse / ebakwa = kwana
ox =DU / child =PL
'two oxen' / 'children' | PL |
| b. | e -kwa =ke = ekatse
3 -mother =3 =DU
'his mother and associated person' | APL |
| c. | karetu = kwana
cart -PL
'the cart and everything (the oxen, the load, etc)' | SPL |
| d. | cuatro kilometro = dyane
four kilometer =XPL
'about four kilometers' | XPL |
- (Guillaume 2008, pp. 74, 177, 482, 535)

Buwal violates the hierarchy in (358c), which says that if the approximative plural is identical to the regular plural then the associative plural will be too. Buwal has an approximative plural that is identical to the regular plural (360a-b) and yet has an associative plural that uses dedicated associative plural morphology (360c).

(360) *XPL_{PP} → APL_{PP}

- a. menjevek =égē PL
 remedy =PL
 ‘remedies’
- b. rāgbák á xā gbák =égē XPL
 twenty PREP over two =PL
 ‘around twenty two’
- c. ātā Martan APL
 APL Martin
 ‘Martin and his associates’

(Viljoen 2013, pp. 273, 275–6)

Finally, Turkish contradicts (358d), which says that if the approximative plural is identical to the regular plural then the simulative plural will be too. Turkish uses the regular plural as the approximative plural (361a-b) and has dedicated morphology for the simulative plural (361c).

(361) *XPL_{PP} → SPL_{PP}

- a. ev -ler PL
 house -PL
 ‘houses’
- b. bura -lar -da XPL
 here -PL -LOC
 ‘around here’
- c. bulut m- ulut SPL
 cloud M- cloud
 ‘clouds and the like’

(Göskel and Kerslake 2005, p. 150)

(Gillon et al. 2014)

There is thus no implicational relationship between the expression of the approximative plural and either of the other heterogeneous plurals. This finding suggests that the associative and simulative plural constructions are more closely related to each other than either

of them are to approximative plurals. In the following section I offer an account of the implicational hierarchy found between associative and similitive plurals that also captures this interrelatedness.

5.1.3 Accounting for the implicational hierarchy

In this section, I offer an account of the implicational hierarchy in (362).

(362) If the similitive plural shows the *plural pattern*, the associative plural shows the *plural pattern* too ($SPL_{PP} \rightarrow APL_{PP}$).

The implicational hierarchy in (362) is a restriction on the concurrent exponence of three kinds of plural marking: the regular plural, the similitive plural, and the associative plural. In morphology, morphological patterns (of e.g., suppletion or syncretism) in three cell paradigms are described using letters. When a pattern is found in all three cells, that is an AAA pattern. For example, the English degree adjective *loud* shows an AAA suppletion pattern because the positive, comparative and superlative adjectives all share a root (363).

		POS	CMPR	SPRL	
(363)	English	loud	loud-er	loud-est	AAA

Thus, when the regular, similitive and associative plural all share a single exponent (pattern combination (i) from Section 5.1.2), we can describe this pattern of plural marking as an AAA pattern (364).

	PL	APL	SPL	
(364)	(i)	A	A	A

A distinct element in a three-cell paradigm is expressed with a different letter. For example, the English degree adjective *good* shows an ABB suppletion pattern because the comparative and superlative adjectives share a suppletive root distinct from the positive adjective (see the first pattern in (365)). Sometimes every element in a paradigm is distinct, resulting in an ABC pattern. For example, the Welsh degree adjective *da* (‘good’) shows an ABC suppletion pattern—each adjective has a different root (see the second pattern in (365)).

		POS	CMPR	SPRL	
(365)	English	good	be(tt)-er	be-st	ABB
	Welsh	da	gwell	gor-au	ABC

In the three-cell paradigm of plural marking under discussion, the associative and the simulative plural may share an exponent, with the regular plural distinct, or all three plurals may be distinct ((366); = pattern (ii)). This is an ABB or ABC pattern, respectively. Alternatively, the regular and associative plural may share an exponent and the simulative plural can be distinct ((367), = pattern (iii)). This is an AAB pattern.¹⁹

		PL	APL	SPL	
(366)	(ii)	A	B	B/C	

		PL	APL	SPL	
(367)	(iii)	A	A	B	

Interestingly, work on morphological patterns of exponence has shown that ABA patterns are virtually absent cross-linguistically (e.g. Bobaljik 2012). For example, in degree

¹⁹Note that patterns (i) and (iii) are not really three-way distinctions, given that I treat the *plural pattern* in the associative plural as the realization of Num⁰ (the regular plural) that has incorporated into the Associative head (see Chapter 3). These are thus AB and AA patterns respectively. I discuss this issue below.

adjectives, an ABA pattern would involve a suppletive comparative form that reverted back to the non-suppletive root in the superlative (368).

	POS	CMPR	SPRL
(368)	UNATTESTED	good	be(tt)-er
		good-est	ABA

The lack of ABA patterns cross-linguistically is known as the *ABA-generalization. It has been shown that ABA patterns do not arise in degree adjectives (Bobaljik 2012), case syncretism (Caha 2009), case and number suppletion in pronouns (Smith et al. 2019) and pronoun clusivity (inclusive/exclusive marking on 1PL pronouns) (Moskal 2018), among other things. Crucially, when viewed as a three-way pattern of exponence, the disallowed pattern of plural marking—the regular and similitive plural share an exponent, and the associative plural is distinct ((369); = pattern (iv))—in fact looks like a *ABA pattern. In other words, if the regular plural is *not* recruited as the associative plural in a particular language — but rather dedicated morphology is used — it is not possible for the similitive plural to revert back to using the regular plural. It, too, must use dedicated similitive plural morphology (either ABB or ABC).

	PL	APL	SPL
(369)	*(iv)	A	B
		A	A

To account for the *ABA generalization, many authors argue that certain feature inventories stand in a containment or nesting relation, and are therefore sensitive to the *Elsewhere* ordering of spell-out rules (Kiparsky 1973): more specific rules take precedence over more general ones. For example, Bobaljik (2012) argues that the morphosyntactic features of degree adjectives are in the containment relation in (370). The positive adjective has the feature [ADJ], the comparative adjective has the features [ADJ] and [CMPR] and the superla-

tive adjective has the features [ADJ], [CMPR] and [SPRL]. The morphosyntactic features that make up the superlative properly contain the morphosyntactic features that make up the comparative, which in turn properly contain the morphosyntactic feature that makes up the basic form of the adjective.

- (370) a. [ADJ] = *positive*
 b. [[ADJ] CMPR] = *comparative*
 c. [[[ADJ] CMPR] SPRL] = *superlative*

Nesting features rule out *ABA patterns (like **good–better–goodest*) because ABA violates the *Elsewhere Condition* in (371), which says that more specific Vocabulary Insertion rules must win out over more general ones.

(371) *Elsewhere Condition*

Where several entries meet the conditions for insertion, the item matching the greatest number of features in the terminal morpheme must apply.

For example, in order to have a suppletive root in the second cell of the degree adjective paradigm, there must be a Vocabulary Insertion rule that specifies the feature [CMPR] (372a). This rule says that the exponent of the root $\sqrt{\text{GOOD}}$ is *be(tt)-* in the context of the [CMPR] feature. Elsewhere (that is, in the absence of the [CMPR] feature), the exponent of the root $\sqrt{\text{GOOD}}$ is *good*.

- (372) a. $\text{GOOD} \rightarrow \text{be(tt)-} / __\] \text{ CMPR}$ (= B)
 b. $\text{GOOD} \rightarrow \text{good}$ (= A)

When it comes to the superlative adjective, the *Elsewhere Condition* ensures that the most specific compatible rule in (372) applies. Crucially, since the feature [CMPR] is properly contained inside the superlative (see (370c)), it is the Vocabulary Insertion rule in

(372a) that must apply. That is, the exponent of the root $\sqrt{\text{GOOD}}$ is *be(tt)-* in the superlative as well. The more general rule in (372b) cannot apply, given the Elsewhere Condition, and thus an ABA pattern is blocked.

As we saw above, the implicational hierarchy that I have established for similitive and associative plurals can be viewed as another instance of the *ABA generalization (369). That is, there is no language in which associative plural morphology is distinct from the regular plural and in which similitive plural morphology reverts back to the regular plural. I therefore propose that the implicational hierarchy between the similitive and associative plural is the result of nesting feature relations, just as in other *ABA patterns. The regular plural has the feature [PL] (373a).²⁰ I claim that the associative plural has the feature [HET] (for *heterogeneous*).²¹ The similitive plural has the additional feature [SIM] (for *similarity*), which properly contains [HET] (373c). Note that the containment relation in (373c) is semantically motivated. Both associative and similitive plurals are heterogeneous plurals (their plural set consists of an element X and other things that are not X), hence they both bear a [HET] feature. The similitive plural has the additional restriction that the non-X elements in this plurality be similar, in some context-dependent way, to X. Associative plurals do not have this similarity requirement.

- | | | |
|-------|----------------|-----------------------------|
| (373) | a. [PL] | = <i>regular plural</i> |
| | b. [HET] | = <i>associative plural</i> |
| | c. [[HET] SIM] | = <i>similitive plural</i> |

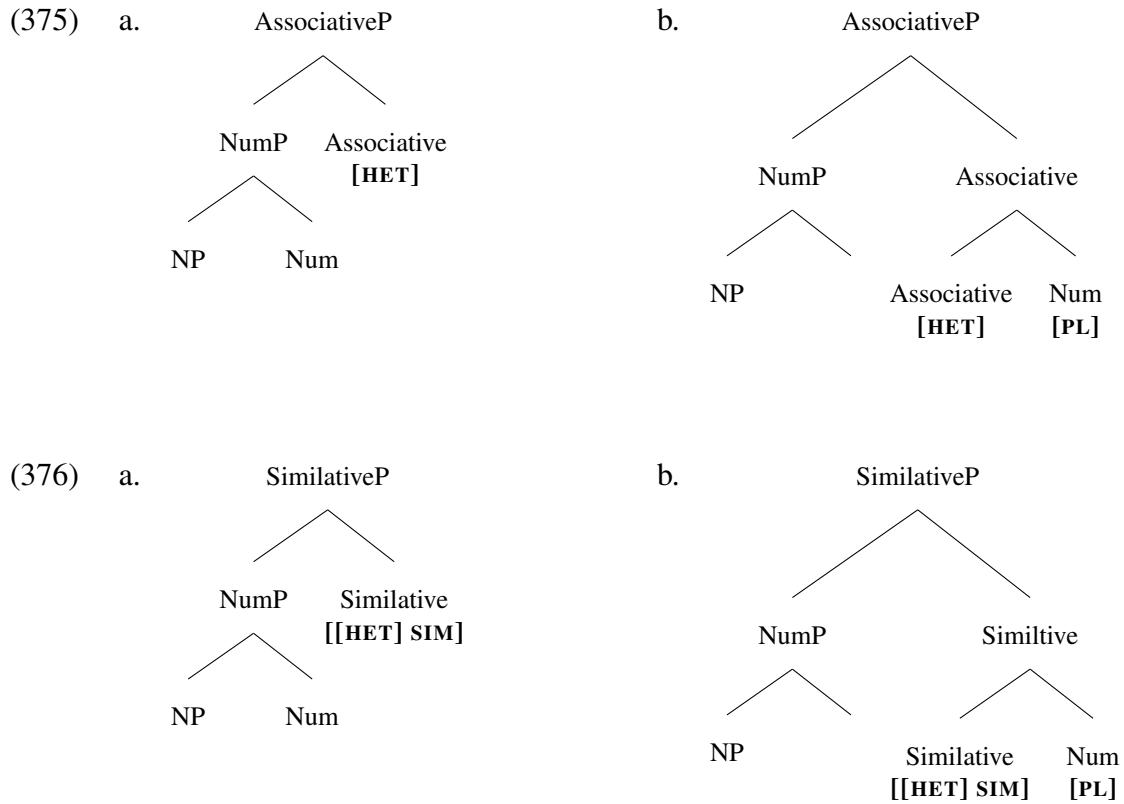
²⁰As mentioned in Chapter 3, the use of privative features like [PL] and [SG] is for ease of exposition only. The analysis to follow will not be affected if one were to use bivalent features instead.

²¹[Het] is what I have been calling [Assoc] up until now.

Let's first consider how the feature bundles in (373) yield the attested pattern combinations (i)–(iii). Pattern (i) (AAA) arises when there is a single Vocabulary Insertion rule targeting the [PL] feature, illustrated in (374).

(374) [PL] $\rightarrow \alpha$

As it stands, however, neither the associative nor simulative plural has a [PL] feature. There is a reason for that. If associative and simulative plurals had an inherent [PL] feature, we would expect that at least some languages with free-standing definite articles could have identical exponents for regular and associative/simulative plurals simply because there is a Vocabulary Insertion rule (374) and nothing more specific. Recall, however, that our typological generalizations show that there are no such languages: languages that show the *plural pattern* in their associative or simulative plural lack free-standing definite articles. We can resolve this issue if the heterogeneous plurals can only have a [PL] feature via head movement, as I argued for associative plurals in Chapter 3. There I argued that a *plural pattern* associative plural was derived via head movement of Num⁰ to an Associative head, triggered by Agree with a [i#:__] on Associative⁰. Since the [PL] feature is a property of Num⁰, then prior to incorporation the associative plural only has the feature [HET] (375a). Extending the head movement analysis of the *plural pattern* to simulative plurals, the simulative plural only has the feature bundle [[HET] SIM] prior to incorporation (376a).



Pattern (i) (AAA) thus requires head movement of Num⁰ and for the language to have the single Vocabulary Insertion rule in (374).

Pattern (ii) (ABB/ABC) arises when a language has an additional Vocabulary Insertion rule that targets [HET] and there is no head movement of Num⁰.

Ma Manda is an example of an ABB language (377), which is characterized by (378). The rule in (378a) applies to the regular plural, yielding (377a). Crucially, this rule cannot apply to either of the two heterogeneous plurals because they do not have a [PL] feature. The rule in (378b) must apply to both the associative and similative plural (377b-c) because both bear a feature [HET] and there is no more specific Vocabulary Insertion rule that applies.

(377) *Ma Manda* [Papuan]

- a. daamin -nek -**ye**
 ancestor -1NSG.POSS -NSG
 ‘our ancestors’
- b. Pandi **kadek**
 Pandi APL
 ‘Pandi and his group’
- c. mi **kadek**
 water SPL
 ‘water and such’

(Pennington 2016, pp. 249–50, 577)

- (378) a. [PL] → *ye*
 b. [HET] → *kadek*

Another pattern (ii) is an ABC pattern, illustrated by Bargam (379). This pattern arises when there is an additional Vocabulary Insertion rule that specifies the [SIM] feature (380c). Thus, the rule in (380a) applies to the regular plural, yielding (379a). Once again, this rule cannot apply to either of the two heterogeneous plurals because they do not have a [PL] feature. The rule in (380b) applies to the associative plural (379b). However, the rule in (380b) cannot apply to the simulative plural because there is a more specific rule, (380c), that can apply instead.

(379) *Bargam* [Trans-New Guinea]

- a. abay -**niz**
 brother/sister.in.law -PL
 ‘brothers/sisters in law’
- b. Anna =**nen**
 Anna =APL
 ‘Anna and those with her’

- c. sansan **nagah**
leaves SPL
'leaves and things'

(Hepner 2006, pp. 31, 58, 145)

- (380)
- a. [PL] → *niz*
 - b. [HET] → *nen*
 - c. [HET] → *nagah* /] SIM

Turning to pattern (iii) (A(A)B), this pattern requires Num⁰ to incorporate into Associative⁰ in the associative plural, but not to incorporate in the simulative plural. Turkish is an example of an A(A)B language (381). This pattern arises when, in addition to the rule targeting the [PL] feature, there is a Vocabulary Insertion rule that specifies the [SIM] feature (382b). The rule in (382a) applies to the regular plural, yielding (381a). Since Num⁰ has incorporated into Associative⁰ in an A(A)B language, the rule in (382a) also applies to the associative plural (381). However, Num⁰ does not incorporate into Simulative⁰ here, hence the rule in (381a) cannot apply to the simulative plural. The rule in (382b), which targets the [HET] feature in the context of SIM, applies to the simulative plural, yielding (381c).

(381) *Turkish* [Turkic]

- a. ev **-ler**
house -PL
'houses'
- b. Ahmet **-ler**
Ahmet -PL
'Ahmet and his family'
- c. bulut **m-** ulut
cloud M- cloud
'clouds and the like'

(Görgülü 2011)

(Gillon et al. 2014)

- (382) a. [PL] $\rightarrow$ *lAr*
 b. [HET] $\rightarrow$ REDUP / __] SIM

Finally, the feature representation in (373) prohibits the unattested pattern (iv) (ABA): here, the simulative plural is the same as the regular plural, and the associative plural is distinct. For this pattern to arise we would need head movement of Num⁰ to Simulative⁰, no head movement of Num⁰ to Associative⁰, and the two Vocabulary Insertion rules in (383). (383a) specifies the exponent of the regular plural. For the associative plural to be distinct, there would have to be a Vocabulary Insertion rule that specifies the feature [HET] (383b).

- (383) a. [PL] $\rightarrow$ α (= A)
 b. [HET] $\rightarrow$ β (= B)

To derive an ABA pattern in this hypothetical language, only the rule in (383a) can have applied to the simulative.

- (384) Features on Simulative⁰: [PL], [[HET] SIM]
 Vocabulary Insertion rule applied: [PL] $\rightarrow$ α (= A)

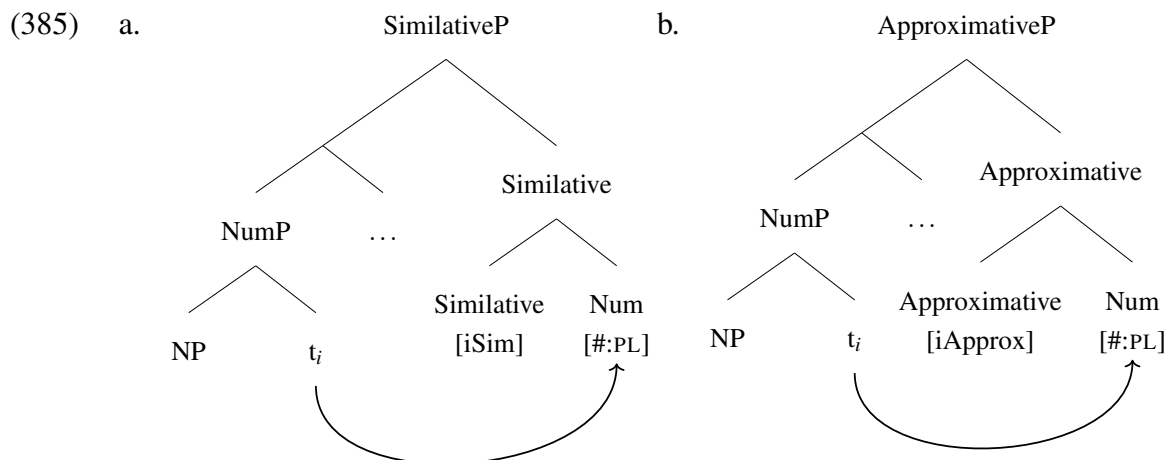
However, because the associative plural feature ([HET]) is properly contained inside the simulative plural, rule (383b) also applies. This means that the realization of the simulative plural could in principle *contain* regular plural morphology in addition to a distinct morpheme—akin to the associative plural in e.g., Central Alaskan Yup'ik—but will never be identical to it (i.e., *ABA). In the current sample of simulative plural languages, simulative plural morphology is either entirely morphologically distinct or entirely identical to the regular plural. I leave it to future research to determine whether or not there are simulative plurals made up of the regular plural plus distinct morphology, and whether this pattern is

found in languages with distinct associative plural morphology (as I predict should be possible). What is important here is that what is not attested (the similitive plural is identical to the regular plural when the associative plural is distinct) is correctly ruled out by the proposed analysis.

To sum up, I have shown that the implicational hierarchy that I have established regarding the possible exponence of similitive and associative plurals – namely, that if the similitive plural shows the *plural pattern*, the associative plural shows the *plural pattern* too – can be analyzed as another instance of the *ABA generalization, where morphological constraints on the realization of nested features ensure that distinct associative plural marking cannot revert back to the regular plural in the similitive.

5.1.4 Accounting for the absence of (free-standing) definite articles in heterogeneous plurals

The second generalization to be captured is that, like associative plurals, similitive and approximative plurals that show the *plural pattern* also lack free-standing definite articles. I derived this generalization for associative plurals by arguing that when regular plural morphology is used as (or as part of) the associative plural, Num⁰ has incorporated into Associative⁰. We may think, then, that head movement of Num⁰ is also responsible for the absence of free-standing definite articles in languages with similitive and approximative plurals that show the *plural pattern*. On this analysis, there would be a functional projection (SimilitiveP or ApproximativeP), analogous to the AssociativeP projection that is theorized for associative plurals, the head of which realizes similitive and approximative plural morphology, respectively. When the similitive or approximative plural is the same as the regular plural, Num⁰ incorporates into Sim⁰/Approx⁰ under Agree (385).



This would derive the absence of free-standing definite articles in *plural pattern* similative plural languages the same way that it derives the absence of free-standing definite articles in *plural pattern* associative plural languages: the presence of D in languages with free-standing definite articles blocks incorporation of Num⁰.

However, the current sample of languages offers up another perspective, at least for similative plurals. All of the *plural pattern* similative plurals in the sample also have an associative plural, meaning we are dealing with all three cells in the plural marking paradigm—regular, associative and similative. Moreover, as we saw in the previous section, there is a morphological restriction against ABA patterns due to feature containment: if the similative plural is the same as the regular plural (A), the associative plural must be too (A/*B) (cf. (386—387)). This is because the morphosyntactic features of the associative plural are properly contained inside the morphosyntactic features of the similative plural, yielding a ban on the ABA pattern.

(386)

	PL	APL	SPL
(i)	A	A	A

(387)

	PL	APL	SPL
*(iv)	A	B	A

Crucially, this entails that any language with a *plural pattern* simulative plural and an associative plural will lack free-standing definite articles because feature containment ensures that the associative plural also shows the *plural pattern*, and *plural pattern* associative plurals lack free-standing definite articles. Thus, the absence of definite articles in *plural pattern* simulative plurals could be a by-product of (i) having an associative plural and (ii) the containment relations of the morphosyntactic features of each plural blocking ABA patterns (or more precisely, forcing an AAA pattern).

Both options—head movement or a by-product of having an associative plural (and the *ABA generalization)—are plausible, but there are ways to tease apart the analyses. First, we saw in Chapter 3.3 that *plural pattern* associative plurals have peculiar properties in some languages that a head movement analysis was able to explain. Most notably, plural possessor agreement is impossible with an associative plural in Turkish and Hungarian (two *plural pattern* languages) but singular possessor agreement is licit. This is illustrated again in (388). (388a) involves singular possessor agreement with the associative plural, and this is grammatical. Replacing singular possessor agreement for plural is ungrammatical (388b).

- (388) a. (benim) abla-**m**-lar / (senin) abla-**n**-lar
 (1 SG.GEN) elder.sister-1 SG.POSS-PL / (2 SG.GEN) elder.sister-2 SG.POSS-PL
 ‘{**my** / **your.SG**} elder sister and her friends’
- b. *(bizim) abla-**miz**-lar / *(sizin) abla-**nin**-lar
 (1 PL.GEN) elder.sister-1 PL.POSS-PL / (2 PL.GEN) elder.sister-2 PL.POSS-PL
 ‘{**our** / **your.PL**} elder sister and her friends’

Additionally, I showed that the problem with (388) is not that associative plurals are incompatible with plural possessors. This is evidenced by the Japanese example in (389), which involves a plural possessor and is perfectly acceptable. The issue with (388) is plural possessor *agreement*. I showed that plural possessor agreement—which involves

plural number features on a functional head between Num⁰ and Associative⁰—intervenes for head movement of Num⁰.

- (389) a. **boku no** oneechan -tachi / **boku -tachi no** oneechan -tachi
 1P POSS elder.sister -PL / 1P -PL POSS elder.sister PL
 ‘{**my** / **our**} elder sister and her friends’
- b. **kimi no** oneechan -tachi / **kimi -tachi no** oneechan -tachi
 2P POSS elder.sister -PL / 2P -PL POSS elder.sister PL
 ‘{**your.SG** / **your.PL**} elder sister and her friends’

Thus, if *plural pattern* similitive and approximative plural languages also involve head movement of Num⁰, we expect to see blocking effects like this with similitive and approximative plurals, too. If we fail to find any such blocking effects, the absence of free-standing definite articles would be better explained as a by-product of having an associative plural and the containment relations of the relevant morphosyntactic features. A more thorough investigation will reveal whether or not such blocking effects are found.

We can also tease apart the two analyses by trying to find *plural pattern* similitive plural languages that do not have associative plurals. If the absence of free-standing definite articles in *plural pattern* similitive plural languages is a by-product of having an associative plural and the *ABA restriction, the generalization should only be found in languages that also have an associative plural. In other words, at least some languages that use the regular plural as the similitive plural, but which lack an associative plural, should have free-standing definite articles. This is because nothing *a priori* prohibits a language from having a free-standing definite article and using the regular plural as the similitive plural. Conversely, if we do not find *plural pattern* similitive plurals with free-standing definite articles even in the absence of an associative plural, this would suggest that the generalization is best captured by a head-movement analysis. So far I am not aware of languages that have *plural pattern* similitive plurals but that lack an associative plural. A more extensive

investigation should however be conducted before making a definitive conclusion.

Finally, another way to tease apart the two analyses is to try to find a language that uses the regular plural as the approximative plural that also has a free-standing definite article. A language like this would be revealing because there is no implicational hierarchy on the exponence of approximative plurals. That is, there is no *ABA restriction with approximative plurals. Thus, if such a language exists this could be interpreted as indicating that the *ABA restriction *is* responsible for the lack of free-standing definite articles in *similative* plural languages. On the other hand, if we fail to find a *plural pattern* approximative plural with a free-standing definite article, this would indicate the opposite: the *ABA-generalization is likely not responsible for the absence of free-standing definite articles in *plural pattern* similative plurals. So far no such language has been found, but at any rate, the above discussion indicates what kind of language we should be searching for in subsequent investigations.

5.1.5 Summary

This chapter has expanded the investigation into heterogeneous plurals to include similative and approximative plurals. I showed that the typological generalization that I established for *plural pattern* associative plurals also holds of *plural pattern* similative and approximative plurals. I also showed in this chapter that there is a robust implicational hierarchy regarding the exponence of associative and similative plurals: If a language shows the plural pattern in the similative plural, it shows the plural pattern in the associative plural too (if it has one). In other words, there is no language that uses dedicated morphology for the associative plural and regular plural morphology for the similative plural. I showed that this resembles a *ABA restriction on the exponence of the three plurals (regular, associative,

similative). I also proposed an account where nesting features in the heterogeneous plurals account for the implicational hierarchy.

Bibliography

- Abels, Klaus (2003). “Successive Cyclicity, Anti-locality, and Adposition Stranding”. PhD thesis. University of Connecticut.
- Abondolo, Daniel Mario (1985). “Hungarian Inflectional Morphology”. PhD thesis. Columbia University.
- Abramovitz, Rafael Meghani (2021). “Topics in the Grammar of Koryak”. PhD thesis. Massachusetts Institute of Technology.
- Ackema, Peter and Ad Neeleman (2018). *Features of person: From the inventory of persons to their morphological realization*. MIT Press.
- Aikhenvald, Alexandra (1995). “Person marking and discourse in North Arawak”. In: *Studia Linguistica* 49.2, pp. 152–195.
- Aikhenvald, Alexandra (2003). *A grammar of Tariana, from Northwest Amazonia*. Cambridge: Cambridge University Press.
- Aikhenvald, Alexandra (2008). *The Manambu Language of East Sepik, Papua New Guinea*. Oxford: Oxford University Press.
- Aikhenvald, Alexandra (2010). “Language contact and pragmatic notions: Tariana in its multilingual context”. In: *Information structure in indigenous languages of the Americas: Syntactic approaches*. Ed. by Rodrigo Gutiérrez-Bravo José Camacho and Liliana Sánchez. Walter de Gruyter, pp. 17–38.

- Aikhenvald, Alexandra (2014). "Number and noun categorization: A view from north-west Amazonia". In: *Number-Constructions and Semantics: Case studies from Africa, Amazonia, India and Oceania*. Ed. by Anne Storch and Gerrit J. Dimmendaal. Amsterdam: John Benjamins, pp. 33–56.
- Alamin, Suzan (2014). "Noun Phrase Constructions in Nubian Languages: A Comparative Study". In: *A Journal of Nubian Studies* 1, pp. 203–220.
- Albers, Ulrike (2020). "A description of bare noun phrases in Reunion Creole". In: *Journal of Pidgin and Creole languages* 35.1, pp. 1–36.
- Alexiadou, Artemis (2011). "Plural mass nouns and the morpho-syntax of number". In: *Proceedings of the 28th West Coast Conference on Formal Linguistics*. Cascadilla Proceedings Project Somerville, MA, pp. 33–41.
- Allison, Sean David (2012). "Aspects of a grammar of Makary Kotoko". PhD thesis. University of Colorado.
- Ameka, Felix (1991). "Ewe: its grammatical constructions and illocutionary devices". PhD thesis. Australian National University.
- Arnott, David W. (1970). *The nominal and verbal systems of Fula*. Oxford: Oxford University Press.
- Asher, R. E. and T. C. Kumari (1997). *Malayalam*. Routledge.
- Ba, Ibrahima (2013). "The Clause Structure of Pulaar". MA thesis. University of Kansas.
- Baertlein, Elizabeth and Martin Ssekitto (2014). "Luganda nouns: Inflectional morphology and tests". In: *Linguistic Portfolios* 3, pp. 21–26.
- Baker, Mark (1985). "The Mirror Principle and morphosyntactic explanation". In: *Linguistic Inquiry* 16.3, pp. 373–415.
- Baker, Mark (1996). *The polysynthesis parameter*. New York: Oxford University Press.
- Bamgbose, Ayo (1966). *A grammar of Yoruba*. Cambridge: Cambridge University Press.

- Bartos, Huba (1999). “Morphosyntax and interpretation. The syntactic background of Hungarian inflexional phenomena”. PhD thesis. Eötvös University.
- Bartos, Huba (2000). “Az inflexiók jelenségek szintaktikai háttere [The syntactic background of inflectional phenomena]”. In: *Strukturális magyar nyelvtan*. Ed. by Ferenc Kiefer, pp. 653–762.
- Becker, Laura (2021). *Articles in the World’s Languages*. Berlin: Mouton de Gruyter.
- Bennett, Michael Eric (1986). “Aspects of the simple clause in Malagasy: A stratificational approach”. PhD thesis. Michigan State University.
- Berg, Rene van den (1989). *A grammar of the Muna language*. Dordrecht: Foris.
- Besnier, Niko (2000). *Tuvulan*. New York: Routledge.
- Besten, Hans den (1996). “Associative DP”. In: *Linguistics in the Netherlands* 13, pp. 13–24.
- Bhatia, Tej K. (1993). *Punjabi*. New York: Routledge.
- Bickel, Balthasar (2017). “Belhare”. In: *The Sino-Tibetan languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 696–711.
- Bird, Charles, John Hutchinson, and Mamadou Kanté (1977). *An ka Bamanankan Kalan: Introductory Bambara*. Indiana: Indiana University Linguistics Club.
- Biswas, Priyanka (2014). “Bangla associative plural -ra: A cross-linguistic comparison with Chinese *men* and Japanese *-tachi*”. In: *Proceedings of the 31st West Coast Conference of Formal Linguistics (WCCFL 31)*. Ed. by Robert E. Santana-LaBarge. Somerville, MA: Cascadilla Press, pp. 56–65.
- Boas, Franz (1911). “Introduction”. In: *Handbook of American Indian Languages*. Ed. by Franz Boas. Washington DC: Bureau of American Ethnology, pp. 1–83.
- Bobaljik, Jonathan David (2008). “Missing persons: A case study in morphological universals”. In: *The Linguistic Review* 25, pp. 203–230.

- Bobaljik, Jonathan David (2012). *Universals in comparative morphology: Suppletion, superlatives and the structure of words*. Cambridge, MA: MIT Press.
- Bobaljik, Jonathan David and Susi Wurmbrand (2001). “Seven prefix-suffix asymmetries in Itelmen”. In: *Proceedings from the 37th Annual Meeting of the Chicago Linguistic Society*. Ed. by Mary Andronis et al., pp. 205–220.
- Bogolmolets, Ksenia (2020). “Lexical accent in languages with complex morphology”. PhD thesis. University of Connecticut.
- Booij, Geert and Jerzy Rubach (1984). “Morphological and prosodic domains in lexical phonology”. In: *Phonology Yearbook* 1, pp. 1–27.
- Borer, Hagit (2005). *Structuring Sense: In name only*. Oxford: Oxford University Press.
- Bošković, Zeljko (1994). “D-structure, theta-criterion, and movement into theta-positions”. In: *Linguistic Analysis* 24, pp. 247–286.
- Bošković, Zeljko (2005). “On the locality of left branch extraction and the structure of NP”. In: *Studia Linguistica* 59, pp. 1–45.
- Bošković, Zeljko (2006). “Case and agreement with genitive of quantification in Russian”. In: *Agreement Systems*. Ed. by Cedric Boeckx. Amsterdam: John Benjamins, pp. 99–121.
- Bošković, Zeljko (2008a). “On successive cyclic movement and the freezing effect of feature checking”. In: *Sounds of silence: Empty elements in syntax and phonology*. Ed. by Henk van Riemsdijk Jutta Hartmann Veronika Hegedus. Amsterdam: Elsevier, pp. 195–233.
- Bošković, Zeljko (2008b). “What will you have, DP or NP?” In: *Proceedings of the 37th North East Linguistics Socety (NELS 37)*. Ed. by Emily Elfner and Martin Walkow. Amherst, MA: GLSA (Graduate Linguistic Student Association), pp. 101–114.

- Bošković, Zeljko (2012a). “Adjectival escapades”. In: *Formal Approaches to Slavic Linguistics: The Indiana Meeting*. Ed. by Steven Franks et al. Ann Arbor, Michigan: Michigan Slavic Publications, pp. 1–25.
- Bošković, Zeljko (2012b). “On NPs and clauses”. In: *Discourse and grammar: From sentence types to lexical categories*. Ed. by Günther Grewendorf and Thomas Ede Zimmermann. Berlin: Mouton de Gruyter, pp. 179–242.
- Bošković, Zeljko (2013). “Phases beyond Clauses”. In: *The nominal structure in Slavic and beyond*. Ed. by Lilia Schürcks, Anastasia Giannakidou, and Urtzi Etxeberria. Boston: De Gruyter Mouton, pp. 75–116.
- Bošković, Zeljko (2014). “Now I’m a Phase, Now I’m Not a Phase: On the Variability of Phases with Extraction and Ellipsis”. In: *Linguistic Inquiry* 45.1, pp. 27–89.
- Bošković, Zeljko (2016). “On second position clitics cross-linguistically”. In: *Formal Studies in Slovenian Syntax*. Ed. by Franc Lanko Marušič and Rok Žaucer. Amsterdam: John Benjamins.
- Bošković, Zeljko (2018). “On affixal articles: An argument from Bantu for Bulgarian, Romanian, and Icelandic”. In: *Katerino Mome: Studies in Bulgarian morphosyntax in honor of Catherine Rudin*. Ed. by Steven Franks et al. Bloomington: Slavica.
- Bošković, Zeljko (2023). “Multiple wh-fronting in a typological setting: what is behind multiple wh-fronting?” Ms. University of Connecticut.
- Bošković, Zeljko and Jairo Nunes (2007). “The copy theory of movement A view from PF”. In: *The Copy Theory of Movement*. Ed. by Norbert Corver and Jairo Nunes. Amsterdam: John Benjamins Publishing, pp. 13–74.
- Bromley, Myron (1972). “The grammar of Lower Grand Valley Dani in discourse perspective”. PhD thesis. Yale University.
- Brown, Lea (2001). “A grammar of Nias Selatan”. PhD thesis. University of Sydney.

- Brown, Penelope and Stephen C Levinson (1987). *Politeness: Some universals in language usage*. Vol. 4. Cambridge: Cambridge University Press.
- Bruce, Les (1984). *The Alamlak language of Papua New Guinea (East Sepik)*. Pacific Linguistics, Research School of Pacific and Asian Studies.
- Bugaeva, Anna (2022). *Handbook of the Ainu Language*. Vol. 12. Walter de Gruyter.
- Bugenhagen, Robert D. (1995). *A grammar of Mangap-Mbula: An Austronesian language of Papua New Guinea*. Canberra: Australian National University Department of Linguistics.
- Burbiel, Gustav (2018). *Tatar grammar*. Stockholm: Institute for Bible Translation.
- Burling, Robbins (1961). *A Garo Grammar*. Pune: Deccan College Postgraduate and Research Institute.
- Burling, Robbins (2003). *A Garo grammar*. Pune: Deccan College Postgraduate and Research Institute.
- Bye, Patrik and Peter Svenonius (2012). “Non-concatenative morphology as epiphenomenon”. In: *The Morphology and Phonology of Exponence*. Ed. by Jochen Trommer. Amsterdam: John Benjamins, pp. 427–495.
- Caha, Pavel (2009). “The nanosyntax of case”. PhD thesis. University of Tromsø.
- Carlino, Salvatore (2022). “Theya (Okinawa, Northern Ryukyuan)”. In: *An Introduction to the Japonic Languages*. Brill, pp. 58–90.
- Carlson, Greg et al. (2006). “Weak definite noun phrases”. In: *Proceedings from the 36th Annual Meeting of the North East Linguistic Society*. Ed. by Chris Davis, Amy Rose Deal, and Yuri Zabbal. Amherst, MA: GLSA, pp. 179–196.
- Castillo, Yolanda Rivero (2022). *A description of Papiamentu: A creole language of the Caribbean area*. Boston: Brill.
- Chandralal, Dileep (2010). *Sinhala*. Amsterdam: John Benjamins Publishing.

- Charney, Jean O. (1993). "A grammatical sketch of the Comanche language". PhD thesis. University of Colorado at Boulder.
- Chierchia, Gennaro (1998a). "Plurality of Mass Nouns and the Notion of "Semantic Parameter"". In: *Events and grammar*. Ed. by Susan Rothstein. Springer, pp. 53–103.
- Chierchia, Gennaro (1998b). "Reference to kinds across languages". In: *Natural Language Semantics* 6, pp. 339–405.
- Chomsky, Noam (2000). "Minimalist enquiries: The framework". In: *Step by step*. Cambridge, Mass: MIT Press, pp. 89–155.
- Cinque, Guglielmo (2018). "On the Merge position of additive and associative plurals". In: *From sounds to structures: Beyond the veil of Maya*. Ed. by Roberto Petrosino, Pietro Cerrone, and Harry van der Hulst. Berlin: Mouton de Gruyter, pp. 499–513.
- Colarusso, John (1992). *A Grammar of the Kabardian Language*. Calgary: University of Calgary Press.
- Collins, James N. (2019). "Definiteness determined by syntax". In: *Natural Language & Linguistic Theory* 37, pp. 1367–1420.
- Corbett, Greville (1987). "The Morphology/Syntax Interface: Evidence from Possessive Adjectives in Slavonic". In: *Language* 63 (2), pp. 299–345.
- Corbett, Greville (2000). *Number*. Cambridge: Cambridge University Press.
- Corbett, Greville and Marianne Mithun (1996). "Associative forms in a typology of number systems: Evidence from Yup'ik". In: *Journal of Linguistics* 32, pp. 1–17.
- Corver, Norbert (1992). "On deriving certain left branch extraction asymmetries: A case study in parametric syntax". In: *North East Linguistics Society*. Vol. 22, pp. 67–84.
- Crazzolara, J.P. (1960). *A Study of the Logbara (Ma'di) language*. Oxford: Oxford University Press.

- Creissels, Denis (2007). “A sketch of Bambara argument structure”. In: *Workshop on Grammar and Processing of Verbal Arguments*, pp. 1–42.
- Creissels, Denis (2013). “A sketch of Mandinka”.
- Creissels, Denis (2022). “Number in Tswana”. In: *Number in the world’s languages: a comparative handbook*. Ed. by Paolo Acquaviva and Michael Daniel. Boston: Walter de Gruyter, pp. 107–127.
- Cysouw, Michael (2003). *The paradigmatic structure of person marking*. Oxford, UK: Oxford University Press.
- Daniel, Michael (2004). “From headless possessives to associative plurals”. In: *Possesives and beyond: Semantics and Syntax*. Ed. by Ji-yung Kim, Yury Lander, and Barbara H. Partee. Amherst, Mass: GLSA, pp. 387–409.
- Daniel, Michael (2020). “Associative plural as indexical category”. In: *Language Sciences* 81, pp. 101–256.
- Daniel, Michael and Yury Lander (2011). “The Caucasian languages”. In: *The languages and linguistics of Europe*. Ed. by Bernd Kortmann and Johan van der Auwera. Mouton de Gruyter, pp. 125–158.
- Daniel, Michael and Edith Moravcsik (2013). “The associative plural”. In: *The World Atlas of Language Structures Online (WALS)*. Ed. by Matthew S. Dryer and Martin Haspelmath. Leipzig: Max Planck Institute for Evolutionary Anthropology. URL: <https://wals.info/chapter/36>.
- Danon, Gabi (2011). “Agreement and DP-Internal Feature Distribution”. In: *Syntax* 14.4, pp. 297–317.
- Davidson, Matthew (2002). “Studies in Southern Wakashan (Nootkan) grammar”. PhD thesis. University of New York at Buffalo.

- Dayal, Veneeta (2017). “Determining (In)definiteness in the Absence of Articles”. In: *Proceedings of TripleA 3: Fieldwork Perspectives on the Semantics of African, Asian and Austronesian Languages*. Ed. by Vera Hohaus & Wanda Rothe, pp. 85–99.
- Dékány, Eva (2011). “A profile of the Hungarian DP: The interaction of lexicalization, agreement and linearization with the functional sequence”. PhD thesis. University of Tromsø.
- Dékány, Eva (2021). *The Hungarian nominal functional sequence*. Springer.
- den Dikken, Marcel (2023). “Associative plurals and their associates: An integrated approach to the nominal plural markers of Hungarian”. In: *Journal of Uralic Linguistics* 2.1, pp. 50–73.
- Derbyshire, Desmond (1979). *Hixkaryana*. Lingua Descriptive Studies. North-Holland Publishing Company.
- Despić, Miloje (2011). “Syntax in the absence of determiner phrase”. PhD thesis. University of Connecticut.
- Despić, Miloje (2013). “Intensifiers, Focus, and Clitics: Is Pronoun Position Truly an Argument for D in SC?” In: *The Nominal Structure in Slavic and Beyond*. Ed. by Lilia Schürcks, Anastasia Giannakidou, and Urtzi Etxeberria. Vol. 116. Studies in Generative Grammar. Berlin: De Gruyter Mouton, pp. 39–74.
- Despić, Miloje (2015). “Phases, reflexives, and definiteness”. In: *Syntax* 18.3, pp. 201–234.
- Despić, Miloje (2019). “On kinds and anaphoricity in languages without definite articles”. In: *Definiteness across languages*. Ed. by Ana Aguilar-Guevara, Julia Pozas Loyo, and Violeta Vázquez-Rojas Maldonado. Berlin: Language Science Press, pp. 259–291.
- Dienst, Stefan (2014). *A grammar of Kulina*. Vol. 66. Walter de Gruyter.

- Dikken, Marcel den (2003). “On the morphosyntax of *wh*-movement”. In: *Multiple Wh-Fronting*. Ed. by Cedric Boeckx and Kleanthes K. Grohmann. Amsterdam: John Benjamins Publishing, pp. 77–98.
- Driem, George van (1987). *A grammar of Limbu*. Berlin: Mouton de Gruyter.
- Dryer, Matthew (2013). “Definite Articles”. In: *The World Atlas of Language Structures Online (WALS)*. Ed. by Matthew S. Dryer and Martin Haspelmath. Leipzig: Max Planck Institute for Evolutionary Anthropology. URL: <http://wals.info/chapter/37>.
- Dubinsky, Stanley and Mila Tasseva-Kurkchieva (2014). “On the NP/DP languages frontier: Bulgarian as a transitional case”. Poster presented at the 89th Annual Meeting of the Linguistic Society of America, Portland, OR.
- Dunn, Michael (1999). “A grammar of Chukchi”. PhD thesis. Australian National University.
- Dwyer, David (1985). “Evolutionary morphology of definite articles in Southwestern Mande”. In: *Current approaches to African linguistics*. Ed. by Gerrit J. Dimmendaal. Dordrecht: Foris, pp. 149–170.
- É. Kiss, Katalin (2002). *The syntax of Hungarian*. Cambridge: Cambridge University Press.
- Ebihara, Shiho (2010). “Amdo Tibetan”. In: ed. by Yauhiro Yamakoshi. Tokyo: Research Institute for Languages, Cultures of Asia, and Africa (ILCAA), pp. 41–78.
- Ebihara, Shiho (2014). “The inclusive-exclusive distinction in spoken and written Tibetan”. In: *Research Annual Report 51*, pp. 85–102.
- Elbert, Samuel H. and Mary Kawena Pukui (1979). *Hawaiian Grammar*. Honolulu: University of Hawaii Press.
- Elbourne, Paul (2001). “On the semantics of pronouns and definite articles”. In: *Proceedings of the 20th West Coast Conference on Formal Linguistics (WCCFL 20)*. Vol. 20. Somerville, MA: Cascadilla Press, pp. 164–177.

- Elson, Mark (1976). “The definite article in Bulgarian and Macedonian”. In: *The Slavic and East European Journal* 20.3.
- Enfield, N. J. (2007). *A grammar of Lao*. New York: de Gruyter.
- Epps, Patience (2008). *A grammar of Hup*. Mouton de Gruyter.
- Erschler, David (2016). “Ossetic”. In: *Word-Formation*. Ed. by Peter Müller et al. Vol. 5. Mouton de Gruyter, pp. 3157–3179.
- Erschler, David (2019). “A new argument for existence of the DP in languages without articles”. In: *Journal of Linguistics* 55, pp. 879–887.
- Fedden, Sebastian and Greville Corbett (2017). “Extreme Classification”. In: *Cognitive Linguistics* 29.4, pp. 633–675.
- Feldman, Harry (1986). *A grammar of Awtuw*. Dept. of Linguistics, Research School of Pacific Studies, The Australian National University, 1986.
- Fenger, Paula (2020). “Words within words: The Internal Syntax of Verbs”. PhD thesis. University of Connecticut.
- Feu, Veronica Du (1996). *Rapanui*. New York: Routledge.
- Filchenko, Andrey Yury (2007). “A grammar of Eastern Khanty”. PhD thesis. Rice University.
- Forbes, Clarissa (2013). “Associative plurality in the Gitksan nominal domain”. In: *Proceedings of the 2013 annual conference of the Canadian Linguistic Society*.
- Frajzyngier, Zygmunt (1997). “Grammaticalization of number: from demonstratives to nominal and verbal plural”. In: *Linguistic Typology* 1, pp. 193–242.
- Franks, Steven (1994). “Parametric properties of numeral phrases in Slavic”. In: *Natural Language and Linguistic Theory* 12, pp. 570–649.
- Franks, Steven (1998). “Clitics in Slavic”. Presented at the Comparative Slavic Morphosyntax Workshop, Spencer Creek, Indiana.

- Franks, Steven (2001). “The internal structure of Slavic NPs, with special reference to Bulgarian”. In: *Generative Linguistics in Poland: Syntax and Morphosyntax*. Ed. by Adam Przepiórkowski and Piotr Barański, pp. 53–69.
- Franks, Steven (2007). “Deriving discontinuity”. In: *Proceedings of FDSL 6.5*, pp. 103–120.
- Franks, Steven and Tracy King (2000). *A handbook of Slavic Clitics*. Oxford: Oxford University Press.
- Fried, Robert Wayne (2010). “A grammar of Bao’an Tu, a Mongolic language of northwest China”. PhD thesis. State University of New York at Buffalo.
- Friedman, Victor A. (2020). “Lak”. In: ed. by Maria Polinsky. Oxford: Oxford University Press, pp. 201–242.
- Gaby, Alice (2018). *A grammar of Kuuk Thaayorre*. Boston: De Gruyter.
- Garland, Jennifer (2006). “Morphological typology and the complexity of nominal morphology in Sinhala”. In: *Santa Barbara papers in Linguistics*. Ed. by Robert Englebretson and Carol Genetti. Vol. 17. California: UCSB Department of Linguistics, pp. 1–19.
- Genetti, Carol (2017). “Dolakha Newar”. In: *The Sino-Tibetan Languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 436–452.
- Gerland, Doris (2014). “Definitely not possessed: Possessive suffixes with definiteness marking function”. In: *Frames and Concept types: Applications in language and philosophy*. Ed. by Thomas Gamerschlag et al. New York: Springer, pp. 269–292.
- Ghomeshi, Jila and Diane Massam (2020). “Number is different in Nominal and Pronominal phrases”. In: *Linguistic Inquiry* 51.3, pp. 597–610.

- Gillon, Carrie, Solveiga Armoskaite, and Deniz Aysegul Kutlu (2014). “M-Reduplication in colloquial Turkish”. Poster presented at the 17th Annual Meeting of the International Conference on Turkish Linguistics, University of Rouen, France.
- Gorelova, Liliya M. (2002). *Manchu Grammar*. Brill.
- Görgülü, Emrah (2011). “Plural marking in Turkish: Additive or associative?” In: *Working Papers of the Linguistics Circle* 21, pp. 70–80.
- Göskel, Asli and Celia Kerslake (2005). *Turkish: a comprehensive grammar*. New York: Routledge.
- Greenberg, Joseph (1977). “Niger-Congo nouns class markers: prefixes, Suffixes, both or neither”. In: *Papers from the Eighth Conference on African Linguistics*. Ed. by Martin Mould and Thomas Hinnebusch, pp. 97–104.
- Greenberg, Joseph (1978). “How does a language acquire gender markers?” In: *Universals of human language*. Ed. by Joseph Greenberg, Charles Ferguson, and Edith Moravcsik. Vol. 3. Stanford, Calif. : Stanford University Press, pp. 47–82.
- Grohmann, Kleanthes K (2000). “Prolific peripheries: A radical view from the left”. PhD thesis. University of Maryland.
- Grossman, Rebecca (1992). “A discourse grammar of Bandi”. PhD thesis. The University of Texas at Arlington.
- Gruzdeva, Ekaterina (2020). “Demonstratives in Nivkh: A semantic and pragmatic analysis”. In: *Studia Orientalia Electronica* 8 (1), pp. 1–60.
- Gruzdeva, Ekaterina (2022). “Number in Nivkh”. In: *Number in the world’s languages: a comparative handbook*. Ed. by Paolo Acquaviva and Michael Daniel. Boston: Walter de Gruyter, pp. 375–425.
- Guillaume, Antoine (2008). *A grammar of Cavineña*. New York: Mouton de Gruyter.

- Gunrow-Harsta, Karen (2011). “Adjectives and adjectivals in Magar”. In: *Himalayan Linguistics* 10, pp. 101–126.
- Hagman, Roy S. (1977). *Nama Hottentot Grammar*. Bloomington, IN: Research center for language and semiotic studies.
- Hajdú, P. (1983). “The main characteristic features of the Uralic languages”. In: *Acta Linguistica Academiae Scientiarum Hungaricae* 33, pp. 101–112.
- Harbour, Daniel (2011). “Valence and atomic number”. In: *Linguistic inquiry* 42.4, pp. 561–594.
- Harbour, Daniel (2014). “Paucity, abundance, and the theory of number”. In: *Language*, pp. 185–229.
- Hargreaves, David (2017). “Kathmandu Newar”. In: *The Sino-Tibetan Languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 453–467.
- Hari, Maria (1973). “Nepali Texts”. In: *Clause, Sentence, and Discourse Patterns in selected languages of Nepal*. Ed. by Irvine Davis. Oklahoma: Summer Institute of Linguistics, pp. 3–52.
- Harizanov, Boris (2014). “The role of prosody in the linearization of clitics: Evidence from Bulgarian and Macedonian”. In: *Proceedings of the 22nd Formal Approaches to Slavic Linguistics*, pp. 109–130.
- Haspelmath, Martin (1993). *A grammar of Lezgian*. Mouton de Gruyter.
- Haspelmath, Martin (1997). *Indefinite pronouns*. Oxford: Oxford University Press.
- Haspelmath, Martin (2023). “Types of clitics in the world’s languages”. In: *Linguistic Typology at the Crossroads* 3 (2), pp. 1–59.
- Heath, Jeffrey (1999). *Grammar of Koyrobora Senni*. Köln: Rüdiger Köppe Verlag.
- Heim, Irene and Angelika Kratzer (1998). *Semantics in generative grammar*. Oxford: Blackwell.

- Hellwig, Birgit (2011). *A grammar of Goemai*. Boston: De Gruyter.
- Hepner, Mark (2006). *Bargam grammar sketch*. Ukarumpa, Papua New Guinea: Summer Institute of Linguistics.
- Hildebrandt, Kristine (2007). “Prosodic and Grammatical Domains in Limbu”. In: *Himalayan Linguistics* 8, pp. 1–34.
- Himmelman, Nikolaus P. (2001). “Demonstratives in Narrative discourse: A taxonomy of universal uses”. In: *Studies in Anaphora*. Ed. by Barbara Fox. Philadelphia: John Benjamins, pp. 205–254.
- Hinds, John (2003). *Japanese: descriptive grammar*. Routledge.
- Hirose, Tomio (2004). “N-Plural vs. D-Plural”. In: *Proceedings of the 23rd West Coast Conference on Formal Linguistics (WCCFL 23)*. Ed. by Vineeta Chand et al. Somerville, MA: Cascadilla Press, pp. 332–345.
- Horton, Alonzo E (1949). “A grammar of Luvala”. In.
- Hutchinson, John (1981). *A reference grammar of the Kanuri language*. University of Wisconsin-Madison, African Studies Program.
- Hyman, Larry (2008). “Directional asymmetries in the morphology and phonology of words, with special reference to Bantu”. In: *Linguistics* 46 (2), pp. 309–350.
- Ileri, Muhammed (2023). “Haplology and local dislocation in Turkish: Evidence from associative plural constructions”. In: *Proceedings of the Workshop on Turkic and Languages in Contact with Turkic*. Vol. 8, pp. 118–132.
- Jambrović, Samuel (2022). “Singular Referential Names as Nonrigid Designators and Bound Variables”. In: *NELS 52: Proceedings of the fifty-second annual meeting of the North East Linguistic Society*. Ed. by Özge Bakay et al. Graduate Linguistics Student Association, pp. 73–86.
- Janhunen, Juha et al. (2008). *Wutun*. Lincom Europa.

- Jasenovic, Ludmilla (1983). *Serbo-Croatian: Basic Structure*. Maryland: National Cryptologic School.
- Jendraschek, Gerd (2012). “A grammar of Iatmul”. PhD thesis. University of Regensburg.
- Jespersen, Otoo (1924). *The philosophy of grammar*. London: George Allen and Unwin Ltd.
- Jiang, Li (2012). “Nominal arguments and language variation”. PhD thesis. Harvard University.
- Jiang, Li (2017). “Mandarin associative plural -men and NPs with -men”. In: *International Journal of Chinese Linguistics* 4.2, pp. 191–256.
- Johanson, Lars (1998). “The structure of Turkic”. In: *The Turkic languages*. Ed. by Lars Johanson and Éva Csató. New York: Routledge, pp. 30–66.
- Kayne, Richard (1994). *The antisymmetry of syntax*. Cambridge, MA: MIT Press.
- Kester, Ellen-Petra and Christina Schmitt (2007). “Papiamentu and Brazilian Portuguese A comparative study of bare nominals”. In: *Noun Phrases in Creole languages*. Ed. by Marlyse Baptista and Jacqueline Guéron. John Benjamins Publishing, pp. 107–144.
- Kibe, Nobuko, Hajime Oshima, and Masahiro Yamada (2018). “Plural forms in Yoron-Ryukyuan and address nouns in Ryukuan languages”. In: *Japanese/Korean Linguistics* 25.
- Kim, Chonghyuck (2005). “The Korean plural marker tul and its implications”. PhD thesis. University of Delaware.
- Kim, Kyumin and Paul B. Melchin (2018). “Modifying plurals, classifiers, and co-occurrence: The case of Korean”. In: *Glossa* 3 (1), pp. 1–29.
- Kiparsky, Paul (1973). ““Elsewhere” in phonology”. In: *A festschrift for Morris Halle*. Ed. by Paul Kiparsky and Stephen Anderson. New York: Holt, Rinehart and Winston, pp. 93–106.

- Klingvall, Eva (2018). “Agreement and reconstruction correlate in Swedish: Evidence from tough-constructions”. In: *Natural Language & Linguistic Theory* 36, pp. 1165–1205.
- Klumpp, James and Donald A. Burquest (1983). “Relative clauses in Piapoco”. In: *International Journal of American Linguistics* 49.4, pp. 388–399.
- Ko, Dongho and Gyudong Yurn (2011). *A description of Najkhin Nanai*. South Korea: Seoul National University Press.
- Koehn, Edward and Sally Koehn (1986). “Apalaɪ”. In: *Handbook of Amazonian languages*. Ed. by Desmond Derbyshire and Geoffrey Pullum. Vol. 1. Berlin: Mouton de Gruyter, pp. 33–127.
- Kouneli, Maria (2019). “Plural marking on mass nouns: Evidence from Greek”. In: *Gender and Noun Classification*. Ed. by Éric Mathieu, Myriam Dali, and Gita Zareikar. Oxford: Oxford University Press, pp. 234–248.
- Kouwenberg, Silvia (2007). “Bare nouns in Berbice Dutch Creole”. In: *Noun phrases in creole languages: A multi-faceted approach*. Ed. by Marlyse Baptista and Jacqueline Guéron. Amsterdam: John Benjamins.
- Kramer, Ruth (2009). “Definite markers, phi-features, and agreement: A morphosyntactic investigation of the Amharic DP”. PhD thesis. University of California, Santa Cruz.
- Kramer, Ruth (2016). “A split analysis of plurality: Number in Amharic”. In: *Linguistic Inquiry* 47.3, pp. 527–559.
- Kratochvíl, František (2014). “Number in Abui and Sawila”. In: *Number and quantity in East Nusantara*. Ed. by Frantisek Kratochvil Margaretha Anna Flora Klammer. Asia-Pacific Linguistics Canberra, pp. 123–151.
- Kratzer, Angelika (2009). “Making a pronoun: Fake indexicals as windows into the properties of pronouns”. In: *Linguistic Inquiry* 40.2, pp. 187–237.

- Kunduracı, Aysun (2013). “Pseudo 3rd person marker and possessive constructions in Turkish”. In: *Proceedings of the 2013 annual conference of the Canadian Linguistics Association*, pp. 1–14.
- Kurafuji, Takeo (2004). “Plural morphemes, definiteness, and the notion of semantic parameter”. In: *Language and Linguistics* 5.1, pp. 211–242.
- Landau, Idan (2020). “A Scope Argument against T-to-C Movement in Sluicing”. In: *Syntax* 23.4, pp. 375–393.
- LaPolla, Randy (2017). “Overview of Sino-Tibetan morphosyntax”. In: *The Sino-Tibetan languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 40–69.
- Lee, Sung-Woo (2011). “Eastern Tamang grammar sketch”. Unpublished MA thesis, Graduate Institute of Applied Linguistics, Dallas and SIL Nepal.
- Lewis, Beccy (Apr. 2021). “The affixal article in Mandinka”. Paper presented at ACAL 51–52. University of Florida.
- Lewis, Geoffrey (2000). *Turkish grammar*. Oxford University Press.
- Li, Yen-Hui Audrey (1999). “Plurality in a classifier language”. In: *Journal of East Asian Linguistics* 8, pp. 75–99.
- Lima Silva, Wilson de (2012). “A descriptive grammar of Desano”. PhD thesis. The University of Utah.
- Lyons, John (1968). *Introduction to Theoretical Linguistics*. Cambridge, UK: Cambridge University Press.
- MacWhinney, Brian (1976). “Hungarian research on the acquisition of morphology and syntax”. In: *Journal of child language* 3, pp. 397–410.
- Mainwaring, G. B. (1876). *Róng (Lepcha) Language*. Calcutta: Baptist Mission Press.

- Maldonado, Mora and Carmen Saldana Gascon (2022). “Interpretations of plurality in personal pronouns differ across person categories: An experimental study”. In: *Proceedings of the 23rd Amsterdam Colloquium*, pp. 388–398.
- Mallikadevi, Narayanan (2023). “An analysis of the production of plural nouns in Sinhala”. In: *Strad Resrarch* 10.8, pp. 544–551.
- Maltseva, Vera S. and Elvira V. Sultrekova (2023). “Khakas”. In: *Journal of Endangered Languages* 13, pp. 237–252.
- Marelj, Marijana (2011). “Bound-Variable Anaphora and Left Branch Condition”. In: *Syntax* 14, pp. 205–229.
- Marušič, Franc and Rok Žaucer (2006). “The ‘definite article’ TA in colloquial Slovenian”. In: *Formal Approaches to Slavic Linguistics*. Vol. 14, pp. 189–204.
- Masayoshi, Shibanti (1990). *The languages of Japan*. Cambridge: Cambridge University Press.
- Matasović, Ranko (2010). “A short grammar of East Circassian (Kabardian)”.
- Matisoff, James (1973). *A grammar of Lahu*. Berkeley: University of California Press.
- Matushansky, Ora (2005). “Call me Ishmael”. In: *Proceedings of Sinn und Bedeutung*. Vol. 9, pp. 226–240.
- Mauri, Caterina and Andrea Sansó (2017). “Names & co: A diachronic typology of associative and similitive plurals”. Paper presented at the 50th annual meeting of the Societas Linguistica Europaea, University of Zurich, September 10-13.
- Mauri, Caterina and Andrea Sansó (2019). “Nouns & co. Converging evidence in the analysis of associative plurals”. In: *STUF - Language Typology and Universals* 72.4, pp. 603–626. DOI: [doi : 10 . 1515 / stuf - 2019 - 0023](https://doi.org/10.1515/stuf-2019-0023). URL: [https : // doi . org / 10 . 1515 / stuf - 2019 - 0023](https://doi.org/10.1515/stuf-2019-0023).

- Mauri, Caterina and Andrea Sansó (2021). “Heterogeneous sets: a diachronic typology of associative and similitive plurals”. In: *Linguistic Typology*, pp. 1–40.
- McCracken, Chelsea L. (2012). “A grammar of Belep”. PhD thesis. Rice University.
- McGregor, William (1990). *A functional grammar of Gooniyandi*. Amsterdam: John Benjamins.
- McPherson, Laura (2013). *A grammar of Tommo So*. Berlin: Mouton de Gruyter.
- Merlan, Francesca C. (1994). *A grammar of Wardaman*. New York: de Gruyter.
- Michailovsky, Boyd (2017a). “Hayu”. In: *The Sino-Tibetan languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 680–695.
- Michailovsky, Boyd (2017b). “Kiranti Languages”. In: *The Sino-Tibetan Languages*. Ed. by Graham Thurgood and Randy LaPolla. Routledge, pp. 646–679.
- Miyaoka, Osahito (2012). *A Grammar of Central Alaskan Yupik (CAY)*. De Gruyter Mouton.
- Mojapelo, Mampaka Lydia (2007). “Definiteness in Northern Sotho”. PhD thesis. University of Stellenbosch.
- Moravcsik, Edith (2003). “A semantic analysis of associative plurals”. In: *Studies in Language* 27, pp. 469–503.
- Moravcsik, Edith (2020). “The place of ad hoc categories within the typology of plural expressions”. In: *Language Sciences* 81. Ad hoc categorization and language: the construction of categories in discourse, p. 101298. ISSN: 0388-0001.
- Moskal, Beata (2018). “Excluding exclusively the exclusive: Suppletion patterns in clusivity”. In: *Glossa* 3.1.
- Mühlhäusler, Peter (2020). *Pitkern-Norf’k: the language of Pitcairn Island and Norfolk Island*. Bloomington, IN: Research center for language and semiotic studies.

- Mülhäuser, Peter (1981). “The development of the category of number in Tok Pisin”. In: *Generative studies on Creole languages*. Ed. by Peter Muysken. Dordrecht: Foris, pp. 35–84.
- Nagayama, Yukari (2016). “Nominalization in Alutor”. In: *Studia Orientalia Electronica* 117, pp. 119–136.
- Nakagawa, Natsuko (2022). “Nambu (Aomori, Eastern Japanese)”. In: *An introduction to Japonic languages*. Ed. by Michinori Shimoji. Boston: Brill, pp. 188–226.
- Nakanishi, Kimiko and Elizabeth Ritter (Feb. 2008). “Plurality in languages without a count-mass distinction”. Paper presented at the Mass/Count Workshop. University of Toronto.
- Nakanishi, Kimiko and Satoshi Tomioka (2004). “Japanese plurals are exceptional”. In: *Journal of East Asian Linguistics* 13, pp. 113–140.
- Nedjalkov, Igor (1997). *Evenki*. New York: Routledge.
- Nercesian, Verónica (2011). “Gramática del wichí, una lengua chaqueña: Interacción fonología-morfología-sintaxis en el léxico”. PhD thesis. Universidad de Buenos Aires.
- Nevins, Andrew (2011). “Marked Targets versus Marked Triggers and Impoverishment of the Dual”. In: *Linguistic Inquiry* 42.3, pp. 413–444.
- New, Keely (2021). “A post-suppositional account of associative plurals in Burmese”. In: *Semantics and Linguistic Theory*. Vol. 31, pp. 166–183.
- Nichols, Johanna (2011). *Ingush Grammar*. Berkeley: University of California Press.
- Nikoleva, Irina and Maria Tolskaya (2001). *A grammar of Udihe*. Berlin: Mouton de Gruyter.
- Noonan, Michael and Kristine Hildebrandt (2017). “Chantyal”. In: *The Sino-Tibetan languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 494–508.

- Noyer, Robert (1992). “Features, Positions and Affixes in autonomous morphological structure”. PhD thesis. MIT.
- Oda, Hiromune (2022). “A more fine-grained distinction of NP/DP languages and Parameters in Minimalism”. PhD thesis. University of Connecticut.
- Olawsky, Knut J. (2003). “What is a word in Dagbani?” In: *Word: a crosslinguistics typology*. Ed. by R. M. W. Dixon and Alexandra Y. Aikhenvald. Cambridge: Cambridge University Press, pp. 205–226.
- Pancheva, Roumyana and Barbara Tomaszewicz (2012). “Cross-linguistic differences in superlative movement out of nominal phrases”. In: *Proceedings of the 30th West Coast Conference on Formal Linguistics (WCCFL 30)*, pp. 292–302.
- Parker, Clinton (2023). “A grammar of the Shughni language”. PhD thesis. McGill University.
- Partee, Barbara H. (1987). “Noun phrase interpretation and type-shifting principles”. In: *Studies in discourse representation theory and the theory of generalized quantifiers*. Ed. by Jeroen Groenendijk, Dick de Jongh, and Martin Stokhof. Dordrecht: Foris, pp. 115–141.
- Pasch, Helma (1996). “Sango”. In: *Contact Languages: A wider perspective*. Ed. by Sarah G. Thomason. Amsterdam: John Benjamins, pp. 209–270.
- Patrick, Peter (2013). “Jamaican Creole”. In: *World Atlas of varieties of English*. Ed. by Bernd Kortmann and Kerstin Lunkenheimer. Mouton de Gruyter.
- Paudyal, Netra (2013). “Aspects of Chintang Syntax”. PhD thesis. University of Zurich.
- Paul, Ileana (2009). “On the presence versus absence of determiners in Malagasy”. In: *Determiners*. Ed. by Jila Ghomeshi, Ileana Paul, and Martina Wiltschko. Amsterdam: John Benjamins Publishing, pp. 215–241.

- Paul, Ileana (2015). “When bare nouns scope wide. The case of Malagasy”. In: *Natural Language & Linguistic Theory* 34, pp. 271–305.
- Paul, Ileana (2018). “The features of proper determiners in four varieties of Malagasy”. In: *McGill Working Papers in Linguistics* 25, pp. 324–331.
- Paul, Ileana, Key Coretes, and Lareina Milambiling (2015). “Definiteness without D: The case of *ang* and *ng* in Tagalog”. In: *Canadian Journal of Linguistics* 60.3, pp. 361–390.
- Payne, Doris (2007). “Source of the Yagua nominal classification system”. In: *International Journal of American Linguistics* 73.4, pp. 447–474.
- Payne, Thomas E (1997). *Describing morphosyntax: A guide for field linguists*. Cambridge University Press.
- Pennington, Ryan (2016). “A grammar of Ma Manda: A Papuan language of Morobe Province, Papua New Guinea”. PhD thesis. James Cook University.
- Perekhvalskaya, Elena and Juha Janhunen (2024). “Udihe”. In: *The Tungusic Languages*. Ed. by José Andrés Alonso de la Fuente Alexandra Vovin and Juha Janhunen. New York: Routledge, pp. 326–363.
- Peterson, John (2011). *A grammar of Kharia: A South Munda language*. Boston: Brill.
- Peterson, John (2014). “Number in Kharia”. In: *Number—constructions and semantics: case studies from Africa, Amazonia, India and Oceania*. Ed. by Anne Storch. John Benjamins Publishing, pp. 77–110.
- Plaisier, Heleen (2006). “A Grammar of Lepcha”. PhD thesis. University of Leiden.
- Pleshak, Polina (2021). “The exponence of caseless NPs in Moksha”. In: *Proceedings of the Linguistic Society of America* 6.1, pp. 628–640.
- Plungian, Vladimir (1995). *Dogon*. München: LINCOM EUROPA.
- Polina, Nasledskova (2019). *The structure of Kina Rutul noun phrase*. Higher School of Economics Research Paper No. WP BRP 82/LNG/2019.

- Poppe, Nicholas (1954). *A grammar of written Mongolian*. Wiesbaden: Harrassowitz Verlag.
- Poppe, Nicholas (1960). *Buriat Grammar*. Bloomington, ID: Indiana University Publications.
- Potsdam, Eric (2013). “CP-negation and the domain of quantifier raising”. In: *Linguistic inquiry* 44.4, pp. 674–684.
- Poudel, Tikaram (2017). “The semantics of the ergative in Nepali”. In: *Gipan*. Ed. by Balaram Prasain, Karnakhar Khatiwada, and Tara Mani Rai. Vol. 3. Kathmandu: Central Department of Linguistics, Tribhuvan University, pp. 61–86.
- Preminger, Omer (2011). “Agreement as a Fallible Operation”. PhD thesis. MIT.
- Rákosi, Gyogy (2017). “The definite article and anaphoric possessors in Hungarian”. In: *Linguistica Brunensia* 65, pp. 21–33.
- Rákosi, György (2020). “With or without the definite article: On the syntax of anaphoric possessor strategies in Hungarian”. In: *Approaches to Hungarian 16*. Ed. by Vera Hegedűs and Irene Vogel. Amsterdam: John Benjamins, pp. 315–403.
- Reed, Irene (1977). *Yupik Eskimo Grammar*. Alaska University, Fairbanks: Alaska Native Language Center.
- Reh, Mechthild (1983). “Krongo: a VSO language with postpositions”. In: *Journal of African Languages and Linguistics* 5, pp. 45–55.
- Reuland, Eric (2011). *Anaphora and language design*. Cambridge, Mass: MIT Press.
- Riccardi, Theodore (2003). “Nepali”. In: *The Indo-Aryan Languages*. Ed. by George Cardona and Dhanesh Jain. Routledge, pp. 590–636.
- Ritter, Elizabeth (1992). “Cross-linguistic evidence for Number Phrase”. In: *Canadian Journal of Linguistics* 37, pp. 197–218.

- Ritter, Elizabeth and Martina Wiltschko (2018). “Distinguishing speech act roles from grammatical person features”. In: *Proceedings of the 2018 annual conference of the Canadian Linguistic Association*, pp. 1–15.
- Ritter, Elizabeth and Martina Wiltschko (2019). “Nominal speech act structure: Evidence from the structural deficiency of impersonal pronouns”. In: *Canadian Journal of Linguistics/Revue canadienne de linguistique* 64.4, pp. 709–729.
- Roberts, Ian (2010). *Agreement and head movement: Clitics, incorporation and defective goals*. MIT Press.
- Roberts, James (1994). “Nontonal floating features as grammatical morphemes”. In: *Work Papers of the Summer Institute of Linguistics* 38, pp. 87–99.
- Rounds, Carol (2009). *Hungarian : an Essential Grammar*. Routledge.
- Rullman, Hotze (2004). “First and Second Person Pronouns as Bound Variables”. In: *Linguistic Inquiry* 35.1, pp. 159–168.
- Runić, Jelena (2014). “A new look at clitics, clitic doubling, and argument ellipsis: Evidence from Slavic”. PhD thesis. University of Connecticut.
- Sadowska, Iwona (2012). *Polish: A Comprehensive Grammar*. New York: Routledge.
- Safford, William Edwin (1903). “The Chamorro language of Guam”. In: *American Anthropologist* 5.2, pp. 289–311.
- Samarin, William J. (1963). *A grammar of Sango*. Connecticut: Hartford Seminary Foundation.
- Sandman, Erika (2012). “Bonan grammatical features in Wutun Mandarin”. In: *Per Urales ad Orientem: Iter Polyphonicum Multilingue*. Ed. by Tiina Hyytiäinen Juha Janhunen. Helsinki: Suomalais-ugrilainen Seura, pp. 375–387.
- Sandman, Erika (2016). “A grammar of Wutun”. PhD thesis. University of Helsinki.

- Sapir, Edward (1922). “The Takelma language of Southwestern Oregon”. In: *Handbook of American Indian languages*. Ed. by Franz Boas. Vol. 2. Washington: Government Printing Office, pp. 1–296.
- Sauerland, Uli, Jan Anderssen, and Kazuko Yatsushiro (2005). “The plural is semantically unmarked”. In: *Linguistic Evidence* 4, pp. 409–430.
- Schachter, Paul and Fe Otanes (1972). *Tagalog reference grammar*. Berkley: University of California Press.
- Schwarz, Florian (2014). “How weak and how definite are Weak Definites?” In: *Weak Referentiality*. Ed. by Ana Aguilar-Guevara, Bert Le Bruyn, and Joost Zwarts. Amsterdam: John Benjamins Publishing, pp. 213–235.
- Shen, Zheng (2014). “On the relative readings with NP internal focus of superlatives”. In: *Proceedings of the 31st West Coast Conference on Formal Linguistics*, pp. 409–418.
- Shimoji, Michinori and Naoyuki Hirosawa (2022). “Shiiba (Miyazaki, Kyūshū Japanese)”. In: *An Introduction to the Japonic Languages*. Brill, pp. 293–329.
- Shiohara, Asako and Anthony Jukes (2018). “Two definite markers in Manado Malay”. In: *Perspectives on information structure in Austronesian languages*. Ed. by Sonja Riesberg, Asako Shiohara, and Atsuko Utsumi. Berlin: Freie Universitat, pp. 115–135.
- Simon, Camille (2021). “The grammaticalization of plurality in the languages of Amdo”. In: *Himalayan Linguistics* 20.3.
- Smith, Peter et al. (2019). “Case and number suppletion in pronouns”. In: *Natural Language & Linguistic Theory* 37, pp. 1029–1101.
- Smith, Ryan Walter (2020). “Similative plurals and the nature of alternatives”. PhD thesis. The University of Arizona.
- Soe, Myint (1999). “A grammar of Burmese”. PhD thesis. University of Oregon.
- Solnit, David B. (1997). *Eastern Kayah Li*. Hawai’i: University of Hawai’i Press.

- Steeman, Sander (2011). “A grammar of Sandawe: A khoisan language of Tanzania”. PhD thesis. University of Leiden.
- Steeman, Sander (2014). “A grammar of Nungon: a Papuan language of the Morobe Province, Papua New Guinea”. PhD thesis. James Cook University.
- Stegovec, Adrian (2022). “Number in Slovenian”. In: *Number in the World’s Languages: A Comparative Handbook*. Ed. by Paolo Acquaviva and Michael Daniel. Berlin: De Gruyter Mouton, pp. 187–260.
- Stjepanović, Sandra (1998). “Extraction of adjuncts out of NPs”. Paper presented at the Comparative Slavic Morphosyntax Workshop.
- Sultan, Valerie (2006). “Information Packaging in Sinhala: A preliminary study of adverbial clauses in focus constructions”. In: *Santa Barbara papers in Linguistics*. Ed. by Robert Englebretson and Carol Genetti. Vol. 17. California: UCSB Department of Linguistics, pp. 128–149.
- Syea, Anand (2007). “The development of the noun phrase in Mauritian Creole and the mechanisms of language development”. In: *The making of Mauritian Creole*. Ed. by Guillaume Fon Sing and Philip Baker. London: Battlebridge Publications, pp. 93–112.
- Szabolcsi, Anna (1994). “The Noun Phrase”. In: *Syntax and Semantics* 27, pp. 179–274.
- Talić, Aida (2017). “From A to N and Back: Functional and Bare Projections in the Domain of N and A”. PhD thesis. University of Connecticut.
- Tamamoto, Koji (2022). “Kin (Okinawa, Northern Ryukyuan)”. In: *An Introduction to the Japonic Languages*. Brill, pp. 91–123.
- Tanaka, Hideki (1996). “On the Palatalization of [dz] and [s] in the Nambu Dialect”. In: *Tsukuba English Studies* 15, pp. 159–186.
- Taylor, Ann (1990). “Clitics and configurationality in Ancient Greek”. PhD thesis. University of Pennsylvania.

- Taylor, Frank (1921). *A first grammar of the Adamawa dialect of the Fulani language (Fulfulde)*. Oxford: Clarendon Press.
- Terraza, Jimena and Lorena Cayré Baito (2014). “Phonological, grammatical, and written words in Wichí”. In: *Morphology* 24, pp. 199–221.
- Thach, Sharon and David Dwyer (1981). *Kpelle: A reference handbook of phonetics, grammar, lexicon and learning procedures*. Michigan State University African Studies Center.
- Thompson, Chad Lawrence (1989). “Voice and obviation in Athabaskan and other languages”. PhD thesis. University of Oregon.
- Thompson, Hanne-Ruth (2012). *Bengali*. Amsterdam: John Benjamins Publishing.
- Tomioka, Satoshi (2021). “Japanese -tati and Generalized Associative Plurals”. In: *The Oxford Handbook of Grammatical Number*. Ed. by Patricia Cabredo Hofherr and Jenny Doetjes. Oxford: Oxford University Press, pp. 1–21.
- Tranel, Bernard (1993). “Tone sandhi and vowel deletion in Margi”. In: *Studies in African Linguistics* 23 (2), pp. 111–183.
- Treis, Yvonne (2014). “Number in Kambaata”. In: *Number-Constructions and Semantics: Case studies from Africa, Amazonia, India and Oceania*. Ed. by Anne Storch and Gerrit J. Dimmendaal. Amsterdam: John Benjamins, pp. 111–133.
- Ueda, Yasuki and Tomoko Haraguchi (2008). “Plurality in Japanese and Chinese”. In: *Nanzan Linguistics: special issue* 3, pp. 229–242.
- Uriagereka, Juan (1988). “On government”. PhD thesis. University of Connecticut.
- Vassilieva, Maria (2005). “Associative and pronominal plurality”. PhD thesis. Stony Brook University.
- Verhaar, John (1995). *Toward a reference grammar of Tok Pisin: An experiment in corpus linguistics*. Honolulu: University of Hawai’i Press.

- Viljoen, Melanie Helen (2013). “A grammatical description of the Buwal language”. PhD thesis. La Trobe University.
- Vogel, Irene (1989). “Prosodic constituents in Hungarian”. In: *Acta Linguistica Hungarica* 39 (1/4), pp. 333–351.
- Vuillermet, Marine (2012). “A grammar of Ese Ejja, a Takana language of the Bolivian Amazon”. PhD thesis. Lumière University Lyon 2.
- Vydrin, Valentin (2020). “Dan”. In: *The Oxford handbook of African languages*. Ed. by Rainer Vossen and Gerrit J. Dimmendaal. Oxford: Oxford University Press, pp. 451–462.
- Vydrin, Valentin (2022). “Number in Eastern Dan”. In: *Number in the world’s languages: a comparative handbook*. Ed. by Paolo Acquaviva and Michael Daniel. Boston: Walter de Gruyter, pp. 27–62.
- Wang, Ruonan (2023). “Honorifics without [HON]”. In: *Natural Language & Linguistic Theory*, pp. 1–61.
- Watanabe, Akira (2004). “Indeterminates and determiners”. In: *MIT Working Papers in Linguistics* 46, pp. 390–405.
- Wechsler, Stephen (2010). “What ‘you’ and ‘I’ mean to each other: Person indexicals, self-ascription, and theory of mind”. In: *Language* 86.2, pp. 332–365.
- Welmers, William Everett (1946). “A descriptive grammar of Fanti”. In: *Language* 22.3, pp. 3–78.
- Wheatley, Julian K. (2017). “Burmese”. In: *The Sino-Tibetan languages*. Ed. by Graham Thurgood and Randy LaPolla. 2nd ed. New York: Routledge, pp. 856–876.
- Wiedenhof, Jeroen (2015). *A grammar of Mandarin*. Philadelphia: John Benjamins Publishing.

- Wilhoit, Laura (1999). “A principles and parameters approach to Loma grammar”. PhD thesis. The University of Texas at Arlington.
- Wilson, Janet (2002). “From preposition to plural marker and more: Grammaticalization of Kuche bānà”. In: *Studies in African Linguistics* 31.1/2, pp. 155–175.
- Wiltschko, Martina (2008). “The syntax of non-inflectional plural marking”. In: *Natural Language and Linguistic Theory* 26, pp. 639–694.
- Winkler, Elizabeth (1997). *Kpelle-English dictionary, with accompanying English-Kpelle glossary*. Kentucky: English Faculty Book Gallery.
- Wolf, Matthew (2006). “For an autosegmental theory of mutation”. In: *University of Massachusetts, Occasional Papers in Linguistics 32: Papers in Optimality Theory III*. Ed. by Leah Bateman et al. Amherst: GSLA, pp. 315–403.
- Yadroff, Michael and Loren Billings (1998). “The syntax of approximative inversion in Russian (and the general architecture of nominal expressions)”. In: *Formal Approaches to Slavic Linguistics: The Connecticut Meeting*. Ed. by Zeljko Bošković, Steven Franks, and Willian Snyder. Ann Arbor, Michigan: Michigan Slavic Publications, pp. 319–338.
- Yousef, Saeed and Hayedeh Torabi (2018). *Persian: A comprehensive grammar*. New York: Routledge.
- Zanon, Ksenia (2015). “On hybrid coordination and quantifier raising in Russian”. PhD thesis. University of Indiana, Bloomington.
- Zikmundová, Veronika and Gao Wa (2024). “Spoken Manchu”. In: *The Tungusic Languages*. Ed. by José Andrés Alonso de la Fuente Alexandra Vovin and Juha Janhunen. New York: Routledge, pp. 464–482.
- Zlatić, Larisa (1997). “The structure of the Serbian noun phrase”. PhD thesis. The University of Texas at Austin.